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# MATHEMATICAL miniatures



Svetoslav Savchev  
Titu Andreescu

# **Mathematical Miniatures**

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# Mathematical Miniatures

**Svetoslav Savchev**

**Titu Andreescu**



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# Preface

It was in the summer of 1993 in Turkey, at the International Mathematical Olympiad, when a Bulgarian and a Romanian-American decided to write a book on problem-solving. This decision brought together two people separated by half of the world, not speaking each other's language and not knowing each other very well. Six months later they started working together in America.

We are Svetoslav Savchev, editor of the Bulgarian journal *Matematika*, and Titu Andreescu, former editor-in-chief of the journal *Revista Matematică Timișoara* and coach of the Romanian Mathematical Olympiad Team, now Director of the American Mathematics Competitions and head coach of the US Mathematical Olympiad Team.

The material we started with was vast, coming from all kinds of mathematical competitions, books, research papers, personal discussions and communications, and our own work. Such mathematical substance was sure to go beyond the pragmatic purposes of a traditional problem book. The most attractive pieces of mathematics refused to fit into the rigid schemes of an instruction manual, to merely exemplify typical problem-solving techniques. Their depth and appeal endow such statements with individuality demanding a separate treatment. And our original program, in which they were destined to play an auxiliary role, impressed us increasingly as being too narrow.

Little by little, the idea came into being to isolate certain statements or groups of related statements into independent sections. By doing so we also hoped to emphasize the true source of their natural charm—the connection with authentic mathematical experience. And indeed, is it “just another problem” that was posed for a certain contest but originated from the ingenious work of professionals, some of them rather prominent? Or, the

other way around, is it merely an olympiad question that readily develops into a well-shaped result of broader interest?

In this way yet another part of the book was born, essays on topics not only useful but also beautiful, with a lot of esthetic appeal. These essays are very diverse, enlivened by fresh and nonstandard ideas. We believe they are miniatures of genuine mathematical work in the proper sense.

So, part of the book can be regarded as a tool chest of problem-solving techniques, and another part is something of a modest anthology of mathematical verse at a certain level.

Proper referencing was not an easy matter. A number of statements and proofs in the book were invented by us but no reference to ourselves is made throughout the text. Elsewhere we mentioned the source or the author to the best of our knowledge, and we would appreciate receiving exact references for ones not mentioned. Sections 17, 20–49 were written by Svetoslav Savchev, sections 1–16, 18–19, 50 and “Instead of an Afterword” by Titu Andreescu. The Bulgarian author suggested the idea of diversifying the text by adding a warm-up problem session and coffee breaks after each five sections. He also contributed the majority of the coffee break problems. Some of these questions are quite challenging.

When submitting the manuscript, we did not imagine the amount of work still to come. Members of the editorial board of the Anneli Lax New Mathematical Library carefully acquainted themselves with the text and suggested numerous changes. Their task was a very difficult one, and they carried it out with meticulous care. Various improvements are due to friendly criticism by many people. We wish to extend our gratitude to everyone whose continued cooperation over the past nine years influenced, in one way or another, the final version of the book.

The times of our joint work were far and away not the easiest ones for our families. Yet our mothers, wives, and children were remarkably patient, understanding, and cheerful. One last word of thanks goes to them all. Emma Andreescu and Christo Savchev passed away when the manuscript was almost ready. May this book be a tribute to their bright memory.

Svetoslav Savchev

Titu Andreescu

# Contents

|                                          |     |
|------------------------------------------|-----|
| Preface .....                            | vii |
| Warm-up Problem Set .....                | 1   |
| 1. A Telescoping Sum .....               | 4   |
| 2. Lagrange's Identity .....             | 6   |
| 3. Perfect Squares .....                 | 8   |
| 4. Least Common Multiples .....          | 11  |
| 5. Trig Substitutions .....              | 13  |
| Coffee Break 1 .....                     | 17  |
| 6. Popoviciu's Theorem .....             | 19  |
| 7. Catalan's Identity .....              | 21  |
| 8. Several Inequalities .....            | 24  |
| 9. Vectors .....                         | 29  |
| 10. Mathematical Induction at Work ..... | 32  |
| Coffee Break 2 .....                     | 36  |
| 11. A Highly Divisible Determinant ..... | 39  |
| 12. Hermite's Identity .....             | 41  |
| 13. Complete Sequences .....             | 45  |
| 14. Three Polynomials .....              | 48  |
| 15. More about Induction .....           | 51  |

|                                             |     |
|---------------------------------------------|-----|
| Coffee Break 3 .....                        | 56  |
| 16. A Classical Identity .....              | 59  |
| 17. Multiplicative Functions .....          | 62  |
| 18. The “Arbitrary” Proizvolov .....        | 66  |
| 19. Hölder’s Inequality .....               | 70  |
| 20. Symmetry .....                          | 73  |
| Coffee Break 4 .....                        | 77  |
| 21. He Knows I Know He Knows .....          | 80  |
| 22. A Special Inequality .....              | 83  |
| 23. Two Inductive Constructions .....       | 85  |
| 24. Some Old-Fashioned Geometry .....       | 88  |
| 25. Extremal Arguments .....                | 93  |
| Coffee Break 5 .....                        | 98  |
| 26. The AMS Inequality .....                | 101 |
| 27. Helly’s Theorem for One Dimension ..... | 104 |
| 28. Two Approaches .....                    | 107 |
| 29. Radical Axis .....                      | 112 |
| 30. The Pigeonhole Principle .....          | 116 |
| Coffee Break 6 .....                        | 121 |
| 31. The Three Jug Problem .....             | 124 |
| 32. Rectifying Trajectories .....           | 128 |
| 33. Numerical Systems .....                 | 131 |
| 34. More on Polynomials .....               | 136 |
| 35. Geometric Transformations .....         | 141 |
| Coffee Break 7 .....                        | 146 |
| 36. The <i>Game of Life</i> Problem .....   | 149 |
| 37. Tetrahedra with a Point in Common ..... | 152 |
| 38. Should We Count? .....                  | 155 |
| 39. Let’s Count Now! .....                  | 158 |
| 40. Some Elementary Number Theory .....     | 163 |

|                                |     |
|--------------------------------|-----|
| Coffee Break 8.....            | 168 |
| 41. Euclid's Game.....         | 171 |
| 42. Perfect Powers .....       | 175 |
| 43. The $2n - 1$ Problem ..... | 179 |
| 44. The $2n + 1$ Problem ..... | 184 |
| 45. The $3n$ Problem .....     | 188 |
| Coffee Break 9.....            | 190 |
| 46. Pairwise Sums.....         | 193 |
| 47. Integer Progressions.....  | 197 |
| 48. Incomparable Sets .....    | 200 |
| 49. Morse's Sequence .....     | 203 |
| 50. A Favorite of Erdős.....   | 209 |
| Instead of an Afterword.....   | 213 |
| Glossary.....                  | 215 |
| About the Authors.....         | 223 |



# Warm-up Problem Set

1. An artist paints two congruent dragons on two congruent circular paper discs. The center of the first disc coincides with one of the dragon's eyes, which is not the case with the second disc. Prove that the second disc can be cut into two pieces from which a disc of the same radius can be assembled, containing the same dragon, but so that his eye coincides with the center of the new disc.
2. A row of minuses is written on a blackboard. Two players take turns in replacing either a single minus by a plus or two adjacent minuses by pluses. The one who cannot make a move loses. Can the player who starts force a win?
3. Several weights are given, each of which is not heavier than 1 lb. It is known that they cannot be divided into two groups such that the weight of each group is greater than 1 lb. Find the maximum possible total weight of these weights.
4. In a parliament, each parliamentarian has at most three enemies. Prove that the parliament can be divided into two chambers in such a way that no parliamentarian has more than one enemy in his or her chamber.
5. The *two-move chess game* has the same rules as the regular one, with only one exception: each player has to make *two* consecutive moves at a time. Prove that White (who goes first) has a nonlosing strategy.

## Solutions

1. Put the second disc over the first so that both dragons coincide. They are contained in the common part of the discs, and the second dragon's eye is at the center of the first disc. In the picture, a seahorse is shown rather than a dragon but this is hardly a drawback: the argument works for any two congruent figures.



The portions of the two discs that do not overlap are congruent. Cut off the nonoverlapping portion of the second disc and place it on the corresponding one of the first disc. You get a new disc of the same radius, with the second dragon's eye at its center.

2. The player who goes first can always win. If  $n$  is even, he might, for instance, change the two minuses in the middle on his first move. Then, after every change made by his opponent, he can change the plus sign(s) symmetric to it in the center.

If  $n$  is odd, it suffices to change the minus in the middle and then act as in the first case.

3. Imagine you have a balance. Start putting the weights into the pan, one at a time, and stop at the first moment when the total weight becomes greater than 1 lb. The last weight that was put into the pan is not heavier than 1 lb, so there is no more than 2 lb on the balance. Besides, the remaining weights are not heavier than 1 lb by hypothesis, so the total weight is not greater than 3 lb. This is indeed the maximum possible total weight, as a set of three 1-lb weights shows.

4. Form two chambers in an arbitrary way and denote by  $e$  the total number of pairs of enemies who are members of the same chamber. If there is a parliamentarian  $A$  who has at least two enemies in his chamber, send him to the other one, where he may have no more than one enemy. So at least two pairs of enemies disappear, while no more than one new pair can be formed. It follows that  $e$  decreases at least by 1 after this operation. Repeating the same several times, we will arrive at a situation when  $e$  cannot be decreased any more (because it is a nonnegative integer). This means that each parliamentarian will have at most one enemy in his or her chamber.

5. Assume on the contrary that this is not true. Then, *no matter how White plays, Black (White's opponent) has an opposing winning strategy*. In particular, this applies to the game that starts as follows: White moves the knight from **b1** to **a3** on his first move and then from **a3** to **b1** on his second move. By hypothesis, Black has a way to force a win. But, on the other hand, he is now in the initial position of White, and the same hypothesis shows that he is condemned to lose. A contradiction!

We are not the first (nor the last) to remember the words of G. H. Hardy, one of the greatest English mathematicians:

*“Reductio ad absurdum, which Euclid loved so much, is one of a mathematician’s finest weapons. It is a far finer gambit than any chess gambit: a chess player may offer the sacrifice of a pawn or even a piece, but a mathematician offers the game.”*

# 1

## A Telescoping Sum

Here we consider a sum of the form

$$\sum_{k=1}^n [f(k) - f(k+1)], \quad (1)$$

where  $f$  is a numerical function. Such sums are called *telescoping*, because all of their terms, except for the extremes, cancel out. By choosing  $f$  appropriately, one can often write a sum  $\sum_{k=1}^n a_k$  in the form (1), and hence “telescope” it to

$$[f(1) - f(2)] + [f(2) - f(3)] + \cdots + [f(n) - f(n+1)] = f(1) - f(n+1).$$

*Evaluate in closed form*

$$\sum_{k=1}^n \frac{k}{(k+1)!}.$$

In this example,

$$\frac{k}{(k+1)!} = \frac{(k+1) - 1}{(k+1)!} = \frac{k+1}{(k+1)!} - \frac{1}{(k+1)!} = \frac{1}{k!} - \frac{1}{(k+1)!},$$

and our  $f(k)$  equals  $1/k!$ . There is hardly anything more to do. The sum we are evaluating becomes

$$\left(\frac{1}{1!} - \frac{1}{2!}\right) + \left(\frac{1}{2!} - \frac{1}{3!}\right) + \cdots + \left(\frac{1}{n!} - \frac{1}{(n+1)!}\right) = 1 - \frac{1}{(n+1)!}. \quad \blacksquare$$

It is surprising to find out how often we can proceed in virtually the same way.

Compute the sum

$$\sum_{k=1}^n \frac{k+1}{(k-1)! + k! + (k+1)!}.$$

A little algebra work on the expression and it becomes something quite familiar:

$$\begin{aligned} \frac{k+1}{(k-1)! + k! + (k+1)!} &= \frac{k+1}{(k-1)![1 + k + k(k+1)]} \\ &= \frac{k+1}{(k-1)!(k+1)^2} \\ &= \frac{1}{(k-1)!(k+1)} = \frac{k}{(k+1)!}. \end{aligned}$$

By the previous problem, the sum is equal to  $1 - [1/(n+1)!]$ . ■

Incidentally, the very same summation yields an unexpected solution to the following 1986 Polish Olympiad problem.

*Prove that, for each  $n \geq 3$ , the number  $n!$  can be represented as the sum of  $n$  distinct divisors of itself.*

Assume that 1 can be represented in the form

$$1 = \frac{1}{m_1} + \frac{1}{m_2} + \cdots + \frac{1}{m_n}, \quad (2)$$

where  $m_1, m_2, \dots, m_n$  are distinct divisors of  $n!$ . Then multiplying by  $n!$  yields

$$n! = \frac{n!}{m_1} + \frac{n!}{m_2} + \cdots + \frac{n!}{m_n},$$

and  $n!/m_1, n!/m_2, \dots, n!/m_n$  are also distinct divisors of  $n!$ . So we just have to find a representation of the form (2).

Since we already know that

$$\sum_{k=1}^{n-1} \frac{k}{(k+1)!} = 1 - \frac{1}{n!},$$

the relation (2) is satisfied by the numbers

$$m_1 = 2!, \quad m_2 = \frac{3!}{2}, \quad \dots, \quad m_{n-1} = \frac{n!}{n-1}, \quad m_n = n!,$$

which are distinct divisors of  $n!$  for  $n \geq 3$ . ■

# 2

## Lagrange's Identity

The simplest form of Lagrange's identity is

$$(a^2 + b^2)(c^2 + d^2) = (ac + bd)^2 + (ad - bc)^2. \quad (1)$$

Although (1) is so elementary and unsophisticated, it has interesting applications. To illustrate this, we will consider a few examples.

*Let  $m$  and  $n$  be distinct positive integers. Represent  $m^6 + n^6$  as the sum of two perfect squares different from  $m^6$  and  $n^6$ .*

Many of our friends struggled with this problem. The first thing that comes to mind is to write  $m^6 + n^6$  as  $(m^2)^3 + (n^2)^3$  and then factor it into  $(m^2 + n^2)(m^4 - m^2n^2 + n^4)$ . Now a nontrivial step follows: let us write the second factor as  $(m^2 - n^2)^2 + (mn)^2$  and employ (1).

The choice  $a = m$ ,  $b = n$ ,  $c = m^2 - n^2$ ,  $d = mn$  yields the trivial  $(m^3)^2 + (n^3)^2$ , whereas the one in which the sign of  $d$  is changed gives the desired representation:  $(m^3 - 2mn^2)^2 + (n^3 - 2m^2n)^2$ . ■

The second example is a classic.

*Let  $P(x)$  be a polynomial with real coefficients so that  $P(x) \geq 0$  for all real  $x$ . Prove that there exist polynomials with real coefficients,  $Q_1(x)$  and  $Q_2(x)$ , such that*

$$P(x) = Q_1^2(x) + Q_2^2(x) \quad \text{for all } x.$$

From the hypothesis,  $P(x)$  is of the form  $c \prod_{k=1}^n (x^2 + p_k x + q_k)$ , where  $c, p_k, q_k$  are real numbers such that  $c \geq 0$  and  $p_k^2 - 4q_k \leq 0$  for all  $k = 1, 2, \dots, n$ .

Writing each  $x^2 + p_k x + q_k$  in the form

$$\left(x + \frac{p_k}{2}\right)^2 + \left(\frac{\sqrt{4q_k - p_k^2}}{2}\right)^2 = U_k^2(x) + V_k^2(x),$$

we have to prove that

$$[U_1^2(x) + V_1^2(x)] [U_2^2(x) + V_2^2(x)] \cdots [U_n^2(x) + V_n^2(x)]$$

is representable as a sum of the squares of two polynomials with real coefficients. This follows immediately from (1) by induction. ■

We close with a problem from the 1985 British Olympiad:

*Show that the equation  $x^2 + y^2 = z^5 + z$  has infinitely many relatively prime integer solutions.*

A short proof can be obtained by using Lagrange's identity and the following well-known facts:

- 1° There are infinitely many primes of the form  $4k + 1$ .
- 2° Each prime of the form  $4k + 1$  is representable as the sum of two squares (in a unique way).

Take any prime  $p$  of the form  $4k + 1$ . By 2°, it can be represented as the sum of two squares. The same holds for  $p^4 + 1$ , and (1) shows that  $p^5 + p = p(p^4 + 1)$  is also representable as a sum of two squares. Let  $p^5 + p = u^2 + v^2$ ; then  $x = u$ ,  $y = v$ ,  $z = p$  is a solution of the given equation. Because  $p$  is a prime number,  $x$ ,  $y$  and  $z$  are relatively prime. Now it suffices to note that (see 1°) there are infinitely many primes of the form  $4k + 1$ . ■

Clearly the same argument holds for the Diophantine equation

$$x^2 + y^2 = z^{2l+1} + z,$$

where  $l$  is a positive integer.

# 3

## Perfect Squares

We begin the section with a Romanian proposal for the 1986 IMO:

*Let  $a_0 = a_1 = 1$  and  $a_{n+1} = 7a_n - a_{n-1} - 2$  for all positive integers  $n$ . Prove that  $a_n$  is a perfect square for all  $n$ .*

Checking the first few cases, we find  $a_2 = 2^2$ ,  $a_3 = 5^2$ ,  $a_4 = 13^2$ , squares of every other term of the Fibonacci sequence defined by the equalities  $F_1 = F_2 = 1$  and  $F_{n+1} = F_n + F_{n-1}$  for  $n \geq 2$ . This suggests that  $a_n = F_{2n-1}^2$  for all  $n \geq 1$ , which we will prove by induction.

The claim is true for  $n = 1, 2, 3, 4$ . Assuming  $a_k = F_{2k-1}^2$  for  $k \leq n$ , and subtracting  $a_n = 7a_{n-1} - a_{n-2} - 2$  from  $a_{n+1} = 7a_n - a_{n-1} - 2$ , we obtain

$$\begin{aligned} a_{n+1} &= 8a_n - 8a_{n-1} + a_{n-2} \\ &= 8F_{2n-1}^2 - 8F_{2n-3}^2 + F_{2n-5}^2. \end{aligned}$$

It is not difficult to check that  $F_{m-2} = 3F_m - F_{m+2}$  for all  $m \geq 2$ . Thus, replacing  $F_{2n-5}$  by  $3F_{2n-3} - F_{2n-1}$  yields

$$\begin{aligned} a_{n+1} &= 8F_{2n-1}^2 - 8F_{2n-3}^2 + (3F_{2n-3} - F_{2n-1})^2 \\ &= (3F_{2n-1} - F_{2n-3})^2 = F_{2n+1}^2, \end{aligned}$$

completing the proof. ■

A similar argument proves the more general result:

*If  $k$  is an integer greater than 1,  $a_0 = a_1 = 1$  and*

$$a_{n+1} = (k^2 - 2)a_n - a_{n-1} - 2(k - 2) \quad \text{for } n \geq 1,$$

*then  $a_n$  is a perfect square for all  $n$ . More precisely,  $a_n = b_n^2$ , where  $b_n$  is the general term of the sequence given by  $b_0 = b_1 = 1$  and  $b_{n+1} = kb_n - b_{n-1}$  for  $n \geq 1$ .*

Special cases of this result have yielded a number of olympiad problems. A recent example is the following USA proposal for the 1998 IMO:

Let  $(x_n)_{n \geq 0}$  be a sequence such that  $x_0 = x_1 = 5$  and

$$x_n = \frac{x_{n-1} + x_{n+1}}{98}$$

for all positive integers  $n$ . Prove that  $(x_n + 1)/6$  is a perfect square for all  $n$ .

Next we have a harder problem about a nonlinear sequence yielding perfect squares. It was published by **Georgi Demirov** in the Bulgarian high-school journal *Matematika*, No. 7, 1989.

Let  $k$  be an integer greater than 1,  $a_0 = 4$ ,  $a_1 = a_2 = (k^2 - 2)^2$  and

$$a_{n+1} = a_n a_{n-1} - 2(a_n + a_{n-1}) - a_{n-2} + 8 \quad \text{for } n \geq 2.$$

Prove that  $2 + \sqrt{a_n}$  is a perfect square for all  $n$ .

The Fibonacci numbers are involved here again but it is much harder to guess how they are related to the solution.

Let  $\lambda, \mu$  be the roots of the equation  $t^2 - kt + 1 = 0$ . Notice that  $\lambda + \mu = k$ ,  $\lambda\mu = 1$ . Amending the Fibonacci sequence by setting  $F_0 = 0$ , we claim that

$$a_n = (\lambda^{2F_n} + \mu^{2F_n})^2 \quad \text{for } n = 0, 1, 2, \dots$$

This is readily checked for  $n = 0, 1, 2$ . Assume it holds for all  $k \leq n$ . Note that the given recurrence can be written as

$$a_{n+1} - 2 = (a_n - 2)(a_{n-1} - 2) - (a_{n-2} - 2),$$

and that  $a_k = (\lambda^{2F_k} + \mu^{2F_k})^2$  is equivalent to  $a_k - 2 = \lambda^{4F_k} + \mu^{4F_k}$ . Using the induction hypothesis for  $k = n - 2, n - 1, n$ , we obtain

$$\begin{aligned} a_{n+1} - 2 &= (\lambda^{4F_n} + \mu^{4F_n}) (\lambda^{4F_{n-1}} + \mu^{4F_{n-1}}) - (\lambda^{4F_{n-2}} + \mu^{4F_{n-2}}) \\ &= \lambda^{4(F_n + F_{n-1})} + \mu^{4(F_n + F_{n-1})} + \lambda^{4(F_{n-1} + F_{n-2})} \mu^{4F_{n-1}} \\ &\quad + \mu^{4(F_{n-1} + F_{n-2})} \lambda^{4F_{n-1}} - (\lambda^{4F_{n-2}} + \mu^{4F_{n-2}}) \\ &= \lambda^{4F_{n+1}} + \mu^{4F_{n+1}} + (\lambda\mu)^{4F_{n-1}} (\lambda^{4F_{n-2}} + \mu^{4F_{n-2}}) \\ &\quad - (\lambda^{4F_{n-2}} + \mu^{4F_{n-2}}). \end{aligned}$$

Since  $\lambda\mu = 1$ , it follows that

$$a_{n+1} = 2 + \lambda^{4F_{n+1}} + \mu^{4F_{n+1}} = (\lambda^{2F_{n+1}} + \mu^{2F_{n+1}})^2$$

and the induction is complete.

Now

$$2 + \sqrt{a_n} = 2 + \lambda^{2F_n} + \mu^{2F_n} = (\lambda^{F_n} + \mu^{F_n})^2.$$

Since

$$(\lambda^{m-1} + \mu^{m-1})(\lambda + \mu) = (\lambda^m + \mu^m) + \lambda\mu(\lambda^{m-2} + \mu^{m-2}),$$

we have

$$\lambda^m + \mu^m = k(\lambda^{m-1} + \mu^{m-1}) - (\lambda^{m-2} + \mu^{m-2}),$$

leading to an easy proof by induction that  $\lambda^m + \mu^m$  is an integer for all nonnegative integers  $m$ . The solution is complete. ■

# 4

## Least Common Multiples

Our first problem reads:

*Several positive integers are given not exceeding a fixed integer constant  $m$ . Prove that if the least common multiple of every two of them is greater than  $m$ , then the sum of the reciprocals of these numbers is less than  $\frac{3}{2}$ .*

Given  $n$  numbers, denote them by  $x_1, x_2, \dots, x_n$ . For a given  $i$ , there are  $\lfloor m/x_i \rfloor$  multiples of  $x_i$  among  $1, 2, \dots, m$ . None of them is a multiple of  $x_j$  for  $j \neq i$ , since the least common multiple of  $x_i$  and  $x_j$  is greater than  $m$  by hypothesis. So there are  $\lfloor m/x_1 \rfloor + \lfloor m/x_2 \rfloor + \dots + \lfloor m/x_n \rfloor$  distinct elements of  $\{1, 2, \dots, m\}$ , which are divisible by one of the numbers  $x_1, x_2, \dots, x_n$ . None of these elements can be 1 (unless  $n = 1$ , in which case the claim is obvious). Hence

$$\left\lfloor \frac{m}{x_1} \right\rfloor + \left\lfloor \frac{m}{x_2} \right\rfloor + \dots + \left\lfloor \frac{m}{x_n} \right\rfloor \leq m - 1.$$

Taking into account that  $m/x_i < \lfloor m/x_i \rfloor + 1$  for each  $i$ , we obtain

$$m \left( \frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n} \right) < m + n - 1.$$

We now prove that  $n \leq (m+1)/2$ , which will imply

$$\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n} < 1 + \frac{n-1}{m} < \frac{3}{2}.$$

Indeed, note that the greatest odd divisors of  $x_1, x_2, \dots, x_n$  are all distinct. Otherwise, if some two of the given numbers shared the same greatest odd divisor, one of them would be a multiple of the other, contradicting the hypothesis. Hence  $n$  does not exceed the number of odd integers among  $1, 2, \dots, m$ , and the desired inequality  $n \leq (m+1)/2$  follows. ■

Let us now proceed to one of the 1979 Canadian Olympiad problems. It reads:

*Let  $a, b, c, d$  and  $e$  be integers satisfying  $1 \leq a < b < c < d < e$ . Prove that*

$$\frac{1}{[a, b]} + \frac{1}{[b, c]} + \frac{1}{[c, d]} + \frac{1}{[d, e]} \leq \frac{15}{16},$$

where  $[x, y]$  denotes the least common multiple of  $x$  and  $y$ .

Here we shall prove the following generalization:

*If  $a_0 < a_1 < \dots < a_n$  are arbitrary positive integers, then*

$$\frac{1}{[a_0, a_1]} + \frac{1}{[a_1, a_2]} + \dots + \frac{1}{[a_{n-1}, a_n]} \leq 1 - \frac{1}{2^n}.$$

This is problem M865 from the Russian mathematical journal *Kvant*, No. 5, 1984, proposed by **Boris Ivlev**.

We use induction on  $n$ . The case  $n = 1$  is trivial. Suppose the statement is true for  $n$  numbers, and take  $n + 1$  positive integers  $a_0 < a_1 < \dots < a_n < a_{n+1}$ . If  $a_{n+1} \geq 2^{n+1}$  then  $[a_n, a_{n+1}] \geq 2^{n+1}$ , and by the induction hypothesis,

$$\begin{aligned} \frac{1}{[a_0, a_1]} + \frac{1}{[a_1, a_2]} + \dots + \frac{1}{[a_{n-1}, a_n]} + \frac{1}{[a_n, a_{n+1}]} \\ \leq 1 - \frac{1}{2^n} + \frac{1}{2^{n+1}} = 1 - \frac{1}{2^{n+1}}. \end{aligned}$$

Assume now that  $a_{n+1} < 2^{n+1}$ . We shall use the identity

$$[p, q] \cdot (p, q) = pq,$$

where  $(p, q)$  is the greatest common divisor of  $p$  and  $q$ . The difference of  $p$  and  $q$  is divisible by  $(p, q)$ , so, if  $p > q$ , we have  $(p, q) \leq p - q$ . Hence

$$\frac{1}{[a_k, a_{k+1}]} = \frac{(a_k, a_{k+1})}{a_k a_{k+1}} \leq \frac{a_{k+1} - a_k}{a_k a_{k+1}} = \frac{1}{a_k} - \frac{1}{a_{k+1}}$$

for all  $k = 0, 1, \dots, n$ . Therefore

$$\sum_{k=0}^n \frac{1}{[a_k, a_{k+1}]} \leq \sum_{k=0}^n \left( \frac{1}{a_k} - \frac{1}{a_{k+1}} \right) = \frac{1}{a_0} - \frac{1}{a_{n+1}}.$$

Since  $1/a_0 \leq 1$  and  $1/a_{n+1} > 1/2^{n+1}$ , the proof is complete. ■

# 5

## Trig Substitutions

We know many people who do not think much of Dr. Trig. They say that his work is too computational and outdated. But every now and then quite a few of our friends remember him and often change their opinion, at least for a moment. For example, most of them came, eventually, to think about Dr. Trig after attempting, unsuccessfully, to solve the equation

$$6x + 8\sqrt{1 - x^2} = 5(\sqrt{1 + x} + \sqrt{1 - x})$$

in the interval  $(0.6, 1)$ . This is a problem by **Vladimir Kostov** published in *Matematika*, No. 10, 1986.

Let  $\varphi_0 = \cos^{-1} 0.6$ . The substitution  $x = \cos \varphi$ , where  $\varphi \in (0, \varphi_0)$  in view of  $\cos \varphi_0 = 0.6 < x = \cos \varphi < 1$ , reduces the equation to

$$6 \cos \varphi + 8 \sin \varphi = 5 \left( \sqrt{1 + \cos \varphi} + \sqrt{1 - \cos \varphi} \right). \quad (1)$$

Since  $(0, \varphi_0) \subset (0, \pi/2)$  and  $\sin \varphi_0 = 0.8$ , we obtain

$$\begin{aligned} 10(0.6 \cos \varphi + 0.8 \sin \varphi) &= 5\sqrt{2} \left( \cos \frac{\varphi}{2} + \sin \frac{\varphi}{2} \right), \\ 10(\cos \varphi_0 \cos \varphi + \sin \varphi_0 \sin \varphi) &= 5\sqrt{2} \cdot \sqrt{2} \cos \left( \frac{\pi}{4} - \frac{\varphi}{2} \right), \\ \cos(\varphi_0 - \varphi) &= \cos \left( \frac{\pi}{4} - \frac{\varphi}{2} \right). \end{aligned}$$

Note that  $\varphi_0/2 < \pi/4$ , because

$$\cos \frac{\varphi_0}{2} = \sqrt{\frac{1 + \cos \varphi_0}{2}} = \sqrt{0.8} > \cos \frac{\pi}{4}.$$

Now, since  $0 < \varphi_0 - \varphi < \pi/2$  and  $0 < \pi/4 - \varphi/2 < \pi/2$ , the equation  $\cos(\varphi_0 - \varphi) = \cos(\pi/4 - \varphi/2)$  is equivalent to  $\varphi_0 - \varphi = \pi/4 - \varphi/2$ . It

follows that the only solution of (1) in  $(0, \varphi_0)$  is  $\varphi = 2\varphi_0 - \pi/2$ . Therefore the initial equation has a unique solution in  $(0.6, 1)$ . It is

$$x = \cos\left(2\varphi_0 - \frac{\pi}{2}\right) = \sin 2\varphi_0 = 2 \sin \varphi_0 \cos \varphi_0 = 0.96. \blacksquare$$

The USA officials for the 1993 IMO also thought of Dr. Trig's sneaky techniques when submitting the following problem to the IMO:

*Solve the system of equations*

$$\frac{3x - y}{x - 3y} = x^2, \quad \frac{3y - z}{y - 3z} = y^2, \quad \frac{3z - x}{z - 3x} = z^2$$

*in real numbers.*

From the first equation it follows that, if  $x$  is 0, then so is  $y$ , making  $x^2$  indeterminate; hence  $x$ , and similarly  $y$  and  $z$ , cannot be 0. Solving the equations respectively for  $y$ ,  $z$  and  $x$ , we obtain the equivalent system

$$y = \frac{3x - x^3}{1 - 3x^2}, \quad z = \frac{3y - y^3}{1 - 3y^2}, \quad x = \frac{3z - z^3}{1 - 3z^2},$$

where  $x, y, z$  are real numbers different from 0.

There exists a unique number  $u$  in the interval  $(-\pi/2, \pi/2)$  such that  $x = \tan u$ . Then

$$\begin{aligned} y &= \frac{3 \tan u - \tan^3 u}{1 - 3 \tan^2 u} = \tan 3u, \\ z &= \frac{3 \tan 3u - \tan^3 3u}{1 - 3 \tan^2 3u} = \tan 9u, \\ x &= \frac{3 \tan 9u - \tan^3 9u}{1 - 3 \tan^2 9u} = \tan 27u. \end{aligned}$$

The last equality yields  $\tan u = \tan 27u$ , so  $u$  and  $27u$  differ by an integer multiple of  $\pi$ . Therefore,  $u = k\pi/26$  for some integer  $k$  satisfying  $-\pi/2 < k\pi/26 < \pi/2$ . Besides,  $k$  must not be 0, since  $x \neq 0$ . Hence the possible values of  $k$  are  $\pm 1, \pm 2, \dots, \pm 12$ , each of them generating the corresponding triple

$$x = \tan \frac{k\pi}{26}, \quad y = \tan \frac{3k\pi}{26}, \quad z = \tan \frac{9k\pi}{26}.$$

It is immediately checked that all of these triples are solutions of the initial system.  $\blacksquare$

We continue with problem M703 from *Kvant*, No. 9, 1981.

Find all triples  $(x, y, z)$  of real numbers satisfying the system of equations

$$\begin{cases} 3\left(x + \frac{1}{x}\right) = 4\left(y + \frac{1}{y}\right) = 5\left(z + \frac{1}{z}\right) \\ xy + yz + zx = 1. \end{cases}$$

Simplifying and inverting in the first line of equations, we obtain

$$\frac{x}{3(1+x^2)} = \frac{y}{4(1+y^2)} = \frac{z}{5(1+z^2)}, \quad (2)$$

from which it is clear that  $x, y, z$  all have the same sign. Also, if  $(x, y, z)$  is a solution of the system, then so is  $(-x, -y, -z)$ . Thus we may restrict ourselves to finding the positive solutions.

The equalities (2) and Trig's formula

$$\sin \varphi = \frac{2 \tan \frac{\varphi}{2}}{1 + \tan^2 \frac{\varphi}{2}}$$

suggest the idea of representing the unknowns in the form

$$x = \tan \frac{\alpha}{2}, \quad y = \tan \frac{\beta}{2}, \quad z = \tan \frac{\gamma}{2},$$

where  $0 < \alpha, \beta, \gamma < \pi$ . This is possible, because the tangent function assumes all values in the interval  $(0, \pi/2)$ . Then

$$\sin \alpha = \frac{2x}{1+x^2}, \quad \sin \beta = \frac{2y}{1+y^2}, \quad \sin \gamma = \frac{2z}{1+z^2},$$

and (2) takes the form

$$\frac{\sin \alpha}{3} = \frac{\sin \beta}{4} = \frac{\sin \gamma}{5}. \quad (3)$$

Also, the second equation becomes

$$\frac{1}{z} = \frac{x+y}{1-xy} \quad \text{or} \quad \cot \frac{\gamma}{2} = \tan \frac{\alpha+\beta}{2}.$$

Note that  $z \neq 0$  and  $xy \neq 1$ . Since  $\alpha, \beta, \gamma$  are numbers from  $(0, \pi)$ , this implies

$$\frac{\alpha+\beta}{2} = \frac{\pi}{2} - \frac{\gamma}{2}, \quad \text{or} \quad \alpha + \beta + \gamma = \pi.$$

The latter means that  $\alpha, \beta, \gamma$  are the angles of some triangle  $T$ . By (3) and the law of sines, its sides, opposite to  $\alpha, \beta, \gamma$ , respectively, are in the ratio

3 : 4 : 5. Now it is clear that  $T$  is right-angled with  $\gamma = \pi/2$ ,  $\sin \alpha = \frac{3}{5}$  and  $\sin \beta = \frac{4}{5}$ . Then

$$x = \tan \frac{\alpha}{2} = \frac{1}{3}, \quad y = \tan \frac{\beta}{2} = \frac{1}{2}, \quad z = \tan \frac{\gamma}{2} = 1.$$

So the system has two solutions:  $(\frac{1}{3}, \frac{1}{2}, 1)$  and  $(-\frac{1}{3}, -\frac{1}{2}, -1)$ . ■

Trig's techniques can be used in other situations as well. To illustrate this, let us consider a problem used in the training of the USA team for the 1996 IMO:

*Consider  $a_1, a_2, \dots, a_n$  in the interval  $[-2, 2]$  such that their sum is zero. Prove that*

$$|a_1^3 + a_2^3 + \dots + a_n^3| \leq 2n.$$

Taking into account that  $a_k$  is in the interval  $[-2, 2]$  leads to the substitution  $a_k = 2 \cos b_k$ ,  $k = 1, 2, \dots, n$ . Then the triple angle formula  $\cos 3b = 4 \cos^3 b - 3 \cos b$  gives  $2 \cos 3b_k = a_k^3 - 3a_k$ ,  $k = 1, 2, \dots, n$ . Using the hypothesis  $\sum_{k=1}^n a_k = 0$ , we obtain

$$2 \sum_{k=1}^n \cos 3b_k = \sum_{k=1}^n a_k^3,$$

and, noting that  $|\cos x| \leq 1$  for all  $x$ , the conclusion follows. ■

A similar argument proves helpful in solving a recent problem from the final round of the 2002 Romanian National Olympiad:

*Find all numbers  $a, b, c, d, e$  in the interval  $[-2, 2]$  satisfying simultaneously the equations*

$$\begin{cases} a + b + c + d + e = 0 \\ a^3 + b^3 + c^3 + d^3 + e^3 = 0 \\ a^5 + b^5 + c^5 + d^5 + e^5 = 10. \end{cases}$$

It turns out that the solutions are

$$(a, b, c, d, e) = \left( 2, \frac{-1 + \sqrt{5}}{2}, \frac{-1 + \sqrt{5}}{2}, \frac{-1 - \sqrt{5}}{2}, \frac{-1 - \sqrt{5}}{2} \right)$$

and all of its permutations.

Thank you, Dr. Trig.

# Coffee Break 1

1. Let  $n$  be a nonnegative integer. Prove that the numbers  $n + 2$  and  $n^2 + n + 1$  cannot both be perfect cubes.
2. What regular  $n$ -gons can be inscribed in a noncircular ellipse?
3. Prove that any three real numbers  $x, y, z$  satisfy the inequality

$$|x| + |y| + |z| - |x + y| - |y + z| - |z + x| + |x + y + z| \geq 0.$$

## Solutions

1. If  $n + 2$  and  $n^2 + n + 1$  are both perfect cubes, then so is their product. On the other hand,

$$\begin{aligned}(n + 2)(n^2 + n + 1) &= (n - 1)(n^2 + n + 1) + 3(n^2 + n + 1) \\ &= (n^3 - 1) + (3n^2 + 3n + 3) \\ &= (n + 1)^3 + 1.\end{aligned}$$

This is impossible, since no two positive perfect cubes differ by 1.

2. Assume that a regular  $n$ -gon is inscribed in a noncircular ellipse. Then its circumcircle has at least  $n$  common points with the ellipse: the vertices of the  $n$ -gon. But two conics may intersect at no more than four points, so  $n \leq 4$ . One can clearly inscribe an equilateral triangle and a square in an ellipse, so these two regular polygons are the ones we are after.

3. Let  $z$  be the greatest in absolute value among the numbers  $x, y, z$ . Everything is clear if  $z = 0$ . If  $z \neq 0$ , divide both sides by  $|z| \neq 0$  to get the equivalent inequality

$$\left| \frac{x}{z} \right| + \left| \frac{y}{z} \right| + 1 - \left| \frac{x}{z} + \frac{y}{z} \right| - \left| \frac{y}{z} + 1 \right| - \left| 1 + \frac{x}{z} \right| + \left| \frac{x}{z} + \frac{y}{z} + 1 \right| \geq 0.$$

Note that  $-1 \leq x/z \leq 1$  and  $-1 \leq y/z \leq 1$ , so  $1 + x/z \geq 0$ ,  $1 + y/z \geq 0$ , and this simplifies to

$$\left| \frac{x}{z} \right| + \left| \frac{y}{z} \right| - \left| \frac{x}{z} + \frac{y}{z} \right| - \left( \frac{x}{z} + \frac{y}{z} + 1 \right) + \left| \frac{x}{z} + \frac{y}{z} + 1 \right| \geq 0.$$

The sum of the first three summands is nonnegative in view of the triangle inequality  $|a + b| \leq |a| + |b|$ ; the sum of the remaining two is also nonnegative, because no number exceeds its absolute value.

We have bad news about the general inequality

$$\begin{aligned}\sum_{i=1}^n |x_i| - \sum_{1 \leq i < j \leq n} |x_i + x_j| + \sum_{1 \leq i < j < k \leq n} |x_i + x_j + x_k| \\ - \cdots + (-1)^{n-1} |x_1 + x_2 + \cdots + x_n| \geq 0.\end{aligned}$$

It fails even for  $n = 4$  (a counterexample is  $x_1 = 2, x_2 = x_3 = x_4 = -1$ ).

# 6

## Popoviciu's Theorem

The Romanian mathematician **Tiberiu Popoviciu** proved a quite general theorem about convex functions:

*Let  $f$  be a convex real-valued function defined on an interval  $I$ .  
Then*

$$f(x) + f(y) + f(z) + 3f\left(\frac{x+y+z}{3}\right) \\ \geq 2\left[f\left(\frac{x+y}{2}\right) + f\left(\frac{y+z}{2}\right) + f\left(\frac{z+x}{2}\right)\right]$$

*for all  $x, y, z$  in  $I$ .*

Note that no regularity conditions, such as continuity, differentiability, and so on, are imposed on  $f$ . Here is a proof published in the Romanian high-school journal *Revista Matematică Timișoara*, No. 1, 1991.

The case  $x = y = z$  is obvious, so let us assume that at least two of the numbers  $x, y, z$  are distinct.

Suppose that  $x \leq y \leq z$ . If  $y \leq (x+y+z)/3$ , then a calculation will show that

$$\frac{x+y+z}{3} \leq \frac{x+z}{2} \leq z \quad \text{and} \quad \frac{x+y+z}{3} \leq \frac{y+z}{2} \leq z.$$

Therefore there exist  $s, t \in [0, 1]$  such that

$$\frac{x+z}{2} = s\frac{x+y+z}{3} + (1-s)z, \\ \frac{y+z}{2} = t\frac{x+y+z}{3} + (1-t)z.$$

Adding up these equalities and rearranging yields

$$\frac{x+y-2z}{2} = (s+t)\frac{x+y-2z}{3},$$

hence  $s+t = \frac{3}{2}$ . Here we used the fact that  $x+y-2z \neq 0$ , for if not,  $x \leq y \leq z$  yields  $x = y = z$ , contrary to the initial assumption. The function  $f$  is convex, so

$$\begin{aligned} f\left(\frac{x+z}{2}\right) &= f\left(s\frac{x+y+z}{3} + (1-s)z\right) \\ &\leq sf\left(\frac{x+y+z}{3}\right) + (1-s)f(z). \end{aligned}$$

Similarly,

$$\begin{aligned} f\left(\frac{y+z}{2}\right) &= f\left(t\frac{x+y+z}{3} + (1-t)z\right) \\ &\leq tf\left(\frac{x+y+z}{3}\right) + (1-t)f(z). \end{aligned}$$

Add these up with

$$f\left(\frac{x+y}{2}\right) \leq \frac{1}{2}f(x) + \frac{1}{2}f(y),$$

and the conclusion follows.

The case  $(x+y+z)/3 < y$  can be treated in a similar manner. In this situation some computation will show that

$$x \leq \frac{x+z}{2} \leq \frac{x+y+z}{3}, \quad x \leq \frac{x+y}{2} \leq \frac{x+y+z}{3}. \quad \blacksquare$$

To illustrate the theorem, consider the inequality

$$x^6 + y^6 + z^6 + 3x^2y^2z^2 \geq 2(x^3y^3 + y^3z^3 + z^3x^3),$$

where  $x, y, z$  are positive real numbers. To prove it, just make use of Popoviciu's theorem for

$$x' = 6 \ln x, \quad y' = 6 \ln y, \quad z' = 6 \ln z, \quad \text{and} \quad f(t) = e^t.$$

Of course, you could prove the inequality directly. It is possible but not pleasant.

Applying the theorem to the convex function  $f(x) = |x|$  yields one more proof of problem 3, Coffee Break 1.

# 7

## Catalan's Identity

There is a curious and somewhat unexpected relation among the reciprocals of the first  $2n$  positive integers:

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{2n-1} - \frac{1}{2n} = \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n}.$$

It is called *Catalan's identity*. To prove this, set

$$S_k = 1 + \frac{1}{2} + \cdots + \frac{1}{k} \quad (k = 1, 2, \dots).$$

Then

$$\begin{aligned} 1 - \frac{1}{2} + \frac{1}{3} - \cdots + \frac{1}{2n-1} - \frac{1}{2n} &= S_{2n} - 2 \left( \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2n} \right) \\ &= S_{2n} - S_n \\ &= \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n}. \quad \blacksquare \end{aligned}$$

Here are two applications of this identity. The first one is from the 1995 junior contest of *Matematika*.

*Prove the equality*

$$\frac{1995}{2} - \frac{1994}{3} + \cdots - \frac{2}{1995} + \frac{1}{1996} = \frac{1}{999} + \frac{3}{1000} + \frac{5}{1001} + \cdots + \frac{1995}{1996}.$$

We shall prove the more general identity

$$\frac{2n-1}{2} - \frac{2n-2}{3} + \cdots - \frac{2}{2n-1} + \frac{1}{2n} = \frac{1}{n+1} + \frac{3}{n+2} + \cdots + \frac{2n-1}{2n},$$

where  $n$  is a positive integer (the problem deals with the case  $n = 998$ ). The addends on the left-hand side are of the form  $(2n-k)/(k+1)$ ,  $k = 1, 2, \dots, 2n-1$ , and the ones on the right-hand side are of the form

$(2l - 1)/(n + l)$ ,  $l = 1, 2, \dots, n$ . For all admissible values of  $k$  and  $l$  we have

$$\frac{2n - k}{k + 1} = \frac{(2n + 1) - (k + 1)}{k + 1} = \frac{2n + 1}{k + 1} - 1,$$

$$\frac{2l - 1}{n + l} = \frac{2(n + l) - (2n + 1)}{n + l} = 2 - \frac{2n + 1}{n + l}.$$

Hence the identity becomes

$$\left(\frac{2n + 1}{2} - 1\right) - \left(\frac{2n + 1}{3} - 1\right) + \dots - \left(\frac{2n + 1}{2n - 1} - 1\right) + \left(\frac{2n + 1}{2n} - 1\right)$$

$$= \left(2 - \frac{2n + 1}{n + 1}\right) + \left(2 - \frac{2n + 1}{n + 2}\right) + \left(2 - \frac{2n + 1}{n + 3}\right) + \dots + \left(2 - \frac{2n + 1}{2n}\right),$$

or

$$(2n + 1) \left(\frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \dots - \frac{1}{2n - 1} + \frac{1}{2n}\right) - 1$$

$$= 2n - (2n + 1) \left(\frac{1}{n + 1} + \frac{1}{n + 2} + \dots + \frac{1}{2n}\right).$$

Now a short computation will show that this reduces to Catalan's identity, so we are done. ■

Another application is the first 1979 IMO problem:

*Let  $m$  and  $n$  be positive integers such that*

$$\frac{m}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{1318} + \frac{1}{1319}.$$

*Prove that  $m$  is divisible by 1979.*

By Catalan's identity,

$$\frac{m}{n} = \left(\frac{1}{660} + \frac{1}{661} + \dots + \frac{1}{1318}\right) + \frac{1}{1319}$$

$$= \left(\frac{1}{660} + \frac{1}{1319}\right) + \left(\frac{1}{661} + \frac{1}{1318}\right) + \dots + \left(\frac{1}{989} + \frac{1}{990}\right)$$

$$= \frac{1979}{660 \cdot 1319} + \frac{1979}{661 \cdot 1318} + \dots + \frac{1979}{989 \cdot 990}.$$

Adding the terms in the last sum, we find that

$$\frac{m}{n} = \frac{1979 \cdot k}{660 \cdot 661 \cdot \dots \cdot 1318 \cdot 1319}$$

for some (very large) integer  $k$ . Since 1979 is a prime number and all the factors of the denominator are less than it, we conclude that  $m$  is divisible by 1979. ■

In a similar fashion one can prove the more general statement (published in *Revista Matematică Timișoara*, Nos. 1–2, 1979):

Let  $p$  be a prime greater than 3,  $q = \lfloor 2p/3 \rfloor$  and let  $m$  and  $n$  be positive integers such that

$$\frac{m}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + (-1)^{q-1} \frac{1}{q}.$$

Then  $m$  is divisible by  $p$ .

This result proves crucial when solving the following 1994 Bulgarian IMO selection test problem (also 1996 Putnam problem A5):

If  $p$  is a prime greater than 3 and  $q = \lfloor 2p/3 \rfloor$ , prove that

$$\binom{p}{1} + \binom{p}{2} + \cdots + \binom{p}{q}$$

is divisible by  $p^2$ .

Each binomial coefficient in question is divisible by  $p$ , because  $p$  divides the numerator and not the denominator of  $\binom{p}{r} = p!/[r!(p-r)!]$ . Thus it suffices to show that the sum

$$S = 1 + \frac{p-1}{2!} + \frac{(p-1)(p-2)}{3!} + \cdots + \frac{(p-1)(p-2) \cdots [p-(q-1)]}{q!}$$

is divisible by  $p$ . Expanding the numerators, we represent  $S$  as

$$\frac{pk_2}{2!} + \frac{pk_3}{3!} + \cdots + \frac{pk_q}{q!} + \left(1 - \frac{1}{2} + \frac{1}{3} - \cdots + \frac{(-1)^{q-1}}{q}\right),$$

where  $k_2, k_3, \dots, k_q$  are integers, and the quantity  $Q$  in the parentheses is the one encountered in the previous statement. By this statement, we have  $Q = (pm)/n$  for some integers  $m$  and  $n$ , where  $n$  is not a multiple of  $p$ . Hence the integer  $S$  can be written as

$$S = p \left( \frac{k_2}{2!} + \frac{k_3}{3!} + \cdots + \frac{k_q}{q!} + \frac{m}{n} \right).$$

None of the denominators is divisible by  $p$ , and the conclusion follows. ■

# 8

## Several Inequalities

We begin the section with a problem, a special case of which was given at the Romanian selection test for the 1977 IMO.

*The positive numbers  $a_1, a_2, \dots, a_n$ ,  $b_1 \leq b_2 \leq \dots \leq b_n$  satisfy the inequalities*

$$a_1 \leq b_1, \quad a_1 + a_2 \leq b_1 + b_2, \dots, \\ a_1 + a_2 + \dots + a_n \leq b_1 + b_2 + \dots + b_n.$$

*Prove that*

$$\sqrt{a_1} + \sqrt{a_2} + \dots + \sqrt{a_n} \leq \sqrt{b_1} + \sqrt{b_2} + \dots + \sqrt{b_n}.$$

The solution uses Abel's summation formula. The AM–GM inequality yields

$$\sqrt{a_k} \leq \frac{1}{2} \left( \frac{a_k}{\sqrt{b_k}} + \sqrt{b_k} \right)$$

for each  $k = 1, 2, \dots, n$ , so we have

$$\begin{aligned} & \sqrt{a_1} + \sqrt{a_2} + \dots + \sqrt{a_n} \\ & \leq \frac{1}{2} \left[ \left( \frac{a_1}{\sqrt{b_1}} + \frac{a_2}{\sqrt{b_2}} + \dots + \frac{a_n}{\sqrt{b_n}} \right) + \left( \sqrt{b_1} + \sqrt{b_2} + \dots + \sqrt{b_n} \right) \right]. \end{aligned}$$

In view of this, it suffices to prove that

$$\frac{a_1}{\sqrt{b_1}} + \frac{a_2}{\sqrt{b_2}} + \dots + \frac{a_n}{\sqrt{b_n}} \leq \sqrt{b_1} + \sqrt{b_2} + \dots + \sqrt{b_n}.$$

If we set

$$S_k = a_1 + a_2 + \dots + a_k, \quad T_k = b_1 + b_2 + \dots + b_k$$

for  $k = 1, 2, \dots, n$ , then  $S_k \leq T_k$  by hypothesis, so by Abel's formula we have

$$\begin{aligned} & \frac{a_1}{\sqrt{b_1}} + \frac{a_2}{\sqrt{b_2}} + \dots + \frac{a_n}{\sqrt{b_n}} \\ &= \frac{1}{\sqrt{b_n}} S_n + \left( \frac{1}{\sqrt{b_{n-1}}} - \frac{1}{\sqrt{b_n}} \right) S_{n-1} + \dots + \left( \frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) S_1 \\ &\leq \frac{1}{\sqrt{b_n}} T_n + \left( \frac{1}{\sqrt{b_{n-1}}} - \frac{1}{\sqrt{b_n}} \right) T_{n-1} + \dots + \left( \frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) T_1 \\ &= \sqrt{b_1} + \sqrt{b_2} + \dots + \sqrt{b_n}. \end{aligned}$$

This finishes the proof. ■

**Gheorghe Eckstein** is the author of the next problem. It was given on a Romanian selection test as well, for the 1986 IMO.

*Let  $a, b, x, y, z$  be positive real numbers. Prove the inequality*

$$\frac{x}{ay + bz} + \frac{y}{az + bx} + \frac{z}{ax + by} \geq \frac{3}{a + b}.$$

*First solution.* Setting

$$ay + bz = u, \quad az + bx = v, \quad ax + by = w,$$

we express  $x, y, z$  in terms of  $u, v, w$ :

$$\begin{aligned} x &= \frac{-abu + b^2v + a^2w}{a^3 + b^3}, & y &= \frac{a^2u - abv + b^2w}{a^3 + b^3}, \\ z &= \frac{b^2u + a^2v - abw}{a^3 + b^3}. \end{aligned}$$

If  $L$  is the left-hand side of the given inequality, then a standard manipulation shows that

$$L = \frac{1}{a^3 + b^3} \left[ a^2 \left( \frac{w}{u} + \frac{u}{v} + \frac{v}{w} \right) + b^2 \left( \frac{v}{u} + \frac{w}{v} + \frac{u}{w} \right) - 3ab \right].$$

Since, by the AM–GM inequality,

$$\frac{w}{u} + \frac{u}{v} + \frac{v}{w} \geq 3, \quad \frac{v}{u} + \frac{w}{v} + \frac{u}{w} \geq 3,$$

we get

$$L \geq \frac{3}{a^3 + b^3} (a^2 + b^2 - ab) = \frac{3}{a + b}.$$

*Second solution.* Applying the Cauchy–Schwarz inequality to the triples

$$\begin{array}{ccc} \sqrt{\frac{x}{ay+bz}}, & \sqrt{\frac{y}{az+bx}}, & \sqrt{\frac{z}{ax+by}} \quad \text{and} \\ \sqrt{x(ay+bz)}, & \sqrt{y(az+bx)}, & \sqrt{z(ax+by)} \end{array}$$

yields

$$\begin{aligned} & \left( \frac{x}{ay+bz} + \frac{y}{az+bx} + \frac{z}{ax+by} \right) \\ & \quad \times (x(ay+bz) + y(az+bx) + z(ax+by)) \\ & \geq (x+y+z)^2, \end{aligned}$$

which can be written in the form

$$\frac{x}{ay+bz} + \frac{y}{az+bx} + \frac{z}{ax+by} \geq \frac{(x+y+z)^2}{(a+b)(xy+yz+zx)}.$$

Now it suffices to prove that

$$(x+y+z)^2 \geq 3(xy+yz+zx),$$

and this is equivalent to  $(x-y)^2 + (y-z)^2 + (z-x)^2 \geq 0$ . ■

*Do there exist permutations*

$$a_1, a_2, \dots, a_{50}, \quad b_1, b_2, \dots, b_{50}, \quad c_1, c_2, \dots, c_{50}, \quad d_1, d_2, \dots, d_{50}$$

*of the first fifty positive integers such that*

$$\sum_{i=1}^{50} a_i b_i = 2 \sum_{i=1}^{50} c_i d_i?$$

This problem was published in *Középiskolai Matematikai Lapok*, Nos. 8–9, 1986, problem F. 2587. One’s first impression is that such a statement could hardly have anything in common with inequalities. The following solution shows that this is not exactly true.

Let  $x_1, x_2, \dots, x_{50}, y_1, y_2, \dots, y_{50}$  be any two permutations of the numbers  $1, 2, \dots, 50$ , and  $S = \sum_{i=1}^{50} x_i y_i$ . Since there are finitely many permutations, the sum  $S$  has a minimum and a maximum value. It is well known that:

1° The maximum value of  $S$  is assumed if and only if  $y_i = x_i$  for each  $i = 1, 2, \dots, 50$ ; that is, “when a greater  $y_i$  corresponds to a greater  $x_i$ .”

2° The minimum value of  $S$  is assumed if and only if  $y_i = 51 - x_i$  for  $i = 1, 2, \dots, 50$ ; that is, “when a smaller  $y_i$  corresponds to a greater  $x_i$ .”

We omit the proof, which is straightforward. The difficulty here is, of course, not the proof, but the idea of using this kind of argument.

Taking 1° and 2° into account, it is easy to find the maximum and the minimum values of  $S$ . We have

$$S_{\max} = \sum_{i=1}^{50} i^2 = \frac{50 \cdot 51 \cdot 101}{6} = 42,925,$$

$$S_{\min} = \sum_{i=1}^{50} i(51 - i) = 51 \sum_{i=1}^{50} i - \sum_{i=1}^{50} i^2 = 22,100.$$

Since  $S_{\max} < 2S_{\min}$ , the answer to the initial question is no; there are no permutations with the desired property. ■

*Prove the inequality*

$$ax + by + cz + \sqrt{(a^2 + b^2 + c^2)(x^2 + y^2 + z^2)} \geq \frac{2}{3}(a + b + c)(x + y + z),$$

where  $a, b, c, x, y, z$  are arbitrary real numbers.

We present the nice proof published in *Kvant*, No. 4, 1990 by V. **Dubrovski** and the author of the problem, **Vasile Cârtoaje**.

If  $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = 0$ , then either  $a = b = c = 0$  or  $x = y = z = 0$ , and the inequality holds. Otherwise consider the vectors

$$\mathbf{u} = (a, b, c) \quad \mathbf{v} = (x, y, z), \quad \mathbf{w} = (1, 1, 1).$$

We have

$$\mathbf{u} \cdot \mathbf{v} = ax + by + cz, \quad \mathbf{v} \cdot \mathbf{w} = x + y + z, \quad \mathbf{w} \cdot \mathbf{u} = a + b + c,$$

$$|\mathbf{u}|^2 = a^2 + b^2 + c^2, \quad |\mathbf{v}|^2 = x^2 + y^2 + z^2, \quad |\mathbf{w}|^2 = 3.$$

Dividing both sides by  $\sqrt{(a^2 + b^2 + c^2)(x^2 + y^2 + z^2)}$ , we can write the inequality in the form

$$\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}| \cdot |\mathbf{v}|} + 1 \geq 2 \frac{\mathbf{w} \cdot \mathbf{u}}{|\mathbf{w}| \cdot |\mathbf{u}|} \cdot \frac{\mathbf{v} \cdot \mathbf{w}}{|\mathbf{v}| \cdot |\mathbf{w}|}.$$

Taking the definition of dot product into account, this is equivalent to

$$\cos \gamma + 1 \geq 2 \cos \beta \cos \alpha,$$

where  $\alpha = \angle(\mathbf{v}, \mathbf{w})$ ,  $\beta = \angle(\mathbf{w}, \mathbf{u})$ ,  $\gamma = \angle(\mathbf{u}, \mathbf{v})$ . Thus, the problem reduces to proving the last inequality for the plane angles  $\alpha, \beta, \gamma$  of any trihedral angle (possibly degenerate), one edge of which is collinear with  $\mathbf{w} = (1, 1, 1)$ . Recall that these plane angles satisfy the inequalities

$$\gamma \leq \alpha + \beta \quad (\text{the triangle inequality}) \quad \text{and} \quad \alpha + \beta + \gamma \leq 2\pi.$$

They imply  $\gamma \leq \alpha + \beta \leq 2\pi - \gamma$ . Now, if  $\varphi$  and  $\psi$  are angles in the interval  $[0, \pi]$  and  $\psi \leq \varphi \leq 2\pi - \psi$ , then  $\cos \psi \geq \cos \varphi$ . It follows that  $\cos \gamma \geq \cos(\alpha + \beta)$ . This yields

$$\cos \gamma + 1 \geq \cos(\alpha + \beta) + \cos(\alpha - \beta) = 2 \cos \alpha \cos \beta,$$

and we are done.

Furthermore, the argument also shows that equality occurs if and only if  $\cos(\alpha - \beta) = 1$  and  $\cos \gamma = \cos(\alpha + \beta)$ , that is:  $\alpha = \beta$  and either  $\gamma = 2\alpha$  or  $\gamma + 2\alpha = 2\pi$ . In both cases, geometrically this means that the rays determined by the vectors  $\mathbf{u}$  and  $\mathbf{v}$  are symmetric with respect to the line  $x = y = z$ . A straightforward computation shows that the vector  $\mathbf{v}_1$  symmetric to  $\mathbf{v}$  with respect to  $x = y = z$  has coordinates

$$x_1 = \frac{2(y+z) - x}{3}, \quad y_1 = \frac{2(z+x) - y}{3}, \quad z_1 = \frac{2(x+y) - z}{3},$$

so the condition of equality is that

$$\frac{x_1}{a} = \frac{y_1}{b} = \frac{z_1}{c}. \quad \blacksquare$$

This approach, however, provides a hint towards a purely algebraic solution: It suffices to plug the above values of  $x_1, y_1, z_1$  into the Cauchy–Schwarz inequality

$$ax_1 + by_1 + cz_1 \leq \sqrt{(a^2 + b^2 + c^2)(x_1^2 + y_1^2 + z_1^2)}.$$

# 9

## Vectors

Here we present several problems nicely handled by employing vector techniques.

*Prove that for any eight real numbers  $a, b, c, d, e, f, g, h$  at least one of the numbers*

$$ac + bd, \quad ae + bf, \quad ag + bh, \quad ce + df, \quad cg + dh, \quad eg + fh$$

*is nonnegative.*

Consider the four vectors whose coordinates with respect to some Cartesian coordinate system in the plane are  $(a, b), (c, d), (e, f), (g, h)$ . Regardless of the numbers  $a, b, \dots, h$ , at least two of the vectors form an angle not exceeding  $90^\circ$ . The dot product of these vectors is nonnegative, and coincides with one of the six given sums. ■

*Let  $\overrightarrow{OA}, \overrightarrow{OB}, \overrightarrow{OC}$  be three coterminal rays in space. Prove that the three angles formed by the bisectors of  $\angle AOB, \angle BOC, \angle COA$  are all acute, all right or all obtuse.*

We use the fact that the angle between two vectors is acute, right or obtuse according as their dot product is positive, equal to 0 or negative, respectively. Take the unit vectors  $\mathbf{i}, \mathbf{j}, \mathbf{k}$  collinear with  $\overrightarrow{OA}, \overrightarrow{OB}, \overrightarrow{OC}$ , respectively.

Their pairwise sums  $\mathbf{i} + \mathbf{j}, \mathbf{j} + \mathbf{k}$  and  $\mathbf{k} + \mathbf{i}$  are collinear with the bisectors of  $\angle AOB, \angle BOC$  and  $\angle COA$ . Since  $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = 1$ , the dot product  $(\mathbf{i} + \mathbf{j}) \cdot (\mathbf{j} + \mathbf{k})$  equals

$$(\mathbf{i} + \mathbf{j}) \cdot (\mathbf{j} + \mathbf{k}) = \mathbf{j}^2 + \mathbf{i} \cdot \mathbf{j} + \mathbf{j} \cdot \mathbf{k} + \mathbf{k} \cdot \mathbf{i} = 1 + \mathbf{i} \cdot \mathbf{j} + \mathbf{j} \cdot \mathbf{k} + \mathbf{k} \cdot \mathbf{i}.$$

By symmetry, we obtain the same result about each of the dot products  $(\mathbf{j} + \mathbf{k}) \cdot (\mathbf{k} + \mathbf{i})$  and  $(\mathbf{k} + \mathbf{i}) \cdot (\mathbf{i} + \mathbf{j})$ . In particular, the three products are

of the same sign, implying that the angles formed by the bisectors are all acute, all right, or all obtuse. ■

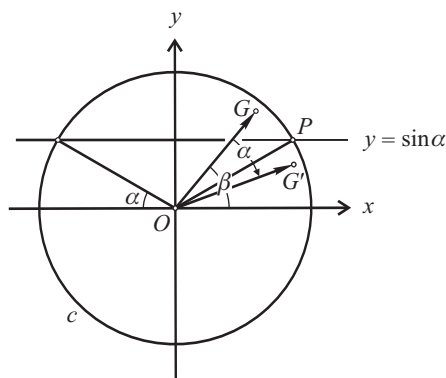
Let  $x_1, x_2, \dots, x_n$  be real numbers such that

$$\sin x_1 + \sin x_2 + \dots + \sin x_n \geq n \sin \alpha,$$

where  $\alpha \in [0, \frac{\pi}{2}]$ . Prove that

$$\sin(x_1 - \alpha) + \sin(x_2 - \alpha) + \dots + \sin(x_n - \alpha) \geq 0.$$

This problem from the 1983 Romanian Olympiad admits of the following solution. Take the points  $P(\cos \alpha, \sin \alpha)$  and  $P_i(\cos x_i, \sin x_i)$  ( $i = 1, 2, \dots, n$ ) on the unit circle  $c$ , with center  $O$ . One may assume without loss of generality (reordering the points if necessary) that  $P_1 P_2 \dots P_n$  is a convex  $n$ -gon inscribed in  $c$ .



This polygon is not depicted in the figure; what is essential is that, being convex, it contains its centroid

$$G \left( \frac{1}{n} \sum_{i=1}^n \cos x_i, \frac{1}{n} \sum_{i=1}^n \sin x_i \right).$$

In particular,  $G$  is contained in  $c$ . On the other hand, the ordinate of  $G$  is by hypothesis not less than  $\sin \alpha$ , so  $G$  also lies on or above the line  $y = \sin \alpha$ . All of this implies that the angle  $\beta$  formed by  $\overrightarrow{OG}$  and the positive direction of the  $x$ -axis is in the interval  $[\alpha, \pi - \alpha]$ .

Rotate the system  $\{P_1, P_2, \dots, P_n\}$  through an angle  $-\alpha$  about  $O$ . This takes it to a new system  $\{P'_1, P'_2, \dots, P'_n\}$  whose centroid  $G'$  is the rotated position of  $G$  (because rotation is a linear operation). We proved

that  $\alpha \leq \beta \leq \pi - \alpha$ , so the ordinate of  $G'$  is nonnegative. Since the ordinate of  $P'_i$  is  $\sin(x_i - \alpha)$  for each  $i = 1, 2, \dots, n$ , this yields

$$\sin(x_1 - \alpha) + \sin(x_2 - \alpha) + \dots + \sin(x_n - \alpha) \geq 0. \blacksquare$$

We conclude with a problem by **R. P. Ushakov** from *Kvant* (No. 1, 1987, problem M1024) which is hard if solved the standard way.

*Given two triangles with angles  $\alpha, \beta, \gamma$  and  $\alpha_1, \beta_1, \gamma_1$ , prove that*

$$\frac{\cos \alpha_1}{\sin \alpha} + \frac{\cos \beta_1}{\sin \beta} + \frac{\cos \gamma_1}{\sin \gamma} \leq \cot \alpha + \cot \beta + \cot \gamma,$$

*with equality if and only if  $\alpha = \alpha_1, \beta = \beta_1, \gamma = \gamma_1$ .*

Fix  $\alpha, \beta, \gamma$  and consider the left-hand side,  $L$ , as a function of  $\alpha_1, \beta_1, \gamma_1$ , subject to the constraints  $\alpha_1 > 0, \beta_1 > 0, \gamma_1 > 0$  and  $\alpha_1 + \beta_1 + \gamma_1 = \pi$ . We have to prove that the maximum value of  $L$  is taken on only for  $\alpha = \alpha_1, \beta = \beta_1, \gamma = \gamma_1$ . To this end, multiply  $L$  by  $\sin \alpha \sin \beta \sin \gamma > 0$  and study the expression

$$L_1 = \sin \beta \sin \gamma \cos \alpha_1 + \sin \gamma \sin \alpha \cos \beta_1 + \sin \alpha \sin \beta \cos \gamma_1.$$

Take a triangle  $A_1B_1C_1$  with  $\angle A_1 = \alpha_1, \angle B_1 = \beta_1, \angle C_1 = \gamma_1$ . Consider the vectors  $\mathbf{a}, \mathbf{b}, \mathbf{c}$  collinear with  $\overrightarrow{B_1C_1}, \overrightarrow{C_1A_1}, \overrightarrow{A_1B_1}$ , respectively, and such that  $|\mathbf{a}| = \sin \alpha, |\mathbf{b}| = \sin \beta, |\mathbf{c}| = \sin \gamma$ . Recall that, for any two vectors  $\mathbf{u}, \mathbf{v}$ , we have  $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}| \cdot |\mathbf{v}| \cos \theta$ , where  $\theta$  is the angle between  $\mathbf{u}$  and  $\mathbf{v}$ . Then

$$\mathbf{a} \cdot \mathbf{b} = -\sin \alpha \sin \beta \cos \gamma_1,$$

$$\mathbf{b} \cdot \mathbf{c} = -\sin \beta \sin \gamma \cos \alpha_1,$$

$$\mathbf{c} \cdot \mathbf{a} = -\sin \gamma \sin \alpha \cos \beta_1.$$

We thus obtain  $L_1 = -(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a})$ . On the other hand,

$$\begin{aligned} -(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}) &= \frac{1}{2} (\mathbf{a}^2 + \mathbf{b}^2 + \mathbf{c}^2 - (\mathbf{a} + \mathbf{b} + \mathbf{c})^2) \\ &\leq \frac{1}{2} (\mathbf{a}^2 + \mathbf{b}^2 + \mathbf{c}^2), \end{aligned}$$

so that  $L_1 \leq \frac{1}{2} (\mathbf{a}^2 + \mathbf{b}^2 + \mathbf{c}^2)$ . Equality is possible and holds if and only if  $\mathbf{a} + \mathbf{b} + \mathbf{c} = \vec{0}$ . In the latter case a triangle can be formed by the vectors  $\mathbf{a}, \mathbf{b}, \mathbf{c}$ , with side lengths  $\sin \alpha, \sin \beta, \sin \gamma$ , and opposite angles  $\alpha_1, \beta_1, \gamma_1$ , respectively. Using the law of sines, it is not hard to infer that  $\alpha = \alpha_1, \beta = \beta_1, \gamma = \gamma_1$ , and the proof is finished.  $\blacksquare$

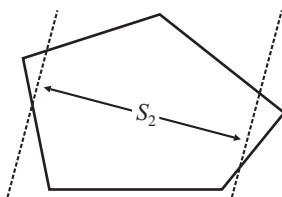
# 10

## Mathematical Induction at Work

The applications of mathematical induction are so diverse that any general comment seems inadequate. So let us proceed to the examples; the reader may then find something interesting.

*Prove that for any  $n$  greater than 3 there exists a convex polygon with  $n$  sides, not all equal, such that the sum of the distances from any interior point to its sides is a constant.*

It is not hard to see that the only triangle with the property that the sum of the distances from any interior point to its sides is a constant is the equilateral triangle. Thus we ask the question for  $n > 3$ .



Consider a convex  $n$ -gon with the given property. Choose a direction not parallel to any of its sides and, by drawing two parallels in this direction, cut off two vertices together with parts of the sides having them as endpoints, as shown in the figure. (Entire sides are not being cut off.)

We obtain a convex polygon with  $n + 2$  sides, and we may assume that the two newly formed sides have different lengths (clearly the cuts can be performed in a convenient way to ensure this). The new polygon also has the stated properties. Indeed, not all of its sides are equal. Furthermore, the sum of the distances from any interior point to its sides is equal to the sum  $S_1$  of the distances from the same point to the sides of the old polygon plus

the sum  $S_2$  of the distances from the point to the two parallels. But  $S_1$  is a constant according to the induction hypothesis, and  $S_2$  is also a constant (the distance between two fixed parallel lines). Hence  $S = S_1 + S_2$  is a constant, too.

The above argument shows that the property in question is transmitted from  $n$  to  $n + 2$ . In order to prove it by induction, we just need to check the base cases  $n = 4$  and  $n = 5$ . For  $n = 4$  it suffices to take any rectangle. The case  $n = 5$  is, however, not obvious if considered directly. But we can start with an equilateral triangle, which is known to satisfy our conditions, and perform the construction described above, yielding the desired pentagon. ■

Here is an example from the 1968 All-Union Olympiad in the former Soviet Union, where the inductive step is rather nontrivial.

*Prove that each positive integer not exceeding  $n!$  can be represented as the sum of  $n$  or fewer distinct divisors of  $n!$ .*

This is clearly true for  $n = 1$ . Assume that the statement holds for some  $n \geq 1$  and consider any positive integer  $k$  not exceeding  $(n + 1)!$ . Let

$$k = q(n + 1) + r, \quad \text{where } 0 \leq r < n + 1.$$

Since  $q \leq n!$ , the induction hypothesis implies the existence of distinct divisors  $d_1, d_2, \dots, d_m$  of  $n!$  such that  $m \leq n$  and  $q = d_1 + d_2 + \dots + d_m$ . Then  $k$  has a representation of the form

$$k = d_1(n + 1) + d_2(n + 1) + \dots + d_m(n + 1) + r.$$

The sum in the right-hand side contains at most  $n + 1$  summands. It remains to note that, since  $r < n + 1$ , these summands are distinct and each of them divides  $(n + 1)!$  (unless  $r = 0$ , in which case we need not count it as a summand). The proof is finished. ■

*Let  $x_1, x_2, \dots, x_n$ ,  $n \geq 4$ , be nonnegative real numbers that add up to 1. Prove that*

$$x_1x_2 + x_2x_3 + \dots + x_nx_1 \leq \frac{1}{4}.$$

A sketch of a solution first: the conclusion follows from the inequality

$$(x_1 + x_2 + \dots + x_n)^2 \geq 4(x_1x_2 + x_2x_3 + \dots + x_nx_1),$$

which holds for  $n \geq 4$  and  $x_k \geq 0$ ,  $k = 1, 2, \dots, n$ . (Naturally, one has to prove this inequality.)

An alternative solution is by induction on  $n \geq 4$ . Assume  $x_1, x_2, \dots, x_n$  arranged about a circle, so that  $S_n = x_1x_2 + x_2x_3 + \dots + x_nx_1$  can be regarded as the sum of the products of the  $n$  pairs of adjacent numbers. Factoring the expression for  $S_4$  and using the AM–GM inequality for the numbers  $x_1 + x_3, x_2 + x_4$ , we get

$$S_4 = (x_1 + x_3)(x_2 + x_4) \leq \frac{1}{4}(x_1 + x_2 + x_3 + x_4)^2 = \frac{1}{4},$$

with equality if and only if  $x_1 + x_3 = x_2 + x_4$ . Thus the statement holds for  $n = 4$ . Assuming that it is true for  $n - 1$  numbers around the circle, we pass to the case of  $n$  numbers. If  $a, b, c, d$  are four of them, taken in order, and  $a \geq b$  (note that one can always find such a quadruple) then

$$\begin{aligned} ab + bc + cd &\leq ab + ac + cd \\ &\leq ab + ac + bd + cd = a(b + c) + (b + c)d. \end{aligned}$$

Hence we may delete  $b$  and  $c$ , replacing them by  $b + c$ , thus getting a collection of  $n - 1$  nonnegative numbers adding up to 1, for which the cyclic sum in question is not less than the corresponding sum for the initially chosen  $n$  numbers. It remains only to apply the induction hypothesis. ■

Induction is a common tool not only for proving statements but also for constructing mathematical objects with given properties. Let us give an example. It is a problem by **Plamen Dzhakov** published in *Matematika*, No. 6, 1981.

*Let  $p$  be a prime and  $b_0$  an integer,  $0 < b_0 < p$ . Prove that there exists a unique sequence of base  $p$  digits  $b_0, b_1, b_2, \dots, b_n, \dots$  with the following property: If the base  $p$  representation of a number  $x$  ends in the group of digits  $b_nb_{n-1} \dots b_1b_0$  then so does the representation of  $x^p$ .*

We are looking for a sequence  $b_0, b_1, b_2, \dots, b_n, \dots$  of base  $p$  digits such that the numbers  $x_n = b_0 + b_1p + \dots + b_np^n$  and  $x_n^p$  are congruent modulo  $p^{n+1}$  for each  $n = 0, 1, 2, \dots$ . Of course, the choice of the first term  $b_0$  is predetermined, and given in the problem statement; let us note that the numbers  $x_0 = b_0$  and  $x_0^p$  are congruent modulo  $p$  by Fermat's little theorem. Suppose that the base  $p$  digits  $b_1, b_2, \dots, b_n$  are already chosen in such a way that  $x_n^p \equiv x_n \pmod{p^{n+1}}$ . We shall prove that there is a

unique digit  $b_{n+1}$  such that

$$(x_n + b_{n+1}p^{n+1})^p \equiv x_n + b_{n+1}p^{n+1} \pmod{p^{n+2}};$$

this proves the existence and the uniqueness at the same time. Since

$$(x_n + b_{n+1}p^{n+1})^p = x_n^p + \binom{p}{1}x_n^{p-1}b_{n+1}p^{n+1} + Cp^{n+2}$$

for some integer constant  $C$ , and since  $\binom{p}{1}$  is divisible by  $p$ , we get

$$(x_n + b_{n+1}p^{n+1})^p \equiv x_n^p \pmod{p^{n+2}}.$$

Hence  $b_{n+1}$  should satisfy the congruence

$$x_n^p - x_n - b_{n+1}p^{n+1} \equiv 0 \pmod{p^{n+2}}. \quad (1)$$

By the induction hypothesis, the number  $x_n^p - x_n$  is divisible by  $p^{n+1}$ . This implies that its  $(n+2)$ nd base  $p$  digit (from right to left) is indeed the only choice for  $b_{n+1}$  such that (1) holds. The inductive proof is complete. ■

## Coffee Break 2

1. Four cars  $A$ ,  $B$ ,  $C$ ,  $D$  traveled at constant speeds on the same road.  $A$  passed  $B$  and  $C$  at 8 AM and 9 AM, respectively, and met  $D$  at 10 AM;  $D$  met  $B$  and  $C$  at 12 AM and 2 PM, respectively. Determine at what time  $B$  passed  $C$ . (Don't frown at this; it has a nice solution.)

2. Let  $P$  be a polynomial with positive coefficients. Prove that if

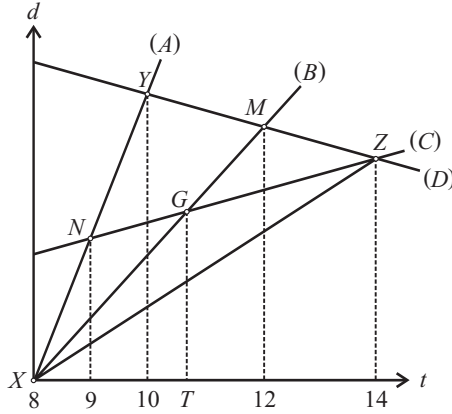
$$P\left(\frac{1}{x}\right) \geq \frac{1}{P(x)}$$

holds for  $x = 1$ , then it holds for every  $x > 0$ .

3. The positive integers are to be partitioned into several subsets  $A_1, A_2, \dots, A_n$  such that, for  $i = 1, 2, \dots, n$ , if  $x \in A_i$  then  $2x \notin A_i$ . What is the minimum value of  $n$ ?

## Solutions

1. Draw the graphs of the movement of the cars (which are straight lines), regarding 8 AM as the initial moment of time.



Suppose that  $B$  passed  $C$  at the moment  $T$ . In the figure, the points  $X, Y, Z, M, N, G$  correspond to meeting or passing between some pair of cars; we omit their exact definitions. It is easy to observe that  $M$  and  $N$  are the midpoints of  $YZ$  and  $XY$ , respectively and, hence,  $XM$  and  $ZN$  are medians in  $\triangle XYZ$ . This implies that their common point  $G$ , corresponding to the passing of  $B$  by  $C$ , is *the centroid of this triangle*. Then the abscissa of  $G$  is the arithmetic mean of the abscissae of  $X, Y, Z$ , that is  $T = \frac{1}{3}(8 + 10 + 14) = 10\frac{2}{3}$ . Thus  $B$  passed  $C$  at 10:40 AM.

2. Let  $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$ , where  $a_k > 0$  for each  $k = 0, 1, \dots, n$ . Then, applying the Cauchy–Schwarz inequality, we obtain for each positive  $x$

$$\begin{aligned} P(x)P\left(\frac{1}{x}\right) &= \left[ \sum_{k=0}^n \left( \sqrt{a_k} x^k \right)^2 \right] \left[ \sum_{k=0}^n \left( \sqrt{a_k} x^{-k} \right)^2 \right] \\ &\geq \left( \sum_{k=0}^n a_k \right)^2 = P^2(1). \end{aligned}$$

Since  $P(1) \geq 1$  by hypothesis, the conclusion follows.

3. The answer is 2. Each positive integer has a unique representation of the form  $2^m k$ , where  $m$  is a nonnegative integer and  $k$  is odd. Put all numbers with odd  $m$  in one group and the ones with even  $m$  in another. These groups meet the requirements, because multiplying by 2 changes the parity of  $m$  in the representation  $2^m k$ .

# 11

## A Highly Divisible Determinant

There is a column in the Romanian high-school journal *Gazeta Matematică* where preparatory problems for international competitions are published. The following problem was posed in this column by **Marius Cavachi** (No. 9, 1987, problem O525):

*Consider 25 arbitrary integers. Prove that they can be arranged in a  $5 \times 5$  matrix whose determinant is divisible by 25.*

A rather more general statement appeared soon after, in *Revista Matematică Timișoara*, No. 1–2, 1988:

*Prove that any  $n^2$  integers can be arranged in an  $n \times n$  matrix whose determinant is divisible by  $n^{\lfloor (n-1)/2 \rfloor}$ .*

For each  $k = 0, 1, \dots, n-1$ , let  $m_k$  be the number of the given integers that are congruent to  $k$  modulo  $n$ . We will prove that “sufficiently many” nonoverlapping pairs can be formed so that the numbers in each pair are congruent to each other modulo  $n$ . There are  $\lfloor m_k/2 \rfloor$  such pairs for each  $k = 0, 1, \dots, n-1$ , hence the total number of the pairs in question is

$$\sum_{k=0}^{n-1} \left\lfloor \frac{m_k}{2} \right\rfloor.$$

Now, since  $m_0 + m_1 + \dots + m_{n-1} = n^2$  and  $\lfloor m_k/2 \rfloor \geq (m_k - 1)/2$  for each  $k = 0, 1, \dots, n-1$ , we obtain

$$\sum_{k=0}^{n-1} \left\lfloor \frac{m_k}{2} \right\rfloor \geq \sum_{k=0}^{n-1} \frac{m_k - 1}{2} = \frac{n^2 - n}{2} \geq n \left\lfloor \frac{n-1}{2} \right\rfloor.$$

This means that it is possible to form at least  $\lfloor (n-1)/2 \rfloor$  pairs of columns with height  $n$  so that the corresponding elements in each pair, the ones in

the same row, are congruent modulo  $n$ . We arrange these pairs in succession to obtain the first  $2\lfloor(n-1)/2\rfloor$  columns of an  $n \times n$  matrix  $A$ , then fill in the remaining column(s) with the numbers that were not included in pairs.

It is not difficult to prove now that the determinant of  $A$  is divisible by  $n^{\lfloor(n-1)/2\rfloor}$ . To do this, subtract each even-numbered column from the previous one. The determinant does not change, and its first  $\lfloor(n-1)/2\rfloor$  odd-numbered columns become divisible by  $n$ . Because any given term in the expansion of the determinant contains some factor from each of these columns, the conclusion follows. ■

# 12

## Hermite's Identity

We begin this section with an identity attributed to Hermite:

$$\lfloor nx \rfloor = \lfloor x \rfloor + \left\lfloor x + \frac{1}{n} \right\rfloor + \cdots + \left\lfloor x + \frac{n-1}{n} \right\rfloor, \quad (1)$$

where  $x$  denotes a real number and  $n$  a positive integer.

There are many proofs for this identity. The one we present here is a personal choice.

Let  $f(x)$  be the difference between the right-hand side and the left-hand side of (1). Then

$$\begin{aligned} f\left(x + \frac{1}{n}\right) &= \left\lfloor x + \frac{1}{n} \right\rfloor + \cdots + \left\lfloor x + \frac{1}{n} + \frac{n-1}{n} \right\rfloor - \left\lfloor n\left(x + \frac{1}{n}\right) \right\rfloor \\ &= \left\lfloor x + \frac{1}{n} \right\rfloor + \cdots + \left\lfloor x + \frac{n-1}{n} \right\rfloor + \lfloor x + 1 \rfloor - \lfloor nx + 1 \rfloor, \end{aligned}$$

and since  $\lfloor x + k \rfloor = \lfloor x \rfloor + k$  for each integer  $k$ , it follows that

$$f\left(x + \frac{1}{n}\right) = f(x)$$

for all real  $x$ . Hence  $f$  is periodic with period  $1/n$ . Thus it suffices to study  $f(x)$  for  $0 \leq x < 1/n$ . But  $f(x) = 0$  for all these values, hence  $f(x) = 0$  for all real  $x$ , and the proof is complete. ■

In our first example we evaluate the sum

$$\sum_{0 \leq i < j \leq n} \left\lfloor \frac{x+i}{j} \right\rfloor,$$

where  $x$  is a real number.

Denote the sum in question by  $S_n$ . Then

$$\begin{aligned} S_n - S_{n-1} &= \left\lfloor \frac{x}{n} \right\rfloor + \left\lfloor \frac{x+1}{n} \right\rfloor + \cdots + \left\lfloor \frac{x+n-1}{n} \right\rfloor \\ &= \left\lfloor \frac{x}{n} \right\rfloor + \left\lfloor \frac{x}{n} + \frac{1}{n} \right\rfloor + \cdots + \left\lfloor \frac{x}{n} + \frac{n-1}{n} \right\rfloor, \end{aligned}$$

and, according to (1),

$$S_n - S_{n-1} = \left\lfloor n \cdot \frac{x}{n} \right\rfloor = \lfloor x \rfloor.$$

Because  $S_1 = \lfloor x \rfloor$ , it follows that  $S_n = n \lfloor x \rfloor$  for all  $n$ . ■

The next problem comes from the 1968 IMO:

*Prove that, for all positive integers  $n$ ,*

$$\left\lfloor \frac{n+1}{2} \right\rfloor + \left\lfloor \frac{n+2}{2^2} \right\rfloor + \cdots + \left\lfloor \frac{n+2^k}{2^{k+1}} \right\rfloor + \cdots = n.$$

We rewrite the equality as

$$\left\lfloor \frac{n}{2} + \frac{1}{2} \right\rfloor + \left\lfloor \frac{n}{2^2} + \frac{1}{2} \right\rfloor + \cdots + \left\lfloor \frac{n}{2^{k+1}} + \frac{1}{2} \right\rfloor + \cdots = n,$$

and use a special case of Hermite's identity ( $n = 2$ ):

$$\left\lfloor x + \frac{1}{2} \right\rfloor = \lfloor 2x \rfloor - \lfloor x \rfloor.$$

This allows us to write the equality as

$$\lfloor n \rfloor - \left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{2} \right\rfloor - \left\lfloor \frac{n}{2^2} \right\rfloor + \cdots + \left\lfloor \frac{n}{2^k} \right\rfloor - \left\lfloor \frac{n}{2^{k+1}} \right\rfloor + \cdots = n.$$

The sum telescopes, and  $\lfloor n/2^{k+1} \rfloor = 0$  for large enough  $k$ 's. ■

We continue with a similar problem:

*Evaluate the difference between the numbers*

$$\sum_{k=0}^{2000} \left\lfloor \frac{3^k + 2000}{3^{k+1}} \right\rfloor \quad \text{and} \quad \sum_{k=0}^{2000} \left\lfloor \frac{3^k - 2000}{3^{k+1}} \right\rfloor.$$

We can write each term of the difference in question as

$$\left\lfloor \frac{1}{3} + v_k \right\rfloor - \left\lfloor \frac{1}{3} - v_k \right\rfloor,$$

where  $v_k = 2000/3^{k+1}$ . Since  $-\lfloor u \rfloor = \lfloor -u \rfloor + 1$  for each nonintegral value of  $u$ , and since  $\frac{1}{3} - v_k$  is never an integer, we have to examine the sum

$$\sum_{k=0}^{2000} \left( \left\lfloor v_k + \frac{1}{3} \right\rfloor + \left\lfloor v_k - \frac{1}{3} \right\rfloor + 1 \right).$$

Taking  $n = 3$  and  $x = v - \frac{1}{3}$  in (1) yields

$$\left\lfloor v + \frac{1}{3} \right\rfloor + \left\lfloor v - \frac{1}{3} \right\rfloor + 1 = \lfloor 3v \rfloor - \lfloor v \rfloor.$$

Hence the desired difference becomes

$$\sum_{k=0}^{2000} \left( \left\lfloor \frac{2000}{3^k} \right\rfloor - \left\lfloor \frac{2000}{3^{k+1}} \right\rfloor \right)$$

and telescopes to

$$\lfloor 2000 \rfloor - \left\lfloor \frac{2000}{3} \right\rfloor + \left\lfloor \frac{2000}{3} \right\rfloor - \left\lfloor \frac{2000}{3^2} \right\rfloor + \cdots = 2000. \blacksquare$$

We close the section with a 1981 USAMO problem:

*If  $x$  is a positive real number and  $n$  a positive integer, prove that*

$$\lfloor nx \rfloor \geq \frac{\lfloor x \rfloor}{1} + \frac{\lfloor 2x \rfloor}{2} + \cdots + \frac{\lfloor nx \rfloor}{n}.$$

It can be solved by using Hermite's identity, but we choose to present a different (and shorter) proof instead. Denote the quantity on the right-hand side by  $S_n$ . Then, setting additionally  $S_0 = 0$ , we get

$$S_n - S_{n-1} = \frac{\lfloor nx \rfloor}{n} \quad \text{for all } n = 1, 2, \dots$$

So,

$$k(S_k - S_{k-1}) = \lfloor kx \rfloor \quad \text{for } k = 1, 2, \dots, n+1.$$

Adding up these  $n+1$  equations gives

$$-S_1 - S_2 - \cdots - S_n + (n+1)S_{n+1} = \lfloor (n+1)x \rfloor + \lfloor nx \rfloor + \cdots + \lfloor x \rfloor.$$

We now proceed by induction. The base  $n = 1$  is clear. Assuming that  $S_k \leq \lfloor kx \rfloor$  for  $1 \leq k \leq n$ , we express  $(n+1)S_n$  from the previous

equation and obtain

$$(n+1)S_{n+1} \leq \lfloor (n+1)x \rfloor + (\lfloor nx \rfloor + \lfloor x \rfloor) + (\lfloor (n-1)x \rfloor + \lfloor 2x \rfloor) \\ + \cdots + (\lfloor x \rfloor + \lfloor nx \rfloor).$$

Using  $n$  times the fact that  $\lfloor u \rfloor + \lfloor v \rfloor \leq \lfloor u+v \rfloor$  for all real  $u, v$  yields  $(n+1)S_{n+1} \leq (n+1) \lfloor (n+1)x \rfloor$ , and the claim is proved. ■

# 13

## Complete Sequences

A sequence  $(a_m)_{m \geq 1}$  with positive integer terms is called *totally complete* if each positive integer can be expressed as a sum of distinct terms of this sequence. The most popular sufficient condition for total completeness was proposed as a problem at the 1960 Kürschák competition. This Hungarian mathematical contest is considered to be the oldest one in the world.

*Every positive integer sequence  $a_1, a_2, \dots, a_n, \dots$  satisfying the conditions  $a_1 = 1$  and*

$$a_{n+1} \leq 1 + a_1 + a_2 + \dots + a_n, \quad n = 1, 2, \dots,$$

*is totally complete.*

Actually, we will prove by induction on  $n$  that each positive integer not exceeding  $a_1 + a_2 + \dots + a_n$  can be expressed as a sum of distinct terms chosen from  $a_1, a_2, \dots, a_n$ .

This is obvious for  $n = 1$ . Assume that it holds for some  $n \geq 1$  and let  $k$  be a positive integer such that  $k \leq a_1 + a_2 + \dots + a_n + a_{n+1}$ . The case  $k \leq a_1 + a_2 + \dots + a_n$  is directly settled by the induction hypothesis, so we may assume that

$$1 + a_1 + a_2 + \dots + a_n \leq k \leq a_1 + a_2 + \dots + a_n + a_{n+1}.$$

By hypothesis, the left-hand side is greater than or equal to  $a_{n+1}$ , thus  $0 \leq k - a_{n+1} \leq a_1 + a_2 + \dots + a_n$ . We are done if  $k - a_{n+1} = 0$ . And, if  $k - a_{n+1} > 0$ , then, by the induction hypothesis,  $k - a_{n+1}$  is expressible as a sum of distinct terms among  $a_1, a_2, \dots, a_n$ . In this case  $k$  can be expressed as a sum of distinct terms chosen from  $a_1, a_2, \dots, a_{n+1}$ , and the induction is complete. ■

Here is an application of this statement, a problem from the 1975 West German Olympiad.

*Two brothers inherited  $n$  pieces of gold with total weight  $2n$ . Each piece has integer weight and the heaviest of them is not heavier than the remaining ones combined. Prove that if  $n$  is even then the brothers can divide the inheritance into two parts with equal weights.*

Denote the given integer weights by  $a_1, a_2, \dots, a_n$  and suppose, without loss of generality, that  $a_1 \leq a_2 \leq \dots \leq a_n$ . By hypothesis,

$$a_1 + a_2 + \dots + a_n = 2n \quad \text{and}$$

$$a_n \leq a_1 + a_2 + \dots + a_{n-1}.$$

Everything is clear if  $a_n = a_1 + a_2 + \dots + a_{n-1}$ . The case  $a_1 \geq 2$  is trivial as well, because then  $a_i \geq 2$  for all  $i$  which, in view of the condition  $a_1 + a_2 + \dots + a_n = 2n$ , implies  $a_1 = a_2 = \dots = a_n = 2$ . Since  $n$  is even, the required division is easy.

So we may assume that  $a_1 = 1$  and  $a_n < a_1 + a_2 + \dots + a_{n-1}$ . Still more,

$$a_n \leq 1 + a_1 + a_2 + \dots + a_{n-1}.$$

In view of the sufficient condition proved above, all we need is to show that

$$a_{k+1} \leq 1 + a_1 + a_2 + \dots + a_k, \quad k = 1, 2, \dots, n-2.$$

Assume that this is not true and let  $a_{k+1} > 1 + a_1 + a_2 + \dots + a_k$  for some  $k \in \{1, 2, \dots, n-2\}$ . Then  $a_{k+1} \geq k+2$ , so  $a_i \geq k+2$  for each  $i = k+1, k+2, \dots, n$ . Also, we have  $a_i \geq 1$  for  $i = 1, 2, \dots, k$ , hence

$$\begin{aligned} 2n &= a_1 + a_2 + \dots + a_n \geq k + (n-k)(k+2) \\ &= -k^2 + (n-1)k + 2n. \end{aligned}$$

This implies  $k^2 - (n-1)k \geq 0$ , so  $k \leq 0$  or  $k \geq n-1$ . Either of these contradicts  $k \in \{1, 2, \dots, n-2\}$ , and we are done. ■

This argument also shows that in the nontrivial case, when not all of the  $a_i$ 's are equal to 2, each positive integer not exceeding  $2n$  is expressible as a sum of several of the numbers  $a_1, a_2, \dots, a_n$ . In this case the assumption that  $n$  is even is redundant.

In their book *Selected Chapters from Number Theory* (Budapest, 1960) Erdős and Surányi considered the following problem:

*Every integer  $n$  can be represented in infinitely many ways as*

$$n = \pm 1^2 \pm 2^2 \pm \cdots \pm k^2 \quad (1)$$

*for a convenient  $k$  and a suitable choice of the signs  $+$  and  $-$ .*

It suffices to prove the statement for nonnegative  $n$ 's, because for negative  $n$ 's we can simply change all the signs. The proof goes by induction. Here are representations of the form (1) for 0, 1, 2 and 3:

$$0 = 1^2 + 2^2 - 3^2 + 4^2 - 5^2 - 6^2 + 7^2;$$

$$1 = 1^2;$$

$$2 = -1^2 - 2^2 - 3^2 + 4^2;$$

$$3 = -1^2 + 2^2.$$

If  $n$  is representable in the form (1) then so is  $n + 4$ , because 4 can be written as

$$4 = (k+1)^2 - (k+2)^2 - (k+3)^2 + (k+4)^2 \quad (2)$$

for any  $k$ . It follows inductively that a representation of the form (1) can be written for any nonnegative integer  $n$ . From (2) it also follows that

$$\begin{aligned} & (k+1)^2 - (k+2)^2 - (k+3)^2 + (k+4)^2 \\ & - (k+5)^2 + (k+6)^2 + (k+7)^2 - (k+8)^2 = 0 \end{aligned}$$

for any  $k$ , hence it can be easily inferred that the number of representations of an integer in the form (1) is infinite. ■

# 14

## Three Polynomials

The idea that brings together the problems we are about to present is this: A polynomial of degree  $n$  is completely determined by the values it takes on at  $n + 1$  distinct points.

Our first problem was published by **Murray Klamkin** in *Pi Mu Epsilon*, Vol. 6, 1964:

*Let  $P(x)$  be a polynomial of degree  $n$  such that  $P(k) = 2^k$  for each  $k = 0, 1, 2, \dots, n$ . Determine  $P(n + 1)$ .*

The well-known identity

$$2^k = \binom{k}{0} + \binom{k}{1} + \cdots + \binom{k}{k}$$

suggests considering the polynomial

$$Q(x) = \binom{x}{0} + \binom{x}{1} + \cdots + \binom{x}{n}.$$

Since  $\binom{k}{m} = 0$  for  $m > k$ , we have  $Q(k) = 2^k$  for  $k = 0, 1, 2, \dots, n$ . The summand of highest degree in  $Q(x)$  is

$$\binom{x}{n} = \frac{x(x-1)(x-2)\cdots(x-n+1)}{n!},$$

so  $Q(x)$  is of degree  $n$ . Since  $P(x)$  and  $Q(x)$  are both polynomials of degree  $n$  and take on the same values at  $n + 1$  distinct points, they are identical. Hence

$$P(n+1) = \binom{n+1}{0} + \binom{n+1}{1} + \cdots + \binom{n+1}{n} = 2^{n+1} - 1. \blacksquare$$

We continue with a Romanian submission for the 1983 IMO:

Let  $F_n$  be the  $n$ th Fibonacci number (that is,  $F_1 = F_2 = 1$ ,  $F_{n+1} = F_n + F_{n-1}$  for  $n \geq 2$ ), and let  $P(x)$  be the polynomial of degree 990 such that  $P(k) = F_k$  for  $k = 992, 993, \dots, 1982$ . Prove that  $P(1983) = F_{1983} - 1$ .

Denote by  $P_n(x)$  the (unique) polynomial of degree  $n$  such that

$$P_n(k) = F_k \quad \text{for } k = n+2, n+3, \dots, 2n+2. \quad (1)$$

We are going to show that  $P_n(2n+3) = F_{2n+3} - 1$  for all  $n \geq 0$ .

Clearly  $P_0(x) = 1$  and the claim is true for  $n = 0$ . Suppose it holds for  $P_{n-1}(x)$ , and consider  $P_n(x)$ . The polynomial

$$Q(x) = P_n(x+2) - P_n(x+1)$$

has degree at most  $n-1$ . In view of (1),

$$Q(k) = P_n(k+2) - P_n(k+1) = F_{k+2} - F_{k+1} = F_k$$

for each  $k = n+1, n+2, \dots, 2n$ . Therefore  $Q(x)$  and  $P_{n-1}(x)$  agree at  $n$  distinct points, and hence  $Q(x) = P_{n-1}(x)$  for all  $x$ . In other words,  $P_n(x+2) = P_n(x+1) + P_{n-1}(x)$  for all  $x$ . Combined with the induction hypothesis  $P_{n-1}(2n+1) = F_{2n+1} - 1$ , this implies

$$P_n(2n+3) = F_{2n+2} + F_{2n+1} - 1 = F_{2n+3} - 1.$$

The induction is complete. ■

Finally, we consider an Italian proposal for the 1997 IMO. The solution is due to the Argentine student **German Muller**.

Let  $p$  be a prime number and let  $f(x)$  be a polynomial of degree  $d$  with integer coefficients such that:

- (i)  $f(0) = 0, \quad f(1) = 1$ ;
- (ii) for every positive integer  $n$ ,  $f(n)$  is congruent to either 0 or 1 modulo  $p$ .

Prove that  $d \geq p-1$ .

Assume, by way of contradiction, that  $d \leq p-2$ . This is already a big step: the polynomial  $f(x)$  is then entirely determined by its values at  $0, 1, \dots, p-2$ . By the Lagrange interpolation formula, we have

$$f(x) = \sum_{k=0}^{p-2} f(k) \frac{x(x-1) \cdots (x-k+1)(x-k-1) \cdots (x-p+2)}{k!(-1)^{p-k}(p-k-2)!}$$

for any  $x$ . Setting  $x = p - 1$  yields

$$f(p-1) = \sum_{k=0}^{p-2} f(k) \frac{(p-1) \cdots (p-k)}{(-1)^{p-k} k!} = \sum_{k=0}^{p-2} f(k) (-1)^{p-k} \binom{p-1}{k}. \quad (2)$$

A simple induction on  $k$  shows that if  $p$  is a prime and  $0 \leq k \leq p-1$ , then

$$\binom{p-1}{k} \equiv (-1)^k \pmod{p}. \quad (3)$$

Indeed, this is certainly true for  $k = 0$ . Choose a  $k$  with  $1 \leq k \leq p-1$  and assume that  $\binom{p-1}{k-1} \equiv (-1)^{k-1} \pmod{p}$ . It is a well-known fact that

$$\binom{p-1}{k} = \binom{p}{k} - \binom{p-1}{k-1}$$

and that  $\binom{p}{k}$  is divisible by  $p$ . Thus the induction hypothesis implies (3).

From (2) and (3) we obtain

$$f(p-1) \equiv (-1)^p \sum_{k=0}^{p-2} f(k) \pmod{p}.$$

This can be rewritten as

$$f(0) + f(1) + \cdots + f(p-1) \equiv 0 \pmod{p} \quad (4)$$

for any prime  $p$  (odd or not).

The above argument, and hence the relation (4), holds for *each* polynomial with integer coefficients of degree not exceeding  $p-2$ . Now it suffices to show that (4) contradicts the conditions (i) and (ii).

From (ii),  $f(0) + f(1) + \cdots + f(p-1) \equiv k \pmod{p}$ , where  $k$  denotes the number of those  $n \in \{0, 1, \dots, p-1\}$  for which  $f(n) \equiv 1 \pmod{p}$ . On the other hand, (i) implies  $k \leq p-1$  and  $k \geq 1$ , so

$$f(0) + f(1) + \cdots + f(p-1) \not\equiv 0 \pmod{p}.$$

The proof is complete. ■

# 15

## More about Induction

The principle of mathematical induction is not a magical tool; that is, not every statement  $P(n)$  about a positive integer,  $n$ , can be proven by first checking the base case and then showing that, for any such  $n$ ,  $P(n)$  implies  $P(n+1)$ .

Take, for example, the inequality

$$\frac{1}{2} \cdot \frac{3}{4} \cdot \dots \cdot \frac{2n-1}{2n} < \frac{1}{\sqrt{3n}},$$

where  $n$  is a positive integer. After checking the base case ( $n = 1$ ) we will want to show that  $P(n)$  implies  $P(n+1)$ . This comes down to the inequality

$$\frac{2n+1}{2(n+1)} < \frac{\sqrt{3n}}{\sqrt{3(n+1)}},$$

which is not true. (Well, it barely fails:  $4n^2 + 4n + 1$  is not less than  $4n^2 + 4n$ .)

We are stuck but not hopeless. The point here is that the inductive assumption is not good enough to ensure success in our attempt to pass the inductive step. In search of a better assumption, let us refine our inequality a bit:

$$\frac{1}{2} \cdot \frac{3}{4} \cdot \dots \cdot \frac{2n-1}{2n} \leq \frac{1}{\sqrt{3n+1}}.$$

The base case is still satisfied, and now the inductive step passes, because

$$\frac{2n+1}{2(n+1)} < \frac{\sqrt{3n+1}}{\sqrt{3(n+1)+1}}$$

holds for any  $n \geq 1$ . ■

Going for more by induction (trying to prove a stronger statement) is at times more rewarding. This is what Pólya calls *the researcher's paradox*. We indeed have to prove more; but the induction hypothesis is stronger, so it gives us more, too.

The above example is not unique. Consider, for instance, the inequality

$$\frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} < \frac{3}{4}.$$

Straightforward induction does not work. So let us try a stronger statement of the type

$$S_n + a_n \leq \frac{3}{4} \quad (n \geq 2),$$

where  $a_n > 0$  and  $S_n$  is the sum on the left-hand side. Induction will be possible if the  $a_n$ 's satisfy two conditions.

First,  $a_2$  has to be chosen so that the base case holds; that is,

$$\frac{1}{4} + a_2 \leq \frac{3}{4} \quad \text{or} \quad a_2 \leq \frac{1}{2}. \quad (1)$$

Next, the inductive step requires that the condition  $S_n + a_n \leq \frac{3}{4}$  implies  $S_{n+1} + a_{n+1} \leq \frac{3}{4}$ . Since  $S_{n+1} = S_n + [1/(n+1)^2]$ , this will be ensured if  $a_n$  and  $a_{n+1}$  satisfy

$$\frac{1}{(n+1)^2} - a_n + a_{n+1} \leq 0$$

or

$$a_n - a_{n+1} \geq \frac{1}{(n+1)^2} \quad (n \geq 2). \quad (2)$$

Now it is not hard to guess that the numbers  $a_n = 1/n$  satisfy (2) for  $n \geq 2$ , for, as is well known,

$$\frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)},$$

and this is greater than  $1/(n+1)^2$ . More than that,  $a_2 = \frac{1}{2}$  satisfies (1), so the numbers  $a_n = 1/n$  will do the job. We do not even need to repeat the inductive argument. It will work for any choice of  $a_2, a_3, \dots, a_n, \dots$  such that (1) and (2) hold. ■

The inequality

$$\frac{1}{2\sqrt{1}} + \frac{1}{3\sqrt{2}} + \cdots + \frac{1}{(n+1)\sqrt{n}} < 2$$

can be treated in a similar fashion. Refining it to

$$\sum_{k=1}^n \frac{1}{(k+1)\sqrt{k}} < 2 - \frac{2}{\sqrt{n+1}}$$

makes the base case still hold and the inductive step successful:

$$\frac{1}{(n+2)\sqrt{n+1}} < \frac{2}{\sqrt{n+1}} - \frac{2}{\sqrt{n+2}}$$

reduces to

$$\frac{1}{\sqrt{n+2}} < 2(\sqrt{n+2} - \sqrt{n+1}) = \frac{2}{\sqrt{n+2} + \sqrt{n+1}},$$

which is clearly true. ■

As a matter of fact, induction is not the only approach here. Some skillful algebraic manipulations make the left-hand side less than

$$\begin{aligned} \sum_{k=1}^n \frac{2}{(k+1)\sqrt{k} + k\sqrt{k+1}} &= 2 \sum_{k=1}^n \frac{1}{\sqrt{k+1}\sqrt{k}} \cdot \frac{1}{\sqrt{k+1} + \sqrt{k}} \\ &= 2 \sum_{k=1}^n \frac{\sqrt{k+1} - \sqrt{k}}{\sqrt{k+1} \cdot \sqrt{k}} \\ &= 2 \sum_{k=1}^n \left( \frac{1}{\sqrt{k}} - \frac{1}{\sqrt{k+1}} \right), \end{aligned}$$

which telescopes to  $2 - (2/\sqrt{n+1})$ .

This idea does not work just for inequalities. For example, here is one more solution of the last problem in Section 1:

*Prove that, for each  $n \geq 3$ , the number  $n!$  can be represented as the sum of  $n$  distinct divisors of itself.*

Strengthening the statement, by imposing the condition that *one of the  $n$  divisors should be 1*, puts us in a winning position. The question here is how we came to think of this. Well, there is just about one way to go in using the induction hypothesis  $n! = d_1 + d_2 + \cdots + d_n$  (where  $d_1, d_2, \dots, d_n$  are the  $n$  divisors arranged in increasing order); namely, multiplying the above relation by  $n+1$ . This yields

$$\begin{aligned} (n+1)! &= (n+1)d_1 + (n+1)d_2 + \cdots + (n+1)d_n \\ &= d_1 + nd_1 + (n+1)d_2 + \cdots + (n+1)d_n. \end{aligned}$$

We split  $(n+1)d_1$  into  $d_1 + nd_1$ , thus getting  $n+1$  summands, as needed. Of them, only the second one might not be a divisor of  $(n+1)!$ . We would

like to ensure that it is such a divisor, too. Hence the idea of insisting that  $d_1 = 1$ . ■

We close the section with a combinatorial geometry problem.

*Several nonintersecting diagonals are drawn in a convex polygon  $P$ . Prove that there exist two vertices of  $P$  that are not endpoints of any of these diagonals.*

Straightforward induction does not work here. Indeed, the claim is obviously true for  $n = 3$  and  $n = 4$ , but the inductive step fails. All we can do is to assume that everything works for all  $k$  less than  $n$ , to take an  $n$ -gon with nonintersecting diagonals drawn, and to consider the two polygons  $P_1$  and  $P_2$  into which  $P$  is divided by one of the diagonals, say  $AB$ . The sides of  $P_1$  and  $P_2$  indeed number less than  $n$  and one could claim that each of them has two “free” vertices. We are tempted to infer from here that  $P$  also has two free vertices. But it can happen—bad luck!—that the free vertices in both  $P_1$  and  $P_2$  are exactly  $A$  and  $B$ . If so, we are not able to proceed further, because they are connected with a diagonal in  $P$ .

But there is a way out, and it is inductive. We shall prove the following stronger statement:

*If several nonintersecting diagonals are drawn in an  $n$ -sided polygon  $P$  then there exist two *nonadjacent* vertices of  $P$  that are not endpoints of any of these diagonals.*

This clearly holds for  $n = 4$ , where  $P$  is a quadrilateral and no more than one diagonal may be drawn. Assume that the stronger statement is true for all  $k < n$ , and consider an  $n$ -gon  $P$  in which several nonintersecting diagonals are drawn. Let  $AB$  be one of them. It partitions  $P$  into two polygons  $P_1$  and  $P_2$  with smaller numbers of sides. Each of them contains two nonadjacent free vertices according to the induction hypothesis. No trouble can occur now, since at least one of the free vertices in  $P_1$  is neither  $A$  nor  $B$ ; indeed,  $A$  and  $B$  are adjacent in  $P_1$ . Label this vertex  $C$ . Naturally, the same argument applies to  $P_2$ , so it has a free vertex  $D$  distinct from  $A$  and  $B$ . It is obvious that  $C$  and  $D$  are nonadjacent free vertices of  $P$  and the proof of the stronger statement is completed. Of course, the initial statement follows immediately.

Rigorously speaking, we omitted the case when one of the polygons  $P_1$  and  $P_2$ , say  $P_1$ , is a triangle  $ABC$ . The induction hypothesis does not include this case. But it is evident that then  $C$  is a free vertex for both  $P_1$  and  $P$ , and the proof can go on as above.

Let us note finally that this proof does not depend on the convexity or nonconvexity of  $P$  (provided the diagonals drawn lie in its interior). ■

## Coffee Break 3

1. Prove that each nonnegative integer can be represented in the form  $a^2 + b^2 - c^2$ , where  $a, b, c$  are positive integers with  $a < b < c$ .
2. Consider all quadratic equations  $x^2 + px + q = 0$ , where the coefficients  $p$  and  $q$  range, independently of one another, over the interval  $[-1, 1]$ . Find all possible values of the real roots of these equations.
3. All the edges of a tetrahedron subtend the same angle at a point interior to the tetrahedron. Find all possible values of this angle.

## Solutions

1. Let  $k$  be a nonnegative integer. If  $k$  is even, say  $k = 2n$ , consider the identity

$$2n = (3n)^2 + (4n - 1)^2 - (5n - 1)^2.$$

Since  $3n < 4n - 1 < 5n - 1$  for  $n > 1$  and

$$0 = 3^2 + 4^2 - 5^2, \quad 2 = 5^2 + 11^2 - 12^2,$$

we are done with this case.

If  $k$  is odd, we use the identity

$$2n + 3 = (3n + 2)^2 + (4n)^2 - (5n + 1)^2,$$

where, for  $n > 2$ ,  $3n + 2 < 4n < 5n + 1$ . Since

$$1 = 4^2 + 7^2 - 8^2,$$

$$3 = 4^2 + 6^2 - 7^2,$$

$$5 = 4^2 + 5^2 - 6^2,$$

$$7 = 6^2 + 14^2 - 15^2,$$

we have exhausted the case  $k$  odd as well.

2. Let  $p, q \in [-1, 1]$  and let  $x_0$  be a real root of the quadratic equation  $x^2 + px + q = 0$ . Then

$$x_0 = \frac{-p \pm \sqrt{p^2 - 4q}}{2} \leq \frac{1 + \sqrt{1 + 4 \cdot 1}}{2} = \frac{1 + \sqrt{5}}{2}.$$

Note that  $x_0$  can actually equal  $(1 + \sqrt{5})/2$  if  $p = q = -1$  (and only in this case). Now comes the key observation:

If  $y$  is a real root of an equation of the form  $t^2 + pt + q = 0$  with  $p, q \in [-1, 1]$ , then so is any number  $z$  not exceeding  $y$  in absolute value.

Indeed, let  $y^2 + py + q = 0$  for some  $p, q \in [-1, 1]$ , and let  $z = \alpha y$ , where  $|\alpha| \leq 1$ . The equation  $t^2 + \alpha pt + \alpha^2 q = 0$  has its coefficients  $\alpha p$  and  $\alpha^2 q$  in the interval  $[-1, 1]$ , because  $|\alpha| \leq 1$ . In addition,  $z$  is a root of this equation, since

$$\begin{aligned} z^2 + \alpha pz + \alpha^2 q &= (\alpha y)^2 + \alpha p(\alpha y) + \alpha^2 q \\ &= \alpha^2(y^2 + py + q) = 0. \end{aligned}$$

All this implies that the values in question are the numbers in the interval

$$\left[ -\frac{1+\sqrt{5}}{2}, \frac{1+\sqrt{5}}{2} \right].$$

Yet another appearance of the remarkable golden ratio!

3. Having in mind the analogous situation in the plane, one's first guess is that there is only one possible value for the angle in question, and this is indeed the case. Let  $O$  be a point inside the tetrahedron  $ABCD$  such that

$$\angle AOB = \angle BOC = \angle COA = \angle DOA = \angle DOB = \angle DOC = \varphi.$$

Choose the points  $A_1, B_1, C_1, D_1$  on the rays  $\overrightarrow{OA}, \overrightarrow{OB}, \overrightarrow{OC}, \overrightarrow{OD}$ , respectively, so that

$$OA_1 = OB_1 = OC_1 = OD_1.$$

The six triangles  $OA_1B_1, OB_1C_1, OC_1A_1, OD_1A_1, OD_1B_1$  and  $OD_1C_1$  are congruent; moreover, the new tetrahedron  $A_1B_1C_1D_1$  is *regular*, and  $O$  is its circumcenter. This means that  $\varphi$  has indeed only one possible value: it must equal the angle subtended by an edge of a regular tetrahedron at its circumcenter. A standard computation yields  $\varphi = \arccos\left(-\frac{1}{3}\right)$ .

# 16

## A Classical Identity

Students are often invited to find patterns for the sums

$$S_n(m) = 1^m + 2^m + \cdots + n^m, \quad m = 1, 2, 3, \dots,$$

and to verify them by mathematical induction. Doing this, they come to know the identity

$$1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2. \quad (1)$$

This relation is truly surprising and one might ask if there are any other positive integer sequences  $(a_n)$  that satisfy an analogous relation.

*The only infinite sequence  $a_1, a_2, \dots, a_n, \dots$  of positive numbers (not necessarily integers) such that the equality*

$$a_1^3 + a_2^3 + \cdots + a_n^3 = (a_1 + a_2 + \cdots + a_n)^2 \quad (2)$$

*holds for every positive integer  $n$  is the sequence given by  $a_n = n$  for  $n = 1, 2, 3, \dots$*

More exactly, we shall see that if some sequence of positive numbers  $a_1, a_2, \dots, a_n, \dots$  satisfies (2) for  $n = 1, 2, \dots, k$ , then  $a_k = k$ .

To prove this, we proceed by induction. Evidently this is true for  $k = 1$ , because  $a_1^3 = a_1^2$  and  $a_1 > 0$  lead to  $a_1 = 1$ . Assuming that the statement holds for  $1, 2, \dots, k$ , we will prove that it also holds for  $k + 1$ .

So, let the equality (2) be satisfied for  $n = 1, 2, \dots, k, k + 1$ . Then the induction hypothesis shows that  $a_1 = 1, a_2 = 2, \dots, a_k = k$ , and the condition (2) for  $n = k + 1$  becomes

$$1^3 + 2^3 + \cdots + k^3 + a_{k+1}^3 = (1 + 2 + \cdots + k + a_{k+1})^2.$$

Expanding the right-hand side, we get

$$\begin{aligned} (1^3 + 2^3 + \cdots + k^3) + a_{k+1}^3 \\ = (1 + 2 + \cdots + k)^2 + 2(1 + 2 + \cdots + k)a_{k+1} + a_{k+1}^2. \end{aligned}$$

The first addends on the two sides can be cancelled out in view of the formula (1), so

$$a_{k+1}^3 = 2(1 + 2 + \cdots + k)a_{k+1} + a_{k+1}^2.$$

Replacing  $1 + 2 + \cdots + k$  by  $[k(k+1)]/2$  and dividing by  $a_{k+1} \neq 0$ , we obtain

$$a_{k+1}^2 - a_{k+1} - k(k+1) = 0.$$

This is a quadratic in  $a_{k+1}$ , with roots  $-k$  and  $k+1$ . Since  $a_{k+1}$  is positive, we deduce that  $a_{k+1} = k+1$ . This finishes the inductive proof. ■

It is natural to conjecture that all solutions of the equation

$$x_1^3 + x_2^3 + \cdots + x_n^3 = (x_1 + x_2 + \cdots + x_n)^2$$

in distinct positive integers are permutations of the numbers  $1, 2, \dots, n$ . Proving this directly turns out to be an easy task. In the spirit of the previous section, we prove by induction the following refined statement:

*If  $a_1 < a_2 < \cdots < a_n$  are positive integers, then*

$$a_1^3 + a_2^3 + \cdots + a_n^3 \geq (a_1 + a_2 + \cdots + a_n)^2,$$

*with equality if and only if  $a_k = k$  for each  $k = 1, 2, \dots, n$ .*

For  $n = 1$ ,  $a_1 \geq 1$  implies  $a_1^3 \geq a_1^2$ , and  $a_1^3 = a_1^2$  if and only if  $a_1 = 1$ . Suppose the claim is true for some  $n = k$ , and consider  $k+1$  positive integers  $a_1 < a_2 < \cdots < a_k < a_{k+1}$ . We have  $a_{k+1} \geq a_k + 1$ , hence

$$\frac{(a_{k+1} - 1)a_{k+1}}{2} \geq \frac{a_k(a_k + 1)}{2} = 1 + 2 + \cdots + a_k.$$

Note that the sum  $1 + 2 + \cdots + a_k$  contains all positive integers not exceeding  $a_k$ , so it is greater than or equal to  $a_1 + a_2 + \cdots + a_k$ , a sum of distinct integers among  $1, 2, \dots, a_k$ . So

$$\frac{(a_{k+1} - 1)a_{k+1}}{2} \geq a_1 + a_2 + \cdots + a_k$$

which, multiplied by  $2a_{k+1}$ , gives

$$(a_{k+1}^2 - a_{k+1}) a_{k+1} \geq 2(a_1 + a_2 + \cdots + a_k) a_{k+1},$$

that is,

$$a_{k+1}^3 \geq 2(a_1 + a_2 + \cdots + a_k) a_{k+1} + a_{k+1}^2.$$

On the other hand, by the induction hypothesis,

$$a_1^3 + a_2^3 + \cdots + a_k^3 \geq (a_1 + a_2 + \cdots + a_k)^2.$$

Adding up the last two inequalities yields

$$a_1^3 + a_2^3 + \cdots + a_k^3 + a_{k+1}^3 \geq (a_1 + a_2 + \cdots + a_k + a_{k+1})^2,$$

hence the inequality is true for  $n = k + 1$ .

It is not difficult to infer from the above argument that equality holds only if

$$a_{k+1} = a_k + 1 \quad \text{and} \quad a_1^3 + a_2^3 + \cdots + a_k^3 = (a_1 + a_2 + \cdots + a_k)^2.$$

In view of the last equation, the induction hypothesis implies that if equality holds, then  $a_1 = 1, a_2 = 2, \dots, a_k = k$ . Then  $a_{k+1} = a_k + 1$  gives  $a_{k+1} = k + 1$ . That is, our sequence is  $a_i = i, i = 1, 2, \dots, k + 1$ . Conversely, equality holds for  $a_1 = 1, a_2 = 2, \dots, a_k = k, a_{k+1} = k + 1$  in view of the classical formula (1). The inductive proof is complete. ■

# 17

## Multiplicative Functions

A function  $f$  defined over the positive integers is called *multiplicative* if  $f(1) = 1$  and

$$f(mn) = f(m)f(n)$$

for every two relatively prime numbers  $m$  and  $n$ . In most number theory books multiplicative functions are presented as objects of purely theoretical value, and it is rare that they solve a “living problem.” Here we consider examples where this happens.

A general observation is needed. Let  $f$  be a multiplicative function and  $n$  a positive integer. Assume, in what follows, that  $n$  has prime factorization  $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ . Then

$$\sum_{d|n} f(d) = \prod_{i=1}^k [1 + f(p_i) + f(p_i^2) + \cdots + f(p_i^{\alpha_i})], \quad (1)$$

where the sum ranges over all positive divisors of  $n$ . This sum is sometimes called *the sum function* of  $f$ . Informally, (1) shows how the sum function is expressed in terms of the values  $f$  takes on at the prime power divisors of  $n$ .

To see that (1) is true, expand the product on the right, replacing each of the 1's by  $f(1)$ . This yields a sum with addends all products of the form  $f(p_1^{\beta_1}) f(p_2^{\beta_2}) \cdots f(p_k^{\beta_k})$ , where  $0 \leq \beta_i \leq \alpha_i$  for  $i = 1, 2, \dots, k$ , each of the possible products occurring exactly once. We have

$$f(p_1^{\beta_1}) f(p_2^{\beta_2}) \cdots f(p_k^{\beta_k}) = f(p_1^{\beta_1} p_2^{\beta_2} \cdots p_k^{\beta_k}),$$

since  $f$  is multiplicative. On the other hand, the products  $p_1^{\beta_1} p_2^{\beta_2} \cdots p_k^{\beta_k}$  satisfying  $0 \leq \beta_i \leq \alpha_i$  for  $i = 1, 2, \dots, k$  are precisely the positive divisors of  $n$ . Hence the right-hand side of (1) equals  $\sum_{d|n} f(d)$ . ■

Our first problem is from Sierpinski's classic book *250 Problems in Elementary Number Theory* (it is proved there by induction):

*Prove that the number of divisors of the form  $4k + 1$  of each positive integer is not less than the number of its divisors of the form  $4k + 3$ .*

From a certain point of view, this statement requires no proof at all: The number of ways in which a positive integer can be represented as a sum of two squares is four times the excess of the number of its divisors of the form  $4k + 1$  over the number of the ones of the form  $4k + 3$ . So this excess is nonnegative, but, of course, we do not count this as a proof.

To solve the problem, consider the function

$$f(n) = \begin{cases} 0, & \text{if } n \text{ is even;} \\ 1, & \text{if } n \equiv 1 \pmod{4}; \\ -1, & \text{if } n \equiv 3 \pmod{4}. \end{cases}$$

It follows directly from this definition that  $f(n)$  is multiplicative. Now we apply (1). The even divisors of  $n$  do not influence its left-hand side. Each divisor of the form  $4k + 1$  contributes a 1, and each divisor of the form  $4k + 3$  contributes a  $-1$ . Consequently, it suffices to prove that  $\sum_{d|n} f(d)$  is nonnegative for each positive integer  $n$ .

Take any prime divisor  $p_i$  of  $n$ . If  $p_i \equiv 1 \pmod{4}$ , then the same congruence holds for all powers of  $p_i$ , so the  $i$ th factor in the right-hand side of (1) is positive. If  $p_i$  is congruent to 3 modulo 4, then so are its odd powers while the even powers are congruent to 1 modulo 4. In this case the  $i$ th factor in the right-hand side has the form  $1 - 1 + 1 - 1 + \cdots$ , and it equals 1 or 0 according as  $\alpha_i$  is even or odd. Summing up, we conclude that the sum in question is nonnegative. ■

Following this proof, it is also not difficult to establish all cases when the sum is 0, that is,  $n$  has exactly as many divisors of the form  $4k + 1$  as of the form  $4k + 3$ . This happens if and only if at least one prime divisor of  $n$  of the type  $4k + 3$  enters its prime factorization with an odd power. (It is exactly these  $n$ 's that cannot be represented as a sum of two squares.)

The last problem in the previous section implies that the Diophantine equation

$$x_1^3 + x_2^3 + \cdots + x_n^3 = (x_1 + x_2 + \cdots + x_n)^2 \quad (2)$$

has a unique solution in positive integers (up to permutation) with  $x_1, x_2, \dots, x_n$  *distinct*. If repetitions of the  $x$ 's are allowed, then (2) has many other positive integer solutions. The so-called Liouville theorem provides a lot of examples.

*Let  $d_1, d_2, \dots, d_l$  be all positive divisors of an arbitrary positive integer  $n$  (1 and  $n$  included). For each  $i = 1, 2, \dots, l$  denote by  $a_i$  the number of positive divisors of  $d_i$ . Then*

$$a_1^3 + a_2^3 + \dots + a_l^3 = (a_1 + a_2 + \dots + a_l)^2.$$

For example, if  $n = 12$  we have  $d_1 = 1, d_2 = 2, d_3 = 3, d_4 = 4, d_5 = 6, d_6 = 12$ ;  $a_1 = 1, a_2 = 2, a_3 = 2, a_4 = 3, a_5 = 4, a_6 = 6$  and

$$1^3 + 2^3 + 2^3 + 3^3 + 4^3 + 6^3 = 324 = (1 + 2 + 2 + 3 + 4 + 6)^2.$$

The function  $f(n)$  that does the job here is defined merely as *the number of divisors of  $n$* . We know that it is multiplicative. Actually, one more fact is necessary: that the product of two or more multiplicative functions is also multiplicative. This is because if  $f$  and  $g$  are multiplicative, then  $(fg)(1) = f(1)g(1) = 1 \cdot 1 = 1$  and

$$(fg)(mn) = f(mn)g(mn) = f(m)f(n)g(m)g(n) = (fg)(m)(fg)(n)$$

for arbitrary relatively prime positive integers  $m, n$ .

This time we apply (1) twice: to  $f(n)$  and  $f^3(n)$ . Since

$$\begin{aligned} a_1 + a_2 + \dots + a_l &= \sum_{d|n} f(d), \\ a_1^3 + a_2^3 + \dots + a_l^3 &= \sum_{d|n} f^3(d), \end{aligned}$$

we obtain

$$\sum_{i=1}^l a_i = \prod_{i=1}^k [1 + f(p_i) + f(p_i^2) + \dots + f(p_i^{\alpha_i})], \quad (3)$$

$$\sum_{i=1}^l a_i^3 = \prod_{i=1}^k [1 + f^3(p_i) + f^3(p_i^2) + \dots + f^3(p_i^{\alpha_i})]. \quad (4)$$

The action of  $f$  on prime powers is simple:  $p^r$  has  $r + 1$  divisors, so  $f(p^r) = r + 1$ . Then the square of the  $i$ th factor in the right-hand side of (3) is  $[1 + 2 + \dots + \alpha_i + (\alpha_i + 1)]^2$ , and the  $i$ th factor in the right-hand side of (4) is  $1^3 + 2^3 + \dots + \alpha_i^3 + (\alpha_i + 1)^3$ . Now just recall that  $1^3 + 2^3 + \dots + s^3 = (1 + 2 + \dots + s)^2$  for all  $s$ . ■

This proof gives some insight into *why* Liouville's theorem is true: because the classic identity from the previous section is.

As for the equation (2), it would have been remarkable if its only solutions were the ones provided by Liouville's theorem. Unfortunately, this is not the case. . .

# 18

## The “Arbitrary” Proizvolov

A nice problem posed by **Vyacheslav Proizvolov** became popular around the world after the 1985 All-Union Olympiad:

*The numbers  $1, 2, \dots, 2n$  are partitioned arbitrarily into two groups of  $n$  integers each. Let  $a_1 < a_2 < \dots < a_n$  be the numbers in the first group, and  $b_1 > b_2 > \dots > b_n$  the numbers in the second one. Prove that*

$$|a_1 - b_1| + |a_2 - b_2| + \dots + |a_n - b_n| = n^2.$$

This can be proved in a number of ways but perhaps that the shortest one is the following.

Note that for any  $i = 1, 2, \dots, n$  exactly one of the numbers  $a_i, b_i$  belongs to  $\{1, 2, \dots, n\}$  and the other one to  $\{n+1, n+2, \dots, 2n\}$ . Indeed, if, for example,  $a_i \leq n$  and  $b_i \leq n$  then  $\{1, 2, \dots, n\}$  must contain the  $n+1$  distinct numbers  $a_1, a_2, \dots, a_i$  (since  $a_1 < a_2 < \dots < a_i$ ) and  $b_i, b_{i+1}, \dots, b_n$  (since  $b_i > b_{i+1} > \dots > b_n$ ), which is impossible. Assuming  $a_i \geq n+1$  and  $b_i \geq n+1$  leads to a contradiction in almost exactly the same way.

It follows that

$$\begin{aligned} & |a_1 - b_1| + |a_2 - b_2| + \dots + |a_n - b_n| \\ &= [(n+1) + (n+2) + \dots + 2n] - [1 + 2 + \dots + n] \\ &= n^2. \quad \blacksquare \end{aligned}$$

Another of Proizvolov's problems was published in *Matematika*, No. 5, 1986:

*An arbitrary set of  $m+n$  numbers is divided into two arbitrary groups,  $a_1, a_2, \dots, a_m, b_1, b_2, \dots, b_n$ , and the numbers in each*

group arranged in ascending order:

$$a_1 < a_2 < \cdots < a_m, \quad b_1 < b_2 < \cdots < b_n.$$

Then the same numbers are again divided into two groups,  $c_1, c_2, \dots, c_m, d_1, d_2, \dots, d_n$ , and the numbers in each group are arranged in ascending order:

$$c_1 < c_2 < \cdots < c_m, \quad d_1 < d_2 < \cdots < d_n.$$

Prove the equality

$$\begin{aligned} & |a_1 - c_1| + |a_2 - c_2| + \cdots + |a_m - c_m| \\ &= |b_1 - d_1| + |b_2 - d_2| + \cdots + |b_n - d_n|. \end{aligned} \quad (1)$$

The family name Proizvolov comes from the Russian word *proizvolnii*, which means *arbitrary*, prompting a running joke in the office of *Matematika*. Each time we found another problem of this type that he had proposed, it was suggested that because of his name Proizvolov might feel obliged to think only about problems involving *arbitrary* numbers, groups, and so on; otherwise he ignores the problem straightaway.

A solution to this second problem goes by induction on  $m + n$ . The claim is clear if  $m + n = 2$ . Assume that it holds for  $m + n < k$  and consider  $m + n = k$  arbitrary numbers  $a_1, a_2, \dots, a_m, b_1, b_2, \dots, b_n$ . Let

$$\begin{aligned} A &= \{a_1, a_2, \dots, a_m\}, & B &= \{b_1, b_2, \dots, b_n\}, \\ C &= \{c_1, c_2, \dots, c_m\}, & D &= \{d_1, d_2, \dots, d_n\}. \end{aligned}$$

Now we distinguish two cases.

1°  $A \cap C \neq \emptyset$  or  $B \cap D \neq \emptyset$ . Let, for example,  $A \cap C \neq \emptyset$ . This implies  $a_i = c_j$  for some indices  $i, j \in \{1, 2, \dots, m\}$ . Without loss of generality, let  $i \leq j$ . If  $i = j$ , one of the terms in (1) is zero, and all we have to do is to apply the induction hypothesis directly. If  $i < j$  then  $c_i < \cdots < c_j = a_i < a_{i+1} < \cdots < a_j$ , hence

$$\begin{aligned} & |a_i - c_i| + |a_{i+1} - c_{i+1}| + \cdots + |a_{j-1} - c_{j-1}| + |a_j - c_j| \\ &= (a_i - c_i) + (a_{i+1} - c_{i+1}) + \cdots + (a_{j-1} - c_{j-1}) + (a_j - c_j). \end{aligned}$$

Changing the order of the summands does not change the value of the sum, so it equals

$$\begin{aligned} & (a_{i+1} - c_i) + (a_{i+2} - c_{i+1}) + \cdots + (a_j - c_{j-1}) + (a_i - c_j) \\ &= |a_{i+1} - c_i| + |a_{i+2} - c_{i+1}| + \cdots + |a_{j-1} - c_{j-2}| + |a_j - c_{j-1}| \end{aligned}$$

(since  $a_i - c_j = 0$ ). Then the desired result follows from the induction hypothesis for the  $m + n - 1 = k - 1$  numbers

$$a_1 < a_2 < \cdots < a_{i-1} < a_{i+1} < \cdots < a_m; \quad b_1 < b_2 < \cdots < b_n;$$

$$c_1 < c_2 < \cdots < c_{j-1} < c_{j+1} < \cdots < c_m; \quad d_1 < d_2 < \cdots < d_n.$$

2°  $A \cap C = B \cap D = \emptyset$ . It is clear in this case that  $a_1 \in D$  and  $b_1 \in C$ . Without loss of generality, let  $a_1 < b_1$ . Then  $a_1$  must also equal  $d_1$ . We will show that  $b_1$  must equal  $c_1$ . Indeed, if it were true that  $b_1 = c_i$  for some  $i > 1$  then  $b_1 > c_1$  and, since  $c_1 = b_j$  for some  $j > 1$ , we get the impossible inequality  $b_1 > b_j$ . Therefore  $b_1 = c_1$  which (together with  $a_1 = d_1$ ) implies  $|a_1 - c_1| = |b_1 - d_1|$ . Thus it suffices to apply the induction hypothesis to

$$a_2 < a_3 < \cdots < a_m; \quad b_2 < b_3 < \cdots < b_n;$$

$$c_2 < c_3 < \cdots < c_m; \quad d_2 < d_3 < \cdots < d_n. \quad \blacksquare$$

We close this section with a problem intended as a Romanian proposal for the 1984 IMO:

*The squares of a chessboard are labeled arbitrarily 1 through 64. There are 63 knights on the chessboard, each labeled with the number of the square it occupies, the square 64 being vacant. After some moves of these knights, square 64 becomes unoccupied again. If  $n_k$  represents the number of the square where the knight labeled  $k$  ended up, prove that*

$$\sum_{k=1}^{63} |n_k - k| \leq 1984,$$

*and 1984 is the smallest upper bound for the sum on the left-hand side.*

The solution starts with the observation that

$$\sum_{k=1}^{63} |n_k - k| = \pm 63 \pm 63 \pm 62 \pm 62 \pm \cdots \pm 2 \pm 2 \pm 1 \pm 1,$$

where 63 terms are positive and 63 terms are negative, so an upper bound for the sum in question is

$$(63+63+62+62+\cdots+33+33+32)-(32+31+31+\cdots+1+1) = 1984.$$

Now we have to prove that 1984 can be reached for some initial labeling. It is known that a knight can visit all the squares of the chessboard in 64

moves, ending up on its original square. Labeling the board in this order and moving the vacant square from 64 to 1, and then to 2, 3, ..., 64, and continuing in this manner, one can actually reach

$$n_1 = 32, n_2 = 33, \dots, n_{32} = 63; \quad n_{33} = 1, n_{34} = 2, \dots, n_{63} = 31,$$

a position for which the sum is 1984. ■

# 19

## Hölder's Inequality

Hölder's inequality is far less popular than its sister, the Cauchy–Schwarz inequality. While some students reach the level of using Cauchy–Schwarz as a verb, only a few of them can claim Hölder as an entry in their own mathematical glossary.

It is rare that one uses the general form of Hölder's inequality, which can be found in the Glossary. Here we restrict attention to a special but representative case.

*If  $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n, c_1, c_2, \dots, c_n$  are positive numbers then*

$$\left( \sum_{i=1}^m a_i b_i c_i \right)^3 \leq \left( \sum_{i=1}^m a_i^3 \right) \left( \sum_{i=1}^m b_i^3 \right) \left( \sum_{i=1}^m c_i^3 \right). \quad (1)$$

We start with a proof of (1). Consider the numbers

$$A = \sum_{i=1}^m a_i^3, \quad B = \sum_{i=1}^m b_i^3, \quad C = \sum_{i=1}^m c_i^3$$

and

$$x_i = \frac{a_i}{\sqrt[3]{A}}, \quad y_i = \frac{b_i}{\sqrt[3]{B}}, \quad z_i = \frac{c_i}{\sqrt[3]{C}}, \quad i = 1, 2, \dots, m.$$

Then

$$\sum_{i=1}^m x_i^3 = \sum_{i=1}^m \frac{a_i^3}{A} = 1 = \sum_{i=1}^m y_i^3 = \sum_{i=1}^m z_i^3.$$

For each  $i = 1, 2, \dots, m$ , the AM–GM inequality yields

$$x_i y_i z_i = \sqrt[3]{x_i^3 y_i^3 z_i^3} \leq \frac{1}{3} (x_i^3 + y_i^3 + z_i^3).$$

Summing over  $i = 1, 2, \dots, m$ , we obtain

$$\begin{aligned} \sum_{i=1}^m x_i y_i z_i &\leq \frac{1}{3} \sum_{i=1}^m (x_i^3 + y_i^3 + z_i^3) \\ &= \frac{1}{3} \left( \sum_{i=1}^m x_i^3 + \sum_{i=1}^m y_i^3 + \sum_{i=1}^m z_i^3 \right) = \frac{1}{3}(1 + 1 + 1) = 1. \end{aligned}$$

Since

$$x_i y_i z_i = \frac{a_i b_i c_i}{\sqrt[3]{ABC}},$$

we obtain

$$\sum_{i=1}^m a_i b_i c_i \leq \sqrt[3]{ABC},$$

yielding (1). ■

We will now present two problems from *Revista Matematică Timișoara*, Nos. 1–2, 1989. What brings them together is not the names of their authors (they were published jointly by **Vasile Cârtoaje** and **Mircea Lascu**) but the nontrivial use of Hölder's inequality in their solutions.

*Prove that for any real numbers  $x, y, z$ ,*

$$3(x^2 - x + 1)(y^2 - y + 1)(z^2 - z + 1) \geq (xyz)^2 - xyz + 1.$$

We first show that

$$3(t^2 - t + 1)^3 \geq t^6 + t^3 + 1$$

for all real  $t$ . Indeed, the difference between the left-hand and the right-hand side is  $3[(t-1)^2 + t]^3 - (t^6 + t^3 + 1)$ , which equals

$$\begin{aligned} &3(t-1)^6 + 9(t-1)^4 t + 9(t-1)^2 t^2 + 3t^3 - t^6 - t^3 - 1 \\ &= 3(t-1)^4 [(t-1)^2 + 3t] + (3t^2 - 3t)^2 - (t^3 - 1)^2. \end{aligned}$$

We note that the difference of the last two squares factors into

$$(3t^2 - 3t - t^3 + 1)(3t^2 - 3t + t^3 - 1).$$

The first of these two factors is  $-(t-1)^3$  and the second one can be written as  $(t-1)(t^2 + 4t + 1)$ . So we arrive at

$$(t-1)^4 [3(t-1)^2 + 9t - (t^2 + 4t + 1)] = (t-1)^4 \left[ 2 \left( t - \frac{1}{2} \right)^2 + \frac{3}{2} \right],$$

which is nonnegative.

Thus

$$\begin{aligned} 27(x^2 - x + 1)^3(y^2 - y + 1)^3(z^2 - z + 1)^3 \\ \geq (x^6 + x^3 + 1)(y^6 + y^3 + 1)(z^6 + z^3 + 1). \end{aligned}$$

Assume first that  $x, y, z$  are nonnegative. Hölder's inequality (1), with  $a_1 = x^2, a_2 = x, a_3 = 1$ , and similarly  $b_1 = y^2, b_2 = y, b_3 = 1, c_1 = z^2, c_2 = z, c_3 = 1$ , yields

$$(x^6 + x^3 + 1)(y^6 + y^3 + 1)(z^6 + z^3 + 1) \geq [(xyz)^2 + xyz + 1]^3.$$

So in the case of nonnegative  $x, y, z$  we have proved the stronger result

$$3(x^2 - x + 1)(y^2 - y + 1)(z^2 - z + 1) \geq (xyz)^2 + xyz + 1. \quad (2)$$

To settle the general case for arbitrary real  $x, y, z$ , note first that  $|x|, |y|, |z|$  satisfy (2). Now it suffices to apply the inequalities

$$|t|^2 - |t| + 1 \leq t^2 - t + 1 \leq |t|^2 + |t| + 1. \quad \blacksquare$$

Consider the triangles  $A_k B_k C_k$ , where  $\angle A_k \geq 60^\circ, k = 1, 2, 3$ .  
Prove that

$$2a_1 a_2 a_3 \geq b_1 b_2 b_3 + c_1 c_2 c_3.$$

Since  $\angle A_k \geq 60^\circ$ , the law of cosines in triangle  $A_k B_k C_k$  gives

$$a_k^2 \geq b_k^2 - b_k c_k + c_k^2, \quad k = 1, 2, 3. \quad (3)$$

But

$$b^2 - bc + c^2 \geq \left(\frac{b+c}{2}\right)^2$$

for all  $b$  and  $c$ , hence

$$2a_k \geq b_k + c_k, \quad k = 1, 2, 3. \quad (4)$$

The relations (3) and (4) yield

$$2a_k^3 \geq b_k^3 + c_k^3, \quad k = 1, 2, 3.$$

Multiplying these together and using Hölder's inequality, we obtain

$$8a_1^3 a_2^3 a_3^3 \geq (b_1^3 + c_1^3)(b_2^3 + c_2^3)(b_3^3 + c_3^3) \geq (b_1 b_2 b_3 + c_1 c_2 c_3)^3,$$

and the desired inequality follows.  $\blacksquare$

# 20

## Symmetry

One usually associates symmetry with properties of geometric figures. There are, however, many diverse settings where a certain more general idea of symmetry proves quite helpful. Without trying to be exhaustive, we shall consider several of them.

Let us begin with an algebraic example.

*Prove the inequality*

$$\frac{x_1^3}{x_1^2 + x_1x_2 + x_2^2} + \frac{x_2^3}{x_2^2 + x_2x_3 + x_3^2} + \cdots + \frac{x_n^3}{x_n^2 + x_nx_1 + x_1^2} \geq \frac{1}{3}(x_1 + x_2 + \cdots + x_n)$$

*for all positive numbers  $x_1, x_2, \dots, x_n$ .*

Symmetric algebraic inequalities are usually easier to handle but this is not the case here: the left-hand side  $A$  is not symmetric. To introduce symmetry, consider also the expression

$$B = \frac{x_2^3}{x_1^2 + x_1x_2 + x_2^2} + \frac{x_3^3}{x_2^2 + x_2x_3 + x_3^2} + \cdots + \frac{x_1^3}{x_n^2 + x_nx_1 + x_1^2}.$$

It turns out that  $A$  and  $B$  differ only externally. We will show that  $A = B$ . Indeed, the difference  $A - B$  equals

$$\begin{aligned} \frac{x_1^3 - x_2^3}{x_1^2 + x_1x_2 + x_2^2} + \frac{x_2^3 - x_3^3}{x_2^2 + x_2x_3 + x_3^2} + \cdots + \frac{x_n^3 - x_1^3}{x_n^2 + x_nx_1 + x_1^2} \\ = (x_1 - x_2) + (x_2 - x_3) + \cdots + (x_n - x_1) = 0. \end{aligned}$$

Hence  $A = \frac{1}{2}(A+B)$  and instead of working with the original asymmetric inequality, we replace it by the symmetric inequality

$$\frac{1}{2}(A+B) \geq \frac{1}{3}(x_1 + x_2 + \cdots + x_n).$$

This is already not difficult. Let us first note that

$$\frac{a^3 + b^3}{a^2 + ab + b^2} \geq \frac{a + b}{3}$$

for all positive numbers  $a, b$ . The last inequality follows directly after clearing out denominators and rearranging in the form  $2(a+b)(a-b)^2 \geq 0$ . Thus we obtain

$$\begin{aligned} A + B &\geq \frac{1}{3}(x_1 + x_2) + \frac{1}{3}(x_2 + x_3) + \cdots + \frac{1}{3}(x_n + x_1) \\ &= \frac{2}{3}(x_1 + x_2 + \cdots + x_n), \end{aligned}$$

which finishes the proof. ■

One of the most popular examples of symmetry is the way it is said Gauss found the sum of the first 100 positive integers in elementary school. The same idea solves the next problem.

*Let  $p$  be an odd prime. For each  $i = 1, 2, \dots, p-1$  denote by  $r_i$  the remainder when  $i^p$  is divided by  $p^2$ . Evaluate the sum*

$$r_1 + r_2 + \cdots + r_{p-1}.$$

Denote the sum in question by  $S$ . Combine the first summand with the last, the second one with the next-to-last, and so on, to get

$$2S = (r_1 + r_{p-1}) + (r_2 + r_{p-2}) + \cdots + (r_{p-1} + r_1). \quad (1)$$

We have  $r_i + r_{p-i} \equiv i^p + (p-i)^p \pmod{p^2}$  by the definition of the numbers  $r_1, r_2, \dots, r_{p-1}$ . Furthermore, because  $p$  is odd,

$$i^p + (p-i)^p = p^p - \binom{p}{1}p^{p-1}i + \binom{p}{2}p^{p-2}i^2 - \cdots + \binom{p}{p-1}pi^{p-1}.$$

Since  $p$  is a prime, each binomial coefficient above is divisible by  $p$ , which yields the conclusion that  $r_i + r_{p-i}$  is divisible by  $p^2$ . But  $0 < r_i < p^2$ ,  $0 < r_{p-i} < p^2$ , because  $p$  is a prime (so neither one equals 0), and now we may claim that

$$r_i + r_{p-i} = p^2 \quad \text{for } i = 1, 2, \dots, p-1. \quad (2)$$

The equalities (1) and (2) show that

$$S = \frac{p-1}{2} \cdot p^2 = \frac{p^3 - p^2}{2}. \quad \blacksquare$$

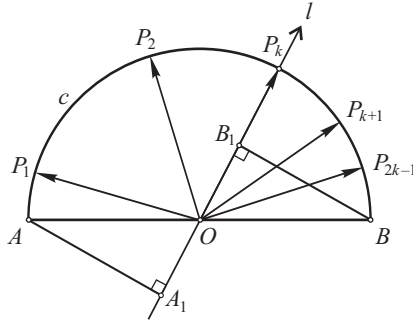
Symmetry works in geometry as well but sometimes using it is highly nontrivial. For almost thirty years many solutions to the first 1973 IMO problem have been published, but none of them compares to the one by the late **Ivan Prodanov**, the Bulgarian team leader at that Olympiad.

Let  $c$  be a semicircle of unit radius and  $P_1, P_2, \dots, P_n$  points on  $c$ , where  $n \geq 1$  is an odd integer. Prove that

$$|\overrightarrow{OP_1} + \overrightarrow{OP_2} + \dots + \overrightarrow{OP_n}| \geq 1,$$

where  $O$  is the center of  $c$ .

The key idea is to show that the orthogonal projection of the vector sum  $\overrightarrow{OP_1} + \overrightarrow{OP_2} + \dots + \overrightarrow{OP_n}$  onto some line has length not less than 1. Let  $n = 2k - 1$ . From considerations of symmetry, the line  $l$  containing the middle vector  $\overrightarrow{OP_k}$  is a natural candidate for such a line (here we use the fact that  $n$  is odd!).



It is technically convenient to consider  $l$  as an axis with positive direction determined by  $\overrightarrow{OP_k}$ . As is well known, the projection of the sum of several vectors is equal to the sum of their projections. Hence it suffices to prove that the sum of the signed lengths  $\overline{OP_1}, \overline{OP_2}, \dots, \overline{OP_{2k-1}}$  of the projections of  $\overrightarrow{OP_1}, \overrightarrow{OP_2}, \dots, \overrightarrow{OP_{2k-1}}$  onto  $l$  is greater than or equal to 1. Denote the diameter of  $c$  by  $AB$  and the orthogonal projections of  $A$  and  $B$  onto  $l$  by  $A_1$  and  $B_1$ . We have  $\overline{OP_k} = 1$  and also

$$\overline{OP_1} + \overline{OP_2} + \dots + \overline{OP_{k-1}} \geq (k-1)\overline{OA_1},$$

$$\overline{OP_{k+1}} + \overline{OP_{k+2}} + \dots + \overline{OP_{2k-1}} \geq (k-1)\overline{OB_1}.$$

This is because  $\overline{OP_j} \geq \overline{OA_1}$  for  $j = 1, \dots, k-1$  and  $\overline{OP_j} \geq \overline{OB_1}$  for  $j = k+1, \dots, 2k-1$ . Since  $\overline{OA_1} + \overline{OB_1} = 0$ , the proof is complete. ■

There is a simple and efficient idea involving symmetry. Roughly speaking, instead of dealing with several mathematical objects separately, it is sometimes helpful to take their “average.” To save further explanations, let us proceed to a 1984 Moscow Olympiad problem.

*Let  $x_1, x_2, \dots, x_n$ ,  $n \geq 3$ , be real numbers that add up to 1. Prove that there is a permutation  $y_1, y_2, \dots, y_n$  of these numbers such that*

$$y_1y_2 + y_2y_3 + \cdots + y_ny_1 \leq \frac{1}{n}.$$

Rather than constructing such a permutation, we prefer an implicit proof. For each cyclic permutation  $\pi = (y_1, y_2, \dots, y_n)$  of  $x_1, x_2, \dots, x_n$ , let

$$S_\pi = y_1y_2 + y_2y_3 + \cdots + y_ny_1,$$

and consider the sum  $S = \sum S_\pi$ , where  $\pi$  ranges over all cyclic permutations. Because of the obvious symmetry, each product of the form  $x_ix_j$ ,  $i \neq j$ , occurs as a summand in  $S$  the same number of times as any other. The number of all cyclic permutations is  $(n-1)!$ , so  $S$  contains  $n \cdot (n-1)! = n!$  summands. Since there are  $\binom{n}{2}$  products  $x_ix_j$  with  $1 \leq i < j \leq n$ , each of them must occur  $(n!)/\binom{n}{2}$  times in  $S$ , and we obtain

$$S = \frac{n!}{\binom{n}{2}} \sum_{1 \leq i < j \leq n} x_ix_j = (n-2)! \left[ \left( \sum_{k=1}^n x_k \right)^2 - \sum_{k=1}^n x_k^2 \right].$$

By hypothesis,  $x_1 + x_2 + \cdots + x_n = 1$ , and then the root mean square inequality yields  $\sum_{k=1}^n x_k^2 \geq 1/n$ . Hence

$$S \leq (n-2)! \left( 1 - \frac{1}{n} \right) = \frac{(n-1)!}{n},$$

implying that the least of the  $(n-1)!$  cyclic sums  $S_\pi$  does not exceed

$$\frac{1}{(n-1)!} \cdot \frac{(n-1)!}{n} = \frac{1}{n}. \quad \blacksquare$$

## Coffee Break 4

1. The entry in the  $i$ th row and the  $j$ th column of an  $n \times n$  matrix equals  $a_i + b_j$ , where  $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$  are distinct real numbers. The products of the numbers in each row of the matrix are equal. Prove that the products of the numbers in each column are also equal.
2. Consider the parabolas  $y = x^2 - ax - 1$ , where  $a$  is a real number. Each of them intersects the coordinate axes at three points, which are the vertices of a triangle. Prove that the circumcircles of these triangles have a common point.
3. Prove that

$$\binom{2p}{p} \equiv 2 \pmod{p^2}$$

for any prime number  $p$ .

## Solutions

1. The product  $(a_i + b_1)(a_i + b_2) \cdots (a_i + b_n)$  of the elements in the  $i$ th row can be regarded as the value of the polynomial

$$f(x) = (x + b_1)(x + b_2) \cdots (x + b_n)$$

for  $x = a_i$ . We are given that

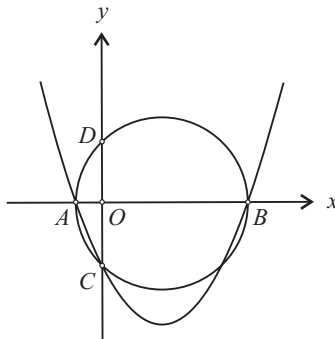
$$f(a_1) = f(a_2) = \cdots = f(a_n) = c$$

for some constant  $c$ . This means that the distinct numbers  $a_1, a_2, \dots, a_n$  are roots of the monic polynomial  $f(x) - c$ , which is of degree  $n$ , hence these are *all* of its roots. It follows that

$$\begin{aligned} (x + b_1)(x + b_2) \cdots (x + b_n) - c &= f(x) - c \\ &= (x - a_1)(x - a_2) \cdots (x - a_n). \end{aligned}$$

Now it suffices to set  $x = -b_j$ : the product of the entries in the  $j$ th column equals  $(-1)^{n+1}c$  for each  $j = 1, 2, \dots, n$ .

2. The quadratic function  $y = x^2 - ax - 1$  has two real roots  $x_1, x_2$ , since its discriminant  $a^2 + 4$  is always positive. Their product is  $-1$ , so we may assume that  $x_1 < 0 < x_2$ . The parabola intersects  $Ox$  at  $A(x_1, 0)$ ,  $B(x_2, 0)$ ; it also intersects  $Oy$  at  $C(0, -1)$ . Denote by  $D$  the second common point of  $Oy$  and the circumcircle of  $\triangle ABC$ . By the power-of-a-point theorem,  $AO \cdot BO = CO \cdot DO$ . But  $AO = -x_1$ ,  $BO = x_2$ ,  $CO = 1$ , so if  $y$  is the ordinate of  $D$ , the last equality becomes  $-x_1x_2 = y$ . Since  $x_1x_2 = -1$ , it turns out that  $y = 1$ , regardless of  $a$ . Hence the circumcircle of  $\triangle ABC$  passes through the fixed point  $D(0, 1)$ .



3. A short proof uses the popular Vandermonde identity

$$\sum_{i=0}^k \binom{m}{i} \binom{n}{k-i} = \binom{m+n}{k}.$$

Set  $m = n = k = p$  to get

$$\binom{2p}{p} = \binom{p}{0} \binom{p}{p} + \binom{p}{1} \binom{p}{p-1} + \cdots + \binom{p}{p-1} \binom{p}{1} + \binom{p}{p} \binom{p}{0}.$$

The first and the last term on the right-hand side equal 1. Since  $p$  is a prime, it divides each binomial coefficient  $\binom{p}{k}$  for  $1 \leq k \leq p-1$ . So each of the remaining terms is divisible by  $p^2$ , and hence  $\binom{2p}{p}$  is congruent to 2 modulo  $p^2$ , as required.

# 21

## He Knows I Know He Knows

Here is one of the Bulgarian proposals for the 1991 IMO:

*Each of the boys  $A$  and  $B$  tells the teacher a positive integer but neither of them knows the other's number. The teacher writes two distinct positive integers on the blackboard and announces that one of them is the sum of the numbers they told him. Then he asks  $A$ , "Can you guess the sum of the two numbers?" If the answer is "No," the teacher asks  $B$  the same question, and so on.*

*Suppose that the boys are intelligent and truthful. Prove that one of their answers eventually will be "Yes."*

At least at a first glance, this seems incredible. But it is true!

Suppose  $A$  and  $B$  have told their teacher the numbers  $a$  and  $b$ , respectively, and suppose the teacher has written the numbers  $S_1$  and  $S_2$  on the blackboard. Let  $0 < S_1 < S_2$ . Denote the difference of  $S_2$  and  $S_1$  by  $d$  and assume that all consecutive answers of the two boys are "No."

Denote by  $\alpha_k$  ( $\beta_k$ ) a conclusion that  $A$  ( $B$ ) could draw after the  $k$ th consecutive "No" of  $B$  ( $A$ ).

$A$ 's first "No" makes it clear for  $B$  that  $0 < a < S_1$ . Indeed, if  $a$  were greater than or equal to  $S_1$ , then  $a + b > S_1$ , so  $A$  would have realized that  $a + b = S_2$  and his first answer would have been "Yes." Thus

$$\beta_1 : 0 < a < S_1.$$

Now, since the first answer of  $B$  is "No,"  $A$  can infer that

$$\alpha_1 : d < b < S_1.$$

Indeed, first of all  $A$  figures out (as  $B$  did above) that  $0 < b < S_1$ . Then he could argue as follows: “ $B$  knows that  $0 < a < S_1$  (this is  $\beta_1$ ) and if  $b$  were less than or equal to  $d$ , then he would have known that

$$a + b \leq a + d < S_1 + (S_2 - S_1) = S_2.$$

Thus the only possibility for  $a + b$  would have been  $a + b = S_1$ , and  $B$  would have answered ‘Yes.’ But he did not, hence  $d < b < S_1$ .”

We proceed further in a similar fashion. After  $A$ ’s second “No,”  $B$  is aware that

$$\beta_2 : 0 < a < S_1 - d$$

(otherwise, if  $S_1 - d \leq a < S_1$ ,  $A$  would have been able to conclude on the basis of  $\alpha_1$  that  $a + b > (S_1 - d) + d = S_1$  and his second answer would have been “Yes”). Similarly,  $B$ ’s second “No” implies

$$\alpha_2 : 2d < b < S_1,$$

because  $B$  already knows  $\beta_2$ , and if  $b$  did not exceed  $2d$ , then he would have been able to conclude that  $a + b < (S_1 - d) + 2d = S_2$ , thus yielding a “Yes” as his second answer.

This argument could be repeated *ad infinitum*, and after the  $k$ th consecutive “No” of the two we get

$$\beta_k : 0 < a < S_1 - (k - 1)d, \quad \alpha_k : kd < b < S_1.$$

It is already clear what follows: since  $d$  is positive,  $kd$  will become greater than  $S_1$  for some  $k$ . At this moment  $A$  will come to know that  $b < S_1$  and, on the other hand, that  $b > kd > S_1$ . So the answer cannot be “No” forever, a contradiction! ■

Shortly after the IMO the proposer found out that this problem may be regarded as a special case of a more general and still more striking statement due to **John Conway**. Essentially, it should read:

*There are  $n$  boys sitting around a round table, each with a cap on his head. A positive integer is written on each boy’s cap. No boy knows the number on his own cap (and he cannot see it) but everyone can see everybody else’s number. The teacher writes  $k$  distinct positive integers on the blackboard and announces that one of them is the sum of the numbers written on all caps. Then he asks one of the boys, “Can you guess the sum of the*

*numbers?" If the answer is "No," then the teacher asks his neighbor to the right the same question, and so on.*

*Suppose that  $k \leq n$ , and that all boys are intelligent and truthful. Prove that one of their consecutive answers eventually will be "Yes."*

# 22

## A Special Inequality

The problem considered in this section was on the 1984 Bulgarian Olympiad. They say, however, that it became popular at some International Mathematical Olympiad in the late seventies.

*Let  $x_1, x_2, \dots, x_n$  and  $y_1, y_2, \dots, y_n$  be nonnegative real numbers such that  $x_i + y_i = 1$  for each  $i = 1, 2, \dots, n$ . Prove that*

$$(1 - x_1 x_2 \cdots x_n)^m + (1 - y_1^m)(1 - y_2^m) \cdots (1 - y_n^m) \geq 1,$$

*where  $m$  is an arbitrary positive integer.*

A direct approach is, of course, possible. But our proof of this purely algebraic relation appeals to a simple probabilistic model suggested by the circumstance that  $x_i + y_i = 1$ .

Consider  $n$  unfair coins. Let  $x_i$  be the probability that a toss of the  $i$ th coin is a head ( $i = 1, 2, \dots, n$ ). Then the probability that a toss of this coin is a tail equals  $1 - x_i = y_i$ .

The probability of  $n$  heads in tossing all the coins once is  $x_1 x_2 \cdots x_n$ , because the events are independent. Hence the probability of at least one tail is  $1 - x_1 x_2 \cdots x_n$ . Consequently, the probability of *at least one tail in each of  $m$  consecutive tosses of all the coins* equals  $(1 - x_1 x_2 \cdots x_n)^m$ .

With probability  $y_i^m$ , each of  $m$  consecutive tosses of the  $i$ th coin is a tail; with probability  $1 - y_i^m$ , we have at least one head. Therefore the probability that *after  $m$  tosses of all the coins each coin has been a head at least once* equals  $(1 - y_1^m)(1 - y_2^m) \cdots (1 - y_n^m)$ .

Denote the events given above in italics by  $A$  and  $B$ , respectively. It is easy to observe that *at least one of them must occur* as a result of  $m$  tosses. Indeed, suppose  $A$  has not occurred. This means that the outcome of some toss has been  $n$  heads, which implies that  $B$  has occurred.

Now we need a line more to finish the proof. Since one of the events  $A$  and  $B$  occurs as a result of  $m$  tosses, the sum of their probabilities is greater than or equal to 1, that is

$$(1 - x_1 x_2 \cdots x_n)^m + (1 - y_1^m)(1 - y_2^m) \cdots (1 - y_n^m) \geq 1. \blacksquare$$

# 23

## Two Inductive Constructions

This story begins with an old and popular problem:

*Let  $n$  be a positive integer. Is it possible to arrange the numbers  $1, 2, \dots, n$  in a row so that the arithmetic mean of any two of these numbers is not equal to some number between them?*

*In other words: Is there a permutation  $a_1, a_2, \dots, a_n$  of the numbers  $1, 2, \dots, n$  such that there are no indices  $i < k < j$  for which  $a_k = \frac{1}{2}(a_i + a_j)$ ?*

The answer is yes, and it suffices to prove this just for the powers of 2; the general case follows from here. For example, one may find a permutation of  $1, 2, \dots, 2048$  with the desired property, and then, by deleting the unnecessary numbers, obtain a solution for  $n = 2000$ . So, we are going to construct a permutation of the desired type for  $n = 2^1, 2^2, \dots, 2^k, \dots$

We proceed by induction. For  $k = 1$  and  $k = 2$ , the permutations  $1, 2$  and  $2, 4, 1, 3$  work. Suppose now that  $a_1, a_2, \dots, a_{2^k}$  is a permutation that yields a solution for the numbers  $1, 2, \dots, 2^k$ . Consider the sequences  $2a_1, 2a_2, \dots, 2a_{2^k}$  and  $2a_1 - 1, 2a_2 - 1, \dots, 2a_{2^k} - 1$ . The first one consists of all even numbers between 1 and  $2^{k+1}$  inclusive, and the second one contains all odd numbers in this range. Hence

$$2a_1, 2a_2, \dots, 2a_{2^k}, 2a_1 - 1, 2a_2 - 1, \dots, 2a_{2^k} - 1 \quad (1)$$

is a permutation of  $1, 2, \dots, 2^{k+1}$ . We shall prove that it meets the requirement.

Any two numbers from different halves of this permutation have different parity, so their arithmetic mean is not an integer. In particular, it cannot be equal to any of the numbers between them. We claim that the same is true when the two numbers are both in the first or in the second

half of (1). Indeed, assume, for example, that there are indices  $i < k < j$  such that

$$\frac{2a_i + 2a_j}{2} = 2a_k.$$

This implies

$$\frac{a_i + a_j}{2} = a_k,$$

which contradicts the assumption that  $a_1, a_2, \dots, a_{2^k}$  is a solution of the problem for  $1, 2, \dots, 2^k$ . The case when the two numbers are from the second half of (1) is treated in the same way. This is the end of the inductive construction, and of the proof as well. ■

Truly impressed by this solution, Bulgarian high-school teacher **Stanko Pavlov** tried to find a two-dimensional variant of the statement. It was not clear initially what the word *between* should mean in the plane. After a number of attempts, he came up with a nice new problem which was published in *Matematika*, No. 6, 1986.

*Prove that there exist infinitely many positive integers  $n$  with the following property: The numbers  $1, 2, \dots, n^2$  can be arranged in an  $n \times n$  square array so that the arithmetic mean of any two of these numbers is not contained in the minimal rectangle that contains them both; that is, the arithmetic mean of  $a_{ij}$  and  $a_{kl}$  is not an entry of the form  $a_{rs}$ , where*

$$\min(i, k) \leq r \leq \max(i, k), \quad \min(j, l) \leq s \leq \max(j, l).$$

This is a considerably more difficult problem. It is quite difficult just to check whether a certain arrangement has the described property.

If  $1, 2, \dots, n^2$  are arranged in an  $n \times n$  array, we shall say for brevity that a number  $z$  is *between* two other numbers  $x$  and  $y$  if it lies in the minimal rectangle containing  $x$  and  $y$ .

As in the above problem, we are going to prove that the numbers of the form  $2^k$  have the stated property. This time, however, such a solution cannot settle the question for *all*  $n$ . Deleting several numbers in a sequence yields a new sequence. But deleting several entries in a square array does not necessarily produce another array; in general, it just produces holes.

We use induction again. For  $k = 1$ , the arrangement

|   |   |
|---|---|
| 1 | 2 |
| 3 | 4 |

gives a solution. Suppose that for some  $k \geq 1$  one can arrange the numbers  $1, 2, \dots, (2^k)^2$  in a  $2^k \times 2^k$  square array  $A = (a_{ij})_{i,j=1}^{2^k}$  so that the arithmetic mean of any two entries is not between them. Then the four  $2^k \times 2^k$  arrays

$$\begin{aligned} A_0 &= (4a_{ij}), & A_3 &= (4a_{ij} - 1), \\ A_2 &= (4a_{ij} - 2), & A_1 &= (4a_{ij} - 3) \end{aligned}$$

contain all numbers between 1 and  $4 \cdot (2^k)^2 = (2^{k+1})^2$  inclusive. Also, the numbers in  $A_i$  are congruent to  $i$  modulo 4 ( $i = 0, 1, 2, 3$ ). Arrange these arrays as shown below to get the  $2^{k+1} \times 2^{k+1}$  array

$$B = \begin{array}{|c|c|} \hline A_0 & A_1 \\ \hline A_2 & A_3 \\ \hline \end{array}$$

containing all the numbers  $1, 2, \dots, (2^{k+1})^2$ . We shall check that the arithmetic mean of any two entries of  $B$  is not between them. Let us call  $A_0, A_1, A_2, A_3$  the four *quadrants*.

If the two entries  $x$  and  $y$  are in the same quadrant, that is, if they belong to one of the arrays  $A_0, A_1, A_2, A_3$ , this is an internal affair of the quadrant and it is easily settled. Indeed, if the average of the entries  $x = 4a_{ij} - r$  and  $y = 4a_{kl} - r$  ( $r = 0, 1, 2, 3$ ) were between them, then a short computation as in the previous proof shows that  $A$  would not be a solution of the problem for  $1, 2, \dots, (2^k)^2$ . This would contradict the induction hypothesis. If  $x$  and  $y$  belong to quadrants in the same horizontal row of  $B$  or to diagonally opposite quadrants, there is nothing to worry about, since their arithmetic mean is not an integer. We only need to check what happens when  $x \in A_0, y \in A_2$  or when  $x \in A_1, y \in A_3$ . In the first case we have

$$x \equiv 0 \pmod{4}, \quad y \equiv 2 \pmod{4},$$

so  $(x + y)/2$  is an odd integer and must belong to either  $A_1$  or  $A_3$ . In the second case both  $x$  and  $y$  are in the “right-hand half” of  $B$ , whereas their arithmetic mean is even and hence belongs to the “left-hand half.” In neither case is  $(x + y)/2$  between  $x$  and  $y$ . Thus  $B$  meets our requirements, and the induction is complete. ■

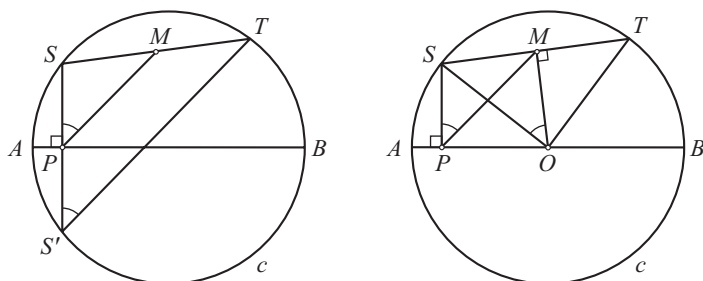
# 24

## Some Old-Fashioned Geometry

There are many geometry problems where a well-chosen auxiliary construction is about the best thing one can think of. Here are several examples.

The first one comes from the 1986 Canadian Olympiad.

*The endpoints of a chord  $ST$  with constant length are moving along a semicircle with diameter  $AB$ . Let  $M$  be the midpoint of  $ST$  and  $P$  the foot of the perpendicular from  $S$  to  $AB$ . Prove that angle  $SPM$  is independent of the location of  $ST$ .*



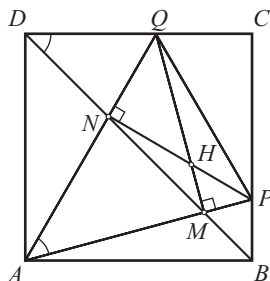
Where there is a semicircle, there is also a circle. So let us complete the given semicircle to form a circle  $c$ . Extend  $SP$  until it meets  $c$  at the point  $S'$ . Clearly  $P$  is the midpoint of  $SS'$  and since  $M$  is the midpoint of  $ST$ , we can claim that  $PM \parallel S'T$ . Then  $\angle SPM = \angle SS'T = \frac{1}{2} \widehat{ST}$ . But the measure of the arc  $\widehat{ST}$  depends only on the length of  $ST$ , and not on the actual location of this chord on the semicircle. ■

Here is another solution, using a different auxiliary construction. Take the center  $O$  of the circle and consider the quadrilateral  $POMS$ . Its opposite angles  $SPO$  and  $SMO$  are right angles, so it is cyclic. It follows

that  $\angle SPM = \angle SOM$ . But clearly  $\angle SOM = \frac{1}{2}\angle SOT = \frac{1}{2}\widehat{ST}$ , which completes the proof. ■

*The points  $P$  and  $Q$  are chosen on the sides  $BC$  and  $CD$  of the square  $ABCD$ , respectively, so that  $\angle PAQ = 45^\circ$ . Construct the perpendicular from  $A$  to the line  $PQ$  with straightedge alone.*

Let the diagonal  $BD$  meet  $AP$  and  $AQ$  at the points  $M$  and  $N$ , respectively. We shall prove that  $PN$  and  $QM$  are altitudes in  $\triangle APQ$ , so their common point  $H$  is the orthocenter of this triangle. Then  $AH \perp PQ$  and since  $M, N, PN, QM$  and  $AH$  can be constructed (in this order) with a straightedge, this will complete the solution.



We have  $\angle QDM = 45^\circ$ , because  $BD$  is a diagonal in the square  $ABCD$ . On the other hand, it is given that  $\angle PAQ = 45^\circ$ , so the line segment  $QM$  subtends equal angles at the points  $D$  and  $A$ . Therefore the quadrilateral  $AMQD$  is cyclic, which implies

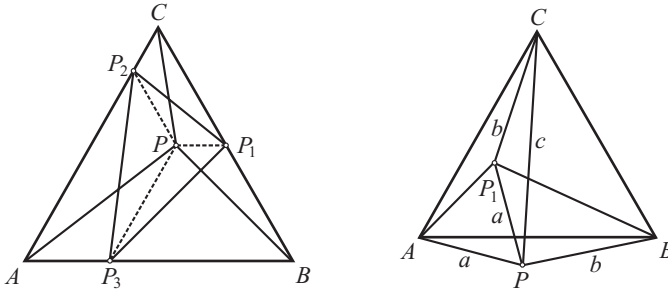
$$\angle AMQ = 180^\circ - \angle ADQ = 90^\circ.$$

We proved that  $QM \perp AP$ ; in a similar fashion,  $PN \perp AQ$ . As pointed out above, this is enough to finish the solution. ■

The following statement is known as Pompeiu's theorem, after the Romanian mathematician **Dimitrie Pompeiu**:

*If  $ABC$  is an equilateral triangle and  $P$  an arbitrary point in its plane, then there exists a triangle with sides equal to the line segments  $PA, PB, PC$ . This triangle is nondegenerate if and only if  $P$  does not lie on the circumcircle of  $ABC$ .*

From a remarkable proof found by the Russian geometer **Zalman Skopetz**, we reproduce just the special case when  $P$  is interior to  $ABC$ .



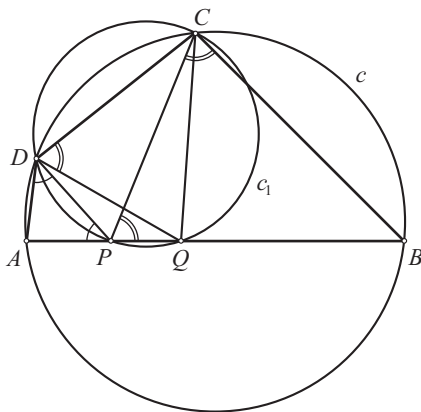
Draw the lines passing through  $P$  and parallel to the sides of  $\triangle ABC$ . Let them meet the sides  $BC, CA, AB$  at  $P_1, P_2, P_3$ , respectively; the exact definitions of  $P_1, P_2, P_3$  can be seen in the figure. We are going to prove that  $P_1P_2P_3$  is a triangle with side lengths  $PA, PB, PC$ . Indeed, consider the trapezoid  $AP_2PP_3$ . It is isosceles, because  $\angle AP_2P = \angle P_2AP_3 = 60^\circ$ . Then its diagonals are equal, that is,  $P_2P_3 = PA$ . In a similar fashion  $P_3P_1 = PB$  and  $P_1P_2 = PC$ , so  $\triangle P_1P_2P_3$  has the desired property. ■

The good news about this proof is that Pompeiu's triangle is obtained as inscribed in the initial triangle  $ABC$ ; the bad news is that we need additional efforts when  $P$  is not interior to  $ABC$ . Here is a simple proof for the general case.

Let  $PA = a, PB = b, PC = c$  and, for example,  $c$  be the greatest among the numbers  $a, b, c$ . Consider the rotation  $R$  through an angle of  $60^\circ$  about  $A$  in counterclockwise direction. Let  $R$  take the given point  $P$  to  $P_1$ . Then we have  $AP = AP_1$  and  $\angle PAP_1 = 60^\circ$ , so  $\triangle APP_1$  is equilateral. Hence  $PP_1 = PA = a$ . Note now that  $R$  takes  $B$  to  $C$ , because  $AB = AC$  and  $\angle BAC = 60^\circ$ . Consequently, the line segment  $P_1C$  is the image of  $PB$  under  $R$ , therefore  $P_1C = PB = b$ . This means that  $\triangle PCP_1$  has side lengths  $a, b, c$ . Because of the assumption that  $c = \max\{a, b, c\}$ , and since  $\angle APP_1 = 60^\circ$ , this triangle is degenerate if and only if  $\angle APC = 60^\circ = \angle ABC$ , in which case  $APBC$  is a cyclic quadrilateral. The latter means that the point  $P$  lies on the arc  $\widehat{BC}$  of the circumcircle of  $\triangle ABC$ . ■

The next problem, by **Igor Sharygin**, is another nice application of an auxiliary circle.

*Let  $ABCD$  be a cyclic quadrilateral such that  $AD + BC = AB$ . Prove that the bisectors of the angles  $ADC$  and  $BCD$  meet on the line  $AB$ .*



Trying to find a clever solution, everybody begins with the point  $P$  on the side  $AB$  for which  $AP = AD$ ,  $BP = BC$  (such a point exists, because  $AD + BC = AB$ ). Unfortunately,  $P$  does *not* have the property we would like it to have: it is not, in general, the intersection point of the two bisectors in question. But introducing  $P$  is still helpful. Consider the circumcircle  $c_1$  of  $\triangle CDP$ ; its second intersection point  $Q$  with  $AB$  is the place where the bisectors of  $\angle ADC$  and  $\angle BCD$  meet! Indeed, assume for definiteness that  $A, P, Q$  and  $B$  are in this order on  $AB$ . Then we have  $\angle CDQ = \angle CPQ$ , since these angles are inscribed in the same arc of  $c_1$ . On the other hand,  $\triangle CPB$  is isosceles, so  $\angle CPQ = \angle CPB = \frac{1}{2}(180^\circ - \angle ABC)$ . Finally,  $\angle ABC = 180^\circ - \angle ADC$ , because  $ABCD$  is a cyclic quadrilateral. All this implies that

$$\angle CDQ = \frac{1}{2}\angle ADC;$$

in the same way,  $\angle DCQ = \frac{1}{2}\angle BCD$ . Therefore  $DQ$  and  $CQ$  are indeed the bisectors of  $\angle ADC$  and  $\angle BCD$ , so we are done. ■

A closely related statement was on the 1985 IMO. It can be proved by using the same auxiliary circle.

*A circle has center on the side  $AB$  of the cyclic quadrilateral  $ABCD$ . The other three sides are tangent to the circle. Prove that  $AD + BC = AB$ .*

Our last problem became extremely popular after appearing on the same 1985 IMO.

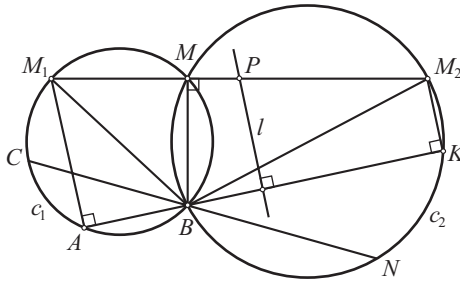
*A circle with center  $O$  passes through the vertices  $A$  and  $C$  of the triangle  $ABC$  and intersects the line segments  $AB$  and  $BC$*

again at distinct points  $K$  and  $N$ , respectively. The circumcircles of the triangles  $ABC$  and  $KBN$  intersect at exactly two distinct points  $B$  and  $M$ . Prove that angle  $OMB$  is a right angle.

There are at least half a dozen solutions, some of them truly nice. We present probably the most elementary one which, in addition, proves more:

The circles  $c_1$  and  $c_2$  intersect at the distinct points  $B$  and  $M$ . Let  $A$  and  $C$  be two points on  $c_1$ . Suppose that  $AB$  and  $CB$  intersect  $c_2$  at  $K$  and  $N$ , respectively, and let  $O$  be the common point of the perpendicular bisectors of  $AK$  and  $CN$ . Then angle  $OMB$  is a right angle.

Note that here  $A$ ,  $C$ ,  $K$  and  $N$  do *not* have to lie on a circle.



To show that  $\angle OMB = 90^\circ$  means to show that  $O$  lies on the line through  $M$  perpendicular to  $BM$ . Then just draw this line! Let it meet  $c_1$  and  $c_2$  at  $M_1$  and  $M_2$ , respectively. Since  $\angle BMM_1 = \angle BMM_2 = 90^\circ$ , it is clear that  $BM_1$  and  $BM_2$  are diameters of  $c_1$  and  $c_2$ , respectively, so  $\angle M_1AB = \angle M_2KB = 90^\circ$ . Thus  $AM_1$  and  $KM_2$  are parallel to one another, and, of course, to the perpendicular bisector  $l$  of  $AK$ , too. This implies that  $l$  intersects  $M_1M_2$  at its midpoint  $P$ . But  $AK$  and  $CN$  are equally nice; the same argument works for the perpendicular bisector of  $CN$ , so the latter also meets  $M_1M_2$  at its midpoint  $P$ . Hence  $O$  coincides with  $P$ , and the proof is complete. ■

# 25

## Extremal Arguments

To shorten the explanations about the title, let us consider a simple example.

*Several counters are placed on a chessboard. At each move one of the counters goes to any vacant square it has not previously occupied. Each counter visits every square of the board and returns to its starting position at its final move. Prove that there is a moment when none of the counters occupies its initial position.*

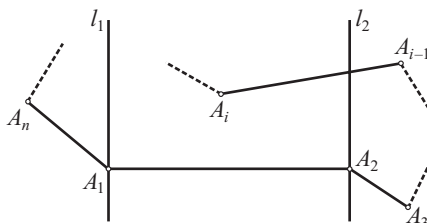
Consider the counter  $A$  that first came back to its initial square. The moment preceding its last move is exactly the one we need. Indeed, each counter must have left its initial position at this moment, otherwise  $A$  would not have been able to visit all squares. And none of the counters had come back yet, since the first one to do this was  $A$ , which did this only at its next move. ■

In many problems it proves helpful to consider “extremes”: the smallest or the largest of several numbers, the leftmost or rightmost among several points on a line, and so on. Combined with the hypothesis, their extremal properties often yield a solution.

*Prove that in each closed polygonal path there exist a side  $AB$  and a vertex  $C$ , different from  $A$  and  $B$ , such that the foot of the perpendicular from  $C$  to the line  $AB$  lies on the closed line segment  $AB$ , and not on its extension.*

Denote the polygonal path by  $L = A_1A_2 \dots A_n$  and consider its longest side; assume that it is  $A_1A_2$ . Draw the lines  $l_1$  and  $l_2$  through  $A_1$  and  $A_2$ , respectively, perpendicular to  $A_1A_2$ , as in the accompanying figure. It suffices to prove that some vertex of  $L$  lies between  $l_1$  and  $l_2$ . If this is true for  $A_3$  or  $A_n$ , the proof is finished. Otherwise  $A_n$  lies to the left of  $l_1$  and  $A_3$  lies to the right of  $l_2$ . Indeed, if both  $A_3$  and  $A_n$  were to lie

on the same side of  $l_1$  or of  $l_2$ , then either  $A_2A_3$  or  $A_nA_1$  would exceed  $A_1A_2$ , a contradiction. Since the (open) polygonal line  $A_3 \dots A_n$  connects  $A_3$  with  $A_n$ , at least one of its sides intersects  $l_2$ . Denote a side of this kind by  $A_{i-1}A_i$ ,  $A_{i-1}$  being to the right of  $l_2$ . Then  $A_i$  must lie between  $l_1$  and  $l_2$ , because  $A_{i-1}A_i \leq A_1A_2$ , and  $A_1A_2$  equals the distance between the parallel lines  $l_1$  and  $l_2$ . ■



The next problem became known as the *dwarfs problem*. Proposed at the All-Union Olympiad and later to the readers of *Kvant*, it was recognized as one of the best problems of the year 1977.

*Seven dwarfs sit around a circular table waiting for Snow White. Each of them has a big cup, and some of the cups contain milk. The first dwarf pours out all his milk into the other six cups, dividing it into six equal portions. The dwarf to his right follows suit. And so they continue until the seventh dwarf pours out all his milk into the other six cups (dividing it into six equal portions). Then it turns out that each of the dwarfs has exactly as much milk as he had at the beginning. Find the amount of milk originally contained in each cup if the total amount was 3 pints.*

One is likely to guess the answer: The dwarfs, taken in the order they poured the milk, have had

$$\frac{6}{7}, \frac{5}{7}, \frac{4}{7}, \frac{3}{7}, \frac{2}{7}, \frac{1}{7}, 0 \quad (1)$$

pints. That (1) is a solution is easy to check and not so surprising. It seems plausible that the amounts of milk remain unchanged after each dwarf's pourings, that is, the initial distribution is just being "rotated." The shortest way to prove that there are no other solutions is probably the following.

Since the original distribution was restored after one cycle, we may imagine the process of pourings repeated periodically. So let us consider a cycle of seven pourings starting with the dwarf  $D$  who had the *least* amount

of milk in his cup before he started dividing and pouring it. Denote this amount by  $a$  pints. Then any other dwarf had at least  $a$  pints of milk before dividing it, and  $\frac{1}{6}$  of this quantity was given to  $D$ . Thus, after all dwarfs did their job,  $D$  had got at least  $6 \cdot a/6 = a$  pints from the remaining six. By hypothesis, this amount should have been equal to the one he initially had, i.e., *exactly* equal to  $a$  pints. The latter is possible if and only if there were *exactly*  $a$  pints in each dwarf's cup before he started his work. It follows immediately from here that the dwarfs, taken in the order they poured the milk, have had  $a, \frac{5}{6}a, \frac{4}{6}a, \frac{3}{6}a, \frac{2}{6}a, \frac{1}{6}a, 0$  pints, respectively. Since the total amount of milk was 3 pints, we find that  $a = \frac{6}{7}$ , so (1) is indeed the unique solution. ■

The argument goes through just as well if the first dwarf has the other extreme, namely the greatest amount of milk.

It is worth noting that, regardless of the initial distribution, the amounts in the seven cups tend to the 7-tuple (1) if the pourings are repeated sufficiently many times.

*Let  $f: \mathbf{N} \rightarrow \mathbf{N}$ ,  $g: \mathbf{N} \rightarrow \mathbf{N}$  be functions such that  $f$  is surjective,  $g$  is injective and*

$$f(n) \geq g(n) \quad \text{for all } n \in \mathbf{N}.$$

*Prove that  $f(n) = g(n)$  for each  $n \in \mathbf{N}$ .*

This problem, proposed by **Gheorghe Eckstein**, was on a Romanian 1986 IMO selection test.

Suppose, by way of contradiction, that  $f(n) \neq g(n)$  for some  $n \in \mathbf{N}$ . Then it follows from the hypothesis that  $f(n) > g(n)$ . So the subset of  $\mathbf{N}$

$$A = \{g(n) \mid f(n) > g(n)\}$$

is nonempty, and hence it has a least element. Denote this element by  $m_0$  and let  $m_0 = g(n_0)$  for some  $n_0 \in \mathbf{N}$ . Note that  $n_0$  is unique, because  $g$  is injective. Since  $f$  is surjective, there exists an  $n_1 \in \mathbf{N}$  for which  $m_0 = f(n_1)$ . Clearly  $n_0 \neq n_1$ , because

$$f(n_0) > g(n_0) = m_0 = f(n_1).$$

The hypothesis implies that  $m_0 = f(n_1) \geq g(n_1)$ , and we have two possibilities.

If  $f(n_1) = g(n_1)$ , then  $g(n_1) = m_0 = g(n_0)$ , which contradicts the uniqueness of  $n_0$ .

If  $f(n_1) > g(n_1)$ , then  $g(n_1) \in A$ . Since  $f(n_1) = m_0$ , it turns out that  $g(n_1)$  is less than the least element  $m_0$  of  $A$ .

In both cases our assumption leads to a contradiction. ■

Here is a beautiful problem from the 1990 edition of the Kürschák competition.

*We wander on the coordinate plane in accordance with the following rules. From a given point  $P(x, y)$  we are allowed to move to one of the four points  $(x, y + 2x)$ ,  $(x, y - 2x)$ ,  $(x - 2y, y)$ ,  $(x + 2y, y)$ . There is only one restriction: If we have moved from  $P$  to  $Q$ , then we cannot go back to  $P$  immediately.*

*Prove that, starting from the point  $(1, \sqrt{2})$ , we cannot return to it any more.*

The key idea is: *If  $P$  is not on one of the axes or on one of the lines  $y = x$  and  $y = -x$ , then exactly one of the possible moves leads closer to the origin, and the other three lead away from it.* This can be verified geometrically (the four points that are “one move away” from  $P$  are the vertices of a rhombus). We prefer to do the verification algebraically. The squares of the distances from the origin  $O$  to  $(x, y + 2x)$ ,  $(x, y - 2x)$ ,  $(x - 2y, y)$ ,  $(x + 2y, y)$ , respectively, are

$$x^2 + (y + 2x)^2 = x^2 + y^2 + 4x(x + y),$$

$$x^2 + (y - 2x)^2 = x^2 + y^2 + 4x(x - y),$$

$$(x - 2y)^2 + y^2 = x^2 + y^2 + 4y(y - x),$$

$$(x + 2y)^2 + y^2 = x^2 + y^2 + 4y(y + x).$$

If  $xy \neq 0$  and  $x \neq \pm y$ , we may assume without loss of generality that  $x^2 > y^2 > 0$ . Then

$$x(x + y) + x(x - y) = 2x^2 > 0,$$

$$x(x + y) \cdot x(x - y) = x^2(x^2 - y^2) > 0,$$

$$y(y - x) \cdot y(y + x) = y^2(y^2 - x^2) < 0.$$

These relations imply that both of the numbers  $x(x + y)$  and  $x(x - y)$  are positive (having a positive sum and a positive product), whereas exactly one of the numbers  $y(y - x)$  and  $y(y + x)$  is positive and the other one is negative. This proves the statement given above in italics.

Now let us suppose that it is possible after a certain series of steps  $P_0 \rightarrow P_1 \rightarrow P_2 \rightarrow \dots \rightarrow P_n \rightarrow P_0$  to come back to the point  $P_0(1, \sqrt{2})$

again. The coordinates of the initial point are nonzero and their ratio is irrational, so this was also true after each move. In particular, none of the points  $P_0, P_1, P_2, \dots, P_n$  lies on either of the axes or on either of the lines  $y = x$  and  $y = -x$ .

Consider the point  $P_i$  for which the distance  $OP_i$  is a maximum. Then  $OP_{i-1} < OP_i$ , hence the only possible move from  $P_i$  to a point closer to  $O$  is  $P_i \rightarrow P_{i-1}$ . (Indices are taken modulo  $n + 1$ .) However, we also have  $OP_{i+1} < OP_i$ , so  $P_{i-1}$  and  $P_{i+1}$  coincide; that is, we have moved back from  $P_i$  to  $P_{i-1}$  immediately. But such a move is forbidden — a contradiction! ■

## Coffee Break 5

1. Assume that the following statements are true:

- (a) Not all *Mathematical Miniatures* readers are US residents;
- (b) All *Chicago Bulls* fans who are not US residents, are not *Mathematical Miniatures* readers.

Do (a) and (b) imply

- (c) Not all *Mathematical Miniatures* readers are *Chicago Bulls* fans?

2. A circular segment is given whose arc is shorter than a semicircle, and a point  $A$  is fixed on its chord. For which point  $X$  of the arc of the segment is the distance  $AX$  minimal?

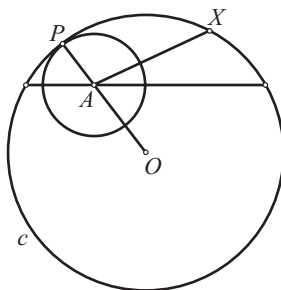
3. Each of  $n$  male citizens knows a different piece of gossip. They are allowed to exchange the gossip they know by phone. During a call, just one of the men speaks and tells the other all the gossip he knows. Determine the minimum number of calls required to enable each man to know all the gossip.

## Solutions

1. The answer is yes. By (a), there are *Mathematical Miniatures* readers who are not US residents. Let  $A$  be one of them. We claim that  $A$  is not a *Chicago Bulls* fan, implying that (c) also holds true.

Assume on the contrary that  $A$  is a *Bulls* fan. Since he is not a US resident, (b) implies that  $A$  is not a *Mathematical Miniatures* reader, which is not true.

2. It is curious that many people cannot guess the answer; the point in question is *not*, for example, the point where the perpendicular to the chord erected at  $A$  intersects the arc. We can, of course, use calculus; but here is another solution.



The given segment is a part of a circle. Denote this circle by  $c$  and its center by  $O$ . Suppose that the ray  $\overrightarrow{OA}$  intersects  $c$  at  $P$ . Then  $P$  is the point we are after. Indeed, consider the circle centered at  $A$  with radius  $AP$ . It is internally tangent to  $c$ , because  $O$ ,  $A$  and  $P$  are collinear. Hence each point  $X$  of the given arc different from  $P$  is outside the new circle, and this implies that  $AX > AP$ .

3. Consider the first moment when some man comes to know all the gossip. Clearly, it was preceded by at least  $n - 1$  calls. This is because everyone else must have told at least one other person his bit of news. So everyone else must have been engaged in at least one call during which he spoke. On the other hand, at the same moment each of the remaining  $n - 1$  men is lacking some piece of gossip, so at least  $n - 1$  more calls are needed. Thus the minimum number of the calls is at least  $(n - 1) + (n - 1) = 2n - 2$ . It is easy to show that  $2n - 2$  is indeed the required minimum. For example, one of the men may call all the others (this makes  $n - 1$  calls), collect

the gossip they have, and then call them back to give them all the gossip (another  $n - 1$  calls).

# 26

## The AMS Inequality

An intriguing cyclic inequality was proposed by the Moscow mathematician **D. P. Mavlo** and published in *Matematika*, No. 4, 1987:

*Let  $a, b, c$  be positive numbers. Prove that*

$$\frac{1}{a(1+b)} + \frac{1}{b(1+c)} + \frac{1}{c(1+a)} \geq \frac{3}{1+abc},$$

*and that equality occurs if and only if  $a = b = c = 1$ .*

This was intended to be one of the difficult problems in the issue, and it really was. Most of the proofs sent in, including the author's, relied upon calculus. It was a surprise, however, to get a short and elementary solution like the one below. It was devised by the Bulgarian student **Zhivko Georgiev**, a ninth-grader at that time.

Adding  $3/(1+abc)$  to both sides (who would ever think of this?) yields the equivalent inequality

$$\frac{1}{a(1+b)} + \frac{1}{b(1+c)} + \frac{1}{c(1+a)} + \frac{3}{1+abc} \geq \frac{6}{1+abc}. \quad (1)$$

Notice that

$$\begin{aligned} \frac{1}{a(1+b)} + \frac{1}{1+abc} &= \frac{1+a+ab+abc}{a(1+b)(1+abc)} \\ &= \frac{1}{1+abc} \left[ \frac{1+a}{a(1+b)} + \frac{b(1+c)}{1+b} \right] \end{aligned}$$

and, analogously,

$$\frac{1}{b(1+c)} + \frac{1}{1+abc} = \frac{1}{1+abc} \left[ \frac{1+b}{b(1+c)} + \frac{c(1+a)}{1+c} \right],$$

$$\frac{1}{c(1+a)} + \frac{1}{1+abc} = \frac{1}{1+abc} \left[ \frac{1+c}{c(1+a)} + \frac{a(1+b)}{1+a} \right].$$

If we now set

$$x = \frac{1+a}{a(1+b)}, \quad y = \frac{1+b}{b(1+c)}, \quad z = \frac{1+c}{c(1+a)},$$

the left-hand side of (1) becomes

$$\frac{1}{1+abc} \left[ x + \frac{1}{y} + y + \frac{1}{z} + z + \frac{1}{x} \right]. \quad (2)$$

We have  $u + 1/u \geq 2$  for  $u > 0$  by the AM–GM inequality, with equality if and only if  $u = 1$ . Hence the expression (2) is not less than

$$\frac{1}{1+abc} [2 + 2 + 2] = \frac{6}{1+abc}.$$

Equality occurs if and only if  $1+a = a(1+b)$ ,  $1+b = b(1+c)$ ,  $1+c = c(1+a)$ , which implies  $a = b = c = 1$ . ■

The story of this remarkable solution has an unexpected development. A little later *Kvant* (Nos. 11–12, 1988, problem M1136) published another cyclic inequality (take a closer look at the notation):

*Prove the inequality*

$$3 + (A + M + S) + \left( \frac{1}{A} + \frac{1}{M} + \frac{1}{S} \right) + \left( \frac{A}{M} + \frac{M}{S} + \frac{S}{A} \right) \\ \geq \frac{3(A+1)(M+1)(S+1)}{AMS+1}$$

for all positive numbers  $A, M, S$ .

This problem was dedicated to the centennial of the American Mathematical Society, and does not seem to be an easy one. The solution published in *Kvant's* No. 5, 1989, sounds more like a hint. Basically, it says, “The proof reduces to a rather artificial (but also ingenious!) algebraic manipulation. Multiply the difference of the left-hand and the right-hand sides by  $AMS(AMS+1)$ , regroup the summands, and you will obtain

$$AM(M+1)(SA-1)^2 + MS(S+1)(AM-1)^2 + SA(A+1)(MS-1)^2,$$

a nonnegative quantity.” This proof was found by **P. H. Diananda** from Singapore, a renowned expert in cyclic inequalities.

On the other hand, it is not hard to prove the AMS inequality by reducing it to the previous problem.

Represent  $3 + (A + M + S)$  as  $(1 + A) + (1 + M) + (1 + S)$ . Next, notice that

$$\begin{aligned}(1 + A) + \frac{1}{M} + \frac{A}{M} &= (1 + A) \left(1 + \frac{1}{M}\right) \\ &= \frac{(1 + A)(1 + M)}{M}.\end{aligned}$$

Then work with the remaining summands analogously. The left-hand side becomes

$$\frac{(1 + A)(1 + M)}{M} + \frac{(1 + M)(1 + S)}{S} + \frac{(1 + S)(1 + A)}{A}.$$

Now it suffices to divide both sides by  $(A + 1)(M + 1)(S + 1)$ , getting an inequality equivalent to the previous one. ■

# 27

## Helly's Theorem for One Dimension

*During a certain lecture, each of the students in the hall fell asleep exactly once. For each pair of students, there was some moment when both were sleeping simultaneously. Prove that, at some moment, all the students were sleeping simultaneously.*

Denote by  $A$  the student who was the first to wake up during the lecture. Any other student  $B$  fell asleep before  $A$  woke up, otherwise the two could not have slept simultaneously. And since  $B$  woke up later than  $A$ , the moment when  $A$  woke up has the desired property. ■

Playing around with this problem makes it easier to remember an important theorem—Helly's theorem on the line:

*If each pair of a finite collection of intervals on a line has a common point, then all these intervals have a common point.*

To prove this, note that the given intervals may be regarded as intervals on the time axis during which the students from the previous problem were sleeping. The moment when everyone was sleeping corresponds to a common point of all the intervals. ■

The theorem holds for any intervals: bounded or unbounded; open, closed or half-open. This is not surprising, since a subset of the real line is convex if and only if it is an interval (and Helly's name is naturally associated with convexity). So the argument above is not exhaustive, but it is not our goal to provide a rigorous proof, and we pass on to an application.

An easy consequence of Helly's theorem is a statement about convex figures in the plane, each pair of which has a point in common. The two-dimensional form of Helly's theorem requires that every *three* of the figures have a common point, and concludes that this is true for all the figures. A weaker assertion holds in our case:

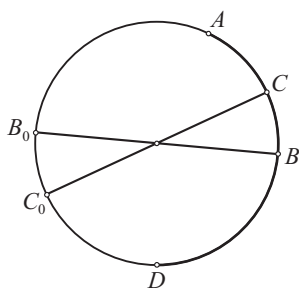
*Finitely many convex figures are given in the plane, and each pair of them has a common point. Prove that for each line  $\ell$  in the plane there exists a line parallel to  $\ell$  that intersects all these figures.*

Take a line  $m$  perpendicular to  $\ell$  and project all the figures onto it. The projections are intervals, because the figures are convex. Now, a point that is common to two figures projects onto a point common to their projections. Thus we obtain a finite collection of intervals on the line  $m$  to which Helly's theorem may be applied. It follows that there is a point  $P$  common for all these intervals. Then the line  $\ell'$  through  $P$  perpendicular to  $m$  intersects all of the given figures, and  $\ell'$  is parallel to  $\ell$ . ■

How about arcs on the circle instead of intervals on the line? One can easily find three arcs on a circle, any two of which have a common point but not all three of them. However, a weaker statement is true:

*Finitely many arcs are given on a circle. Each pair of them has a common point. Then there exists a line through the center of the circle that intersects all these arcs.*

An arc longer than a half-circle contains an endpoint of any diameter. This is also the case if the arc is a half-circle containing at least one of its endpoints. So let us assume without loss of generality that each arc is contained in an open half-circle. In particular, the common part of two arcs is a single arc (or a point). Clearly, we may also assume that no arc contains another one.

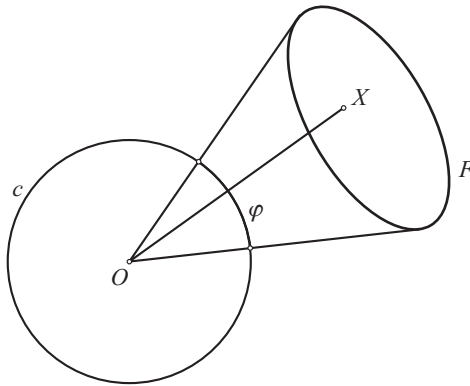


Consider two arcs  $\widehat{AB}$  and  $\widehat{CD}$  whose intersection  $\widehat{CB}$  is minimal. That is, if another intersection of two arcs is contained in  $\widehat{CB}$ , then it coincides with  $\widehat{CB}$ . Note that then  $\widehat{CB}$  (open, closed or half-open) *either is entirely contained in any given arc  $\alpha$  or does not intersect  $\alpha$* . We claim that each diameter with one endpoint on  $\widehat{CB}$  has the stated property.

To prove this, consider the diameters  $BB_0$  and  $CC_0$ . It suffices to prove that if an arc  $\alpha$  does not intersect  $\widehat{CB}$ , then  $\alpha$  contains the diametrically opposite arc  $\widehat{C_0B_0}$ . Indeed, observe first that  $A$  and  $D$  are on the closed arcs  $\widehat{CB_0}$  and  $\widehat{BC_0}$ , respectively. Since  $\alpha$  intersects  $\widehat{AB}$  and  $\widehat{CD}$  but not  $\widehat{CB}$ , there must be points of  $\alpha$  contained in each of the closed arcs  $\widehat{AC}$  and  $\widehat{DB}$ . Taking into account again that  $\alpha$  does not intersect  $\widehat{CB}$ , we infer that  $\alpha$  contains the whole arc  $\widehat{C_0B_0}$ , as needed. ■

This is an analog of the one-dimensional version of Helly's theorem for a circle. Now we are able to say something more about convex figures in the plane, any two of which intersect.

*Each pair of a finite number of convex figures in the plane has a common point. Prove that for each point  $O$  in this plane there exists a line through  $O$  that intersects all these figures.*



Take an arbitrary circle  $c$  centered at  $O$  and project centrally all the figures onto  $c$  with center  $O$ . In other words, if  $F$  is one of the given figures, we construct an arc  $\varphi$  on  $c$  corresponding to  $F$  as follows. Draw all rays  $\overrightarrow{OX}$ , where  $X$  ranges over  $F$ , and denote by  $\varphi$  the set of their intersection points with  $c$ . It is clear that  $\varphi$  is an arc, since  $F$  is convex. Note that such a correspondence is not defined for  $O$  itself if it belongs to  $F$ . But we may consider only figures not containing  $O$ , because otherwise any line through  $O$  intersects such a figure.

Thus we obtain a finite family of arcs on the circle  $c$ , and the conditions of the problem imply that each pair of them has a common point. Hence each arc contains one of the endpoints of some diameter in  $c$ . The line containing this diameter intersects each of the given figures. ■

# 28

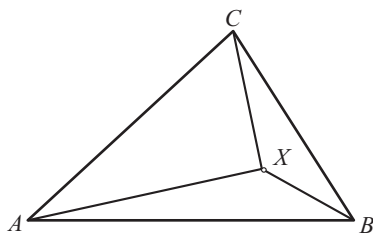
## Two Approaches

It is a really old problem which says that:

*If  $X$  is an arbitrary point interior to the triangle  $ABC$ , then the sum  $AX + BX + CX$  is greater than the semiperimeter of the triangle and less than its perimeter.*

The first part is easy: apply the triangle inequality to  $\triangle ABX$ ,  $\triangle BCX$  and  $\triangle CAX$  to obtain

$$AX + BX > AB, \quad BX + CX > BC, \quad CX + AX > CA.$$



Adding these inequalities gives

$$2(AX + BX + CX) > AB + BC + CA,$$

or

$$AX + BX + CX > \frac{1}{2}(AB + BC + CA).$$

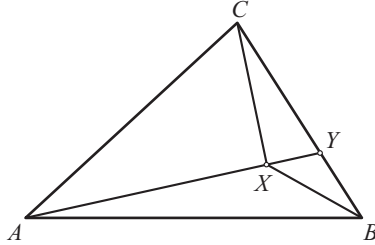
The second part seems easy, too, but its proof is not straightforward. It rests on a fact interesting in itself:

*If  $X$  is an arbitrary point interior to the triangle  $ABC$ , then*

$$AX + BX < AC + BC.$$

To prove this, extend  $AX$  until it meets  $BC$  at  $Y$ . Then, by the triangle inequality,  $BX < XY + BY$  and  $AY < AC + YC$ . Hence

$$\begin{aligned} AX + BX &< AX + XY + BY = AY + BY \\ &< AC + YC + BY = AC + BC. \end{aligned}$$



Now we are ready to prove that  $AX + BX + CX$  is less than the perimeter of  $\triangle ABC$ . Since  $AX + BX < AC + BC$  and, by symmetry,  $BX + CX < BA + CA$ ,  $CX + AX < CB + AB$ , we get

$$2(AX + BX + CX) < 2(AB + BC + CA),$$

or

$$AX + BX + CX < AB + BC + CA. \blacksquare$$

Thus the perimeter of  $\triangle ABC$  turns out to be an upper bound for the sum  $AX + BX + CX$  but not its maximum value, because we have the strict inequality  $AX + BX + CX < AB + BC + CA$ . As a matter of fact, this inequality holds for “almost” any point  $X$  on the boundary of  $\triangle ABC$  as well, but we are not going to explain why right now. Let us find the maximum of  $AX + BX + CX$  instead, letting  $X$  range over the interior and the boundary of  $ABC$ .

*The maximum of  $AX + BX + CX$  is the length sum of the two longest sides of the triangle  $ABC$ .*

The proof we are going to present here starts by recalling a simple fact that seems indispensable in many geometry problems:

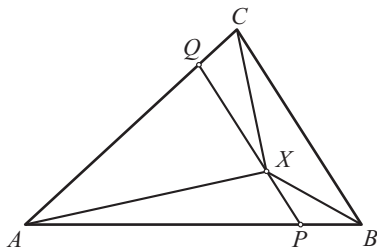
*The distance between a vertex of a triangle and a point on the opposite side does not exceed the length of the longer of the other two sides of the triangle.*

Now let the side lengths of  $\triangle ABC$  be  $BC = a$ ,  $CA = b$ ,  $AB = c$ , and let  $a \leq b \leq c$ . We have to prove that

$$AX + BX + CX \leq b + c$$

for any point  $X$  in  $\triangle ABC$ , and that the equality is possible.

Suppose first that  $X$  lies on a side of the triangle, say  $AB$ . We obtain  $AX + BX = AB = c$ ,  $CX \leq \max(AC, BC) = AC = b$ , and hence  $AX + BX + CX \leq b + c$ . The cases when  $X$  is on  $BC$  or  $CA$  are analogous. Now let us take  $X$  in the interior of the triangle. One gets a nice proof by drawing the parallel through  $X$  to  $BC$ , the shortest side. Let this parallel meet  $AB$  and  $AC$  at  $P$  and  $Q$ , respectively.



First, the triangle inequality, applied to  $\triangle BPX$  and  $\triangle CQX$ , yields  $BX < BP + PX$ ,  $CX < CQ + QX$ . Therefore

$$BX + CX < BP + CQ + PQ.$$

The fact mentioned above implies  $AX \leq \max(AP, AQ)$ . Next, the triangles  $APQ$  and  $ABC$  are similar, so the longest side of  $\triangle APQ$  is  $AP$ . Thus  $AX \leq AP$ , and since  $BX + CX < BP + CQ + PQ$ , we get

$$AX + BX + CX < AP + BP + CQ + PQ = AB + CQ + PQ.$$

Finally,  $PQ$  is the shortest side in  $\triangle APQ$ , so  $PQ \leq AQ$ . Then

$$\begin{aligned} AX + BX + CX &< AB + CQ + PQ \leq AB + CQ + AQ \\ &= AB + AC = b + c. \end{aligned}$$

It remains to notice that if  $X$  coincides with the vertex  $A$ , then

$$AX + BX + CX = 0 + AB + AC = b + c.$$

We thus proved that  $b + c$  is indeed the maximum of  $AX + BX + CX$ . ■

This is a proof following the ancient Greek tradition—inventive and elaborate. There is, however, a “modern” approach that does not require much inventiveness.

To apply it, we need to recall one more fact from the geometry of the triangle.

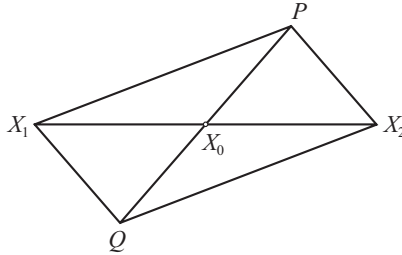
Consider any points  $X_1, X_2$  in the plane, and let  $X_0$  be the midpoint of the line segment  $X_1X_2$ . Then for any point  $P$  we have

$$PX_0 \leq \frac{1}{2}(PX_1 + PX_2). \quad (1)$$

Indeed, we can double the median  $PX_0$  of the (possibly degenerate) triangle  $X_1X_2P$  to obtain the parallelogram  $PX_1QX_2$ . The triangle inequality then yields

$$2PX_0 = PQ \leq PX_1 + X_1Q = PX_1 + PX_2,$$

and (1) follows. The equality occurs when  $P$  is on the line  $X_1X_2$  but not on the open line segment  $X_1X_2$ .



Apply (1), replacing  $P$  by  $A, B$  and  $C$ , the vertices of  $\triangle ABC$ . Adding up the resulting inequalities and setting  $s(X) = AX + BX + CX$  for any point  $X$  in the plane, we get

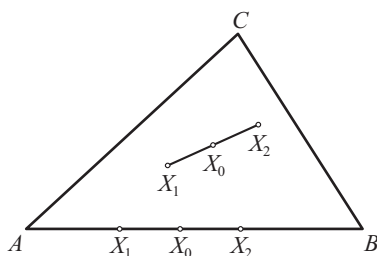
$$s(X_0) < \frac{1}{2}[s(X_1) + s(X_2)]. \quad (2)$$

Note that the inequality is strict, because  $A, B$  and  $C$  cannot be all on the line  $X_1X_2$ .

From a more advanced point of view, it is a standard job to find the maximum value of the function  $s(X) = AX + BX + CX$  when  $X$  ranges over  $\triangle ABC$ . Note first that  $s(X)$  does have a maximum, because it is continuous, and  $\triangle ABC$  is a closed bounded region in the plane. The maximum cannot be assumed at a point  $X_0$  in the interior of the triangle. Indeed, in this case one can find a line segment  $X_1X_2$  in  $\triangle ABC$  with midpoint  $X_0$ . Then (2) shows that  $s(X_0) < s(X_1)$  or  $s(X_0) < s(X_2)$ , and hence  $s(X_0)$  is not the maximum of  $s(X)$ . The same argument shows that a point of maximum cannot lie in the interior of a side either.

It remains to compare the values

$$s(A) = b + c, \quad s(B) = c + a, \quad s(C) = a + b.$$



Since, by assumption,  $a \leq b \leq c$ , the largest of them is  $b + c$ , and the proof is complete. ■

Compare this solution with the previous one. It is a short and standard argument (and can be reduced to a couple of sentences for a reader familiar with convexity), avoiding the additional construction and the sophisticated reasoning. But it lacks the beauty of the old-fashioned purely geometric proof.

Essentially speaking, the second approach yields a proof of a rather general result. A strictly convex function defined on a convex set can take on its maximum value only at points “extremal” for the set. These are points not lying in the interior of any line segment contained in the given set.

Mathematicians in ancient Greece were like skillful and diligent workers who, with only hammers and chisels, put much of their strength, time and ingenuity into breaking a rock into pieces. Quite unlike a modern mathematician, who would just blow up the rock like an experienced miner and bring out the treasures from beneath it.

# 29

## Radical Axis

Each two nonconcentric circles in the plane have a *radical axis*, the locus of points of equal powers with respect to the two circles. This is a line perpendicular to the centerline of the circles. Each three circles with non-collinear centers determine three radical axes, each of them corresponding to a pair of circles. These three axes have a unique common point called the *radical center* of the circles.

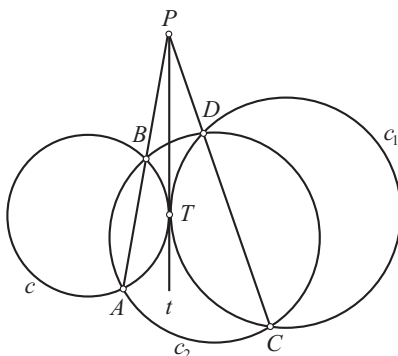
These facts are extremely useful in certain geometric settings. Let us also note that the radical axis of two intersecting circles is the line containing their common chord, and the radical axis of two tangent circles is their common tangent.

Our first example is simple and classic.

*Construct, using straightedge and compass, a circle passing through two given points and tangent to a given circle.*

Suppose we want to construct a circle  $c$  passing through the given points  $A$  and  $B$  and tangent to the given circle  $c_1$ . It suffices to determine the point of tangency  $T$  of  $c$  and  $c_1$ .

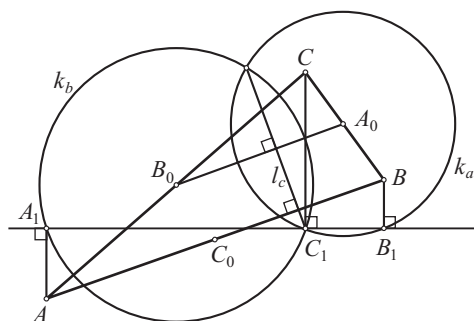
Now, the common tangent  $t$  of  $c$  and  $c_1$  is their radical axis, and the line  $AB$  is the radical axis of any two circles having  $AB$  as their common chord. So let us draw an arbitrary third circle  $c_2$  through  $A$  and  $B$ , and let it meet  $c_1$  at  $C$  and  $D$ . Then  $CD$  is the radical axis of  $c_1$  and  $c_2$ , and since  $AB$  is the radical axis of  $c$  and  $c_2$ , the common point  $P$  of  $AB$  and  $CD$  is the radical center of  $c$ ,  $c_1$  and  $c_2$ . Knowing  $P$ , we may just draw the tangent from  $P$  to  $c_1$ , thus getting  $T$ , the point of tangency we need. It remains to note that all constructions described above can be performed using straightedge and compass. ■



Denote by  $A_1, B_1, C_1$  the orthogonal projections of the vertices  $A, B, C$  of a given triangle  $ABC$  onto a line in its plane. Prove that the perpendiculars from  $A_1, B_1, C_1$  to the lines  $BC, CA, AB$ , respectively, concur.

This is a well-known statement, but the proof below is not so well known.

First note that the perpendicular bisector of the line segment  $C_1A_1$  contains the midpoint  $B_0$  of the side  $CA$ ; hence  $B_0$  is equidistant from  $A_1$  and  $C_1$ . Analogously, we have  $A_0B_1 = A_0C_1$  and  $C_0A_1 = C_0B_1$ , where  $A_0$  and  $C_0$  are the midpoints of  $BC$  and  $AB$ .

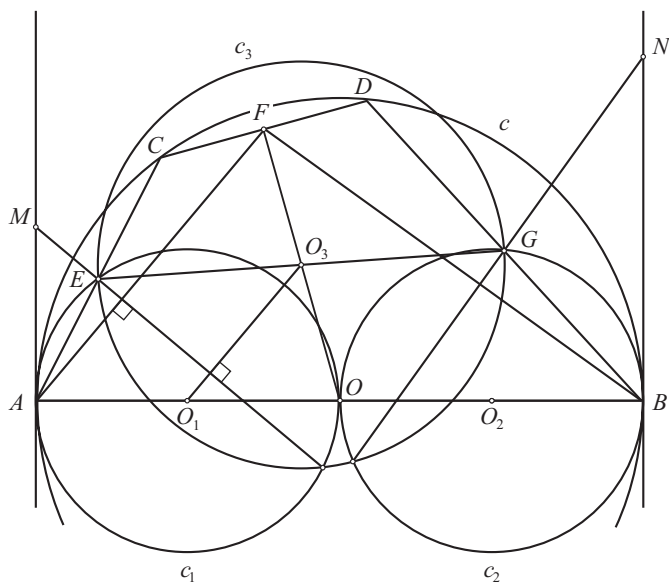


Now consider the three circles  $k_a, k_b, k_c$  centered at  $A_0, B_0, C_0$  with radii  $A_0B_1, B_0C_1, C_0A_1$ , respectively (only  $k_a$  and  $k_b$  are exhibited in the figure). The radical axis  $l_c$  of  $k_a$  and  $k_b$  passes through their common point  $C_1$  and it is perpendicular to their centerline  $A_0B_0$ . But  $A_0B_0 \parallel AB$ , implying  $l_c \perp AB$ . Consequently,  $l_c$  coincides with the perpendicular from  $C_1$  to the line  $AB$ . By symmetry, the perpendiculars from  $A_1$  and  $B_1$  to  $BC$  and  $CA$  coincide with the radical axes  $l_a$  and  $l_b$  of the pairs of circles  $k_b, k_c$  and  $k_c, k_a$ , respectively. But these three radical axes concur at the

radical center of  $k_a, k_b, k_c$  (the centers  $A_0, B_0, C_0$  are noncollinear). The proof is complete. ■

The point whose existence was just proven is called the *orthopole* of  $\triangle ABC$  with respect to the given line.

Let  $C$  and  $D$  be distinct points on a semicircle with diameter  $AB$ . Denote by  $E, F, G$  the midpoints of the line segments  $AC, CD, DB$ , respectively. The perpendicular from  $E$  to  $AF$  meets the tangent to the semicircle at  $A$  at the point  $M$ , and the perpendicular from  $G$  to  $BF$  meets the tangent to the semicircle at  $B$  at the point  $N$ . Prove that  $MN$  is parallel to  $CD$ .



This is really not an easy problem. It was given at a selection test for the Romanian IMO team in the early sixties. Over the years, there have been a number of people able to solve it. But no easy proof turned up; each used some kind of computation. Here is the proof by the coach who proposed it, a truly beautiful geometric argument. It was published in *Revista Matematică Timișoara*, No. 2, 1985.

Denote by  $O$  the midpoint of  $AB$ , and by  $c, c_1, c_2, c_3$  the circles with diameters  $AB, AO, BO, EG$ , respectively. Let  $O_1, O_2, O_3$  be the centers of  $c_1, c_2, c_3$ . Since  $E$  is the midpoint of the chord  $AC$  in  $c$ , we have

$\angle AEO = 90^\circ$ , and so  $E \in c_1$ . Likewise,  $G \in c_2$ . Next, the quadrilateral  $OGFE$  is a parallelogram, because  $O, G, F, E$  are the midpoints of the sides of  $ABDC$ , so the line segments  $EF$  and  $OG$  are each parallel to  $AD$  and half as long. It follows that the diagonals  $EG$  and  $OF$  of  $OGFE$  bisect each other at  $O_3$ .

By hypothesis, the line  $ME$  passes through the common point  $E$  of the circles  $c_1$  and  $c_3$  and is perpendicular to  $AF$ . Now  $O_1O_3 \parallel AF$ , since  $O_1$  and  $O_3$  are the midpoints of  $AO$  and  $FO$ . So  $ME \perp O_1O_3$ , and hence  $ME$  is the radical axis of  $c_1$  and  $c_3$ . But  $AM$  is the radical axis of  $c$  and  $c_1$  (it is their common tangent), so  $M$  is the radical center of  $c, c_1$  and  $c_3$ . In particular,  $M$  lies on the radical axis of  $c$  and  $c_3$ .

Similarly,  $N$  is the radical center of  $c, c_2$  and  $c_3$ , and therefore  $N$  also lies on the radical axis of  $c$  and  $c_3$ . So  $MN$  is the radical axis of  $c$  and  $c_3$ , and hence  $MN \perp OO_3$ , which is to say,  $MN \perp OF$ . It remains to note that  $OF \perp CD$ , and the claim follows. ■

# 30

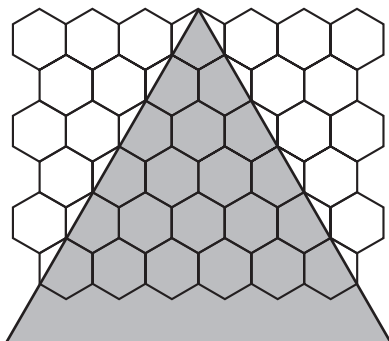
## The Pigeonhole Principle

Here we present several applications of the pigeonhole principle, one of the simplest and most powerful mathematical weapons. Most of them are published in the book *Dirichlet's Principle* by Ivan Prodanov.

The first example is from the 1983 Dutch Olympiad:

*Suppose 111 points are given within an equilateral triangle of side 15. Prove that it is always possible to cover at least 3 of these points by a round coin of diameter  $\sqrt{3}$  (part of which may lie outside the triangle).*

This is interesting, because one can choose the pigeonholes in different ways (there is little doubt, however, that the 111 points are going to be the pigeons). Tile the plane with regular hexagons of side length  $\sqrt{3}/2$ , as shown in the figure. Just a portion of the equilateral triangle  $T$  is illustrated.



The circumcircle of each such regular hexagon has diameter exactly  $\sqrt{3}$ . This looks good, doesn't it? Now let us count the hexagons sharing a common interior point with  $T$ : one in the uppermost row, two in the second

row, and so on; the hexagons lying below the tenth row have nothing in common with  $T$ . So the total number of hexagons we are interested in is

$$1 + 2 + 3 + \cdots + 10 = 55.$$

Since  $111 = 2 \cdot 55 + 1$ , the pigeonhole principle implies that at least three of the given points reside in the same hexagon. They are obviously contained in the circumcircle of this hexagon, and we are done. ■

*Let  $a_1, a_2, \dots, a_n$  be arbitrary real numbers. Prove that there exists a real number  $x$  such that each of the numbers*

$$x + a_1, x + a_2, \dots, x + a_n$$

*is irrational.*

Let  $\alpha$  be any irrational number. Then one of the numbers

$$\alpha, 2\alpha, \dots, n\alpha, (n+1)\alpha$$

will do the job. Indeed, suppose this is not true. Then there is at least one rational number in each row of the  $(n+1) \times n$  table

$$\begin{array}{cccc} \alpha + a_1 & \alpha + a_2 & \cdots & \alpha + a_n \\ 2\alpha + a_1 & 2\alpha + a_2 & \cdots & 2\alpha + a_n \\ \vdots & \vdots & \ddots & \vdots \\ (n+1)\alpha + a_1 & (n+1)\alpha + a_2 & \cdots & (n+1)\alpha + a_n \end{array}$$

Thus the table contains at least  $n+1$  rational numbers and since it has  $n$  columns, at least two of these rationals are contained in the same column. In other words, there exist positive integers  $i, j$  and  $k$ ,  $i \neq j$ , such that  $i\alpha + a_k$  and  $j\alpha + a_k$  are rational. But then the number

$$(i-j)\alpha = (i\alpha + a_k) - (j\alpha + a_k)$$

is also rational, and this contradicts the irrationality of  $\alpha$ . ■

In fact, one can do much better. It is possible to shift any *infinite* sequence  $a_1, a_2, \dots, a_n, \dots$  and get a sequence

$$x + a_1, x + a_2, \dots, x + a_n, \dots$$

consisting only of irrationals. The proof is in some sense even easier than the previous one. It rests on the fact that the integers and the rationals form countable sets, and the reals do not.

A positive integer  $k$  greater than 1 is given. Prove that there exist a prime  $p$  and a strictly increasing sequence of positive integers  $a_1, a_2, \dots, a_n, \dots$  such that the terms of the sequence

$$p + ka_1, p + ka_2, \dots, p + ka_n, \dots$$

are all primes.

This problem turned out to be one of the harder ones given on the Bulgarian selection test for the 1987 IMO. The pigeonhole principle provides an elegant solution. There is nothing to be afraid of, just infinitely many pigeons in finitely many pigeonholes.

For each  $i = 1, 2, \dots, k-1$  denote by  $P_i$  the set of all primes congruent to  $i$  modulo  $k$ . Each prime (except possibly  $k$  itself) is contained in exactly one of the sets  $P_1, P_2, \dots, P_{k-1}$ . Since there are infinitely many primes, at least one of these sets is infinite, say  $P_i$ . Let  $p = x_1 < x_2 < \dots < x_n < \dots$  be its elements arranged in increasing order, and

$$a_n = \frac{x_{n+1} - p}{k} \quad \text{for } n = 1, 2, \dots$$

Then the  $p + ka_n$  simply run through the members of  $P_i$ , beginning at  $x_2$ . The numbers  $a_n$  are positive integers. The prime  $p$  and the strictly increasing sequence  $a_1, a_2, \dots, a_n, \dots$  have the desired properties. ■

The next beautiful problem by Bulgarian tenth-grader **Bozhko Bakalov** was published in *Matematika*'s contest section, No. 6, 1990:

*For a given prime  $p$ , find the greatest positive integer  $n$  with the following property: The edges of the complete graph on  $n$  vertices can be colored with  $p+1$  colors so that*

- (a) *at least two edges have different colors;*
- (b) *if  $A, B, C$  are any three vertices and the edges  $AB, AC$  are of the same color, then  $BC$  has the same color as well.*

Let  $n$  be a number having the given property, and let the edges of the complete graph with  $n$  vertices be colored in  $p+1$  colors denoted  $1, 2, \dots, p+1$  so that (a) and (b) hold. Consider an arbitrary vertex  $A_1$ . Denote by  $x_i$  the number of edges with endpoint  $A_1$  colored in the color  $i$  ( $i = 1, \dots, p+1$ ). Then, of course

$$x_1 + x_2 + \dots + x_{p+1} = n - 1. \quad (1)$$

Invoking the pigeonhole principle, we are going to prove that

$$x_i \leq p - 1 \quad \text{for each } i = 1, 2, \dots, p + 1. \quad (2)$$

Assume that this is not true: Without loss of generality, let the edges  $A_1A_2, A_1A_3, \dots, A_1A_{p+1}$  be of the  $i$ th color. Then, according to (b), each of the edges  $A_kA_l, 2 \leq k < l \leq p + 1$ , is of the  $i$ th color as well. Take any vertex  $B$  of the graph. At least one of the  $p + 1$  edges  $BA_1, \dots, BA_{p+1}$  is colored  $i$ . Otherwise, one of the other  $p$  colors would have to double up, making some pair of edges  $BA_k$  and  $BA_l$  both the same color  $m \neq i$ . But then (b) would imply that  $A_kA_l$  is of the  $m$ th color, a contradiction.

So, the edge  $BA_j$  is of the  $i$ th color for some  $j = 1, 2, \dots, p + 1$ . Then, since  $BA_j$  and  $A_jA_k$  are colored  $i$ , it follows that  $BA_k$  is also colored  $i$ , and this is true for all  $k = 1, 2, \dots, p + 1$ . The vertex  $B$  was chosen arbitrarily, so the same result applies to any other vertex  $C$ . Hence, using (b) for the last time, we obtain that any edge  $BC$  is of the  $i$ th color, and this contradicts (a).

Thus we proved (2) which, combined with (1), gives

$$n - 1 = x_1 + x_2 + \dots + x_{p+1} \leq (p + 1)(p - 1) = p^2 - 1.$$

This means that  $n \leq p^2$ . Note that this conclusion holds regardless of whether or not  $p$  is a prime.

The pigeonhole principle did its job. Now we have to point out an example proving that the edges of the complete graph  $G$  with  $p^2$  vertices can be colored in  $p + 1$  colors so that (a) and (b) hold. The construction is number-theoretical and uses the fact that  $p$  is a prime.

Regard the vertices of  $G$  as ordered pairs  $(i, j)$ , where  $0 \leq i \leq p - 1, 0 \leq j \leq p - 1$ . Let  $(a_1, b_1)$  and  $(a_2, b_2)$  be two vertices of  $G$ . If  $b_1 \neq b_2$ , the difference  $b_1 - b_2$  is relatively prime to  $p$ , because  $p$  is a prime. Then the congruence

$$(b_1 - b_2)x \equiv a_1 - a_2 \pmod{p}$$

has a unique solution  $x = i$  in the set  $\{0, 1, 2, \dots, p - 1\}$ . In this case, color the edge with endpoints  $(a_1, b_1)$  and  $(a_2, b_2)$  with color  $i + 1$ . If  $b_1 = b_2$  then color this edge with color  $p + 1$ . Thus the edges of  $G$  are colored with  $p + 1$  colors. The condition (a) holds for trivial reasons, let us check that the coloring satisfies (b) too.

Suppose that the edges  $AB$  and  $AC$  are of the same color  $i$ , and let the vertices  $A, B$  and  $C$  correspond to the pairs  $(a_1, b_1), (a_2, b_2)$  and

$(a_3, b_3)$ , respectively. If  $i = p + 1$  then  $b_1 = b_2$ ,  $b_1 = b_3$ , so that  $b_2 = b_3$ . Thus the edge  $BC$  is colored  $p + 1$  as well.

In the nontrivial case  $i \neq p + 1$  we have

$$(b_1 - b_2)(i - 1) \equiv a_1 - a_2 \pmod{p},$$

$$(b_1 - b_3)(i - 1) \equiv a_1 - a_3 \pmod{p}.$$

These congruences imply  $(b_2 - b_3)(i - 1) \equiv a_2 - a_3 \pmod{p}$ , meaning that the edge  $BC$  is colored  $i$ , which completes the proof.

So, the greatest number with the desired property is  $p^2$ . ■

A geometric form of the pigeonhole principle is used in our last problem, which was on the 1980 Moscow Olympiad.

*Several arcs of great circles are located on a unit sphere. The sum of their lengths is less than  $\pi$ . Prove that there exists a plane through the center of the sphere that intersects none of these arcs.*

To each point  $P$  on the sphere we assign its *polar circle*  $C_P$  as follows: If  $P$  is regarded as a pole of the sphere then  $C_P$  is its equator. Evidently,  $C_P$  is a great circle, all points of which are equidistant from its pole  $P$ . The term *polar* does not bring just geographic associations to mind. Polar circles have the following basic property of the polar correspondence:

*A point  $Q$  lies on the polar circle  $C_P$  of  $P$  if and only if  $P$  lies on the polar circle  $C_Q$  of  $Q$ .*

Now denote the given arcs by  $\alpha_1, \alpha_2, \dots, \alpha_n$ . Take the endpoints  $A$  and  $B$  of the arc  $\alpha_1$  and consider their polar circles  $C_A$  and  $C_B$ . Let a variable point  $P$  range over  $\alpha_1$ . Then its polar circle  $C_P$  sweeps the portion  $\sigma_1$  of the surface of the sphere contained in one pair of vertical dihedral angles determined by the planes of  $C_A$  and  $C_B$ . Each of these angles has measure numerically equal to the length of  $\alpha_1$ . Then the ratio of the area of  $\sigma_1$  and the area of the whole sphere is  $2\alpha_1/2\pi = \alpha_1/\pi$ .

Construct the regions  $\sigma_2, \dots, \sigma_n$  corresponding to  $\alpha_2, \dots, \alpha_n$ , the remaining arcs. The same conclusions hold for each of them. In view of the condition  $\alpha_1 + \alpha_2 + \dots + \alpha_n < \pi$ , the regions  $\sigma_1, \sigma_2, \dots, \sigma_n$  do not cover the entire sphere, hence there is a point  $X$  on the surface of the sphere that does not belong to any of them. By the polar property given above in italics, the plane of the polar circle of  $X$  does not meet any of the arcs  $\alpha_1, \alpha_2, \dots, \alpha_n$ . ■

## Coffee Break 6

1. The real numbers  $x_1, x_2, \dots, x_n$  satisfy the conditions

$$x_1 + x_2 + \dots + x_n = 0, \quad x_1^2 + x_2^2 + \dots + x_n^2 = 1.$$

Prove that the product of some pair of these numbers is less than or equal to  $-1/n$ .

2. The positive integers  $x_1, x_2, \dots, x_{100}$  satisfy the equation

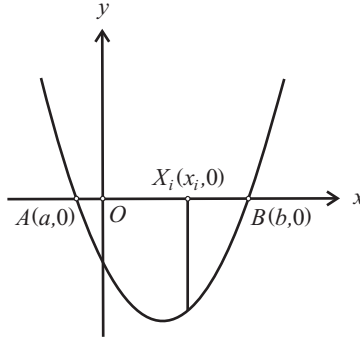
$$\frac{1}{\sqrt{x_1}} + \frac{1}{\sqrt{x_2}} + \dots + \frac{1}{\sqrt{x_{100}}} = 20.$$

Prove that at least two of them are equal.

3. It is well known that the divisibility tests for division by 3 and 9 do not depend on the order of the decimal digits. Prove that 3 and 9 are the only positive integers with this property. More exactly, if an integer  $d > 1$  has the property that  $d|n$  implies  $d|n_1$ , where  $n_1$  is obtained from  $n$  through an arbitrary permutation of its digits, then  $d = 3$  or  $d = 9$ .

## Solutions

1. Let  $a = \min\{x_1, x_2, \dots, x_n\}$  and  $b = \max\{x_1, x_2, \dots, x_n\}$ . Consider the quadratic function  $f(x) = x^2 - (a+b)x + ab$  with roots  $a, b$ .



Since  $a \leq x_i \leq b$  for each  $i = 1, 2, \dots, n$ , we have  $f(x_i) \leq 0$  for all  $i$ , or

$$x_i^2 - (a+b)x_i + ab \leq 0 \quad \text{for } i = 1, 2, \dots, n.$$

Adding these inequalities yields

$$(x_1^2 + x_2^2 + \dots + x_n^2) - (a+b)(x_1 + x_2 + \dots + x_n) + nab \leq 0.$$

But, by hypothesis,  $x_1 + x_2 + \dots + x_n = 0$  and  $x_1^2 + x_2^2 + \dots + x_n^2 = 1$ , so the last inequality simply reduces to  $1 + nab \leq 0$ . From here we deduce  $ab \leq -1/n$ , and the proof is complete.

2. Suppose that the numbers are all distinct and assume, without loss of generality, that  $x_1 < x_2 < \dots < x_{100}$ . It follows that  $x_k \geq k$  for each  $k = 1, 2, \dots, 100$ . Hence

$$20 = \frac{1}{\sqrt{x_1}} + \frac{1}{\sqrt{x_2}} + \dots + \frac{1}{\sqrt{x_{100}}} \leq \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{100}}.$$

On the other hand, we have

$$\frac{1}{\sqrt{k}} < \frac{2}{\sqrt{k} + \sqrt{k-1}} = 2(\sqrt{k} - \sqrt{k-1}),$$

and therefore

$$\begin{aligned} \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{100}} \\ < 2(\sqrt{1} - 0) + 2(\sqrt{2} - \sqrt{1}) + \cdots + 2(\sqrt{100} - \sqrt{99}) \\ = 2\sqrt{100} = 20. \end{aligned}$$

This is a contradiction, so the conclusion follows.

**3.** Let  $d$  be a  $k$ -digit number. Then among the  $(k + 2)$ -digit numbers starting with 10 there is at least one that is divisible by  $d$ . Denote it by  $\overline{10a_1a_2 \dots a_k}$ . The assumption implies that both numbers  $\overline{a_1a_2 \dots a_k10}$  and  $\overline{a_1a_2 \dots a_k01}$  are divisible by  $d$ , and then so is their difference. This difference equals 9 and the proof is finished, since  $d$  may only be some divisor of 9.

# 31

## The Three Jug Problem

You can find this same title in the famous book *Geometry Revisited* by Coxeter and Greitzer. But the story we are going to tell here is quite different. It is about the 1993 Putnam problem B-6:

*Three nonnegative integers are given. We may choose two of them, say  $x$  and  $y$ , and if  $x \leq y$ , replace them by  $2x$  and  $y - x$ . Prove that, after a finite number of such operations, it is possible to obtain 0.*

There is nothing wrong with the statement of the problem but we like another one better:

*Three jugs are given with water in them, each containing an integer number of pints. It is allowed to pour into any jug as much water as it already contains, from any other jug. Prove that after several such pourings it is possible to empty one of the jugs. (Assume that the jugs are sufficiently large; each of them can contain all the water available.)*

This problem was posed for the 1971 All-Union Olympiad. It originated in a paper by the prominent Russian algebraist **Alexei Shirshov**, where he used it for purposes far beyond the scope of our story.

The late Shirshov was a nontraditional mathematician with a highly nontraditional career. What else can you say about a man who graduated from the university with a degree in Russian language and literature, then taught them both in some remote rural area; then survived the entire Second World War fighting in the trenches from the very first to the very last day, succeeding in getting hooked by mathematics in between; then got back to the university at the age of almost thirty, graduated again with a degree

in mathematics, and finally became one of the top mathematicians in his field?

The jugs problem alone is ample evidence of how brilliant he was. Let us share the pleasure of the following solution.

Label the jugs  $A, B, C$  and let them contain  $a, b, c$  pints of water, respectively, where  $0 < a \leq b \leq c$ . It suffices to show how to obtain less than  $a$  pints in one of the jugs, that is, less than the minimum of the numbers  $a, b, c$ . Indeed, we could then decrease the new minimum by repeating this procedure again, and so on. Since a sequence of positive integers cannot decrease indefinitely, the minimum of  $a, b, c$  will eventually become 0, meaning that we will be able to empty one jug entirely.

Let  $b = qa + r$ , where  $0 \leq r < a$ ; note that  $q > 0$ , since  $a \leq b$ . We are going to prove that one can pour exactly  $qa$  pints from  $B$  into  $A$ , thus getting  $r < a$  pints in  $B$  and solving the problem.

Indeed, let  $q = 2^{i_0} + 2^{i_1} + \dots + 2^{i_n}$  be the binary representation of  $q$ ,  $0 \leq i_0 < i_1 < \dots < i_n$ . Then the water in  $B$  may be regarded as consisting of portions with volumes

$$2^{i_0}a, 2^{i_1}a, \dots, 2^{i_n}a, \quad (1)$$

and one more portion of volume  $r$ . In general, some of the numbers  $0, 1, \dots, i_n - 1$  are missing among  $i_0, i_1, \dots, i_n$ . Denote the missing numbers by  $j_1, j_2, \dots, j_k$  and imagine that the portions of volumes

$$2^{j_1}a, 2^{j_2}a, \dots, 2^{j_k}a \quad (2)$$

are present in the biggest jug  $C$ . Then we can pour all portions (1) and (2) into  $A$  in their natural order  $2^0a = a, 2^1a, 2^2a, \dots, 2^{i_n}a$ , taking them from either  $B$  or  $C$ , to get consecutively  $2a, 2^2a, \dots, 2^{i_n+1}a$  pints in  $A$ . As a result, there will remain just  $r$  pints in  $B$ , and we are done.

The only thing remaining is to show that the quantities (2) are actually present in  $C$ . This is true, because even if only  $2^{i_n}a$  were in the sequence (1), then we would still need no more than

$$(2^0 + 2^1 + \dots + 2^{i_n-1})a = (2^{i_n} - 1)a$$

pints in  $C$ . This is less than  $b$ , and it is in turn not greater than  $c$ . ■

Different variants of the jugs problem circulated around the world. Here is one more:

*Several jugs are given (at least two) with a total of  $n$  pints of water in them, each jug containing an integer number of pints.*

*We may pour into any jug as much water as it already contains, from any other jug. It is known that, regardless of the initial distribution of the water in the jugs, it is possible after several such pourings to empty all of them but one, which will contain all the water available. Prove that  $n$  is a power of 2.*

This is a problem from a Bulgarian winter competition in 1989. The solution below, by **Miroslav Petkov**, was awarded a special prize.

Write down  $n$  in the form  $n = 2^m k$ , where  $m \geq 0$  and  $k$  is odd. Assume, by way of contradiction, that  $k > 1$ . One can distribute the  $n$  pints of water in such a way that each jug contains a multiple of  $2^m$  pints (in particular, 0 pints). For example, we can pour  $2^m$  pints into a jug and the remaining  $2^m(k - 1)$  pints into another one. Given such an initial distribution, each pouring from one jug into another involves adding or subtracting an integer number of pints, and always a multiple of  $2^m$ . Hence the number of pints in each jug will be divisible by  $2^m$  after each pouring.

By hypothesis, all the water can be collected in one jug after a certain sequence of pourings. Consider the moment before the last pouring. In order to achieve what was just stated, it is necessary and sufficient that there be 0 pints in all jugs except in two, which contain equal quantities. As pointed out above, these quantities must be of the form  $2^m l$  for some positive integer  $l$ . Therefore the result of the last pouring will be collecting  $2^m l + 2^m l = 2^{m+1} l$  pints in the last jug, so

$$2^m k = n = 2^{m+1} l.$$

But this cannot happen, because  $k$  is odd, a contradiction. ■

The converse is also true, as our last problem shows.

*Several jugs are given with a total of  $2^n$  pints of water in them, each jug containing an integer number of pints. We may pour into any jug as much water as it already contains, from any other jug. Prove that, regardless of the initial distribution of the water, it is possible after several such pourings to empty all jugs but one, which will contain all the water available.*

This time we proceed by induction on  $n$ . The claim is clearly true if  $n = 1$ . Assume it holds for a certain  $n$ ,  $n \geq 1$ , and suppose  $2^{n+1}$  pints are distributed among several jugs, each one containing an integer number of pints. Note that the number of jugs with an odd number of pints is even. Split them in pairs and apply the operation to the jugs in each pair: if they

contain  $a$  and  $b$  pints,  $a \leq b$ , with  $a, b$  odd, after the operation there will be  $2a$  and  $b - a$  pints in them, respectively. Since  $2a$  and  $b - a$  are even, we reach a distribution of the water such that the content of each jug is even.

Let us now call the quantity of 2 pints a “big pint.” Then  $2^n$  big pints are contained in the given jugs, with an integer number of big pints in each one. By the induction hypothesis, all the water can be collected in one jug, which completes the solution. ■

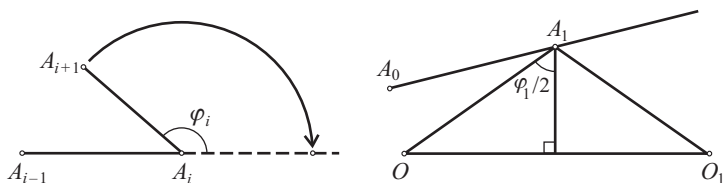
# 32

## Rectifying Trajectories

There are many statements about billiards, where rectification of the trajectory of the ball yields the result immediately. In this section we give other examples of how this nice idea works in completely different situations. The first one is a problem of the 1973 Moscow Olympiad:

*A lion is running along a circular arena of radius 10 m. Moving along a polygonal path, he covers a distance of 30 km. Prove that the sum of the angles through which the lion has turned at his turning points is not less than 2998 radians.*

Denote the route of the lion by  $A_0A_1 \dots A_k$ . Suppose he starts at the point  $A_0$  and ends at  $A_k$ , turning through an angle  $\varphi_i$  at the point  $A_i$  ( $i = 1, 2, \dots, k-1$ ). Note that  $\varphi_i$  is not  $\angle A_{i-1}A_iA_{i+1}$  but its supplement. We shall describe the movement in the following way.



Suppose that the lion has covered the line segment  $A_0A_1$ . Rotate the arena about  $A_1$  so that  $A_1A_2$  becomes an extension of  $A_0A_1$ . Similarly, after the lion gets to  $A_2$ , rotate the arena about  $A_2$  until  $A_2A_3$  becomes an extension of  $A_1A_2$ , and so on. In this way, we may assume that the lion's route is a line segment  $A_0A_k$  of length 30 km.

Let us now follow the movement of the center  $O$  of the arena under the rotations described above. The first rotation (through an angle  $\varphi_1$  about  $A_1$ )

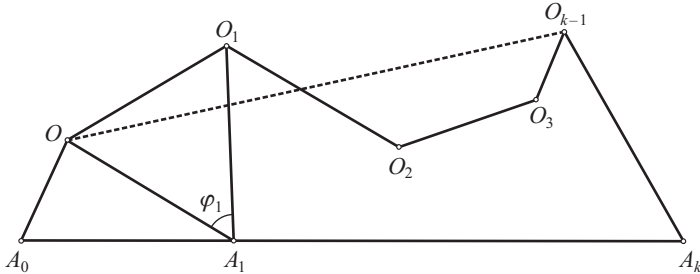
carries  $O$  to a point  $O_1$  such that  $O_1A_1 = OA_1$  and  $\angle OA_1O_1 = \varphi_1$ . Looking at the isosceles triangle  $OO_1A_1$ , we obtain  $OO_1 = 2OA_1 \sin \varphi_1/2$ . Now we have  $\sin \varphi_1/2 \leq \varphi_1/2$ . Moreover, the distance between  $O$  and the rotational center  $A_1$  cannot exceed 10 m (the lion himself could be regarded as the center of rotation, and he is inside the arena). Hence

$$OO_1 = 2OA_1 \sin \frac{\varphi_1}{2} \leq 2 \cdot 10 \cdot \frac{\varphi_1}{2} = 10\varphi_1.$$

The rotated position of the shorter polygonal path  $A_1A_2 \dots A_k$  is contained in the new arena with center  $O_1$ . It follows that the same argument works for the rotated positions  $O_2, O_3, \dots, O_{k-1}$  of the center under all subsequent rotations. We obtain  $O_{i-1}O_i \leq 10\varphi_i$  for  $i = 2, 3, \dots, k-1$ , so

$$OO_{k-1} \leq OO_1 + \dots + O_{k-2}O_{k-1} \leq 10(\varphi_1 + \varphi_2 + \dots + \varphi_{k-1}). \quad (1)$$

Note also that  $A_kO_{k-1} \leq 10$ , because the final position of  $A_k$  is contained in the last rotated position of the arena.



Now consider the points  $A_0$ ,  $A_k$ ,  $O_{k-1}$  and  $O$ . The inequalities  $A_0O \leq 10$  and  $A_kO_{k-1} \leq 10$  yield

$$30,000 = A_0A_k \leq A_0O + OO_{k-1} + O_{k-1}A_k \leq OO_{k-1} + 20.$$

Combined with (1), this yields the desired  $\sum_{i=1}^{k-1} \varphi_i \geq 2998$ . ■

The next beautiful problem was proposed by Hungary for the 1989 IMO:

*At  $n$  distinct points of a circular race course,  $n$  cars are ready to start. Each of them covers the course in an hour. At a given signal every car selects one of the two possible directions and starts immediately. Whenever two cars meet, both of them change directions and go on without loss of speed. Show that at a certain moment each car will be at its starting point.*

Let us label the cars 1 through  $n$ . Suppose that, instead of changing their directions at the meeting points, *the cars simply exchange their labels*. Under this new regulation each car goes round and round at a constant speed in the same direction. An hour later, after several exchanges of labels, it will be back at its starting point.

Coming back to reality, all we can claim is that at this moment each starting point is occupied by *some* car, not necessarily the initial one. But notice that the regulations rule out the possibility of changing the sequence of the cars! Any car will always be between its initial neighbors. Therefore the initial situation is (possibly) only “rotated”; i.e., there exists a nonnegative integer  $d$  such that the car labeled  $i$  is at the starting point of the car labeled  $i + d$  for each  $i = 1, 2, \dots, n$ . (Labels are taken modulo  $n$ .) It is clear that exactly  $n$  hours after the start each car will be at its starting point. ■

# 33

## Numerical Systems

Here are three applications of different numerical systems. It seems as if they were especially invented for these particular problems. The solutions are eloquent enough to allow us to skip further comments.

The first problem was on the 1973 Moscow Olympiad, and its solution was invented (the same year) by members of the high-school mathematical problem-solving group in Rousse, Bulgaria.

*Twelve painters live in 12 red and blue houses built around a circular lane. Each month one of them goes counterclockwise along the lane and, starting from his own house, repaints the houses according to the following rule: If a house is red, he paints it blue and passes to the next house; if a house is blue, he paints it red and goes home. Each painter does this once a year. Prove that if at least one of the houses is red, then a year later each house will have its initial color.*

The number 12 is not essential. Suppose we have  $n$  painters and  $n$  houses labeled 0 through  $n - 1$  counterclockwise around the lane. At each moment the colors of the houses are characterized by a sequence  $c_0, c_1, \dots, c_{n-1}$  of zeros and ones, where  $c_i = 0$  if the  $i$ th house is red, and  $c_i = 1$  if it is blue. This sequence is in turn entirely determined by the base 2 numeral

$$c_0 + c_1 2 + \dots + c_{n-1} 2^{n-1} = \overline{c_{n-1} c_{n-2} \dots c_1 c_0}. \quad (1)$$

For each  $i = 0, 1, \dots, n - 1$  denote by  $a_i$  and  $a_{i+1}$  the numbers of the form (1) before and after the  $i$ th painter's walk. Let us see what change  $a_i$  undergoes after this walk. Note that each painter necessarily reaches a blue house. Suppose the first blue house that the  $i$ th painter meets has

number  $j$ . If  $j > i$ , then  $a_i$  is increased by  $2^i + 2^{i+1} + \dots + 2^{j-1}$  (because the houses labeled  $i, i+1, \dots, j-1$  are red) and finally decreased by  $2^j$  (since the  $j$ th house is blue). As a result, we get

$$a_{i+1} = a_i + (2^i + 2^{i+1} + \dots + 2^{j-1}) - 2^j = a_i - 2^i.$$

If  $j < i$ , then we have to add

$$2^i + \dots + 2^{n-1} + 2^0 + 2^1 + \dots + 2^{j-1}$$

to  $a_i$  and then subtract  $2^j$  from it, which means that

$$a_{i+1} = a_i + (2^n - 2^i + 2^j - 1) - 2^j = a_i + (2^n - 1) - 2^i.$$

One of the two equalities above also holds true for  $j = i$ , according as the  $i$ th painter has painted only his house or completed a full circle around the lane. Note that in all cases

$$a_{i+1} \equiv a_i - 2^i \pmod{(2^n - 1)} \quad \text{for } i = 0, 1, \dots, n-1.$$

Now let  $a$  and  $b$  be the values of the number (1) before and after all painters' walks. It follows that, no matter in what order the painters go out,

$$b \equiv a - (1 + 2^1 + \dots + 2^{n-1}) \equiv a \pmod{(2^n - 1)}.$$

Since  $a \not\equiv 2^n - 1$  (because initially there was at least one red house), and  $b \not\equiv 2^n - 1$  (since the last painted house was painted red), this implies  $b = a$ , which proves the desired statement. ■

There are, of course, other solutions of this problem but we will not discuss them here.

The next problem was on the 1995 USAMO.

*Let  $p$  be an odd prime. The sequence  $(a_n)_{n \geq 0}$  is defined as follows:  $a_0 = 0$ ,  $a_1 = 1, \dots$ ,  $a_{p-2} = p-2$  and, for all  $n \geq p-1$ ,  $a_n$  is the least integer greater than  $a_{n-1}$  that does not form an arithmetic progression of length  $p$  with any of the preceding terms. Prove that, for all  $n$ ,  $a_n$  is the number obtained by writing  $n$  in base  $p-1$  and reading it in base  $p$ .*

We shall say that a subset of  $\mathbf{N}$  is *p-progression-free* if it does not contain an arithmetic progression of length  $p$ . Denote by  $b_n$  the number obtained by writing  $n$  in base  $p-1$  and reading it in base  $p$ . One can easily prove that  $a_n = b_n$  for all  $n = 0, 1, 2, \dots$  by induction, using the

following properties of the set  $B = \{b_0, b_1, \dots, b_n, \dots\}$  (whose proofs we postpone):

1°  $B$  is  $p$ -progression-free;

2° If  $b_{n-1} < a < b_n$  for some  $n \geq 1$ , then the set  $\{b_0, b_1, \dots, b_{n-1}, a\}$  is not  $p$ -progression-free.

Indeed, assume 1° and 2° hold. By the definitions of  $a_k$  and  $b_k$ , we have  $a_k = b_k$  for  $k = 0, 1, \dots, p-2$ . Let  $a_k = b_k$  for all  $k \leq n-1$ , where  $n \geq p-1$ . By 1°, the set

$$\{a_0, a_1, \dots, a_{n-1}, b_n\} = \{b_0, b_1, \dots, b_{n-1}, b_n\}$$

is  $p$ -progression-free, so  $a_n \leq b_n$ . Also, the inequality  $a_n < b_n$  is impossible in view of 2°. Hence  $a_n = b_n$  and we are done.

So it suffices to prove 1° and 2°. Let us note first that  $B$  consists of all numbers whose base  $p$  representation does not contain the digit  $p-1$ . Hence 1° follows from the fact that if  $a, a+d, \dots, a+(p-1)d$  is any arithmetic progression of length  $p$ , then *all* base  $p$  digits occur in the base  $p$  representation of its terms. To see this, represent  $d$  in the form  $d = p^m k$ , where  $(k, p) = 1$ . Then  $d$  ends in  $m$  zeros, and the digit  $\delta$  preceding them is nonzero. It is easy to see that if  $\alpha$  is the  $(m+1)$ st digit of  $a$  (from right to left), then the corresponding digits of  $a, a+d, \dots, a+(p-1)d$  are the remainders of  $\alpha, \alpha+\delta, \dots, \alpha+(p-1)\delta$  modulo  $p$ , respectively. It remains to note that  $\alpha, \alpha+\delta, \dots, \alpha+(p-1)\delta$  is a complete set of residues modulo  $p$ , because  $\delta$  is relatively prime to  $p$ . This finishes the proof of 1°.

We start proving 2° by the remark that  $b_{n-1} < a < b_n$  implies  $a \notin B$ . Since  $B$  consists precisely of the numbers whose base  $p$  representations do not contain the digit  $p-1$ , this very digit *must* occur in the base  $p$  representation of  $a$ . Let  $d$  be the number obtained from  $a$  by replacing each of its digits by 0 if the digit is not  $p-1$ , and by 1 if it is  $p-1$ . Consider the progression

$$a - (p-1)d, a - (p-2)d, \dots, a - d, a.$$

As the definition of  $d$  implies, the first  $p-1$  terms do not contain  $p-1$  in their base  $p$  representation. Hence, being less than  $a$ , they must belong to  $\{b_0, b_1, \dots, b_{n-1}\}$ . Therefore the set  $\{b_0, b_1, \dots, b_{n-1}, a\}$  is not  $p$ -progression-free, and the proof is finished. ■

We close the section with the fifth 1993 IMO problem. This solution was found by Bulgarian contestants **Nikolai Nikolov** and **Yassen Siderov** during the competition.

Determine whether there exists a function  $f: \mathbb{N} \rightarrow \mathbb{N}$  such that:

$$f(1) = 2;$$

$$f(f(n)) = f(n) + n \quad \text{for each } n \in \mathbb{N};$$

$$f(n) < f(n+1) \quad \text{for each } n \in \mathbb{N}.$$

Let  $\{\varphi_n\}_{n=0}^{\infty}$  be the “truncated” Fibonacci sequence defined by

$$\varphi_0 = 1, \varphi_1 = 2, \varphi_{n+1} = \varphi_n + \varphi_{n-1} \quad \text{for } n \geq 1.$$

It is known that each positive integer can be represented in a unique way as the sum of a finite number of terms of  $\{\varphi_n\}_{n=0}^{\infty}$ , no two of which are consecutive. Formally speaking, each positive integer  $n$  has a unique representation of the form

$$n = \varphi_{i_1} + \varphi_{i_2} + \cdots + \varphi_{i_k}, \quad (2)$$

where  $i_r \geq i_{r+1} + 2$ , for all  $r = 1, 2, \dots, k-1$ , and  $i_k \geq 0$ . The equality (2) is called the representation of  $n$  in the Fibonacci base (though  $\{\varphi_n\}_{n=0}^{\infty}$  is not exactly the Fibonacci sequence). Two numbers are compared in this base in the same way as in the usual positional systems. More exactly: Let  $m$  and  $n$  be distinct positive integers, and let  $m = \varphi_{i_1} + \varphi_{i_2} + \cdots + \varphi_{i_r}$ ,  $n = \varphi_{j_1} + \varphi_{j_2} + \cdots + \varphi_{j_s}$  be their representations of the form (2). Then one of these representations is a continuation of the other, and the respective number is greater, or there is an index  $l$  for which  $i_l \neq j_l$ . In the latter case, if  $l$  is the least index with this property, we have  $m < n$  when  $i_l < j_l$  and  $m > n$  when  $i_l > j_l$ .

This information about the Fibonacci base is sufficient to construct a function  $f(n)$  with the desired properties.

Let (2) be the representation of an arbitrary positive integer  $n$  in this base. Set

$$f(n) = \varphi_{i_1+1} + \varphi_{i_2+1} + \cdots + \varphi_{i_k+1}.$$

This function is well defined, because  $n$  has a unique representation of the form (2). Since  $1 = \varphi_0$ , we have  $f(1) = \varphi_{0+1} = \varphi_1 = 2$ . Let  $m = \varphi_{i_1} + \varphi_{i_2} + \cdots + \varphi_{i_r}$ ,  $n = \varphi_{j_1} + \varphi_{j_2} + \cdots + \varphi_{j_s}$  and  $m < n$ . It is clear that  $f(m) < f(n)$  if the second representation is a continuation of the first. Suppose that there exists an index  $l$  such that  $i_l < j_l$  and  $i_k = j_k$  for  $k < l$ . According to the above rule of comparison, we get

$$\varphi_{i_1+1} + \varphi_{i_2+1} + \cdots + \varphi_{i_r+1} < \varphi_{j_1+1} + \varphi_{j_2+1} + \cdots + \varphi_{j_s+1},$$

meaning that  $f(m) < f(n)$ . In particular,  $f(n) < f(n+1)$  for  $n \in \mathbb{N}$ . Finally, if  $n = \varphi_{i_1} + \varphi_{i_2} + \cdots + \varphi_{i_k}$  is any positive integer, then

$$\begin{aligned} f(n) + n &= (\varphi_{i_1+1} + \varphi_{i_2+1} + \cdots + \varphi_{i_k+1}) + (\varphi_{i_1} + \varphi_{i_2} + \cdots + \varphi_{i_k}) \\ &= (\varphi_{i_1+1} + \varphi_{i_1}) + (\varphi_{i_2+1} + \varphi_{i_2}) + \cdots + (\varphi_{i_k+1} + \varphi_{i_k}) \\ &= \varphi_{i_1+2} + \varphi_{i_2+2} + \cdots + \varphi_{i_k+2}, \end{aligned}$$

and the last expression coincides with  $f(f(n))$ . Thus we see that the defined function  $f$  has all the desired properties. ■

Many problems, including difficult and unsolved ones, are waiting for “their” numerical systems.

# 34

## More on Polynomials

Our first two examples are about relations between the roots and the coefficients of a polynomial.

*Let  $f(x)$  be a polynomial of degree  $n$ ,  $n > 1$ , with integer coefficients and  $n$  real roots, not all equal, in the interval  $(0, 1)$ . Prove that if  $a$  is the leading coefficient of  $f(x)$ , then*

$$|a| \geq 2^n + 1.$$

We have  $f(x) = a(x-x_1)(x-x_2)\cdots(x-x_n)$ , where  $x_1, x_2, \dots, x_n$  are the roots of the given polynomial. Note that  $f(0) \neq 0$ ,  $f(1) \neq 0$ , since all the roots lie between 0 and 1. Also,  $f(x)$  takes on an integer value for every integer  $x$ , so  $f(0)$  and  $f(1)$  are nonzero integers. Then

$$\begin{aligned} 1 \leq |f(0)f(1)| &= |a(-1)^n x_1 \cdots x_n \cdot a(1-x_1) \cdots (1-x_n)| \\ &= a^2 x_1(1-x_1)x_2(1-x_2) \cdots x_n(1-x_n), \end{aligned}$$

since the inequalities  $0 < x_k < 1$  ( $k = 1, 2, \dots, n$ ) ensure that each factor is positive. Now, since  $x(1-x) \leq \frac{1}{4}$  for all  $x$  with equality if and only if  $x = \frac{1}{2}$  (which cannot occur for all of the  $x_k$ ), we obtain

$$1 \leq a^2 x_1(1-x_1)x_2(1-x_2) \cdots x_n(1-x_n) < \frac{a^2}{4^n},$$

and this implies  $|a| > 2^n$ . Taking into account that  $a$  is an integer, we get  $|a| \geq 2^n + 1$ . ■

The next statement comes from one of the 1983 Kürschák problems.

*The polynomial*

$$f(x) = x^n + a_1 x^{n-1} + \cdots + a_{n-1} x + 1$$

with nonnegative real coefficients has  $n$  real roots. Prove that:

- (a)  $f(2) \geq 3^n$ ;
- (b)  $f(x) \geq (x+1)^n$  for all  $x \geq 0$ ;
- (c)  $a_k \geq \binom{n}{k}$  for all  $k = 1, 2, \dots, n-1$ .

Clearly  $f(x)$  takes on positive values for  $x \geq 0$ , so its roots, all of them real by hypothesis, are negative. Let them be  $-\alpha_1, -\alpha_2, \dots, -\alpha_n$ , where  $\alpha_1, \alpha_2, \dots, \alpha_n$  are positive. Then we have

$$f(x) = (x + \alpha_1)(x + \alpha_2) \cdots (x + \alpha_n)$$

and, in view of the relations between the roots and the coefficients of a polynomial equation,

$$\alpha_1 \alpha_2 \cdots \alpha_n = 1.$$

As we shall see, the proofs of all three statements rest on this last equality.

(a) Applying the AM–GM inequality, we obtain

$$2 + \alpha_k = 1 + 1 + \alpha_k \geq 3\sqrt[3]{1 \cdot 1 \cdot \alpha_k} = 3\sqrt[3]{\alpha_k}$$

for each  $k = 1, 2, \dots, n$ . Therefore

$$f(2) = (2 + \alpha_1)(2 + \alpha_2) \cdots (2 + \alpha_n) \geq 3^n \sqrt[3]{\alpha_1 \alpha_2 \cdots \alpha_n} = 3^n.$$

(b) This part can be proved essentially the same way, this time using the weighted AM–GM inequality. For all nonnegative numbers  $x$  and all  $k = 1, 2, \dots, n$ , we get

$$\begin{aligned} x + \alpha_k &= (x+1) \left( \frac{x}{x+1} \cdot 1 + \frac{1}{x+1} \alpha_k \right) \\ &\geq (x+1) \cdot 1^{x/(x+1)} \cdot \alpha_k^{1/(x+1)} = (x+1) \alpha_k^{1/(x+1)}. \end{aligned}$$

It follows from here that if  $x \geq 0$ , then

$$f(x) \geq (x+1)^n (\alpha_1 \alpha_2 \cdots \alpha_n)^{1/(x+1)} = (x+1)^n.$$

(c) This is again a consequence of the AM–GM inequality. The coefficient  $a_k$  is the sum of all possible products of numbers among  $\alpha_1, \alpha_2, \dots, \alpha_n$ , taken  $k$  at a time. There are  $\binom{n}{k}$  such products, and each  $\alpha_k$  participates in  $\binom{n-1}{k-1}$  of them. Hence

$$a_k \geq \binom{n}{k} \left[ (\alpha_1 \alpha_2 \cdots \alpha_n)^{\binom{n-1}{k-1}} \right]^{1/\binom{n}{k}} = \binom{n}{k}. \blacksquare$$

We pass on to “real” applications—problems whose statements do not imply that polynomials may be helpful. The first one comes from the 1991 Leningrad Olympiad.

*We say that a sequence  $a_0, a_1, a_2, \dots, a_n$  of real numbers is  $p$ -balanced for some positive integer  $p$  if*

$$\begin{aligned} a_0 + a_p + a_{2p} + \dots &= a_1 + a_{p+1} + a_{2p+1} + \dots = \dots \\ &= a_{p-1} + a_{2p-1} + a_{3p-1} + \dots \end{aligned}$$

*Let the sequence  $a_0, a_1, \dots, a_{49}$  be  $p$ -balanced for  $p = 3, 5, 7, 11, 13, 17$ . Prove that  $a_0 = a_1 = \dots = a_{49} = 0$ .*

Suppose that the sequence  $a_0, a_1, \dots, a_n$  is  $p$ -balanced for some  $p$ , and denote by  $S$  the common value of the sums

$$a_0 + a_p + a_{2p} + \dots, a_1 + a_{p+1} + a_{2p+1} + \dots, \dots, a_{p-1} + a_{2p-1} + a_{3p-1} + \dots.$$

The key step is to introduce the polynomial

$$f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n.$$

Let us replace  $x$  by any nontrivial (different from 1)  $p$ th root  $\omega$  of unity. Since  $\omega^p = 1$ , we have  $\omega^r = \omega^{p+r} = \omega^{2p+r} = \omega^{3p+r} = \dots$  for each  $r = 0, 1, \dots, p-1$ . Hence  $\omega^k$  is equal to one of the basic powers  $1, \omega, \dots, \omega^{p-1}$  for all positive integers  $k$ . So

$$\begin{aligned} f(\omega) &= a_0 + a_1\omega + \dots + a_n\omega^n \\ &= (a_0 + a_p + a_{2p} + \dots) + (a_1 + a_{p+1} + a_{2p+1} + \dots)\omega \\ &\quad + \dots + (a_{p-1} + a_{2p-1} + a_{3p-1} + \dots)\omega^{p-1} \\ &= S(1 + \omega + \dots + \omega^{p-1}) = 0, \end{aligned}$$

and hence  $\omega$  is a root of  $f$ . In our case, the polynomial  $f$  is of degree 49. It follows from the hypothesis that it has two roots corresponding to  $p = 3$ ; four roots corresponding to  $p = 5$ , etc.; 16 roots corresponding to  $p = 17$ . All these roots are distinct, since 3, 5, 7, 11, 13, 17 are distinct primes. Hence  $f$  has at least  $2 + 4 + 6 + 10 + 12 + 16 = 50$  roots. Since its degree is 49, the identity theorem implies that  $f$  is the zero polynomial. Therefore  $a_0 = a_1 = a_2 = \dots = a_{49} = 0$ . ■

For each  $n = 0, 1, 2, \dots$ , there exists a polynomial  $T_n(x)$  such that

$$\cos n\varphi = T_n(\cos \varphi) \quad (1)$$

for all real  $\varphi$ . These belong to the class of the so-called Chebyshev polynomials, rich in amazing properties and amazing applications.

The desired polynomial sequence can be constructed by induction. We recall this construction briefly: Set  $T_0(x) = 1$ ,  $T_1(x) = x$  and

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x), \quad n = 1, 2, \dots \quad (2)$$

Clearly (1) is true for  $n = 0$  and  $n = 1$ . Assuming that it holds for  $k \leq n$ , we obtain

$$\begin{aligned} T_{n+1}(\cos \varphi) &= 2 \cos \varphi T_n(\cos \varphi) - T_{n-1}(\cos \varphi) \\ &= 2 \cos \varphi \cos n\varphi - \cos(n-1)\varphi \\ &= \cos(n+1)\varphi, \end{aligned}$$

as desired.

There are different types of what we call Chebyshev polynomials, for example, the polynomials  $U_0, U_1, \dots, U_n, \dots$  satisfying

$$2 \cos n\varphi = U_n(2 \cos \varphi)$$

for all  $\varphi$ . In some sense, they are even simpler and more convenient than  $T_n$ , having leading coefficients 1 (for  $n > 0$ ). This can be seen from their definition:  $U_0(x) = 2$ ,  $U_1(x) = x$  and

$$U_{n+1}(x) = xU_n(x) - U_{n-1}(x), \quad n = 1, 2, \dots$$

Also,  $U_n$  clearly has integer coefficients for all  $n = 0, 1, 2, \dots$

An impressive application of this second type of Chebyshev polynomials is the proof of the following unexpected result.

*If  $\alpha$  is a real number such that  $\alpha/\pi$  and  $\cos \alpha$  are both rational, then  $\cos \alpha$  is one of the numbers  $0, \pm \frac{1}{2}, \pm 1$ .*

Let  $\alpha = (m/n)\pi$ , where  $m$  and  $n$  are relatively prime integers. Then  $n\alpha = m\pi$ , so  $n\alpha$  is an integer multiple of  $\pi$  and hence  $\cos n\alpha$  is either 1 or  $-1$ . Let us now consider the Chebyshev polynomial  $U_n$ . We have  $2 \cos n\alpha = U_n(2 \cos \alpha)$ , which means that the number  $2 \cos \alpha$ , rational by hypothesis, is a root of the polynomial

$$f(x) = U_n(x) - 2 \cos n\alpha.$$

Clearly,  $f$  has integer coefficients and its leading coefficient is 1. By the rational-root theorem,  $2 \cos \alpha$  must be an integer. On the other hand, we have  $-2 \leq 2 \cos \alpha \leq 2$ , so the only possibilities for  $\cos \alpha$  are indeed  $0, \pm \frac{1}{2}, \pm 1$ . ■

We shall note just two properties of the polynomials  $T_n$ , both immediate consequences of (2) and (1), respectively:

1°  $T_n$  is a polynomial of degree  $n$  with leading coefficient  $2^{n-1}$ ,  $n \geq 1$ .

2° If  $|x| \leq 1$  then  $|T_n(x)| \leq 1$ ,  $n \geq 1$ .

Given  $n \geq 2$  distinct points  $x_1, x_2, \dots, x_n$  in  $[-1, 1]$ , let  $\tau_k$  denote the product of the distances from  $x_k$  to the other points.

Prove that

$$\sum_{k=1}^n \frac{1}{\tau_k} \geq 2^{n-2}$$

and determine when equality holds.

For brevity, let us introduce the notation

$$P_k(x) = (x - x_1) \cdots (x - x_{k-1})(x - x_{k+1}) \cdots (x - x_n).$$

Note that  $|P_k(x_k)|$  is the product of the distances from  $x_k$  to the remaining points, that is,  $|P_k(x_k)| = \tau_k$  for all  $k = 1, 2, \dots, n$ .

Each polynomial of degree  $m$  is entirely determined by its values at  $m + 1$  distinct points. In particular, this applies to the Chebyshev polynomial  $T_{n-1}$  and the points  $x_1, x_2, \dots, x_n$ . More exactly, by Lagrange's interpolation formula,

$$T_{n-1}(x) = \sum_{k=1}^n T_{n-1}(x_k) \frac{P_k(x)}{P_k(x_k)}.$$

As we already noted,  $T_{n-1}$  has leading coefficient  $2^{n-2}$ . On the other hand, the above representation of  $T_{n-1}$  implies that this coefficient is equal to

$$\sum_{k=1}^n \frac{T_{n-1}(x_k)}{P_k(x_k)}.$$

Since  $x_k \in [-1, 1]$ , we have  $|T_{n-1}(x_k)| \leq 1$  for each  $k = 1, 2, \dots, n$ .

Hence

$$2^{n-2} = \sum_{k=1}^n \frac{T_{n-1}(x_k)}{P_k(x_k)} \leq \sum_{k=1}^n \frac{|T_{n-1}(x_k)|}{|P_k(x_k)|} \leq \sum_{k=1}^n \frac{1}{\tau_k},$$

with equality if and only if  $T_{n-1}(x_k)P_k(x_k) \geq 0$  and  $|T_{n-1}(x_k)| = 1$  for each  $k = 1, 2, \dots, n$ . The latter holds if and only if  $x_1, x_2, \dots, x_n$  is a permutation of

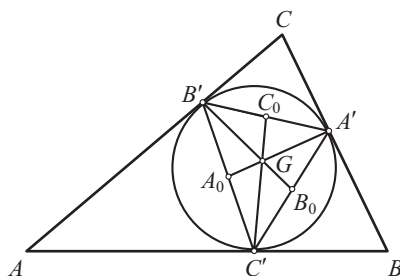
$$1 = \cos \frac{0 \cdot \pi}{n-1}, \cos \frac{\pi}{n-1}, \cos \frac{2\pi}{n-1}, \dots, \cos \frac{(n-1)\pi}{n-1} = -1. \blacksquare$$

# 35

## Geometric Transformations

The variety of applications of geometric transformations is so vast that we can do nothing better than provide several instructive examples. To start with, let us take a Romanian selection test problem for the 1986 IMO, and its “Bulgarian” solution.

*The incircle of the triangle  $ABC$  touches the sides  $AB$ ,  $BC$ ,  $CA$  at the points  $C'$ ,  $A'$ ,  $B'$ , respectively. Prove that the perpendiculars from the midpoints of  $A'B'$ ,  $B'C'$ ,  $C'A'$  to  $AB$ ,  $BC$ ,  $CA$ , respectively, are concurrent.*



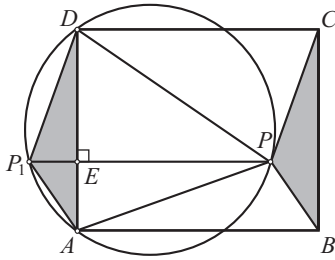
Denote the midpoints of  $A'B'$ ,  $B'C'$ ,  $C'A'$  by  $C_0$ ,  $A_0$ ,  $B_0$ , respectively, and the three perpendiculars in question by  $l_C$ ,  $l_A$ ,  $l_B$ . Consider the centroid  $G$  of  $\triangle A'B'C'$ .

Since  $A_0G : GA' = B_0G : GB' = C_0G : GC' = 1 : 2$ , the dilatation  $h$  with center  $G$  and coefficient  $-2$  takes  $A_0$ ,  $B_0$ ,  $C_0$  to  $A'$ ,  $B'$ ,  $C'$ , respectively. Since dilatations carry straight lines into parallel lines,  $h$  transforms  $l_C$  into the line through  $C'$  perpendicular to  $AB$ . But  $C'$  is the point of tangency of the incircle and  $AB$ , so this line passes through the incenter of  $\triangle ABC$ . The same applies to the images of  $l_A$  and  $l_B$  under

$h$ . Since the images of  $l_A, l_B, l_C$  under  $h$  are concurrent, so are  $l_A, l_B, l_C$  themselves. ■

*An interior point  $P$  is chosen in the rectangle  $ABCD$  with the property that  $\angle BPC + \angle APD = 180^\circ$ . Find the sum of the angles  $BCP$  and  $DAP$ .*

There is a standard solution using trigonometry, but we prefer another one.



Consider the translation determined by the vector  $\overrightarrow{BA}$ . Let it take  $\triangle BCP$  to  $\triangle ADP_1$ . Then, by hypothesis,

$$\angle AP_1D + \angle APD = \angle BPC + \angle APD = 180^\circ,$$

so the quadrilateral  $APDP_1$  is cyclic. Moreover, its diagonals  $AD$  and  $PP_1$  are perpendicular. If  $E$  is their common point, then

$$\begin{aligned} \angle BCP + \angle DAP &= \angle ADP_1 + \angle DAP \\ &= \frac{1}{2} (\widehat{AP_1} + \widehat{DP}) \\ &= \angle DEP = 90^\circ. \quad \blacksquare \end{aligned}$$

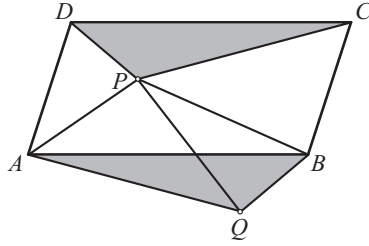
*Let  $P$  be a point interior to the parallelogram  $ABCD$  whose area is  $S$ . Prove the inequality*

$$AP \cdot CP + BP \cdot DP \geq S.$$

Let us note first that

$$S_{ABP} + S_{CDP} = \frac{1}{2} S_{ABCD} = \frac{1}{2} S,$$

where  $S_F$  is the area of the figure  $F$ ; the proof is left to the reader. Construct the point  $Q$  outside  $ABCD$  so that  $AQ = CP$  and  $BQ = DP$ . In fact  $Q$  is the image of  $P$  under the translation  $\overrightarrow{DA}$  and the symmetry across the perpendicular bisector of  $AB$ .



Then  $\triangle ABQ \cong \triangle CDP$ , so the area of the quadrilateral  $AQBP$  equals  $S_{ABP} + S_{CDP} = \frac{1}{2}S$ . On the other hand,  $S_{AQBP} = S_{APQ} + S_{BPQ}$ . Since  $AP \cdot AQ \geq 2S_{APQ}$ ,  $BP \cdot BQ \geq 2S_{BPQ}$ , it is clear that

$$\begin{aligned} AP \cdot CP + BP \cdot DP &= AP \cdot AQ + BP \cdot BQ \\ &\geq 2(S_{APQ} + S_{BPQ}) \\ &= 2S_{AQBP} = S. \blacksquare \end{aligned}$$

A problem from the 1984 Balkan Olympiad can be solved by using the composition of two dilatations.

Let  $A_1A_2A_3A_4$  be a cyclic quadrilateral. Denote by  $H_1, H_2, H_3, H_4$  the orthocenters of the triangles  $A_2A_3A_4, A_3A_4A_1, A_4A_1A_2, A_1A_2A_3$ , respectively. Prove that the quadrilateral  $H_1H_2H_3H_4$  is congruent to  $A_1A_2A_3A_4$ .

Our solution rests on the well-known fact that the centroid  $G$ , the orthocenter  $H$  and the circumcenter  $O$  of any triangle are collinear and that  $\overrightarrow{OH} = 3\overrightarrow{OG}$ .

Denote the circumcenter of  $A_1A_2A_3A_4$  by  $O$ , its centroid by  $G$ , and let  $G_1, G_2, G_3, G_4$  be the centroids of  $A_2A_3A_4, A_3A_4A_1, A_4A_1A_2, A_1A_2A_3$ , respectively. These four triangles have  $O$  as their common circumcenter, so the vector equalities  $\overrightarrow{OH_i} = 3\overrightarrow{OG_i}$  ( $i = 1, 2, 3, 4$ ) hold. They imply that the dilatation with center  $O$  and coefficient  $\frac{1}{3}$  takes  $H_1H_2H_3H_4$  to  $G_1G_2G_3G_4$ .

Consider the centroid  $G_4$  of  $\triangle A_1A_2A_3$ . By definition, we have

$$\overrightarrow{G_4A_1} + \overrightarrow{G_4A_2} + \overrightarrow{G_4A_3} = \vec{0}.$$

Also, by addition of vectors,

$$\overrightarrow{GA_1} = \overrightarrow{GG_4} + \overrightarrow{G_4A_1}, \quad \overrightarrow{GA_2} = \overrightarrow{GG_4} + \overrightarrow{G_4A_2}, \quad \overrightarrow{GA_3} = \overrightarrow{GG_4} + \overrightarrow{G_4A_3},$$

which, when added, give

$$\overrightarrow{GA_1} + \overrightarrow{GA_2} + \overrightarrow{GA_3} = 3\overrightarrow{GG_4} + \left( \overrightarrow{G_4A_1} + \overrightarrow{G_4A_2} + \overrightarrow{G_4A_3} \right) = 3\overrightarrow{GG_4}.$$

Since  $\overrightarrow{GA_1} + \overrightarrow{GA_2} + \overrightarrow{GA_3} + \overrightarrow{GA_4}$  is also the zero vector by definition, we obtain  $\overrightarrow{GA_4} = -3\overrightarrow{GG_4}$ . Similarly,  $\overrightarrow{GA_i} = -3\overrightarrow{GG_i}$  for  $i = 1, 2, 3$ . Hence the dilatation with center  $G$  and coefficient  $-3$  takes  $G_1G_2G_3G_4$  to  $A_1A_2A_3A_4$ .

Summing up, we observe that there is a dilatation taking  $H_1H_2H_3H_4$  to  $A_1A_2A_3A_4$ , with coefficient  $\frac{1}{3} \cdot (-3) = -1$ . (Actually, it is a half turn whose center turns out to be the common midpoint of the line segments  $A_1H_1$ ,  $A_2H_2$ ,  $A_3H_3$ ,  $A_4H_4$ .) In particular, this means that  $H_1H_2H_3H_4$  and  $A_1A_2A_3A_4$  are congruent. ■

*The prism  $ABCA_1B_1C_1$  (not necessarily right) is given. Let  $P$  be the common point of the planes  $(ABC_1)$ ,  $(BCA_1)$ ,  $(CAB_1)$ , and  $P_1$  be the common point of the planes  $(A_1B_1C)$ ,  $(B_1C_1A)$ ,  $(C_1A_1B)$ . Prove that the line segment  $PP_1$  is parallel to  $AA_1$ .*

What does the job here fastest of all is a not very well-known transformation called *skew symmetry*. It is a natural generalization of axial symmetry. Fix a line  $l$  in the plane (the axis of the skew symmetry) and a direction  $\vec{d}$  not parallel to  $l$ . The image of any point  $X$  in the plane under the skew symmetry with axis  $l$  and direction  $\vec{d}$  is obtained as follows. Draw the line through  $X$  parallel to  $\vec{d}$  and find the point  $X_1$  on this line such that the line segment  $XX_1$  has its midpoint on  $l$ . Then  $X_1$  is the skew symmetric image of  $X$ . A similar construction works in space; the difference is that the axis  $l$  is replaced by a *plane of skew symmetry*.

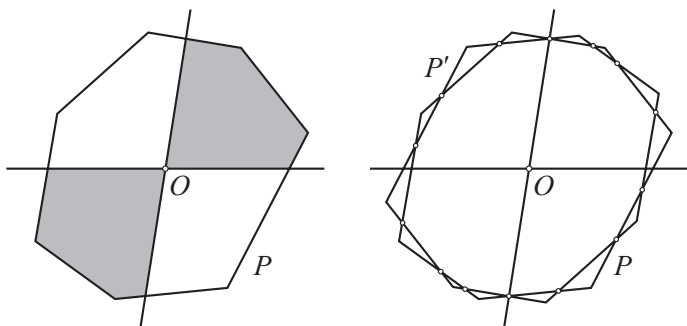
It is easy to see that skew symmetry takes any line to a line and, in space, any plane to a plane.

This is all we need to get a quick solution. Consider the plane containing the midpoints of the edges  $AA_1$ ,  $BB_1$ ,  $CC_1$  as a plane of skew symmetry  $\sigma$  with direction  $\overrightarrow{AA_1}$ . Since  $\sigma$  takes  $A$ ,  $B$ ,  $C$  to  $A_1$ ,  $B_1$ ,  $C_1$ , respectively, the images of the planes  $(ABC_1)$ ,  $(BCA_1)$ ,  $(CAB_1)$  under  $\sigma$  are  $(A_1B_1C)$ ,  $(B_1C_1A)$ ,  $(C_1A_1B)$ , respectively. Then the image of the common point  $P$  of the first three planes is the common point  $P_1$  of the other three. Now it suffices to note that, by definition, the line segment connecting any point with its image under some skew symmetry is parallel to the direction of the symmetry. In our case this yields  $PP_1 \parallel AA_1$ . ■

We close this section with a combinatorial geometry problem from the 1973 All-Union Olympiad.

*A convex  $n$ -gon  $P$  is given with no parallel sides, and a point  $O$  interior to it. Prove that there are no more than  $n$  lines through  $O$  that bisect the area of  $P$ .*

For brevity, we shall call a line *suitable* if it passes through  $O$  and bisects the area of  $P$ . Consider two suitable lines first. They divide  $P$  into four parts, and the two parts inside each pair of vertical angles with vertex  $O$  have the same area.



Let us now consider the  $n$ -gon  $P'$  symmetric to  $P$  with respect to  $O$ . In other words,  $P'$  is obtained by giving  $P$  a half turn about  $O$ . The boundaries of  $P$  and  $P'$  should intersect in the interior of each angle formed by any two suitable lines. Otherwise the area of one of the pieces contained in this angle will be less than the area of the other one. Taking the definition of  $P'$  into account, this contradicts the conclusion about equality of areas we got above. Note that the boundaries of  $P$  and  $P'$  cannot coincide even partially, since  $P$  has no parallel sides.

Suppose that there exist  $k$  suitable lines. They form  $2k$  angles with common vertex  $O$ , and each of these angles contains at least one intersection point of the boundaries of  $P$  and  $P'$ . But no side of  $P$  can contain more than two such intersection points (any line meets the boundary of a convex polygon at most twice). Therefore the number of all intersection points is not less than  $2k$  and no more than  $2n$ . This implies the inequality  $2k \leq 2n$  and it follows that the suitable lines are at most  $n$ .

A standard continuity argument shows that there exists at least one suitable line through each point  $O$  inside  $P$ . ■

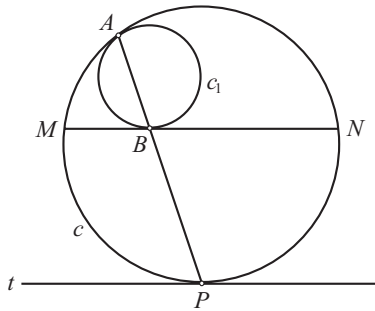
## Coffee Break 7

1. Six points are given in space such that the pairwise distances between them are all distinct. Consider the triangles with vertices at these points. Prove that the longest side in one of these triangles is at the same time the shortest side in another triangle.
2. A circle is inscribed in a given circular segment, touching its arc and chord at  $A$  and  $B$ , respectively. Prove that the line  $AB$  passes through a constant point.
3. The number  $x_n$  is defined as the last digit in the decimal representation of the integer  $\left\lfloor (\sqrt{2})^n \right\rfloor$  ( $n = 1, 2, \dots$ ). Determine whether or not the sequence  $x_1, x_2, \dots, x_n, \dots$  is periodic.

## Solutions

1. Color the shortest side in each triangle black; some of the line segments may be colored more than once. If the remaining edges are colored white, then the complete graph with six vertices has all its edges colored in two colors. As is well known, any such coloring contains a triangle  $T$  whose sides are of the same color. But  $T$  must necessarily be black, because it has at least one black side—the shortest one. The longest side of  $T$  is black; it is therefore the shortest side in some other triangle.

2. Denote the circle containing the arc of the segment by  $c$ , the endpoints of the chord by  $M$  and  $N$ , and the given circle by  $c_1$ . The point  $A$  is the center of a dilatation  $h$  taking  $c_1$  to  $c$ .



The line  $MN$ , which touches  $c_1$  at  $B$ , is taken by  $h$  to the line  $t$  touching  $c$  at the image of  $B$ . Clearly  $t$  is parallel to  $MN$ , and the image of  $B$  under  $h$  is the common point  $P$  of  $BA$  and  $c$ . But a tangent parallel to a chord of a circle touches the circle at the midpoint of one of the arcs determined by the chord. It follows that  $P$  is the midpoint of the arc  $\widehat{MN}$  of  $c$  that does not touch  $c_1$ .

Hence the line  $AB$  passes through the midpoint of  $\widehat{MN}$ , a point independent of the choice of  $c_1$ .

3. Set  $y_n = 0$  if  $x_n$  is even and  $y_n = 1$  otherwise. The new sequence  $y_1, y_2, \dots, y_n, \dots$  is formed by the residues of the numbers  $x_n$  modulo 2. If  $x_1, x_2, \dots, x_n, \dots$  is periodic, then so is  $y_1, y_2, \dots, y_n, \dots$ . We shall prove that  $y_1, y_2, \dots, y_n, \dots$  is not periodic, which implies that the answer to the question is negative.

Let us consider the sequence  $y_1, y_3, y_5, \dots, y_{2n+1}, \dots$ . Its term  $y_{2n+1}$  can be obtained as follows. Write down  $\sqrt{2}$  in base 2, multiply by  $2^n$  (this gives  $(\sqrt{2})^{2n+1}$ ) and discard the fractional part of the result to get

$$\left\lfloor (\sqrt{2})^{2n+1} \right\rfloor.$$

Then take the last (binary) digit of this integer; it is  $y_{2n+1}$ . But multiplying by  $2^n$  in base 2 amounts to simply shifting the binary point  $n$  positions to the right. This implies that  $y_{2n+1}$  is in fact the  $n$ th digit of  $\sqrt{2}$  after the binary point. Since  $\sqrt{2}$  is irrational, we conclude that the sequence  $y_1, y_3, \dots, y_{2n+1}, \dots$  is nonperiodic. It is easy to infer from here that  $y_1, y_2, \dots, y_n, \dots$  is nonperiodic too, and we are done.

# 36

## The Game of Life Problem

Apart from his numerous “serious” mathematical achievements, John Conway became famous by inventing his *Game of Life* in the early seventies. The game was brought to the public’s attention by Martin Gardner through his column in *Scientific American*.

The game is played on an infinite chessboard. Some initial arrangement of “live cells” is given, and each new generation is determined by several rules, or the “genetic laws.” They are quite simple and were chosen carefully after lots of experiments. The only detail we would like to emphasize is that the evolution is “discrete”; that is: Following the rules, one determines whether there should exist life in each particular cell of the chessboard throughout the next generation. Then the old generation vanishes and the next one comes to life. This is exactly the case with the problem we are going to consider.

It is not our intention to discuss the *Game of Life* here. If you want to learn more, read the corresponding three chapters of Martin Gardner’s book *Wheels, Life and Other Mathematical Amusements*.

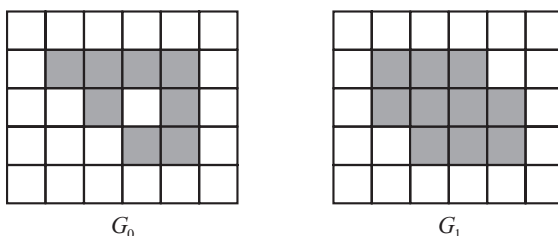
The *Game of Life* was cheered all over the world. Our story is about a problem of the 1973 All-Union Olympiad, obviously inspired by Conway’s invention.

*An infinite chessboard is given whose cells are initially all white. Then a set  $G_0$  of  $n$  cells is chosen, and they are all colored black. At each moment  $t = 1, 2, \dots$  of time, a simultaneous change of color takes place for each cell on the chessboard according to the following rule: Every cell  $C$  gets the color that prevails in the three-cell configuration formed by  $C$  itself and its upper and right neighbors. That is, if two or three of these cells were white, then  $C$  becomes white; if two or three of them were black, then*

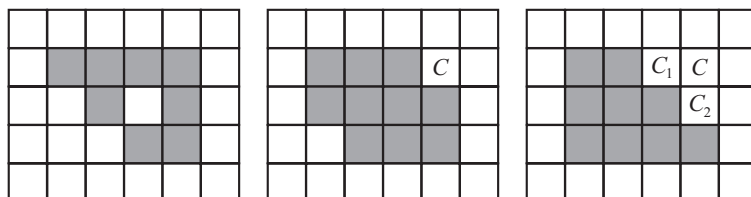
$C$  becomes black. In this way the consecutive steps give rise to new sets  $G_1, G_2, \dots, G_k, \dots$  of black cells.

- (a) Prove that  $G_k = \emptyset$  for some positive integer  $k$  (that is, there will be no black cells on the chessboard after a while).
- (b) Prove that  $G_n = \emptyset$  (that is, all black cells will disappear no later than the moment  $t = n$ ).

The majority of people who try to solve this problem expect that the number of black cells in each next generation  $G_{k+1}$  is *strictly less* than the one in the previous generation  $G_k$ . This is not true. Moreover, it is possible for  $G_{k+1}$  to contain *more* black cells than  $G_k$ , as the example below shows.



This observation is another piece of evidence of how unpredictable the behavior of the generations might be, even with simple rules of the Conway type. Still, another type of monotonicity yields an easy proof of (a).

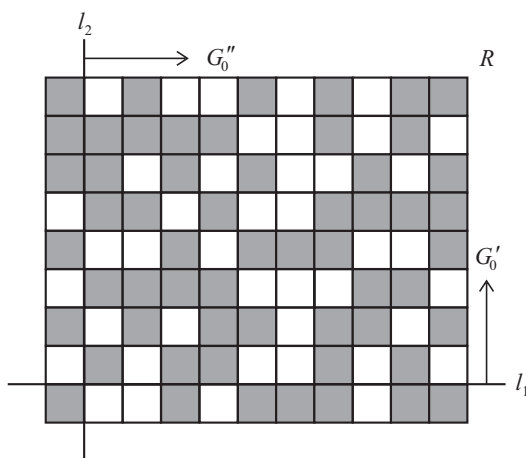


Take the minimal rectangle  $R$  on the chessboard that contains  $G_0$  and note that there can be no life (that is, black cells) outside  $R$  in any of the subsequent generations. Furthermore, the cell  $C$  in the upper right-hand corner of  $R$  will be white in the generation  $G_1$ , regardless of what the other cells in  $G_0$  are. Then it is not hard to see that  $C$  will be white in any subsequent generation. Moreover, so will be its left and downward neighbors  $C_1$  and  $C_2$ . In a similar fashion, each new generation will have an additional “diagonal” of new white cells. Thus the minimal rectangle  $R$  will be exhausted after a finite number of steps, so life will perish.

Part (b) is much harder. That one cannot expect it to be easy may be inferred after finding a large variety of generations  $G_0$  that die out in exactly  $n$  steps. (It is not a bad idea to write a program and play this game on a computer.)

The next proof is an illustrative example of how simple and efficient the method of induction can be.

It is obvious that an initial generation  $G_0$  of just one black cell perishes at the first step, so (b) holds for  $n = 1$ . Assume that any generation  $G_0$  of  $k$  black cells, where  $k < n$ , perishes no later than the  $k$ th step. Now consider an initial generation  $G_0$  of  $n$  black cells.



Take the minimal rectangle  $R$  that contains  $G_0$  again. Denote by  $G_0'$  the set of all black cells of  $G_0$  that do not lie in the bottom row of  $R$ . Note that  $G_0'$  has fewer than  $n$  cells, otherwise  $R$  would not have been minimal. Note also that the cells in the bottom row of  $R$ , black or white, do not influence the future of  $G_0'$ . Then, by the induction hypothesis, the generation  $G_0'$  will disappear at most at the  $(n - 1)$ st step.

But exactly the same argument applies to the set  $G_0''$  of all black cells of  $G_0$  that do not lie in the leftmost column of  $R$ . It will also vanish no later than the  $(n - 1)$ st step.

Consequently, after  $n - 1$  steps there will be no life above the line  $l_1$  and to the right of the line  $l_2$  shown in the figure. Hence life could exist just in the lower left-hand corner of  $R$ ; if so, it will disappear at the  $n$ th step. ■

This is essentially the solution found by **A. Gomilko**, a Russian tenth-grader, during competition time. There is no wonder it won a special prize.

# 37

## Tetrahedra with a Point in Common

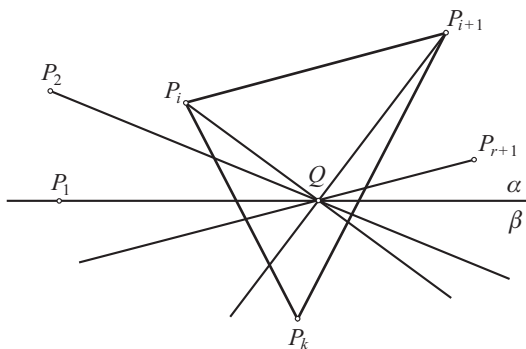
One of the 1983 Kürschák problems reads as follows:

*There are given  $n + 1$  points  $P_1, P_2, \dots, P_n$  and  $Q$  in a plane,  $n \geq 3$ , no three of which are on the same line. It is known that for each pair of distinct points  $P_i$  and  $P_j$  one can find a point  $P_k$  such that  $Q$  is interior to the triangle  $P_i P_j P_k$ . Show that  $n$  must be odd.*

Fix the line  $P_1 Q$  and denote by  $\alpha$  and  $\beta$  the open half-planes with edge  $P_1 Q$ . Without loss of generality, assume that  $\alpha$  contains  $r \geq 1$  of the given points,  $P_2, \dots, P_{r+1}$ , and that

$$\angle P_1 Q P_2 < \angle P_1 Q P_3 < \dots < \angle P_1 Q P_{r+1} < \pi.$$

None of the remaining points  $P_j$  is on the line  $P_1 Q$ , so they are contained in  $\beta$ .



For each  $i = 1, 2, \dots, r$ , there exists a point  $P_k$  such that  $Q$  is interior to the triangle  $P_i P_{i+1} P_k$ . Clearly  $P_k$  lies in the interior of the angle

vertical to  $\angle P_i Q P_{i+1}$ , and hence in the half-plane  $\beta$ . Next, the points  $P_k$  corresponding to different values of  $i = 1, 2, \dots, r$  are different. Indeed, no two of the angles  $\angle P_1 Q P_2, \angle P_2 Q P_3, \dots, \angle P_r Q P_{r+1}$  have a common interior point, and neither do their vertical angles. It follows that  $\alpha$  contains no more points among  $P_1, P_2, \dots, P_n$  than  $\beta$ ; that is,  $r \leq n - 1 - r$ . But if we exchange the roles of  $\alpha$  and  $\beta$ , the very same argument leads to  $n - 1 - r \leq r$ . Therefore  $r = n - 1 - r$ , so  $n = 2r + 1$ . In particular,  $n$  is odd, as required. ■

The proof gives some idea about the structure of a plane point set with the given property. A necessary condition is that any line of the form  $P_i Q$  should split the remaining  $n - 1$  points into two groups with equal number of points. Such configurations exist for any odd  $n$ : for instance, the vertices  $P_1, P_2, \dots, P_n$  and the center  $Q$  of a regular  $n$ -gon.

The idea of this solution probably impressed the organizing committee of the final round of the 1984 Bulgarian Olympiad, because the three-dimensional variant of the Kürschák problem was proposed to the contestants:

*In space,  $n + 1$  points  $P_1, P_2, \dots, P_n$  and  $Q$  are given,  $n \geq 4$ , no four of which are in the same plane. It is known that for each triple of distinct points  $P_i, P_j$  and  $P_k$  one can find a point  $P_l$  such that  $Q$  is interior to the tetrahedron  $P_i P_j P_k P_l$ . Show that  $n$  must be even.*

With slightly more technical trouble, the same idea works in space as well. But it just yields a formal proof, not an exhaustive description of the sets with the given property.

Generally speaking, three-dimensional analogs of simple settings in the plane are much more diverse and often lead to complicated problems. However, the three-dimensional condition proved too restrictive this time...

During the exam **Ivan Dimitrov**, a tenth-grader, conjectured and proved much more. It turned out that the only possibility for  $n$  is in fact  $n = 4$ , something totally unexpected!

Fix  $Q$  and call a point set  $A$  *suitable* if for any three distinct points of  $A$  there exists a fourth one, again from  $A$ , so that these four points are the vertices of a tetrahedron containing  $Q$ . By hypothesis,  $\{P_1, P_2, \dots, P_n\}$  is a suitable set. Note now: If each  $P_i$  is moved along the ray  $\overrightarrow{QP_i}$  to some arbitrary new position  $P'_i$ , then the set  $\{P'_1, P'_2, \dots, P'_n\}$  is also

*suitable*. Hence we may transform our set into a new, “more convenient” one. Assume first that  $P_1, P_2, \dots, P_n$  lie on a sphere centered at  $Q$ . Then the convex hull of  $\{P_1, P_2, \dots, P_n\}$  is a convex polyhedron  $P$  with vertices exactly the points  $P_1, P_2, \dots, P_n$ . In general, some faces of  $P$  may not be triangles. If so, some of the  $P_i$ ’s can be moved “just a little” towards the center  $Q$  along the corresponding rays  $\overrightarrow{QP_i}$  so that no four of the points  $\{P_1, P_2, \dots, P_n\}$  are coplanar, and they still constitute the set of vertices of the convex hull of  $\{P_1, P_2, \dots, P_n\}$ . This is indeed possible, because there is a finite number of points at which any ray  $\overrightarrow{QP_l}$  meets the planes of the form  $P_i P_j P_k$ .

Now consider any face of  $P$ . It is a triangle of the type  $P_i P_j P_k$ , so the hypothesis implies that at least one of the rays  $\overrightarrow{P_l Q}$  (which is opposite to  $\overrightarrow{QP_l}$ ) meets it at some interior point. Also, no ray  $\overrightarrow{P_l Q}$  can intersect more than one face, because  $P$  is convex. Therefore the number  $F$  of faces of  $P$  does not exceed  $n$ , the number of its vertices:

$$F \leq n. \quad (1)$$

Let  $P$  have  $E$  edges. Each of its faces is a triangle, and each edge is a side of exactly two such triangles, so

$$E = \frac{3}{2}F. \quad (2)$$

By Euler’s formula, we have  $n - E + F = 2$ . Combined with (1) and (2), this readily gives  $n \leq 4$ . Hence  $n = 4$ . ■

Thus, any set  $\{P_1, P_2, \dots, P_n, Q\}$  with the given property consists of the vertices of a tetrahedron  $P_1 P_2 P_3 P_4$  and a point  $Q$  interior to it!

# 38

## Should We Count?

One need not know how to count in order to figure out which one of two finite sets has more elements. The problems in this section are based on this simple idea. We start with one from the 1985 Tournament of Towns:

*Consider all sequences with length  $2n + 1$  (where  $n$  is a positive integer) whose terms are each 0 or 1. What fraction of them have more occurrences of 1 among the last  $n + 1$  digits than among the first  $n$ ?*

We shall say that a sequence  $(x_1, x_2, \dots, x_{2n+1})$  of zeros and ones is *good* if 1 occurs more times among  $x_{n+1}, x_{n+2}, \dots, x_{2n+1}$  than among  $x_1, x_2, \dots, x_n$ , and that it is *bad* otherwise.

To each sequence  $\alpha = (x_1, x_2, \dots, x_{2n+1})$  assign the sequence

$$\alpha^* = (1 - x_1, 1 - x_2, \dots, 1 - x_{2n+1}).$$

Suppose that 1 occurs  $a$  times among the numbers  $x_1, x_2, \dots, x_n$  and  $b$  times among  $x_{n+1}, x_{n+2}, \dots, x_{2n+1}$ . Then it occurs  $n - a$  times among the numbers  $1 - x_1, 1 - x_2, \dots, 1 - x_n$  and  $n + 1 - b$  times among  $1 - x_{n+1}, 1 - x_{n+2}, \dots, 1 - x_{2n+1}$ . Now notice that

$$a < b \quad \text{if and only if} \quad n - a \geq n + 1 - b.$$

This means that  $\alpha$  is good if and only if  $\alpha^*$  is bad. It is not hard to see that the correspondence  $\alpha \rightarrow \alpha^*$  is bijective. Therefore there are exactly as many good sequences as bad ones, so exactly half of the sequences we are looking at are good. ■

Next there comes a problem from the 1989 IMO; the solution we present is by Bulgarian contestant **Dobromir Dimitrov**.

We shall call a permutation  $(x_1, x_2, \dots, x_{2n})$  of  $\{1, 2, \dots, 2n\}$  pleasant if  $|x_i - x_{i+1}| = n$  for at least one  $i \in \{1, 2, \dots, 2n-1\}$ . Prove that, for each  $n \geq 1$ , more than half of all permutations are pleasant.

The case  $n = 1$  is trivial, so assume that  $n \geq 2$ . Denote by  $A$  the set of pleasant permutations and by  $B$  the set of all other permutations. We need to prove that  $|A| > |B|$ .

Let  $\sigma = (x_1, x_2, \dots, x_{2n}) \in B$ , that is,  $|x_i - x_{i+1}| \neq n$  for each index  $i \in \{1, 2, \dots, 2n-1\}$ . Then  $|x_{2n-1} - x_{2n}| \neq n$ . It is not hard to see that there exists a unique  $k \in \{1, 2, \dots, 2n-2\}$  such that  $|x_{2n} - x_k| = n$ , so the permutation

$$\sigma^* = (x_1, \dots, x_k, x_{2n}, x_{k+1}, \dots, x_{2n-1})$$

is pleasant. We obtain a well-defined function  $\sigma \rightarrow \sigma^*$  from  $B$  into  $A$ . Note that  $\sigma^*$  contains *exactly one* pair of consecutive elements differing by  $n$ . It follows easily that the defined function is injective, because it takes different permutations from  $B$  to different pleasant permutations. Therefore  $|B| \leq |A|$ . On the other hand, the permutation

$$(1, n+1, 2, n+2, \dots, n, 2n)$$

is pleasant, but it is not the image of any permutation from  $B$  under the function  $\sigma \rightarrow \sigma^*$ . Indeed, this permutation contains more than one pair of consecutive elements differing by  $n$  for  $n \geq 2$ . Consequently,  $\sigma \rightarrow \sigma^*$  is not surjective, which implies  $|A| > |B|$ . ■

Let  $M$  be the number of integer solutions of the equation

$$x^2 - y^2 = z^3 - t^3$$

with the property  $0 \leq x, y, z, t \leq 10^6$ , and let  $N$  be the number of integer solutions of the equation

$$x^2 - y^2 = z^3 - t^3 + 1$$

that have the same property. Prove that  $M > N$ .

This problem was proposed by the Soviet Union for the 1979 IMO.

Write down the two equations in the form

$$x^2 + t^3 = y^2 + z^3, \quad x^2 + t^3 = y^2 + z^3 + 1$$

and, for each  $k = 0, 1, 2, \dots$ , denote by  $n_k$  the number of integer solutions of the equation  $u^2 + v^3 = k$  with the property  $0 \leq u, v \leq 10^6$ . Clearly

$n_k = 0$  for all  $k$  greater than  $l = (10^6)^2 + (10^6)^3$ . Now a key observation follows: We have

$$\begin{aligned} M &= n_0^2 + n_1^2 + \cdots + n_l^2 \quad \text{and} \\ N &= n_0 n_1 + n_1 n_2 + \cdots + n_{l-1} n_l. \end{aligned} \tag{1}$$

To prove, for example, the second of these equalities, note that to any integer solution of  $x^2 + t^3 = y^2 + z^3 + 1$  with  $0 \leq x, y, z, t \leq 10^6$  there corresponds a unique  $k$  ( $1 \leq k \leq l$ ) such that

$$x^2 + t^3 = k, \quad y^2 + z^3 = k - 1. \tag{2}$$

And for any such  $k$ , the pairs  $(x, t)$  and  $(y, z)$  satisfying (2) can be chosen independently of one another in  $n_k$  and  $n_{k-1}$  ways, respectively. Hence, for each  $k = 1, 2, \dots, l$ , there are  $n_{k-1} n_k$  solutions of  $x^2 + t^3 = y^2 + z^3 + 1$  with  $x^2 + t^3 = y^2 + z^3 + 1 = k$ , which implies

$$N = n_0 n_1 + n_1 n_2 + \cdots + n_{l-1} n_l.$$

The proof of the first equality in (1) is essentially the same.

It is not hard to deduce from (1) that  $M > N$ . Indeed, a little algebra work shows that

$$M - N = \frac{1}{2} [n_0^2 + (n_0 - n_1)^2 + (n_1 - n_2)^2 + \cdots + (n_{l-1} - n_l)^2 + n_l^2].$$

Hence  $M - N > 0$ , because  $n_0 \neq 0$  (in fact  $n_0 = 1$ ). ■

Trying to find  $M$  and  $N$  explicitly is, it seems, hopeless, and so is the question about the exact values of the numbers  $n_k$ . We cannot do much better even if we just ask whether a particular  $n_k$  is equal to zero, that is, whether the equation  $u^2 + v^3 = k$  has at least one integer solution.

# 39

## Let's Count Now!

Can you relate the title to the following 1972 Moscow Olympiad problem?

*A positive integer is written in each square of an  $8 \times 8$  chessboard. One is allowed to choose a  $3 \times 3$  or a  $4 \times 4$  square on the chessboard and increase all numbers in it by 1. Is it always possible, applying such operations several times, to arrive at a situation where all squares contain multiples of 10?*

Such a problem usually involves finding a suitable invariant, and its presence in this section may seem strange.

Taking everything modulo 10, we see that the diversity of the possible chessboards is big but not huge. There are “only”  $10^{64}$  of them. Also, we may assume that each  $3 \times 3$  or  $4 \times 4$  square is chosen at most 10 times.

The permitted operation is “invertible”: if a chessboard  $C_1$  can be obtained from another chessboard  $C$  after a finite number of moves, then  $C$  can also be obtained from  $C_1$ . Indeed, suppose a certain  $3 \times 3$  or  $4 \times 4$  square has been changed  $k$  times,  $k \in \{1, 2, \dots, 9\}$ , by adding 1 to each entry. We can restore the original square by repeating the operation  $10 - k$  more times.

Thus the question is: Can we obtain each of the  $10^{64}$  states starting from the “identically zero” chessboard?

The answer is no. Note first that for any  $1 \leq k \leq 8$  there are  $(9 - k)^2$  distinct  $k \times k$  squares on the board. Indeed, any  $k \times k$  square is uniquely determined by its upper right-hand corner, which lies in the upper right-hand  $(9 - k) \times (9 - k)$  square and hence can be chosen in  $(9 - k)^2$  ways. So our operations concern altogether  $(9 - 3)^2 + (9 - 4)^2 = 61$  squares, each one having 10 different states. Thus there are at most  $10^{61}$  distinct

chessboards, which is less than  $10^{64}$ . Hence there exist initial states that cannot be obtained from the “identically zero” chessboard. ■

*Let  $m$  and  $n$  be fixed positive integers. Find the number of sequences*

$$P_1(x_1, y_1), P_2(x_2, y_2), \dots, P_{m+n-1}(x_{m+n-1}, y_{m+n-1})$$

*of  $m+n-1$  points in the coordinate plane such that the following conditions are satisfied:*

- 1° The numbers  $x_i, y_i, i = 1, 2, \dots, m+n-1$ , are integers and  $1 \leq x_i \leq m, 1 \leq y_i \leq n$ .*
- 2° Each of the numbers  $1, 2, \dots, m$  occurs in the sequence  $x_1, x_2, \dots, x_{m+n-1}$ , and each of the numbers  $1, 2, \dots, n$  occurs in the sequence  $y_1, y_2, \dots, y_{m+n-1}$ .*
- 3° For each  $i = 1, 2, \dots, m+n-2$ , the line  $P_i P_{i+1}$  is parallel to one of the coordinate axes.*

This 1992 Bulgarian Olympiad problem is solved by the following combinatorial argument.

Let  $\alpha = P_1(x_1, y_1), P_2(x_2, y_2), \dots, P_{m+n-1}(x_{m+n-1}, y_{m+n-1})$  be a sequence of points in the plane satisfying 1°, 2° and 3°; call such sequences *suitable*. Take  $m$  blue balls labeled 1 through  $m$  and  $n$  red balls labeled 1 through  $n$ . We will arrange them in a row according to the following rule determined by the sequence  $\alpha$ .

Put the blue ball  $x_1$  in the first place, then the red ball  $y_1$ . By 3°, exactly one of the equalities  $x_1 = x_2$  and  $y_1 = y_2$  holds. If  $x_1 \neq x_2$  and  $y_1 = y_2$ , put the blue ball  $x_2$  in the third place. Otherwise, if  $x_1 = x_2$  and  $y_1 \neq y_2$ , put the red ball  $y_2$  there. Proceed further in a similar fashion. If the coordinates of  $P_{i-1}$  and  $P_i$  ( $2 \leq i \leq m+n-1$ ) satisfy  $x_{i-1} \neq x_i$  and  $y_{i-1} = y_i$ , put the blue ball  $x_i$  in the  $(i+1)$ st place; otherwise, if  $x_{i-1} = x_i$  and  $y_{i-1} \neq y_i$ , put the red ball  $y_i$  there.

Each ball will eventually be present in the row. For instance, the red ball labeled  $k$  (for  $1 \leq k \leq n$ ) appears as soon as we encounter a point with ordinate  $k$ . This will certainly happen in view of 2°. Note that the last ball is added the moment we reach the last point  $P_{m+n-1}$ , so the process terminates as soon as both the points  $P_1, P_2, \dots, P_{m+n-1}$  and the  $m+n$  balls are exhausted.

So each suitable sequence  $\alpha$  gives rise to a permutation  $f(\alpha)$  of all  $m+n$  balls with the property that *the first two places are occupied by a blue and by a red ball, respectively*. Call such permutations *suitable* as

well. It is not hard to see that the function  $\alpha \rightarrow f(\alpha)$  is bijective: each suitable permutation is assigned to a unique suitable sequence.

Now it suffices to count the suitable permutations. Since we have  $m$  blue and  $n$  red balls, the first two balls (blue and red, respectively) can be chosen in  $mn$  ways. There are no further constraints on the ordering of the remaining  $m + n - 2$  balls, so they can be arranged in  $(m + n - 2)!$  ways. Thus the answer is  $mn(m + n - 2)!$ . ■

Next there comes the indisputable highlight of the 1997 IMO:

*For each positive integer  $n$ , let  $f(n)$  denote the number of ways of representing  $n$  as a sum of powers of 2 with nonnegative integer exponents. Representations that differ only in the ordering of their summands are considered to be the same. For instance,  $f(4) = 4$ , because the number 4 can be represented in the following four ways:  $4$ ;  $2 + 2$ ;  $2 + 1 + 1$ ;  $1 + 1 + 1 + 1$ . Prove that, for each integer  $n \geq 3$ ,*

$$2^{\frac{1}{4}n^2} < f(2^n) < 2^{\frac{1}{2}n^2}. \quad (1)$$

A typical estimation problem is intended to shed light on the growth of a certain function at infinity. Dealing with initial values of the function is rather a drawback, since they often happen to be accidental deviations from its actual asymptotic behavior. So we sacrifice the requirement  $n \geq 3$  in order to obtain essentially deeper information about  $f(2^n)$  for sufficiently large  $n$ .

Assume that  $2^n$  is represented as the sum of several powers of 2 and denote by  $a_i$  the number of occurrences of  $2^i$  in that sum,  $i = 0, 1, \dots, n$ . Such a representation is uniquely determined by the sequence

$$a_1 2^1, a_2 2^2, \dots, a_n 2^n.$$

Indeed, given the contribution of every *nonzero* power of 2, we are able to restore  $a_1, a_2, \dots, a_n$  as well as  $a_0 = 2^n - (a_1 2^1 + a_2 2^2 + \dots + a_n 2^n)$ .

In turn, each number  $a_i 2^i$  can be identified with its base 2 representation, which contains at most  $n + 1$  binary digits and necessarily ends in  $i$  zeros. For each  $i = 1, 2, \dots, n$ , write down the binary representation of  $a_i 2^i$  in the  $i$ th row of an  $n \times (n + 1)$  array (allowing leading zeros), and then delete the  $i$  zeros in the end. We obtain an  $n \times n$  triangular array filled in with 0's and 1's.

The *weight* of each entry  $\varepsilon$  in this array, that is, its contribution to the sum  $\sum_{i=1}^n a_i 2^i$ , is  $\varepsilon \cdot 2^j$ , where  $j$  is the number of its column, counted



columns. Its total weight is at most

$$\begin{aligned} & [2^1 + 2^2 + \cdots + 2^k] + [2^2 + 2^3 + \cdots + 2^k] + \cdots + [2^{k-1} + 2^k] + 2^k \\ &= k2^{k+1} - [2^1 + 2^2 + \cdots + 2^{k-1} + 2^k] = (k-1)2^{k+1} + 2. \end{aligned}$$

Suppose now that  $k$  satisfies the inequality

$$(k-1)2^{k+1} + 2 \leq 2^n.$$

Then any array with 1's just in its last  $k$  columns is eligible. Since there are  $2^{1+2+\cdots+k} = 2^{\frac{1}{2}k(k+1)}$  such arrays, we get  $f(2^n) \geq 2^{\frac{1}{2}k(k+1)}$ .

For convenience, set  $u(x) = (x-1)2^{x+1} + 2$ . In search of sufficiently large integer solutions of the inequality  $u(x) \leq 2^n$ , observe that it is satisfied by  $x = n - \log_2 n - 1$  (which, in general, is not an integer). Indeed,

$$(n - \log_2 n - 2)2^{n - \log_2 n} + 2 = \left(1 - \frac{\log_2 n}{n} - \frac{2}{n} + \frac{1}{2^{n-1}}\right) 2^n,$$

and the expression in the parentheses is less than 1 for all  $n$ . Taking  $k$  to be the integer part of  $n - \log_2 n - 1$ , we then have  $u(k) \leq 2^n$ , and so

$$f(2^n) \geq 2^{\frac{1}{2}k(k+1)} > 2^{\frac{1}{2}x(x-1)}.$$

This yields a substantially sharper lower bound than asked in (1):

$$f(2^n) > 2^{v_n}, \quad \text{where } v_n = \frac{1}{2}(n - \log_2 n - 1)(n - \log_2 n - 2).$$

Expanding  $v_n$  and ignoring the lower-order positive terms, we obtain the less precise but more convenient lower bound

$$f(2^n) > 2^{\frac{1}{2}n^2 - n \log_2 n - \frac{3}{2}n}. \blacksquare$$

Typically, a solution of this problem starts with a proof of a certain recurrence formula, and then switches to algebraic or calculus manipulations, thus ignoring the combinatorial nature of the matter. Such approaches are lengthy and yield a lower estimate (the more involved part of the problem) of order only  $2^{\lambda n^2}$  for specific  $\lambda$ 's satisfying  $0 < \lambda < \frac{1}{2}$ . Consequently, they are much less precise than the higher-order estimates obtained in the proof above. Also, though the upper estimate is much easier to achieve, it turns out to be closer to the actual asymptotic behavior of the function  $f$  than the lower one.

# 40

## Some Elementary Number Theory

In this section we consider several number theory problems solved with a variety of ideas specific for the field.

Our first problem is a variation on the theme of primes and sums of integer squares:

*Let  $p$  be a prime of the form  $3k + 2$  that divides  $a^2 + ab + b^2$  for some integers  $a$  and  $b$ . Prove that  $a$  and  $b$  are both divisible by  $p$ .*

Let  $p = 3k + 2$  for some  $k \geq 0$ . Since  $p$  divides  $a^2 + ab + b^2$ , it also divides  $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ , so  $a^3 \equiv b^3 \pmod{p}$ . Hence

$$a^{3k} \equiv b^{3k} \pmod{p}. \quad (1)$$

Assume that  $p$  does not divide  $a$ . Then it does not divide  $b$  either, and Fermat's little theorem yields  $a^{p-1} \equiv 1 \pmod{p}$ ,  $b^{p-1} \equiv 1 \pmod{p}$ . Thus, in view of  $p = 3k + 2$ ,

$$a^{3k+1} \equiv b^{3k+1} \pmod{p}. \quad (2)$$

Since  $p$  is relatively prime to  $a$ , (1) and (2) yield that  $a \equiv b \pmod{p}$ . This, combined with  $a^2 + ab + b^2 \equiv 0 \pmod{p}$ , implies  $3a^2 \equiv 0 \pmod{p}$ . Because  $p \neq 3$ , it turns out that  $p$  divides  $a$ , which is a contradiction. ■

The next two examples come from Bulgarian spring and winter competitions.

*Let  $3^n - 2^n$  be a power of a prime for some positive integer  $n$ . Prove that  $n$  is a prime.*

Let  $3^n - 2^n = p^\alpha$  for some prime  $p$  and some  $\alpha \geq 1$ , and let  $q$  be a prime divisor of  $n$ . Assume that  $q \neq n$ ; then  $n = kq$ , where  $k > 1$ .

Since  $p^\alpha = 3^{kq} - 2^{kq} = (3^k)^q - (2^k)^q$ , we observe that  $p^\alpha$  is divisible by  $3^k - 2^k$ . Hence  $3^k - 2^k = p^\beta$  for some  $\beta \geq 1$ . Now we have

$$\begin{aligned} p^\alpha &= (2^k + p^\beta)^q - 2^{kq} \\ &= q2^{k(q-1)}p^\beta + \frac{q(q-1)}{2}2^{k(q-2)}p^{2\beta} + \dots + p^{q\beta}. \end{aligned}$$

Since  $\alpha > \beta$  (because  $p^\beta = 3^k - 2^k$  is less than  $p^\alpha = 3^{kq} - 2^{kq}$ ), it follows that  $p^\alpha$  is divisible by a power of  $p$  at least as great as  $p^{\beta+1}$ . Then the above equality implies that  $p$  divides  $q2^{k(q-1)}$ . On the other hand,  $p$  is obviously odd and hence it divides  $q$ . Being a prime,  $q$  must be then equal to  $p$ . Therefore  $n = kq = kp$  and  $p^\alpha = (3^p)^k - (2^p)^k$  is divisible by  $3^p - 2^p$ , implying  $3^p - 2^p = p^\gamma$  for some  $\gamma \geq 1$ . In particular, we infer that  $3^p \equiv 2^p \pmod{p}$ . Now, observing that  $p \neq 2, 3$ , we reach a contradiction with Fermat's little theorem, by which

$$3^p \equiv 3 \pmod{p}, \quad 2^p \equiv 2 \pmod{p}. \quad \blacksquare$$

*Let  $a_1, a_2, \dots, a_n, \dots$  be a strictly increasing sequence of positive integers such that  $a_{2n} = a_n + n$  for  $n = 1, 2, \dots$ . It is given that if  $a_n$  is a prime then  $n$  is also a prime. Prove that  $a_n = n$  for all  $n$ .*

Checking the first few terms yields  $a_2 = a_1 + 1$ ,  $a_4 = a_2 + 2 = a_1 + 3$ , and since  $a_3$  lies between  $a_2$  and  $a_4$ , we must have  $a_3 = a_1 + 2$ . Thus the sequence begins  $a_1, a_1 + 1, a_1 + 2, a_1 + 3$ , suggesting

$$a_n = a_1 + (n - 1)$$

for all  $n$ . This claim is easily confirmed by induction as follows. Assume that it holds for some  $n \geq 1$  (the base case is already checked). Then

$$a_{2n} = a_n + n = a_1 + n - 1 + n = a_1 + 2n - 1.$$

On the other hand,

$$a_1 + n - 1 = a_n < a_{n+1} < a_{n+2} < \dots < a_{2n} = a_1 + 2n - 1,$$

which forces the conclusion that

$$a_{n+1} = a_1 + n = a_1 + (n + 1) - 1.$$

The induction is complete.

It turns out that all positive integers can be shifted by the same distance  $a_1 - 1$  to the right so that if a number goes to a prime, then the number

is a prime itself. We need to prove that this can happen only in the case  $a_1 - 1 = 0$ , that is,  $a_1 = 1$ . The clue is: *There are arbitrarily long streaks of composite numbers.*

So, assume on the contrary that  $a_1 > 1$ , and denote by  $p$  the least prime greater than  $(a_1 + 1)! + a_1 + 1$ . Then  $p$  is a term of the given sequence, hence we have  $p = a_n = a_1 + n - 1$  for some  $n$ , which has to be a prime by hypothesis. So,  $p - a_1 + 1 = n$  is a prime. Note that  $p - a_1 + 1 < p$ , because  $a_1 > 1$ . Then, being a prime,  $p - a_1 + 1$  must not exceed  $(a_1 + 1)! + a_1 + 1$ , since  $p$  was chosen to be the least prime greater than  $(a_1 + 1)! + a_1 + 1$ . Observing that  $p > (a_1 + 1)! + a_1 + 1$  implies  $p - a_1 + 1 \geq (a_1 + 1)! + 2$ , we obtain

$$(a_1 + 1)! + 2 \leq p - a_1 + 1 \leq (a_1 + 1)! + a_1 + 1.$$

This leads to a contradiction, because each of the numbers

$$(a_1 + 1)! + 2, (a_1 + 1)! + 3, \dots, (a_1 + 1)! + a_1 + 1$$

is composite. The contradiction shows that  $a_1 = 1$ , so  $a_n = n$  for each  $n = 1, 2, \dots$ . ■

An application of the Chinese remainder theorem follows. The solution is short but not easy.

*Let  $P(x)$  be a polynomial with integer coefficients. Suppose that the integers  $a_1, a_2, \dots, a_n$  have the following property: For any integer  $x$  there exists an  $i \in \{1, 2, \dots, n\}$  such that  $P(x)$  is divisible by  $a_i$ . Prove that there is an  $i_0 \in \{1, 2, \dots, n\}$  such that  $a_{i_0}$  divides  $P(x)$  for any integer  $x$ .*

Suppose that the claim is false. Then for each  $i = 1, 2, \dots, n$  there exists an integer  $x_i$  such that  $P(x_i)$  is not divisible by  $a_i$ . Hence, there is a prime power  $p_i^{k_i}$  that divides  $a_i$  and does not divide  $P(x_i)$ . Some of the powers  $p_1^{k_1}, p_2^{k_2}, \dots, p_n^{k_n}$  may have the same base. If so, ignore all but the one with the least exponent. To simplify notation, assume that the sequence obtained this way is  $p_1^{k_1}, p_2^{k_2}, \dots, p_m^{k_m}$ ,  $m \leq n$  ( $p_1, p_2, \dots, p_m$  are distinct primes). Note that each  $a_i$  is divisible by some term of this sequence.

Since  $p_1^{k_1}, p_2^{k_2}, \dots, p_m^{k_m}$  are pairwise relatively prime, the Chinese remainder theorem yields a solution of the simultaneous congruences

$$x \equiv x_1 \pmod{p_1^{k_1}}, \quad x \equiv x_2 \pmod{p_2^{k_2}}, \quad \dots, \quad x \equiv x_m \pmod{p_m^{k_m}}.$$

Now, since  $P(x)$  is a polynomial with integer coefficients, the congruence  $x \equiv x_j \pmod{p_j^{k_j}}$  implies  $P(x) \equiv P(x_j) \pmod{p_j^{k_j}}$  for each index  $j = 1, 2, \dots, m$ . By the definition of  $p_j^{k_j}$ , the number  $P(x_j)$  is never divisible by  $p_j^{k_j}$ ,  $j = 1, 2, \dots, m$ . Thus, for the solution  $x$  given by the Chinese remainder theorem,  $P(x)$  is not divisible by any of the powers  $p_j^{k_j}$ . And because each  $a_i$  is divisible by some  $p_j^{k_j}$ ,  $j = 1, 2, \dots, m$ , it follows that no  $a_i$  divides  $P(x)$  either, a contradiction. ■

We close with a celebrated problem from the 1988 IMO:

*Let  $a$  and  $b$  be positive integers such that  $ab + 1$  divides  $a^2 + b^2$ .*

*Prove that  $(a^2 + b^2)/(ab + 1)$  is the square of an integer.*

The following solution by Bulgarian student **Emanuil Atanasov** was awarded a special prize.

By hypothesis, we have  $(a^2 + b^2)/(ab + 1) = k$  for some positive integer  $k$ . Since this is equivalent to  $a^2 - kab + b^2 = k$ , the statement may be reworded as follows: *If the Diophantine equation*

$$x^2 - kxy + y^2 = k \tag{3}$$

*has a positive integer solution, then  $k$  is a perfect square.*

Suppose  $k$  is not a perfect square and assume, on the contrary, that equation (3) has a positive integer solution. In fact, for any integer solution,  $x$  and  $y$  are nonzero (otherwise  $k$  is a perfect square) and have the same sign (or else  $x^2 - kxy + y^2 \geq x^2 + k + y^2 > k$ ).

Among the positive integer solutions of (3), choose one  $(a, b)$  for which  $a \geq b > 0$  and  $b$  is minimal. Now consider the equality

$$a^2 - kab + b^2 - k = 0$$

as a quadratic in  $a$ . Apart from  $a$ , it has one more solution  $a_1$ , and these solutions satisfy

$$a + a_1 = kb, \quad aa_1 = b^2 - k.$$

The first of these equalities implies that  $a_1$  is an integer; the remark above about the signs shows that  $a_1$  is positive. Hence, the pair  $(a_1, b)$  is a positive integer solution of (3).

On the other hand, the conditions  $aa_1 = b^2 - k$  and  $a \geq b > 0$  easily yield  $a_1 < b$ . Therefore we have a positive integer solution  $(a_1, b)$  of (3) in which the smaller of the numbers is strictly less than  $b$ . This contradicts the minimality of  $b$ . ■

A complement to this beautiful solution is the following assertion:

*If  $k = c^2 > 1$  is a perfect square, then the Diophantine equation (3) has positive integer solutions. All of them are pairs of consecutive terms of the sequence defined by*

$$x_1 = c, \quad x_2 = c^3, \quad x_{n+1} = c^2 x_n - x_{n-1} \quad \text{for } n = 2, 3, \dots$$

The key observation is the same: If  $(x, y)$  is a positive integer solution of (3), then so is  $(y, ky - x)$ . We leave the proof to the reader.

## Coffee Break 8

1. A pile containing 1001 pebbles is given. It is divided into two piles, and the product of the number of pebbles in the two new piles is recorded. The same is repeated with one of the two smaller piles. We perform the same operation to one of the three existent piles, and so on, until all piles contain one pebble each. What are the possible values of the sum of all products listed?
2. The orthogonal projections of a point set in three-dimensional space onto two nonparallel planes are circular discs. Prove that these discs have equal radii.
3. A real number is written in each square of an  $n \times n$  grid so that the sum of all written numbers is positive. Prove that the columns can be permuted in such a way that the sum of the numbers on the main diagonal in the new grid is positive.

## Solutions

1. This sum has just one possible value: 500,500. It is easy to guess the answer: If at each step we remove one pebble from the same pile, the sum will be

$$1000 + 999 + 998 + \cdots + 2 + 1 = 500,500.$$

There are different ways to prove that the same result will be obtained in all cases. Here is probably the shortest one. Imagine that initially each pebble is connected to each other pebble by a thread. When partitioning some pile into two smaller ones, assume that all the threads connecting pebbles that go to different piles are being cut. The number of these threads is exactly equal to the product of the number of pebbles in the smaller piles. Hence in all cases the sum in question will be equal to the total number of threads, that is, to  $1001 \cdot 1000/2 = 500,500$ .

2. Project the discs onto the common line  $l$  of the two planes. We obtain two line segments with lengths equal to the corresponding diameters. But these line segments are obliged to coincide, since each of them may be regarded as the orthogonal projection of the set itself onto  $l$ .

There are sets with the described property that are not spheres, for example, the common part of three right circular cylinders with pairwise perpendicular axes passing through a common point. It is another story that if *any* plane section of a three-dimensional point set is a circular disc, then the set is a ball.

3. Color the squares of the grid with  $n$  colors as shown in the left-hand figure below. Note that the main diagonal squares are exactly the ones colored with 1.

|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| 1   | 2   | 3   |     |     | $n$ |
| $n$ | 1   | 2   | 3   |     |     |
|     | $n$ | 1   | 2   | 3   |     |
|     |     | $n$ | 1   | 2   | 3   |
| 3   |     |     | $n$ | 1   | 2   |
| 2   | 3   |     |     | $n$ | 1   |

|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| 2   | 3   |     |     | $n$ | 1   |
| 1   | 2   | 3   |     |     | $n$ |
| $n$ | 1   | 2   | 3   |     |     |
|     | $n$ | 1   | 2   | 3   |     |
|     |     | $n$ | 1   | 2   | 3   |
| 3   |     |     | $n$ | 1   | 2   |

|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| 3   |     |     | $n$ | 1   | 2   |
| 2   | 3   |     |     | $n$ | 1   |
| 1   | 2   | 3   |     |     | $n$ |
| $n$ | 1   | 2   | 3   |     |     |
|     | $n$ | 1   | 2   | 3   |     |
|     |     | $n$ | 1   | 2   | 3   |

Now shift the first column of the grid to the last place (see the middle figure). The main diagonal of the new grid consists of the squares colored with 2. Repeat the same with this grid to get another one with exactly the squares of color 3 on the main diagonal, and so on. At each of the  $n$  possible cyclic shifts, all main diagonal squares have the same color, and each color appears there once. Since the total sum of all numbers is positive, so is the sum of the numbers of at least one color, say the  $k$ th one. Then at the  $k$ th move we will have a grid with a positive sum of the main diagonal squares.

# 41

## Euclid's Game

This title is our name for a problem from the 1978 All-Union Olympiad:

*Two players alternately remove matches from two piles originally containing  $a$  and  $b$  matches,  $a > b$ . On his turn, each player removes a positive number of matches from one of the piles, provided that this number is a multiple of the number of matches in the other pile. The winner is the one who removes the last match from one of the piles. Determine for which pairs  $(a, b)$  the first player has a winning strategy.*

Denote the first and the second player by  $A$  and  $B$ , respectively. If the two piles contain  $x$  and  $y$  matches at some particular moment, we shall call the ordered pair  $(x, y)$  a position describing the game at that moment. As usual, it will be convenient to call a position *winning* if the player about to move has a strategy that allows him to force a win, and *losing* otherwise. Note that the game always ends.

The positions  $(a, b)$  and  $(b, a)$  are clearly both winning or both losing. Moreover, it is easy to observe that whether or not a position  $(a, b)$  is winning depends only on the ratio  $a/b$ . So it suffices to find out which positive *rational numbers*  $a/b$  correspond to winning positions. Assuming an approach like this, we may say that a legal move in the position  $(a, b)$  amounts to selecting the greater of the numbers  $a/b$  and  $b/a$  and subtracting some positive integer from it, such that the result is nonnegative. The player who reaches 0 is the winner. A trivial observation is that if  $a/b$  is an integer then the position  $(a, b)$  is winning.

Denote by  $\lambda = (1 + \sqrt{5})/2$  the positive root of the equation

$$x^2 - x - 1 = 0.$$

We claim that a position  $(a, b)$  is *losing* if and only if

$$a \neq b \quad \text{and} \quad \frac{a}{b} \in (\lambda - 1, \lambda).$$

Indeed, let us temporarily call a positive integer pair  $(a, b)$  *irregular* if it satisfies these conditions, and *regular* otherwise. Note that the definition of  $\lambda$  implies  $1/\lambda = \lambda - 1$ . Hence, for any  $x > 0$ , the numbers  $x$  and  $1/x$  are both contained or not in the interval  $(\lambda - 1, \lambda)$ .

Suppose now that an irregular pair  $(a, b)$  was obtained at some moment of the game, and let  $a > b$ . Then  $1 < a/b < \lambda < 2$ , so the player to move has just one possible choice: to subtract 1 from  $a/b$ , that is, to make the move  $(a, b) \rightarrow (a - b, b)$ . This yields the pair  $(a - b, b)$ , which is regular, because

$$\frac{a - b}{b} = \frac{a}{b} - 1 < \lambda - 1.$$

Note that the only possible move does not force a win, since  $a - b$  is not 0. Therefore, any legal move (there is actually exactly one) transforms an irregular pair into a regular one, and the game does not end after it.

On the other hand, take a regular pair  $(a, b)$  and suppose again that  $a > b$ . If  $a/b$  is an integer, then the player to move wins on his first move. Otherwise there exists a positive integer  $q$  such that  $(a/b) - q$  belongs to  $(\lambda - 1, \lambda)$  (because  $(\lambda - 1, \lambda)$  has length 1 and  $\lambda$  is irrational) and is different from 1 (because  $a/b$  is not an integer). Thus, for each regular pair there exists either a game-winning move or a move converting it into an irregular one.

Summing up, we conclude that the winning pairs are exactly the regular ones. In view of the constraint  $a > b$ , the answer is: A position  $(a, b)$  is winning if and only if

$$\frac{a}{b} > \lambda = \frac{1 + \sqrt{5}}{2}. \quad \blacksquare$$

There are different ways to explain how the omnipresent “golden section”  $\lambda$  appears in this problem. We proceed with another solution that does not exhibit a winning strategy explicitly.

Write  $a$  in the form  $a = qb + r$ , where  $0 \leq r < b$ . We claim that if  $q \geq 2$ , then the first player has a winning strategy. To confirm this, it suffices to show that at least one of the positions

$$(a - (q - 1)b, b) = (b + r, b) \quad \text{and} \quad (a - qb, b) = (r, b)$$



continued fraction

$$[1; 1, 1, \dots, 1, \dots],$$

and it is well known that this fraction represents the “golden section”  $\lambda$ . ■

Solving problems this way, without pointing out a winning strategy, is not uncommon. Here is one more example.

*Let  $n$  be a fixed positive integer. Two players, taking turns, write positive integers not exceeding  $n$  on the blackboard. The only rule is that they are not allowed to write divisors of numbers already written. The first one who cannot make a move loses. Determine which of the players has a winning strategy.*

The first player can always win. To prove this, consider a new game: The rules are the same but no one is allowed to write 1. If there is a winning strategy in this game for the first player, he applies it immediately. This is possible, since his first move, whatever it is, rules out the possibility for either of the two to write 1 from then on. If such a strategy does not exist, then the first player writes 1, thus leaving the second player in a losing position of the new game. ■

Nice, isn't it?

# 42

## Perfect Powers

Our story starts with the *Kvant* problem M723 (No. 1, 1982):

*Does there exist an infinite set  $A$  of positive integers with the property that the sum of the elements of any finite subset of  $A$  is not a perfect power?*

The author **L. Gurvits** had in mind a solution using the notion of the *density* of a set. Such an advanced approach settles the question without much noise. We provide the necessary preparation here.

If you are scared by this introduction, do not pass on to the next section, but go to the bottom of the next page. An exceptionally simple elementary solution is there.

For each subset  $A$  of the set  $\mathbf{N}$  of positive integers and for each integer  $n \geq 1$ , denote by  $A(n)$  the number of elements of  $A$  not exceeding  $n$ . If the limit  $\lim_{n \rightarrow \infty} A(n)/n$  exists, it is called the density of  $A$ . The density of a set is a measure of its size compared to the whole of  $\mathbf{N}$ . For instance, an infinite arithmetic progression  $a, a + d, \dots, a + dn, \dots$  with common difference  $d$  has density  $1/d$ . This seems reasonable: about  $1/d$  of the positive integers are terms of this progression.

The sets of density 0, also called 0-sets, are particularly interesting for our purpose. For example, the perfect squares form a 0-set. This is because the number of perfect squares not exceeding a given  $n$  is less than  $\sqrt{n}$ , and  $\lim_{n \rightarrow \infty} \sqrt{n}/n = 0$ .

It is less evident that the set of *all* perfect powers (squares, cubes, etc.) is also a 0-set. To prove this, take any positive integer  $n$  and let  $A(n)$  be the number of perfect powers not exceeding it. Since  $a^k \leq n$  if and only if  $a \leq \lfloor n^{1/k} \rfloor$ , the number of  $k$ th powers not exceeding  $n$  is at most  $\lfloor n^{1/k} \rfloor$ . Also, this number is greater than 0 only if  $k \leq \log_2 n$ . Indeed, assuming

that there is at least one perfect  $k$ th power  $a^k \leq n$ , we get  $n \geq a^k \geq 2^k$ , so  $k \leq \log_2 n$ . It follows that

$$A(n) \leq \lfloor \sqrt{n} \rfloor + \lfloor \sqrt[3]{n} \rfloor + \cdots \leq \sqrt{n} + \sqrt[3]{n} + \cdots \leq \sqrt{n} \log_2 n,$$

and hence

$$\lim_{n \rightarrow \infty} \frac{A(n)}{n} \leq \lim_{n \rightarrow \infty} \frac{\sqrt{n} \log_2 n}{n} = \lim_{n \rightarrow \infty} \frac{\log_2 n}{\sqrt{n}} = 0. \blacksquare$$

Let us go back now to the problem we started with. A set with the stated property does exist, and we need one more fact for the proof: that a finite union of 0-sets is also a 0-set. This is a direct consequence of the definition, and we omit the proof.

We are going to prove a stronger result, showing that there is little special about the set of perfect powers:

*If  $A \subset \mathbb{N}$  is a 0-set, then there exists an infinite subset  $B$  of  $\mathbb{N}$  such that the sum of each finite subset of  $B$  is not an element of  $A$ .*

We construct the required infinite set  $B = \{b_1, b_2, \dots, b_n, \dots\}$  by induction. Note first that the 0-set  $A$  cannot coincide with the whole of  $\mathbb{N}$  whose density is 1. So there are positive integers outside  $A$ . Choose any of them and denote it by  $b_1$ . Assume that  $b_1, b_2, \dots, b_n$  are already chosen so that any subset sum of  $\{b_1, b_2, \dots, b_n\}$  does not belong to  $A$ . Let  $s_1, s_2, \dots, s_m$  be the subset sums of  $\{b_1, b_2, \dots, b_n\}$ ; the sum corresponding to the empty subset is also included and assumed to be 0. Now consider the sets

$$A - s_1, A - s_2, \dots, A - s_m,$$

where  $A - s_i = \{a - s_i \mid a \in A \text{ and } a - s_i > 0\}$  for  $i = 1, 2, \dots, m$ . It is easy to see that they all have density 0, since this holds for  $A$ . By the remark above, their union is also a 0-set. Then there is a  $b_{n+1} \in \mathbb{N}$  that is not an element of  $A - s_i$  for any  $i = 1, 2, \dots, m$ . So the choice of  $b_{n+1}$  ensures that the subset sums of  $\{b_1, b_2, \dots, b_{n+1}\}$  do not belong to  $A$ . The inductive construction is complete.  $\blacksquare$

Before this proof was published, **Sergei Konyagin**, a former Russian multiple olympiad winner, invented a fabulous “purely olympic” solution.

Let  $p_n$  be the  $n$ th prime and

$$a_1 = p_1, \quad a_2 = p_1^2 p_2, \quad \dots, \quad a_n = p_1^2 p_2^2 \cdots p_{n-1}^2 p_n, \quad \dots$$

The set  $A = \{a_1, a_2, \dots, a_n, \dots\}$  does the job, because if one takes finitely many  $a_n$ 's and  $a_k$  is the least one of them, then their sum is divisible by the prime  $p_k$  but not by  $p_k^2$ . ■

Solutions like this need no comment at all. Once invented, they seem like the easiest and most natural things in the world. After more than twenty years of work on the famous Hilbert's Tenth Problem, Julia Robinson, Martin Davis and Hilary Putnam did not succeed in getting to the very end. This was done by the former Russian olympiad winner Yuri Matiyasevich in 1970. He was then 22 years old!

Julia Robinson wrote later in her autobiographical notes:

"I have been told that some people think that I was blind not to see the solution myself when I was so close to it. On the other hand, no one else saw it either. There are lots of things, just lying on the beach as it were, that we don't see until someone else picks one of them up. Then we all see that one."

There are many related problems about perfect powers. One could ask questions about them starting just the opposite way:

*Prove that for every positive integer  $n$  there exists a set of  $n$  positive integers such that the sum of the elements of each of its nonempty subsets is a perfect power.*

This is a Korean proposal for the 1992 IMO.

We are tempted to try an inductive construction at first. It would yield an infinite set with the given property but, if true, this does not seem easy.

Actually we will prove a stronger statement:

*For any positive integer set  $\{a_1, a_2, \dots, a_n\}$  there exists a positive integer  $b$  such that the set  $\{ba_1, ba_2, \dots, ba_n\}$  consists of perfect powers.*

We first make sure that the original statement follows from here. Let  $\{x_1, x_2, \dots, x_m\}$  be a finite set of positive integers and  $S_1, S_2, \dots, S_r$  the element sums of its nonempty subsets ( $r = 2^m - 1$ ). Choose a  $b$  so that  $bS_1, bS_2, \dots, bS_r$  are all perfect powers. Then the set  $\{bx_1, bx_2, \dots, bx_m\}$  yields the desired example.

We proceed with the proof of the stronger statement. There is a finite number of primes  $p_1, p_2, \dots, p_k$  that participate in the prime factorization of  $a_1, a_2, \dots, a_n$ . Let

$$a_i = p_1^{\alpha_{i1}} p_2^{\alpha_{i2}} \dots p_k^{\alpha_{ik}} \quad \text{for } i = 1, 2, \dots, n;$$

some of the exponents  $\alpha_{ij}$  may be zeros. A positive integer with prime factorization  $p_1^{u_1} p_2^{u_2} \cdots p_k^{u_k}$  is a perfect  $q$ th power if and only if all the exponents  $u_j$  are divisible by  $q$ . Thus it suffices to find positive integers  $q_1, q_2, \dots, q_n$  greater than 1, and nonnegative integers  $l_1, l_2, \dots, l_k$  such that

$$l_1 + \alpha_{11}, l_2 + \alpha_{12}, \dots, l_k + \alpha_{1k} \text{ are divisible by } q_1;$$

$$l_1 + \alpha_{21}, l_2 + \alpha_{22}, \dots, l_k + \alpha_{2k} \text{ are divisible by } q_2;$$

$$\vdots$$

$$l_1 + \alpha_{n1}, l_2 + \alpha_{n2}, \dots, l_k + \alpha_{nk} \text{ are divisible by } q_n.$$

Now it is clear that we have lots of choices; let, for example,  $q_i$  be the  $i$ th prime number. As far as  $l_1$  is concerned, the above conditions translate into

$$l_1 \equiv -\alpha_{j1} \pmod{q_j}, \quad j = 1, 2, \dots, n.$$

This system of simultaneous congruences has a solution by the Chinese remainder theorem, because  $q_1, q_2, \dots, q_n$  are pairwise relatively prime. Analogously, each of the systems of congruences

$$l_2 \equiv -\alpha_{j2} \pmod{q_j}, \quad j = 1, 2, \dots, n$$

$$l_3 \equiv -\alpha_{j3} \pmod{q_j}, \quad j = 1, 2, \dots, n$$

$$\vdots$$

$$l_k \equiv -\alpha_{jk} \pmod{q_j}, \quad j = 1, 2, \dots, n$$

is solvable by the same reason. Take  $l_1, l_2, \dots, l_k$  such that all these congruences are satisfied. Multiplying each  $a_i$  by  $b = p_1^{l_1} p_2^{l_2} \cdots p_k^{l_k}$  yields a set  $\{ba_1, ba_2, \dots, ba_n\}$  consisting of perfect powers (more exactly,  $ba_i$  is a perfect  $q_i$ th power). ■

# 43

## The $2n - 1$ Problem

One of the most popular elementary applications of the pigeonhole principle is: Among any  $n$  integers one can choose several (possibly one) whose sum is divisible by  $n$ . It is rarely mentioned, however, that  $n - 1$  integers are not enough to ensure this; take, for example,  $x_1 = x_2 = \cdots = x_{n-1} = 1$ .

Many questions may be asked about related and more complicated “additive” problems. Some of them are still unsolved.

To illustrate how quickly the difficulty increases, let us consider the following question.

*For a given positive integer  $n$ , what is the minimal positive integer  $f(n)$  such that among any  $f(n)$  integers one can choose exactly  $n$  whose sum is divisible by  $n$ ?*

The omnipresent **Erdős** was probably the first to find an answer.

Initially it is hard to conjecture anything, but the following example provides a lower bound. Take

$$x_1 = x_2 = \cdots = x_{n-1} = 1, \quad x_n = x_{n+1} = \cdots = x_{2n-2} = 0.$$

These  $2n - 2$  integers do not have the desired property: the sum of any  $n$  of them is a number among  $1, 2, \dots, n - 1$ , so it is not divisible by  $n$ . Consequently,  $f(n) \geq 2n - 1$ . Note that adding just one more (arbitrary) integer  $x_{2n-1}$  to the above sequence would allow us to find  $n$  numbers adding up to a multiple of  $n$ .

Actually, there is ample evidence that  $f(n)$  is exactly  $2n - 1$ . Let, for instance,  $n = 2$ . Then  $2n - 1 = 3$ , and among any three integers there are two whose sum is divisible by 2: it suffices to note that some two of the three are of the same parity. Take  $n = 3$  and  $2n - 1 = 5$  arbitrary integers. If some three of them yield the same remainder modulo 3, their

sum is divisible by 3. If not, each of the three possible remainders occurs at most twice. Thus each of these remainders should be present at least once. But the sum of three numbers that leave different remainders modulo 3 is divisible by 3, so we have proved that  $f(3) = 5$ .

Let us also check that  $f(6) = 2 \cdot 6 - 1 = 11$ . To do this, consider 11 arbitrary integers  $x_1, x_2, \dots, x_{11}$ . Since  $f(2) = 3$  and  $11 > 3$ , one can choose two of them, say  $x_1$  and  $x_2$ , so that  $y_1 = x_1 + x_2$  is divisible by 2. The same can be repeated over and over again, provided that there are still at least three numbers left. We may assume, without loss of generality, that  $y_2 = x_3 + x_4$ ,  $y_3 = x_5 + x_6$ ,  $y_4 = x_7 + x_8$ ,  $y_5 = x_9 + x_{10}$  are all divisible by 2. Now, since  $f(3) = 5$ , we can pick three of the integers  $y_1/2, y_2/2, \dots, y_5/2$  whose sum is divisible by 3. If these are  $y_1/2, y_2/2, y_3/2$ , then it is easy to see that the sum of the six numbers  $x_1, x_2, \dots, x_6$  is divisible by 6. This proves that  $f(6) = 11$ .

Hence, knowing that  $f(2) = 3$  and  $f(3) = 5$ , we were able to prove that  $f(6) = f(2 \cdot 3 - 1) = 11$ . There is no accident in this. Essentially the same way, it is possible to prove that:

*If  $f(m) = 2m - 1$  and  $f(n) = 2n - 1$ , then  $f(mn) = 2mn - 1$ .*

We have to prove that among any  $2mn - 1$  integers one can choose  $mn$  whose sum is a multiple of  $mn$ . Since  $2mn - 1 > 2m - 1 = f(m)$ , it is possible to choose a group of  $m$  numbers whose arithmetic mean is an integer. Another group with this property can be chosen among the remaining  $2mn - m - 1$  numbers, then one more such group among the remaining  $2mn - 2m - 1$  numbers, and so on. This can be repeated  $2n - 1$  times, after which there will be  $2mn - (2n - 1)m - 1 = m - 1$  numbers left. Consider the integer arithmetic means of the  $2n - 1$  groups obtained this way. Since  $f(n) = 2n - 1$ , some  $n$  of these means add up to a multiple of  $n$ . It is clear that the sum of the  $mn$  numbers contained in the corresponding  $n$  groups is divisible by  $mn$ . ■

Thus the problem reduces to the case when  $n$  is a prime, which is the core of the solution.

More exactly, we need to prove that, for each prime  $p$ , among any  $2p - 1$  integers one can find  $p$  with sum divisible by  $p$ . An inductive proof is possible but one needs to state a proper stronger assertion first. Our choice is the following lemma, interesting in itself and perhaps useful in other similar settings.

*Let  $p$  be a prime and  $k$  a positive integer not exceeding  $p$ . Suppose that  $x_1, x_2, \dots, x_{2k-1}$  are integers, no  $k + 1$  of which*

*are congruent modulo  $p$ . Then one can form at least  $k$  sums, each consisting of  $k$  distinct terms of  $x_1, x_2, \dots, x_{2k-1}$ , so that these sums are all different modulo  $p$ .*

Note that, taking  $k = p$ , we obtain the desired result. Indeed, consider any integers  $x_1, x_2, \dots, x_{2p-1}$ . If some  $p+1$  of them are congruent modulo  $p$ , any  $p$  among them work. Otherwise, by the lemma, at least  $p$  of the sums  $x_{i_1} + x_{i_2} + \dots + x_{i_p}$  (with  $1 \leq i_1 < i_2 < \dots < i_p \leq 2p-1$ ) are different modulo  $p$ , so that one of them is 0 modulo  $p$ .

To prove the lemma, we use induction on  $k$ . The base  $k = 1$  is trivial. Suppose the claim holds for a certain  $k$ , take  $k+1 \leq p$ , and let  $x_1, x_2, \dots, x_{2k+1}$  be arbitrary integers, no  $k+2$  of which are congruent modulo  $p$ . Then the largest size of a group of  $x$ 's that are equal modulo  $p$  is  $k+1$ . There can be at most one such group of size  $k+1$ , because  $2(k+1) > 2k+1$ .

Delete one number in each of the two largest groups containing  $x$ 's congruent modulo  $p$ , noting that these two numbers are different modulo  $p$ . Without loss of generality, suppose that  $x_{2k}$  and  $x_{2k+1}$  are deleted. We are then left with  $2k-1$  integers  $x_1, x_2, \dots, x_{2k-1}$ , and by the remark about the largest number of congruent  $x$ 's, no  $k+1$  of them are congruent modulo  $p$ . We also have  $k < p$ , hence the induction hypothesis yields  $k$  sums  $x_{i_1} + x_{i_2} + \dots + x_{i_k}$  (where  $1 \leq i_1 < i_2 < \dots < i_k \leq 2k-1$ ), all different modulo  $p$ . Denote them by  $S_1, S_2, \dots, S_k$  and consider the sums

$$S_1 + x_{2k}, S_2 + x_{2k}, \dots, S_k + x_{2k}; \quad (1)$$

$$S_1 + x_{2k+1}, S_2 + x_{2k+1}, \dots, S_k + x_{2k+1}. \quad (2)$$

The  $k$  sums in (1) are different modulo  $p$ , as well as the ones in (2). Next, each sum in (1) and (2) is formed by  $k+1$  distinct terms of the sequence  $x_1, x_2, \dots, x_{2k+1}$ . So it suffices to prove that (1) and (2) are not the same sets, taken modulo  $p$ . Assume on the contrary that this is not the case. Then (1) and (2) are permutations of one another, so

$$\sum_{i=1}^k S_i + kx_{2k} \equiv \sum_{i=1}^k S_i + kx_{2k+1} \pmod{p}.$$

This implies  $kx_{2k} \equiv kx_{2k+1} \pmod{p}$ . Since  $1 \leq k < p$  and  $p$  is a prime, we obtain  $x_{2k} \equiv x_{2k+1} \pmod{p}$ , a contradiction. ■

Thus we proved that  $f(n) = 2n - 1$  for each positive integer  $n$ .

In addition to this ingenious inductive proof we present an essentially different approach. The disadvantage of having  $p$  a prime when coming up with a statement provable by induction is an advantage now.

Let  $p$  be a fixed prime again and  $x_1, x_2, \dots, x_{2p-1}$  arbitrary integers. We want to prove that at least one of the sums  $x_{i_1} + x_{i_2} + \dots + x_{i_p}$ , where  $1 \leq i_1 < i_2 < \dots < i_p \leq 2p-1$ , is divisible by  $p$ . Assume, by way of contradiction, that this is not true. Then, by Fermat's little theorem, each of these sums satisfies

$$(x_{i_1} + x_{i_2} + \dots + x_{i_p})^{p-1} \equiv 1 \pmod{p}.$$

Consider the sum

$$S = \sum (x_{i_1} + x_{i_2} + \dots + x_{i_p})^{p-1},$$

where the summation ranges over all  $p$ -element subsets  $\{i_1, i_2, \dots, i_p\}$  of  $\{1, 2, \dots, 2p-1\}$ . Each summand is 1 modulo  $p$ . So, taken modulo  $p$ , the sum  $S$  is equal to the number of summands it contains, namely, to the binomial coefficient  $\binom{2p-1}{p}$ . It is not hard to prove that the latter is 1 modulo  $p$ , but all we need is to observe that  $\binom{2p-1}{p}$  is not divisible by  $p$ . This is easy: we have

$$\binom{2p-1}{p} = \frac{(2p-1)(2p-2) \cdots (p+1)}{(p-1)!},$$

and the prime  $p$  divides none of the factors in the numerator.

So, our assumption that the claim is not true implies that the sum  $S$  is not divisible by  $p$ . On the other hand, we prove now that it has to be divisible by  $p$  for any choice of the integers  $x_1, x_2, \dots, x_{2p-1}$ , thus reaching a contradiction.

Indeed, expand  $S$  and look at a typical monomial before collecting like terms. It is of the form  $x_{j_1}^{k_1} x_{j_2}^{k_2} \cdots x_{j_m}^{k_m}$ , with  $j_1, j_2, \dots, j_m, k_1, k_2, \dots, k_m$  positive integers such that

$$1 \leq m \leq p-1, \quad \{j_1, j_2, \dots, j_m\} \subset \{1, 2, \dots, 2p-1\}, \\ k_1 + k_2 + \dots + k_m = p-1.$$

All occurrences of this monomial necessarily come from the expansion of expressions  $(x_{i_1} + x_{i_2} + \dots + x_{i_p})^{p-1}$ , where  $\{i_1, i_2, \dots, i_p\}$  is a  $p$ -element subset of  $\{1, 2, \dots, 2p-1\}$  containing  $j_1, j_2, \dots, j_m$ . There are exactly  $\binom{2p-1-m}{p-m}$  of these and, by symmetry, each of them yields the same number of monomials  $x_{j_1}^{k_1} x_{j_2}^{k_2} \cdots x_{j_m}^{k_m}$ , say  $C$ . Thus  $x_{j_1}^{k_1} x_{j_2}^{k_2} \cdots x_{j_m}^{k_m}$  enters

the normal form of  $S$  with coefficient  $C\binom{2p-1-m}{p-m}$ . Now, the binomial coefficient

$$\binom{2p-1-m}{p-m} = \frac{(2p-m-1)(2p-m-2)\cdots(p-m+1)}{(p-1)!}$$

is divisible by  $p$ . This is because the condition  $1 \leq m \leq p-1$  implies  $p-m+1 \leq p \leq 2p-m-1$ , meaning that  $p$  is a factor in the numerator. This factor cannot cancel out, since the denominator  $(p-1)!$  is not divisible by the prime  $p$ .

Summing up, we conclude that each monomial enters the normal form of the sum  $S$  with coefficient divisible by  $p$ . Hence  $S$  is divisible by  $p$ , a genuine contradiction. ■

There are many related problems, of which we mention just two. Let us call an ordered integer pair  $(x, y)$  divisible by  $n$  if both  $x$  and  $y$  are divisible by  $n$ . The sum of the pairs  $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$  is defined as the pair  $(x_1 + x_2 + \cdots + x_n, y_1 + y_2 + \cdots + y_n)$ .

*For a fixed positive integer  $n$ , find the least number  $g(n)$  such that among any  $g(n)$  ordered pairs of integers one can choose several whose sum is divisible by  $n$ .*

Replacing *several* by *exactly*  $n$ , one obtains yet another problem. Apart from several trivial exceptions, we do not know the answer to either question.

# 44

## The $2n + 1$ Problem

The 1973 Putnam problem B-1 reads:

*Let  $x_1, x_2, \dots, x_{2n+1}$  be a set of integers such that, if any one of them is removed, the remaining ones can be divided into two sets of  $n$  numbers with equal sums. Prove that*

$$x_1 = x_2 = \dots = x_{2n+1}.$$

The problem has been known for a long time; at least, it was posed in the special case  $n = 6$  (that is, for 13 numbers) for seventh- and eighth-graders at the 1949 Moscow Olympiad.

For brevity, call  $P$  the property described in the problem statement. It is “invariant under translation and dilatation”; that is, if  $x_1, \dots, x_{2n+1}$  have the property  $P$ , then so do  $a + x_1, a + x_2, \dots, a + x_{2n+1}$  and  $bx_1, bx_2, \dots, bx_{2n+1}$ , where  $a$  and  $b$  are arbitrary real numbers. Thus we may assume  $x_1, x_2, \dots, x_{2n+1}$  to be positive integers and go by induction, say on the greatest of them. If this is 1, then all of them equal 1 and we are done. Suppose now the statement holds for any collection of  $2n + 1$  positive integers not exceeding  $k$  ( $k \geq 2$ ), and take  $2n + 1$  positive integers  $x_1, x_2, \dots, x_{2n+1}$  with the property  $P$  and not exceeding  $k + 1$ .

Let  $x_1 + x_2 + \dots + x_{2n+1} = S$ . Then  $P$  implies that all the numbers  $S - x_i$  are even, and so  $x_1, x_2, \dots, x_{2n+1}$  are all of the same parity (the parity of  $S$ ). If they are even, then  $x_1/2, x_2/2, \dots, x_{2n+1}/2$  are positive integers having the property  $P$ . They do not exceed  $(k + 1)/2$ , which is less than or equal to  $k$  for  $k \geq 1$ . Thus  $x_1/2, x_2/2, \dots, x_{2n+1}/2$  are equal by the induction hypothesis, and hence so are  $x_1, x_2, \dots, x_{2n+1}$ . And if  $x_1, x_2, \dots, x_{2n+1}$  are odd, the conclusion follows in a similar fashion, considering the numbers  $(x_1 + 1)/2, (x_2 + 1)/2, \dots, (x_{2n+1} + 1)/2$ . ■

This proof depends heavily on the assumption that the given numbers are integers. Our aim here is to prove the following more general result:

*Any  $2n + 1$  real numbers having the property  $P$  are equal.*

The solution uses basic linear algebra. The property  $P$  means that for each  $i = 1, 2, \dots, 2n + 1$  one can put 0 as a coefficient in front of  $x_i$ , then assign a coefficient equal to 1 or  $-1$  to each of the remaining numbers  $x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_{2n+1}$ , so that  $n$  of these coefficients are 1's,  $n$  are  $-1$ 's and the algebraic sum obtained is 0. Formally speaking, for every  $i = 1, 2, \dots, 2n + 1$  there is a sequence  $a_{i1}, a_{i2}, \dots, a_{i,2n+1}$  such that  $a_{ii} = 0$ , exactly  $n$  of the remaining  $a_{ij}$ 's are 1's, exactly  $n$  are  $-1$ 's and

$$a_{i1}x_1 + a_{i2}x_2 + \dots + a_{i,2n+1}x_{2n+1} = 0.$$

In other words,  $(x_1, x_2, \dots, x_{2n+1})$  is a solution of a linear homogeneous system

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1,2n+1}x_{2n+1} = 0 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2,2n+1}x_{2n+1} = 0 \\ \vdots \\ a_{2n+1,1}x_1 + a_{2n+1,2}x_2 + \dots + a_{2n+1,2n+1}x_{2n+1} = 0 \end{cases}$$

whose matrix  $A = (a_{ij})_{i,j=1}^{2n+1}$  has zeros on its main diagonal, ones or negative ones elsewhere, and each of its rows contains  $n$  ones and  $n$  negative ones.

The idea of this approach is to prove that  $A$  has rank  $2n$ . For then the space of all solutions will have dimension  $(2n + 1) - 2n = 1$  and it is enough to know *one* nonzero solution  $(u_1, u_2, \dots, u_{2n+1})$  in order to know them all: they must have the form  $(\lambda u_1, \lambda u_2, \dots, \lambda u_{2n+1})$  for some real  $\lambda$ . And since there is one obvious solution  $(1, 1, \dots, 1)$ , any other will be of the form  $(\lambda, \lambda, \dots, \lambda)$ !

There is no question that  $A$  has determinant zero: adding all columns to the first one yields a column of straight zeros, because each row contains one zero and as many 1's as  $-1$ 's. And how to get a nonzero minor of order  $2n$ ? The one in the upper left-hand corner can help. Moreover:

*Any  $2n \times 2n$  determinant  $\Delta = |a_{ij}|_{i,j=1}^{2n}$  with even main diagonal entries and odd off-diagonal entries is different from 0.*

Indeed,  $\Delta$  is the algebraic sum of all products of the form

$$a_{1i_1} a_{2i_2} \dots a_{2n, i_{2n}},$$

where  $(i_1, i_2, \dots, i_{2n})$  is an arbitrary permutation of  $\{1, 2, \dots, 2n\}$ . (We need not discuss how the signs  $+$  or  $-$  are assigned.) Since  $a_{ij}$  is even if and only if  $i = j$ , a term of the above sum is odd exactly when the corresponding permutation  $(i_1, i_2, \dots, i_{2n})$  satisfies the conditions  $i_k \neq k$  for each  $k = 1, 2, \dots, 2n$ . Such permutations are called *derangements*; so it is enough to prove that the number of derangements of  $\{1, 2, \dots, 2n\}$  is odd.

There are several well-known formulas for the number  $D(l)$  of derangements of the set  $\{1, 2, \dots, l\}$ , any of which could complete the proof:

$$D(l) = l! \left[ 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^l \frac{1}{l!} \right],$$

$$D(l) = lD(l-1) + (-1)^l,$$

$$D(l) = (l-1)[D(l-1) + D(l-2)].$$

But this solution can be done without referring to them. If a permutation  $\sigma$  of  $\{1, 2, \dots, 2n\}$  is a derangement, then so is its inverse  $\sigma^{-1}$ . Forget about  $\sigma$  and  $\sigma^{-1}$  if they are different; deleting them from the set of all derangements will not affect the *parity* of  $D(2n)$ . Take a derangement  $\sigma$  with  $\sigma = \sigma^{-1}$ . Note that  $\sigma$  is uniquely determined by some partition of  $\{1, 2, \dots, 2n\}$  into  $n$  pairs  $\{i_1, j_1\}, \{i_2, j_2\}, \dots, \{i_n, j_n\}$ : it simply takes  $i_k$  to  $j_k$  and vice versa for all  $k = 1, 2, \dots, n$ . The pair  $\{i_1, j_1\}$  can be chosen in  $\binom{2n}{2}$  ways, the next pair  $\{i_2, j_2\}$  in  $\binom{2n-2}{2}$  ways, and so on; there are  $\binom{2}{2} = 1$  possibilities for last pair  $\{i_n, j_n\}$ . Thus there are

$$\binom{2n}{2} \binom{2n-2}{2} \cdots \binom{2}{2}$$

ways to partition  $\{1, 2, \dots, 2n\}$  in the fashion described above; but each of the actual partitions is counted  $n!$  times, because there are  $n!$  ways to permute the same pairs  $\{i_1, j_1\}, \{i_2, j_2\}, \dots, \{i_n, j_n\}$ . Hence the number of derangements in question is

$$\frac{\binom{2n}{2} \binom{2n-2}{2} \cdots \binom{2}{2}}{n!} = \frac{2n(2n-1) \cdots 2 \cdot 1}{2 \cdot 4 \cdots (2n-2) \cdot 2n} \\ = 1 \cdot 3 \cdots (2n-3)(2n-1),$$

and this is an odd integer. Our first solution is complete. ■

Here is a shorter alternate proof. It suffices to show that  $\Delta$  is odd, which will be unaffected if we reduce all entries modulo 2 and perform the computations mod 2. So we need to show that the  $2n \times 2n$  matrix  $M$

with 0's on the diagonal and 1's elsewhere is nonsingular mod 2. Indeed, multiply  $M$  by itself. By the definition of the product of matrices, one infers that each diagonal entry is odd, and each off-diagonal entry even. Hence  $M^2$  is the identity matrix mod 2, so  $M$  is nonsingular. ■

These proofs show that the claim holds not only for reals but for complex numbers as well, and even for more general fields.

These are not the only possible approaches. There is, for example, a proof based on Diophantine approximations. Since it is technical and appeals to a certain nontrivial theorem, we are not going to discuss it here.

# 45

## The $3n$ Problem

Here is another problem that circulated a lot around the world in the 1980s; we call it the  $3n$  problem.

*If the set  $\{1, 2, \dots, 3n\}$  is partitioned into three equal subsets, is it always possible to choose one number out of each set so that one of these numbers is the sum of the other two?*

In an article entitled “Adding Numbers” in *James Cook Mathematical Notes*, Vol. 4, No. 35, October 1984, 4073–4075, Esther and George Szekeres give a positive answer to this question. They say that the question itself is due to C. J. Smyth.

By that time the  $3n$  problem was already popular in Bulgaria, but no one seemed to have a solution, and the Szekeres’ article was unknown. Alexei Sosinski, an editor of *Kvant*, visited *Matematika* in 1985, and thus the  $3n$  question reached the then–Soviet Union. It appeared in *Kvant*’s problem section and, still more interesting, was one of the problems on the final round of the 1986 All-Russian Olympiad.

Answering the question, Esther and George Szekeres prove a more powerful result: If  $\{1, 2, \dots, n\}$  is partitioned into three sets such that it is impossible to choose one number out of each set with one of these three numbers equal to the sum of the other two, then one of the three sets must have no more than  $n/4$  elements.

We present a proof of the  $3n$  problem due to **V. Alexeev**. This is the solution published in *Kvant*, No. 8, 1987.

Suppose that  $\{1, 2, \dots, 3n\}$  is partitioned into three sets  $A$ ,  $B$ ,  $C$ , each set containing  $n$  numbers. For brevity, we shall call a triple  $(a, b, c)$  *good* if  $a \in A$ ,  $b \in B$ ,  $c \in C$  and one of the numbers  $a$ ,  $b$ ,  $c$  is the sum of the remaining two.

Without loss of generality, one may assume that  $1 \in A$  and, if  $k \geq 2$  is the least number outside  $A$ , that  $k \in B$ . Assuming, on the contrary, that there are no good triples, we shall infer that:

$$\text{If } x \in C, \text{ then } x - 1 \in A. \quad (1)$$

Then a contradiction follows from (1). Indeed, if  $C = \{c_1, c_2, \dots, c_n\}$ , then  $A$  contains the numbers  $c_1 - 1, c_2 - 1, \dots, c_n - 1$ , all of which are greater than 1, because 2 belongs to either  $A$  or  $B$ . But  $A$  contains 1 as well, hence it would have at least  $n + 1$  elements.

So all we have to do is to prove (1). Suppose it is not true. Then there is a number  $x \in C$  such that  $x - 1 \notin A$ . Clearly  $x - 1 \notin B$ , since otherwise the triple  $(1, x - 1, x)$  would be good. Hence  $x - 1 \in C$ . Now, based on the fact that  $x \in C$ ,  $x - 1 \in C$ , we are going to prove that  $x - k \in C$ ,  $x - k - 1 \in C$ . (Recall that  $k$  is the least element of  $B$  and the least number outside  $A$ .) Indeed, if  $x - k \in A$ , then  $(x - k, k, x)$  is a good triple; if  $x - k \in B$ , then the triple  $(k - 1, x - k, x - 1)$  is a good one. Similarly, the relations  $x - k - 1 \in A$  and  $x - k - 1 \in B$  yield the good triples  $(x - k - 1, k, x - 1)$  and  $(1, x - k - 1, x - k)$ , respectively. We can repeat this argument as many times as possible, concluding that all numbers  $x - ik$  and  $x - ik - 1$  belong to  $C$  for  $i = 0, 1, 2, \dots$ , provided that they are positive. But  $x - ik$  must be one of the numbers  $1, 2, \dots, k$  for some  $i$ . So it will be an element of either  $A$  or  $B$ , a contradiction showing that  $x \in C$  implies  $x - 1 \in A$  and proving (1). As already pointed out, this completes the proof. ■

Now, you may suspect that the next thing we are going to do after telling about the  $2n - 1$ , the  $2n + 1$  and the  $3n$  problems will be the  $3n + 1$  problem. If so, you are wrong, but the next sections are not less exciting.

## Coffee Break 9

1. The Big Apple got wormy. A worm dug a tunnel of length 101 miles inside it, and went out (starting and ending at the surface). Assuming that The Big Apple is an ideal sphere of radius 51 miles, prove that it can be cut into two congruent pieces one of which is not wormy.
2. Each vertex of a convex polyhedron is the endpoint of an even number of edges. Prove that any plane section of the polyhedron not containing a vertex is a polygon with an even number of sides.
3. Asterisks are placed in some cells of an  $m$  by  $n$  rectangular table, where  $m < n$ , so that there is at least one asterisk in each column. Prove that there exists an asterisk such that there are more asterisks in its row than in its column.

## Solutions

1. Let  $A$  and  $B$  be the beginning and the end of the tunnel. Consider the set of points  $X$  such that  $AX + BX \leq 101$ . It is an ellipsoid of rotation  $E$  with foci  $A$  and  $B$ . Each point  $X$  of the tunnel is contained in  $E$ . Indeed, since  $AX$  and  $XB$  do not exceed the lengths of the worm's routes from  $A$  to  $X$  and from  $X$  to  $B$ , respectively, we have  $AX + XB \leq 101$ . On the other hand, the center  $O$  of the Big Apple is *outside*  $E$ , because  $AO + BO$  equals  $51 + 51 = 102 > 101$ . Everything is clear now. Because  $E$  is convex and  $O$  is outside it, there is a plane through  $O$  that does not intersect  $E$ . It cuts the Big Apple into two equal halves, one of which is not wormy (it does not have any common points with the ellipsoid  $E$  containing the worm's tunnel).

2. The idea is to prove that the surface of the polyhedron can be colored in two colors, red and blue, so that any two adjacent faces (sharing a common edge) are of different colors. Indeed, assume that no edge or diagonal of the polyhedron is parallel to the horizontal plane. Take the horizontal plane through the uppermost vertex  $A_1$  and start bringing it down. Until we meet the second vertex  $A_2$ , it will cut an even number of edges. They are sides of an even number of faces, the ones having  $A_1$  as a vertex, which can be colored alternately red and blue. The same can be repeated with  $A_2$ ; the difference is that this time the pattern of coloring is predetermined, since at least one face with vertex  $A_2$  has already been colored (because  $A_1$  is a vertex of this face, too). Proceeding this way, we are able to color each face of the polyhedron in one of the two colors so that adjacent faces have different colors.

Once this is done, the statement is obvious: in any plane section that does not contain a vertex each red side is adjacent to two blue ones, and vice versa.

3. This can be done in a number of ways, say by induction, but the following solution by **Nikolai Vasilyev** is superior to them all, and not just because of its brevity.

Replace each asterisk in the given table  $A$  by  $1/k$ , where  $k$  is the number of asterisks in its row. We get a new table  $B$  in which the sum of the numbers in each row is 1 or 0 according as the corresponding row

of  $A$  contains at least one asterisk or not. Therefore the sum  $S_B$  of all numbers in  $B$  is an integer not exceeding  $m$ , the number of rows in  $A$ .

In a similar manner, construct a table  $C$  by replacing each asterisk in  $A$  by  $1/l$ , where  $l$  is the number of asterisks in the corresponding column of  $A$ . The sum of the elements in each column of  $C$  is *always exactly* 1, because each column in  $A$  contains at least one asterisk. Hence the sum  $S_C$  of all numbers in  $C$  is  $n$ , the number of its columns.

The purpose of this mysterious construction becomes evident right away. Since  $m < n$ , we get  $S_B \leq m < n = S_C$ . Then at least one entry in  $B$  is less than the corresponding entry in  $C$ . This means that  $A$  contains an asterisk with the following property: If  $k$  and  $l$  are the numbers of asterisks in its row and column, respectively, then  $1/k < 1/l$ . Thus  $k > l$ , so this asterisk is exactly one we need.

# 46

## Pairwise Sums

An old problem of the 1962 Moscow Olympiad reads as follows:

*Ten pairwise sums can be formed from the five real numbers  $x_1, x_2, x_3, x_4, x_5$ ; label them  $a_1, a_2, \dots, a_{10}$ . Prove that, knowing the numbers  $a_1, a_2, \dots, a_{10}$  (but not knowing, of course, the sum of exactly which two  $x$ 's a given  $a_i$  is), one can restore the initial numbers  $x_1, x_2, x_3, x_4, x_5$ .*

Let us assume, without loss of generality, that the five unknown numbers are arranged in ascending order, as well as their pairwise sums:

$$x_1 \leq x_2 \leq \dots \leq x_5; \quad a_1 \leq a_2 \leq \dots \leq a_{10}.$$

Then  $a_1$ , the least of the pairwise sums, is equal to  $x_1 + x_2$ . Similarly, being the greatest of all  $a_i$ 's,  $a_{10}$  must equal  $x_4 + x_5$ . So, if the sum  $S$  of all  $x$ 's were known, we could instantly find  $x_3$ , because

$$x_3 = S - (x_1 + x_2) - (x_4 + x_5).$$

But it is not hard to determine  $S$ . Indeed, each  $x_i$  is an addend in exactly four of the pairwise sums  $a_1, a_2, \dots, a_{10}$ , and hence

$$a_1 + a_2 + \dots + a_{10} = 4S.$$

Knowing  $S$  and  $x_3$ , we can easily recover all of the remaining  $x$ 's, since it is almost obvious that  $x_1 + x_3 = a_2$  and  $x_3 + x_5 = a_9$ . Then

$$x_1 = a_2 - x_3, \quad x_5 = a_9 - x_3.$$

Now  $x_2$  and  $x_4$  can be found from the equalities  $x_1 + x_2 = a_1$  and  $x_4 + x_5 = a_{10}$ . ■

This is the end of the proof and the beginning of an interesting story. What if we consider  $n$  instead of five numbers?

*Suppose that somebody chose  $n$  real numbers, then found their pairwise sums, and wrote them on  $[n(n-1)]/2$  sheets of paper. He then shuffled the sheets and gave them to a friend. Could his friend always restore the initial numbers?*

The answer is not evident at all. You cannot do this if  $n = 2$ , because there is just one pairwise sum  $a_1$  in this case, and there exist infinitely many pairs of real numbers that add up to  $a_1$ . On the other hand, the numbers can be restored if  $n = 3$ ; this is the only case when  $[n(n-1)]/2 = n$ . One could use the idea from the above proof. Denote by  $x_1, x_2, x_3$  the unknown numbers and let  $a_1, a_2, a_3$  be their pairwise sums. Then, if  $x_1 \leq x_2 \leq x_3$  and  $a_1 \leq a_2 \leq a_3$ , it is clear that  $x_1 + x_2 = a_1$ ,  $x_2 + x_3 = a_3$ . Thus the only possibility for  $x_1 + x_3$  is  $x_1 + x_3 = a_2$ , so it turns out that

$$x_1 + x_2 = a_1, \quad x_2 + x_3 = a_3, \quad x_1 + x_3 = a_2.$$

This simple linear system has a unique solution

$$\begin{aligned} x_1 &= \frac{1}{2}(a_1 + a_2 - a_3), \\ x_2 &= \frac{1}{2}(a_3 + a_1 - a_2), \\ x_3 &= \frac{1}{2}(a_2 + a_3 - a_1). \end{aligned}$$

We could try to repeat this argument for four numbers but it fails. Indeed, it is again true what we said about the smallest and the largest of the pairwise sums, but one cannot proceed further. Moreover, the quadruples 1, 4, 6, 7 and 2, 3, 5, 8 yield the same sequence of pairwise sums

$$5, 7, 8, 10, 11, 13.$$

This means that, regardless of our efforts, the unknown numbers cannot be restored if one is given just their pairwise sums.

On the other hand,  $n = 5$  is a “favorable” case, as we saw above. What is your guess? Which positive integers greater than 5, if any, are “favorable”?

An answer was given by **John Selfridge** and **Ernst Straus**. No doubt, you will enjoy their magnificent proof (published in *Pacific Journal of Mathematics* Vol. 8, 1958, pages 847–856). The same idea will occur in Section 49. The statement Selfridge and Straus consider reads as follows:

For a given integer  $n$  greater than 1, let there exist two distinct integer  $n$ -tuples  $a_1, a_2, \dots, a_n$  and  $b_1, b_2, \dots, b_n$  such that their sequences of pairwise sums

$$a_1 + a_2, a_1 + a_3, \dots, a_{n-1} + a_n$$

and

$$b_1 + b_2, b_1 + b_3, \dots, b_{n-1} + b_n$$

coincide up to permutation. Then  $n$  is a power of 2.

We may assume that  $a_i$  and  $b_i$  are nonnegative, since nothing changes if any constant is added to all of the numbers  $a_i$  and  $b_i$ . Consider the generating polynomials of the given sequences:

$$f(x) = x^{a_1} + x^{a_2} + \dots + x^{a_n}, \quad g(x) = x^{b_1} + x^{b_2} + \dots + x^{b_n}.$$

The key idea is to notice that the hypothesis implies

$$\begin{aligned} f^2(x) - g^2(x) &= \left( \sum_{i=1}^n x^{2a_i} + 2 \sum_{1 \leq i < j \leq n} x^{a_i + a_j} \right) \\ &\quad - \left( \sum_{i=1}^n x^{2b_i} + 2 \sum_{1 \leq i < j \leq n} x^{b_i + b_j} \right) \\ &= f(x^2) - g(x^2). \end{aligned}$$

Since  $f(1) = g(1) = n$ ,  $x = 1$  is a root of the difference  $f - g$ . Note that  $f - g$  is a nonzero polynomial, because  $a_1, a_2, \dots, a_n$  and  $b_1, b_2, \dots, b_n$  are distinct sequences. Let  $f(x) - g(x) = (x - 1)^k h(x)$ , where  $k \geq 1$  and  $h(x)$  is a polynomial with  $h(1) \neq 0$ . Then we obtain

$$\begin{aligned} f(x) + g(x) &= \frac{f^2(x) - g^2(x)}{f(x) - g(x)} = \frac{f(x^2) - g(x^2)}{f(x) - g(x)} \\ &= \frac{(x^2 - 1)^k h(x^2)}{(x - 1)^k h(x)} = (x + 1)^k \frac{h(x^2)}{h(x)}. \end{aligned}$$

Now simply set  $x = 1$ :

$$2n = f(1) + g(1) = (1 + 1)^k \frac{h(1^2)}{h(1)} = 2^k,$$

and hence  $n = 2^{k-1}$ . (!!) ■

The result of this masterpiece can be extended to *arbitrary real*  $n$ -tuples, and the same idea works. We just have to replace polynomials by analytic functions, their closest relatives. Thus, if  $n$  is *not* a power of 2, then the sequence of pairwise sums determines the  $n$ -tuple entirely. The only remaining problem is that we do not know an *efficient* way for this, as in the case  $n = 5$ .

There are other proofs of the same assertion, more standard and less spectacular. They will not be discussed here, though some of them apply to arbitrary real numbers, not just integers.

To establish that the powers of 2 are indeed “unfavorable,” we have to provide an example for each  $k = 1, 2, \dots$ , where there exist *different*  $2^k$ -tuples  $a_1, \dots, a_{2^k}$  and  $b_1, \dots, b_{2^k}$  yielding the same collection of pairwise sums. There are lots of examples; we shall point out the one considered in Section 49 involving Morse’s sequence. Denote by  $X_k$  ( $Y_k$ ) the subset of all numbers in  $\{0, 1, 2, \dots, 2^{k+1} - 1\}$  containing an even (odd) number of 1’s in their binary representations. Each of the sets  $X_k$  and  $Y_k$  has  $2^k$  elements. As mentioned in Section 49, an easy induction shows that they have the desired property.

# 47

## Integer Progressions

“Perhaps my favorite problem of all concerns covering congruences,” Paul Erdős wrote in his article, “Some of My Favorite Problems and Results.” He continued: “It was really surprising that it had not been asked before. A system of congruences

$$a_i \pmod{n_i}, \quad n_1 < n_2 < \cdots < n_k, \quad (1)$$

is called a *covering system* if every integer satisfies at least one of the congruences in (1). The simplest covering system is  $0 \pmod{2}$ ,  $0 \pmod{3}$ ,  $1 \pmod{4}$ ,  $5 \pmod{6}$  and  $7 \pmod{12}$ . The main problem is: Is it true that for every  $c$  one can find a covering system all of whose moduli are larger than  $c$ ? I offer \$1000 for a proof or disproof.”

The congruence classes in (1) are arithmetic progressions, infinite in both directions, with common differences  $n_1, n_2, \dots, n_k$ . So, one may regard a covering system as a collection of such progressions whose union contains every integer. Note that, by definition, the common differences of these progressions are all *distinct*.

Many further questions can be asked about covering systems, most of them unsolved and some, in Erdős’ own term, “unattackable.” We mention only the following Erdős–Selfridge conjecture:

*There exists a covering system all of whose moduli are odd.*

Studying covering systems proves to be quite hard, with probably just one remarkable exception: the exact covering congruences.

Erdős called a covering system *exact* if each integer belongs to exactly one class. Surprisingly, **Mirsky** and **Newman** proved that

*There are no exact covering systems.*

In other words, the integers cannot be covered by a finite number of *nonoverlapping* arithmetic progressions, infinite in both directions, whose common differences are all *distinct*.

Actually, Mirsky and Newman established a somewhat stronger result, and their proof comes from *The Book*.

*If the positive integers are partitioned into several (at least two) arithmetic progressions, then the common differences of these progressions cannot be all distinct.*

The arithmetic progression  $\{a + nb \mid n = 0, 1, 2, \dots\}$  has as generating function the geometric progression

$$z^a + z^{a+b} + z^{a+2b} + \dots = \frac{z^a}{1 - z^b}.$$

Assume on the contrary that the positive integers can be partitioned into  $k$  arithmetic progressions  $\{a_j + nb_j\}$  ( $j = 1, 2, \dots, k$ ) whose common differences  $b_1, b_2, \dots, b_k$  are all distinct. Because the generating function of all positive integers is the progression  $z + z^2 + z^3 + \dots = z/(1 - z)$ , we obtain

$$\frac{z}{1 - z} = \frac{z^{a_1}}{1 - z^{b_1}} + \frac{z^{a_2}}{1 - z^{b_2}} + \dots + \frac{z^{a_k}}{1 - z^{b_k}}. \quad (2)$$

As usual, we assume that  $|z| < 1$  (in order to have convergent series) but it will be beneficial to consider *complex* values of the variable  $z$  as well. Let  $b_k$  be the greatest among the common differences  $b_1, b_2, \dots, b_k$ , and let  $z$  tend to  $\varepsilon = e^{2\pi i/b_k}$  in such a way that  $|z|$  remains less than 1. Since  $\varepsilon^{b_k} = 1$  but  $\varepsilon \neq 1$  and  $\varepsilon^{b_j} \neq 1$  for  $1 \leq j < k$ , the left-hand side of (2) tends to a finite limit, and so do all terms in its right-hand side except the last one. The last term, however, tends to  $\infty$ , which is a contradiction. ■

The problem about partitioning the integers into a finite number of progressions is in a certain sense equivalent to the following one:

*Each vertex of a regular polygon is colored in one of a finite number of colors so that the points of the same color are the vertices of some new regular polygon. Prove that at least two of the polygons obtained are congruent.*

This statement was posed at the final round of the 1970 All-Union Olympiad. None of the contestants solved it during the competition. They say it was selected by the jury because of the solution invented by **Alexander Lifschitz**, a jury member and a former olympiad winner. Naturally, this solution emphasizes the common roots of the two problems.

Let the given regular polygon have  $n$  vertices. These can be regarded as the  $n$ th roots of unity

$$1, \varepsilon_n, \varepsilon_n^2, \dots, \varepsilon_n^{n-1}, \quad \text{where } \varepsilon_n = e^{2\pi i/n}, \quad (3)$$

which form a geometric progression.

More generally, the vertices of any regular  $m$ -gon inscribed in the unit circle (of the complex plane) form a geometric progression

$$c, c\varepsilon_m, c\varepsilon_m^2, \dots, c\varepsilon_m^{m-1}, \quad (4)$$

where  $|c| = 1$  and  $\varepsilon_m = e^{2\pi i/m}$ .

What happens if we raise the terms of the progression (4) to some positive integer power  $k$ ? We get a new geometric progression (some of its terms may be equal) whose sum  $S_m^{(k)}$  satisfies

$$S_m^{(k)} = \begin{cases} 0, & \text{if } k \text{ is not divisible by } m, \\ mc^k \neq 0, & \text{if } k \text{ is divisible by } m. \end{cases} \quad (5)$$

Here is what the latter means geometrically. If  $m$  does not divide  $k$ , then raising to the  $k$ th power makes the terms of the progression (4) run through the vertices of a regular  $d$ -gon, where  $d > 1$  is a divisor of  $m$ . The centroid of this  $d$ -gon coincides with the origin, and each of its vertices enters the progression the same number of times. Hence the sum  $S_m^{(k)}$  is 0. But if  $m$  divides  $k$ , then one gets “a regular polygon with one vertex,” that is,  $m$  copies of the same number  $c^k$ .

Assume now, by way of contradiction, that the geometric progression (3) can be partitioned into several progressions of the form (4) so that the corresponding  $m$ 's are all distinct. Let them be

$$m_1 < m_2 < \dots < m_l.$$

Raise the terms of the progression (3) to the power of  $m_1$ . Then, of course,

$$S_n^{(m_1)} = S_{m_1}^{(m_1)} + S_{m_2}^{(m_1)} + \dots + S_{m_l}^{(m_1)}. \quad (6)$$

But one can infer from (5) that all of the above sums equal 0, except  $S_{m_1}^{(m_1)}$ . This is because the only number among  $m_1, m_2, \dots, m_l, n$  that divides  $m_1$  is  $m_1$  itself. Hence the left-hand side of (6) is 0 and its right-hand side is not, a genuine contradiction. ■

# 48

## Incomparable Sets

Two sets are called *incomparable* if neither of them contains the other. Naturally, one can say little about collections of pairwise incomparable sets if they are not contained in a larger one. In the latter case a beautiful theorem of **Sperner** reads:

*The maximum number of pairwise incomparable subsets of an  $n$ -element set is equal to  $\binom{n}{\lfloor n/2 \rfloor}$ , the largest coefficient in the expansion of the binomial  $(1+x)^n$ .*

Many mathematicians, including Erdős, immediately appreciated the beauty and the importance of this result, probably the first nontrivial one in extremal theory of set systems. Several ingenious proofs are known besides Sperner's original proof. We present one by **D. Lubell**, *Journal of Combinatorial Theory*, Vol. 1, 1966, page 299.

Let  $X$  be an  $n$ -element set. A sequence  $C_1, C_2, \dots, C_n$  of subsets of  $X$  is called a chain if

$$C_1 \subset C_2 \subset \dots \subset C_n \quad \text{and} \quad |C_i| = i \quad \text{for each } i = 1, 2, \dots, n.$$

In particular,  $C_n = X$ . The first term of any chain, the one-element set  $C_1$ , can be chosen in  $n$  ways. After this is done, there are  $n-1$  possibilities for  $C_2$  (one adds to  $C_1$  any of the remaining  $n-1$  elements of  $X$ ). In a similar way, given  $C_1$  and  $C_2$ , the set  $C_3$  can be chosen in  $n-2$  ways, and so on. Thus the number of chains in  $X$  is  $n(n-1)(n-2)\cdots 2 \cdot 1 = n!$ .

Let  $S$  be an arbitrary subset of  $X$  and  $|S| = k$ . The number of the chains in  $X$  that contain  $S$ , that is, such that  $C_k = S$ , is  $k!(n-k)!$ . Indeed, we have  $(n-k)!$  sequences  $C_{k+1}, C_{k+2}, \dots, C_n$  of subsets satisfying  $S \subset C_{k+1} \subset C_{k+2} \subset \dots \subset C_n$  and  $|C_i| = i$  for  $i = k+1, \dots, n$  (there are  $n-k$  choices for  $C_{k+1}$ ,  $n-k-1$  choices for  $C_{k+2}$ , and so on). Each

of these sequences may be combined with each chain  $C_1, C_2, \dots, C_k = S$  in the set  $S$ , and the number of these chains is  $k!$ .

The effectiveness of this notion becomes clear right away after the following obvious remark:

*If  $A$  and  $B$  are incomparable subsets of  $X$  then no chain in  $X$  contains them both.*

This simple observation in fact yields a proof of the theorem.

Let  $A_1, A_2, \dots, A_l$  be pairwise incomparable subsets of the  $n$ -element set  $X$ , and let  $|A_i| = k_i$  (with  $i = 1, 2, \dots, l$ ). As proved above, each  $A_i$  is contained in exactly  $k_i!(n - k_i)!$  distinct chains. Moreover, two chains corresponding to different sets  $A_i$  and  $A_j$  are different. Hence there are at least  $k_1!(n - k_1)! + k_2!(n - k_2)! + \dots + k_l!(n - k_l)!$  different chains in  $X$ , which implies

$$k_1!(n - k_1)! + k_2!(n - k_2)! + \dots + k_l!(n - k_l)! \leq n!.$$

Dividing by  $n!$  gives

$$\binom{n}{k_1}^{-1} + \binom{n}{k_2}^{-1} + \dots + \binom{n}{k_l}^{-1} \leq 1.$$

Since

$$\binom{n}{k_i} \leq \binom{n}{\lfloor n/2 \rfloor} \quad \text{for each } i = 1, 2, \dots, l,$$

we obtain

$$1 \geq \binom{n}{k_1}^{-1} + \binom{n}{k_2}^{-1} + \dots + \binom{n}{k_l}^{-1} \geq l \cdot \binom{n}{\lfloor n/2 \rfloor}^{-1},$$

that is,  $l \leq \binom{n}{\lfloor n/2 \rfloor}$ .

We proved that any family of pairwise incomparable subsets of  $X$  has at most  $\binom{n}{\lfloor n/2 \rfloor}$  elements. This is the harder part of the proof. There exists a family with this property containing exactly  $\binom{n}{\lfloor n/2 \rfloor}$  elements: we can take as an example the family of all  $\lfloor n/2 \rfloor$ -element subsets of  $X$ . ■

Since Sperner's theorem was published in 1928, families of pairwise incomparable subsets of a given set are usually called Sperner families. Additional work is needed to determine all maximal Sperner families of an  $n$ -element set  $X$ , that is, the ones of cardinality  $\binom{n}{\lfloor n/2 \rfloor}$ . If  $n$  is even, the maximal Sperner family is unique, and it consists of all  $(n/2)$ -element subsets of  $X$ . And if  $n$  is odd, there are two maximal Sperner families:

one contains all  $\lfloor (n-1)/2 \rfloor$ -element subsets of  $X$ , and the other all of its  $\lfloor (n+1)/2 \rfloor$ -element subsets.

It was Erdős who noticed the connection of Sperner's theorem with certain estimation results. In many cases Sperner-type bounds turn out to be the best possible ones. The following statement can be pointed out as a classic example.

*Let  $x_1, x_2, \dots, x_n$  be real numbers of absolute value at least 1.*

*Then no more than  $\binom{n}{\lfloor n/2 \rfloor}$  of the sums  $\pm x_1 \pm x_2 \pm \dots \pm x_n$  fall in an open interval of length 2.*

Without loss of generality, we may assume that  $x_1, x_2, \dots, x_n$  are of the same sign, say positive. This is because changing the sign of some  $x_k$  does not change the collection of sums  $\pm x_1 \pm x_2 \pm \dots \pm x_n$ .

So, let  $x_k \geq 1$ , with  $i = 1, 2, \dots, n$ . For each subset  $I$  of  $\{1, 2, \dots, n\}$ , set

$$f(I) = \sum_{k \in I} x_k - \sum_{k \notin I} x_k.$$

If  $I$  ranges over all subsets of  $\{1, 2, \dots, n\}$ , then  $f(I)$  ranges over all sums  $\pm x_1 \pm x_2 \pm \dots \pm x_n$ . Observe that if  $I \subset J$  and  $I \neq J$ , then

$$\begin{aligned} f(J) - f(I) &= \left( \sum_{k \in J} x_k - \sum_{k \notin J} x_k \right) - \left( \sum_{k \in I} x_k - \sum_{k \notin I} x_k \right) \\ &= 2 \sum_{k \in J \setminus I} x_k. \end{aligned}$$

Since  $J \setminus I$  is nonempty and  $x_k \geq 1$  for all  $k = 1, 2, \dots, n$ , this implies  $f(J) - f(I) \geq 2$ . So, if  $I \subset J$  and  $I \neq J$ , the sums  $f(I)$  and  $f(J)$  cannot be both contained in an open interval of length 2.

This observation leads to a quick proof of our claim. Suppose that several sums  $\pm x_1 \pm x_2 \pm \dots \pm x_n$  fall in an open interval of length 2. Then their corresponding  $I$ 's form a family of pairwise incomparable subsets of  $\{1, 2, \dots, n\}$ . By Sperner's theorem, the cardinality of such a family does not exceed  $\binom{n}{\lfloor n/2 \rfloor}$ . The example  $x_1 = x_2 = \dots = x_n = 1$  shows that the bound  $\binom{n}{\lfloor n/2 \rfloor}$  is the best possible one. ■

# 49

## Morse's Sequence

For each nonnegative integer  $n$ , let  $\alpha_n = 0$  if the number of 1's in the binary representation of  $n$  is even, and  $\alpha_n = 1$  if this number is odd.

The binary sequence  $(\alpha_n)_{n=0}^{\infty}$  is called *Morse's sequence*. However, this might be a little unfair to the Norwegian mathematician Axel Thue, who first brought serious attention to it in the first decade of the twentieth century, and to G. A. Hedlund, who collaborated with Morse in the 1940s. Here we stick to the name *Morse's sequence* for simplicity. This sequence has unexpected, sometimes amazing applications. Being by far not “random,” it has many properties demanded from random numbers.

Morse's sequence can be defined in different ways. Write a 0 at first. Then perform infinitely many of the following steps: Preserve the block of the sequence that has already been written, and add to the right of it a new block obtained from the old one by replacing 0 by 1 and 1 by 0.

Here is another equivalent description. The pair  $(0, 1)$  is given initially. Then the pair  $(1, 0)$  is added to it, then the two pairs  $(1, 0)$ ,  $(0, 1)$ , and so on; at each step we add as many pairs as we have already written, replacing each pair  $(0, 1)$  by  $(1, 0)$ , and  $(1, 0)$  by  $(0, 1)$ .

Take a nonnegative integer  $n$ . If  $n = \overline{ab \dots c}$  in binary notation, then we have  $2n = \overline{ab \dots c0}$ , containing the same number of binary 1's as  $n$ , and  $2n + 1 = \overline{ab \dots c1}$ , containing one more 1. This yields two basic properties of Morse's sequence:

$$\alpha_{2n} = \alpha_n \quad \text{and} \quad \alpha_{2n+1} = 1 - \alpha_{2n} \quad \text{for } n = 0, 1, 2, \dots \quad (1)$$

The first equality implies that the sequence is determined by its terms with odd indices. The second one means that each term with an odd index is different from the previous (even-indexed) term.

We start by proving that

*Morse's sequence is nonperiodic.*

The proof is indirect. Assume that this is not true and let  $k$  be the length of the smallest period of the sequence. Then there is an index  $\nu$  such that  $\alpha_{n+k} = \alpha_n$  for all  $n \geq \nu$ .

Suppose first that  $k$  is even,  $k = 2l$ . Then, in view of (1),

$$\alpha_n = \alpha_{2n} = \alpha_{2n+k} = \alpha_{2n+2l} = \alpha_{2(n+l)} = \alpha_{n+l}$$

for each  $n \geq \nu$ . This means that  $l < k$  is a period of the given sequence, a contradiction.

Assume now that  $k$  is odd. Choose an even  $m \geq \nu$  and consider the block  $\alpha_m, \alpha_{m+1}, \dots, \alpha_{m+k-1}$  of odd length  $k$ . It cannot contain the same number of zeros and ones, so suppose there are more zeros than ones. Since  $k$  is a period, the block  $\alpha_{m+k}, \alpha_{m+k+1}, \dots, \alpha_{m+2k-1}$  is identical to the first one. Hence the block

$$\alpha_m, \alpha_{m+1}, \dots, \alpha_{m+k-1}, \alpha_{m+k}, \alpha_{m+k+1}, \dots, \alpha_{m+2k-1} \quad (2)$$

of length  $2k$  contains at least two more zeros than ones. On the other hand, (2) can be split into  $k$  pairs  $(\alpha_{2l}, \alpha_{2l+1})$ . Since  $\alpha_{2l+1} = 1 - \alpha_{2l}$  (see (1)), we conclude that there are exactly as many zeros in (2) as ones, a contradiction. ■

Essentially, the 1992 Putnam problem A-5 claims that

*Morse's sequence does not contain three consecutive identical blocks; that is, there do not exist positive integers  $k$  and  $m$  such that*

$$\alpha_{k+j} = \alpha_{k+m+j} = \alpha_{k+2m+j}$$

for all  $0 \leq j \leq m-1$ .

We omit the proof, which does not contain any new ideas.

One of the most astounding applications of Morse's sequence is to the so-called *Tarry-Escott problem*, a statement about partitioning sets of positive integers into subsets with equal sums of "like" powers:

*The set  $\{0, 1, 2, 3, \dots, 2^{k+1} - 1\}$  can be partitioned into two disjoint subsets  $\{x_1, x_2, \dots, x_{2^k}\}$  and  $\{y_1, y_2, \dots, y_{2^k}\}$  such that*

$$\sum_{i=1}^{2^k} x_i^l = \sum_{i=1}^{2^k} y_i^l \quad \text{for all } l = 0, 1, 2, \dots, k.$$

Morse's sequence determines such a partitioning. Denote by  $X_k$  the set of all  $n \in \{0, 1, 2, \dots, 2^{k+1} - 1\}$  with  $\alpha_n = 0$ , and by  $Y_k$  the set of  $n \in \{0, 1, 2, \dots, 2^{k+1} - 1\}$  with  $\alpha_n = 1$ . Our descriptions of Morse's sequence in the beginning of the section imply that  $X_k$  and  $Y_k$  have  $2^k$  elements each. Surprisingly or not, these sets meet the requirements. Take, for instance,  $k = 3$ . The first 16 terms of Morse's sequence are

$$0, 1, 1, 0, 1, 0, 0, 1, 1, 0, 0, 1, 0, 1, 1, 0,$$

implying

$$X_3 = \{0, 3, 5, 6, 9, 10, 12, 15\},$$

$$Y_3 = \{1, 2, 4, 7, 8, 11, 13, 14\}.$$

It is really hard to believe that

$$\begin{aligned} 0^0 + 3^0 + 5^0 + 6^0 + 9^0 + 10^0 + 12^0 + 15^0 &= 1^0 + 2^0 + 4^0 + 7^0 + 8^0 + 11^0 + 13^0 + 14^0, \\ 0^1 + 3^1 + 5^1 + 6^1 + 9^1 + 10^1 + 12^1 + 15^1 &= 1^1 + 2^1 + 4^1 + 7^1 + 8^1 + 11^1 + 13^1 + 14^1, \\ 0^2 + 3^2 + 5^2 + 6^2 + 9^2 + 10^2 + 12^2 + 15^2 &= 1^2 + 2^2 + 4^2 + 7^2 + 8^2 + 11^2 + 13^2 + 14^2, \\ 0^3 + 3^3 + 5^3 + 6^3 + 9^3 + 10^3 + 12^3 + 15^3 &= 1^3 + 2^3 + 4^3 + 7^3 + 8^3 + 11^3 + 13^3 + 14^3. \end{aligned}$$

To show that  $X_k$  and  $Y_k$  yield a solution to the Tarry-Escott problem, we are going to prove a stronger statement by induction, namely:

If  $f(x)$  is a polynomial of degree not exceeding  $k$ , then

$$\sum_{n \in X_k} f(n) = \sum_{n \in Y_k} f(n). \quad (3)$$

This is easy for  $k = 0$ , because  $f(x)$  must be a constant polynomial. Suppose that (3) holds for a certain  $k$ . To carry out the inductive step, note that

$$\begin{aligned} X_{k+1} &= X_k \cup (2^{k+1} + Y_k), \\ Y_{k+1} &= Y_k \cup (2^{k+1} + X_k), \end{aligned} \quad (4)$$

where the notations  $2^{k+1} + Y_k$  and  $2^{k+1} + X_k$  represent the sets obtained by adding  $2^{k+1}$  to each element of  $Y_k$  and  $X_k$ , respectively. Indeed, adding  $2^{k+1}$  to any  $n \in \{0, 1, 2, \dots, 2^{k+1} - 1\}$  increases by 1 the number of its binary 1's, and hence changes the parity of  $\alpha_n$ .

Let  $g$  be any polynomial of degree at most  $k+1$ . Then the polynomial  $f(x) = g(x + 2^{k+1}) - g(x)$  has degree not exceeding  $k$  and, by the

induction hypothesis, it satisfies (3). This can be written as

$$\sum_{n \in X_k} g(n) + \sum_{n \in Y_k} g(2^{k+1} + n) = \sum_{n \in Y_k} g(n) + \sum_{n \in X_k} g(2^{k+1} + n).$$

In view of (4), the latter is simply equivalent to

$$\sum_{n \in X_{k+1}} g(n) = \sum_{n \in Y_{k+1}} g(n),$$

which finishes the proof. ■

A related result about pairwise sums (recall Section 46) will be just stated here, because the proof follows the same pattern.

*The sets  $X_k$  and  $Y_k$ , where  $k \geq 1$ , yield the same sequences of pairwise sums. More exactly, if  $X_k = \{x_1, x_2, \dots, x_{2^k}\}$  and  $Y_k = \{y_1, y_2, \dots, y_{2^k}\}$ , then the sequences*

$$x_1 + x_2, x_1 + x_3, \dots, x_1 + x_{2^k}, x_2 + x_3, \dots, x_{2^k-1} + x_{2^k}$$

$$y_1 + y_2, y_1 + y_3, \dots, y_1 + y_{2^k}, y_2 + y_3, \dots, y_{2^k-1} + y_{2^k}$$

*coincide up to permutation.*

What is even more amazing, this property of Morse's sequence can be extended to the set of all nonnegative integers as follows:

*There is a unique way to partition the nonnegative integers into two sets,  $A$  and  $B$ , such that every positive integer is expressible in the form  $a_1 + a_2$ , where  $a_1, a_2 \in A$  and  $a_1 \neq a_2$ , in exactly the same number of ways as in the form  $b_1 + b_2$ , where  $b_1, b_2 \in B$  and  $b_1 \neq b_2$ .*

This was discovered and proved by the Canadian mathematicians **Joe Lambek** and **Leo Moser** in 1959.

Consider two sets  $A$  and  $B$  of nonnegative integers with generating functions

$$A(x) = \sum_{a \in A} x^a, \quad B(x) = \sum_{b \in B} x^b.$$

Clearly  $A$  and  $B$  cover  $\mathbb{N}$  without overlapping if and only if

$$A(x) + B(x) = x^0 + x^1 + x^2 + \dots + x^n + \dots = \frac{1}{1-x}. \quad (5)$$

The number of ways in which a nonnegative integer  $n$  can be expressed as a sum of two distinct  $a$ 's equals the coefficient of  $x^n$  in the formal series

$$\sum_{\substack{a_i, a_j \in A \\ a_i \neq a_j}} x^{a_i + a_j} = \frac{1}{2} [A^2(x) - A(x^2)].$$

The same argument works for  $B$ , so the nontrivial requirement of the statement translates into

$$A^2(x) - A(x^2) = B^2(x) - B(x^2). \quad (6)$$

Hence a partition of  $\mathbf{N}$  is of the desired type if and only if the generating functions of  $A$  and  $B$  satisfy (5) and (6). Taking this into account, and assuming for definiteness that  $1 \in A$ , we obtain

$$[A(x) - B(x)][A(x) + B(x)] = A(x^2) - B(x^2),$$

or

$$A(x) - B(x) = (1 - x) [A(x^2) - B(x^2)].$$

Repeating the same argument, with  $x$  replaced by  $x^2, x^4, \dots, x^{2^{n-1}}$ , gives

$$A(x) - B(x) = \prod_{i=0}^{n-1} (1 - x^{2^i}) [A(x^{2^n}) - B(x^{2^n})], \quad n = 1, 2, \dots$$

Now, except for the 1 that begins the series  $A(x^{2^n})$ , every power of  $x$  in  $A(x^{2^n}) - B(x^{2^n})$  is of degree at least  $2^n$  and plays no part in the construction of any term  $x^k$  of lesser degree. Since  $2^n$  becomes arbitrarily large with increasing  $n$ , every term  $x^k$  is of degree less than  $2^n$  for some  $n$ . Thus every term  $x^k$  on the right is generated by multiplying out the infinite product  $(1 - x)(1 - x^2) \cdots (1 - x^{2^n}) \cdots$ , and we have

$$A(x) - B(x) = (1 - x)(1 - x^2) \cdots (1 - x^{2^n}) \cdots$$

But the last product, when multiplied out, gives a series of terms of the form  $+x^n$  if  $n$  is the sum of an even number of distinct powers of 2, and  $-x^n$  otherwise. It is easily inferred from here that the only partitioning with the desired property is indeed defined by Morse's sequence  $(\alpha_n)_{n \geq 0}$ :  $A$  is the set of all nonnegative integers  $n$  with  $\alpha_n = 0$ , and  $B$  is the set of all  $n$  with  $\alpha_n = 1$ . ■

Lambek and Moser also partitioned the positive integers into two sets  $C$  and  $D$  which have the same property with regard to multiplication

instead of addition:

$$C = \{1, 6, 8, 10, 12, 14, 15, 18, 20, 21, 22, 26, 27, 28, \dots\},$$

$$D = \{2, 3, 4, 5, 7, 9, 11, 13, 16, 17, 19, 23, 24, 25, 29, 30, \dots\}.$$

The rule of formation is: If  $n = p_1^{k_1} p_2^{k_2} \cdots p_\ell^{k_\ell}$  is the prime factorization of  $n$  then, expressing all the exponents  $k_i$  in binary notation,  $n$  goes into  $C$  if the total number of 1's in all the  $k_i$ 's is even, and into  $D$  if it is odd.

# 50

## A Favorite of Erdős

Reciprocals of positive integers have attracted the attention of mathematicians since time immemorial. They were already used in ancient Egypt, and are usually referred to as Egyptian fractions.

In 1202, Leonardo Pisano proved that each positive rational number  $r$  can be expressed as a sum of distinct Egyptian fractions:

$$r = \frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}, \quad \text{where } 1 \leq a_1 < a_2 < \cdots < a_n. \quad (1)$$

The number of summands,  $n$ , depends on  $r$ . If  $n$  is fixed, then by far not all rationals have representations as in (1). Moreover, even if equal denominators are allowed, each interval  $\Delta$  contains a smaller subinterval  $\delta$  such that no rational in  $\delta$  is expressible in the form (1).

In particular, for each  $n$  there is a constant  $c_n < 1$  such that no sum of  $n$  Egyptian fractions falls in the interval  $(c_n, 1)$ . A natural question arises: Which is the largest  $c_n$  with this property? In other words, how close to 1 can one get, adding up  $n$  Egyptian fractions?

This is a hard problem solved by **Curtiss** in 1922 and, independently, by **Erdős** in 1932. As Ronald Graham said in a talk entitled “Paul Erdős’ Favorite Problems,” this was perhaps the oldest of Erdős’ favorites. It gave rise to many other questions, all quite substantial and some still awaiting their answers, carrying a famous Erdős \$50 to \$250 price-tag.

The closest under-approximation of 1 by Egyptian fractions has been a subject of considerable interest for quite a while. For instance, some half a century ago it appeared as a *Monthly* proposal (*AMM*, Vol. 38, 1932, page 175, problem 3536), but no solution was published. There was an additional issue of the *American Mathematical Monthly* called *The Otto Dunkel Memorial Problem Book* (Vol. 64, No. 7, Part II, 1957), where

the statement is included without a proof. No proof can be found in the Russian version of the *Memorial Problem Book*, edited by the well-known Russian mathematician V. Alexeev.

And if all this is not enough to indicate that the problem is rather hard, it is worth noting that one of the few complete proofs known to us relies on properties of continuous functions on compact sets in  $\mathbf{R}^n$ .

But the concept of continuity has little to do with the simple nature of Egyptian fractions. So it is our pleasure to present an Erdős-style proof—not easy to find but completely elementary.

*The sequence  $u_1, u_2, \dots, u_n, \dots$  is defined by*

$$u_1 = 2 \quad \text{and} \quad u_n = u_1 u_2 \cdots u_{n-1} + 1, \quad n = 2, 3, \dots$$

*Then, for all positive integers  $n$ , the closest under-approximation of 1 by  $n$  Egyptian fractions is*

$$1 - \frac{1}{u_1 u_2 \cdots u_n}.$$

The sequence  $\{u_n\}$  is, of course, well known, and its basic property is a straightforward consequence of the definition:

$$\frac{1}{u_1} + \frac{1}{u_2} + \cdots + \frac{1}{u_n} + \frac{1}{u_1 u_2 \cdots u_n} = 1 \quad \text{for } n = 1, 2, 3, \dots$$

So, we need to prove that if  $x_1, x_2, \dots, x_n$  are positive integers satisfying  $1/x_1 + 1/x_2 + \cdots + 1/x_n < 1$ , then

$$\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n} \leq \frac{1}{u_1} + \frac{1}{u_2} + \cdots + \frac{1}{u_n}.$$

The proof is by induction on  $n$ . Everything is clear for  $n = 1$ . Assume that the claim holds for each  $k = 1, 2, \dots, n-1$ , and consider any positive integers  $x_1, x_2, \dots, x_n$  such that  $1/x_1 + 1/x_2 + \cdots + 1/x_n < 1$ . Let

$$\begin{aligned} \frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_i} &= X_i \\ \frac{1}{u_1} + \frac{1}{u_2} + \cdots + \frac{1}{u_i} &= U_i \end{aligned} \quad \text{for } i = 1, 2, \dots, n.$$

Since  $x_1 x_2 \cdots x_n X_n$  is an integer less than  $x_1 x_2 \cdots x_n$ , we then obtain  $x_1 x_2 \cdots x_n X_n \leq x_1 x_2 \cdots x_n - 1$ , or  $X_n \leq 1 - [1/(x_1 x_2 \cdots x_n)]$ .

Now, assume on the contrary that  $X_n > U_n$ . This assumption, combined with

$$X_n \leq 1 - \frac{1}{x_1 x_2 \cdots x_n}$$

and

$$U_n = 1 - \frac{1}{u_1 u_2 \cdots u_n},$$

implies

$$x_1 x_2 \cdots x_n > u_1 u_2 \cdots u_n. \quad (2)$$

On the other hand,  $X_i < 1$  for  $i = 1, 2, \dots, n-1$ , so, by the induction hypothesis,

$$X_1 \leq U_1, X_2 \leq U_2, \dots, X_{n-1} \leq U_{n-1}. \quad (3)$$

Consider the sum  $\sum_{i=1}^n x_i/u_i$  and apply Abel's summation formula to get

$$\sum_{i=1}^n \frac{x_i}{u_i} = \sum_{i=1}^{n-1} U_i (x_i - x_{i+1}) + U_n x_n. \quad (4)$$

We may assume, without loss of generality, that  $x_1 \leq x_2 \leq \cdots \leq x_n$ , so  $x_i - x_{i+1} \leq 0$  for  $i = 1, 2, \dots, n-1$ . Then, in view of (3), (4) and the assumption  $X_n > U_n$ , we obtain

$$\sum_{i=1}^n \frac{x_i}{u_i} < \sum_{i=1}^{n-1} X_i (x_i - x_{i+1}) + X_n x_n = \sum_{i=1}^n \frac{x_i}{x_i} = n.$$

Now, by the AM–GM inequality,

$$n > \sum_{i=1}^n \frac{x_i}{u_i} \geq n \sqrt[n]{\frac{x_1 x_2 \cdots x_n}{u_1 u_2 \cdots u_n}},$$

so  $u_1 u_2 \cdots u_n > x_1 x_2 \cdots x_n$ , contradicting (2). ■

Who knows—maybe it is such a proof that would have made Erdős exclaim: “This is straight from *The Book*!”



## Instead of an Afterword

Little by little, this book has come to its close. Let us finish with a problem whose solution we find really outstanding.

Certain mathematical facts seem so well established that it is hard to imagine what further development they could have. Nevertheless, a closer look at them often confirms that there are many more questions in mathematics than answers.

Take, for example, any prime  $p$ , and consider the binomial coefficients

$$\binom{p}{1}, \binom{p}{2}, \dots, \binom{p}{p-1}.$$

It is a well-known fact that all of them are divisible by  $p$ . In particular, if  $0 < i < j < p$ , then  $\binom{p}{i}$  and  $\binom{p}{j}$  have a common divisor greater than 1.

It is most natural to ask if the latter holds for any binomial coefficients  $\binom{n}{i}$  and  $\binom{n}{j}$ , where  $n$  is not necessarily a prime. (Of course, it is not required that *all* binomial coefficients

$$\binom{n}{1}, \binom{n}{2}, \dots, \binom{n}{n-1}$$

share the same divisor; we just conjecture that any two of them are *not* relatively prime.) More exactly, our query reads:

*If  $0 < i < j < n$ , is it true that the greatest common divisor of the binomial coefficients  $\binom{n}{i}$  and  $\binom{n}{j}$  is greater than 1?*

The question appears innocent; many people have given it a try and no one has found it easy. Traditional tricks of the trade (divisibility of binomial coefficients) fail to work here, including the widely known Lucas' theorem. Using a computer, one might observe that, for relatively small  $n$ , say  $n \leq 2000$ , the coefficients  $\binom{n}{i}$  and  $\binom{n}{j}$  ( $0 < i < j < n$ ) do share a

nontrivial common divisor (in most cases, their greatest common divisor is a “small” prime). Such evidence is enough to make us believe that the answer is yes but it provides no insight into how to approach the proof. And then, as Selfridge exclaimed, it is “those little binomial coefficient identities” again that can handle the job. Indeed, just take a look at the proof below.

Consider the identity

$$\binom{n}{i} \binom{n-i}{j} = \binom{n}{j} \binom{n-j}{i}, \quad (1)$$

which can be easily checked. Surprisingly for the ones who had come to know the real difficulty of the problem, it immediately yields a quick proof. And indeed, suppose on the contrary that  $\binom{n}{i}$  and  $\binom{n}{j}$  are relatively prime. Then (1) implies that  $\binom{n-j}{i}$  is divisible by  $\binom{n}{i}$ , which is impossible: The second binomial coefficient is clearly greater.

To make the proof rigorous, we still have a little more work to do. Note that (1) holds for any  $i, j$  and  $n$ , but our divisibility argument fails if some of the participating binomial coefficients are 0. This can indeed happen if  $i + j > n$ , but it is a minor trouble. Since  $\binom{n}{k} = \binom{n}{n-k}$  for any  $k = 0, 1, 2, \dots, n$ , we may assume that  $i \leq \lfloor n/2 \rfloor$ ,  $j \leq \lfloor n/2 \rfloor$ . Thus  $i + j \leq n$ , avoiding the technical difficulties. ■

# Glossary

**Abel's summation formula.** If  $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$  are real or complex numbers, and

$$S_i = a_1 + a_2 + \cdots + a_i \quad \text{for } i = 1, 2, \dots, n,$$

then

$$\sum_{i=1}^n a_i b_i = \sum_{i=1}^{n-1} S_i (b_i - b_{i+1}) + S_n b_n.$$

**Arithmetic mean–geometric mean (AM–GM) inequality.** For any nonnegative numbers  $a_1, a_2, \dots, a_n$ ,

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \cdots a_n},$$

with equality if and only if  $a_1 = a_2 = \cdots = a_n$ .

**Bijective function.** A function that is injective (one-to-one) and surjective (onto).

**Binomial coefficient.** The coefficient of  $x^k$  in the expansion of the binomial  $(1+x)^n$ . It is equal to  $n!/[k!(n-k)!]$  and is denoted by  $\binom{n}{k}$ .

**Cauchy–Schwarz inequality.** For any real numbers  $a_1, a_2, \dots, a_n$  and  $b_1, b_2, \dots, b_n$ ,

$$(a_1 b_1 + a_2 b_2 + \cdots + a_n b_n)^2 \leq (a_1^2 + a_2^2 + \cdots + a_n^2)(b_1^2 + b_2^2 + \cdots + b_n^2).$$

Equality occurs if and only if  $b_1 = b_2 = \cdots = b_n = 0$  or if there is a constant  $C$  such that  $a_i = Cb_i$  ( $i = 1, 2, \dots, n$ ).

**Centroid.** The centroid of the system of points  $A_1, A_2, \dots, A_n$  is a point  $G$  such that

$$\overrightarrow{GA_1} + \overrightarrow{GA_2} + \cdots + \overrightarrow{GA_n} = \vec{0}.$$

The centroid of any system exists and is unique. Its coordinates with respect to a coordinate system are the arithmetic means of the corresponding coordinates of the points.

**Centroid of a triangle.** The centroid of a triangle coincides with the intersection point of its medians.

**Chebyshev polynomial.** In this book this is either a polynomial expressing  $\cos n\varphi$  as a polynomial in  $x = \cos \varphi$  or a polynomial expressing  $2 \cos n\varphi$  as a polynomial in  $x = 2 \cos \varphi$ .

**Chinese remainder theorem.** Let  $m_1, m_2, \dots, m_k$  be pairwise relatively prime integers, and let  $a_1, a_2, \dots, a_k$  be arbitrary integers. Then the system of simultaneous congruences

$$x \equiv a_1 \pmod{m_1}, \quad x \equiv a_2 \pmod{m_2}, \quad \dots, \quad x \equiv a_k \pmod{m_k}$$

has a solution.

**Circular segment.** The figure formed by an arc of a circle and the line segment joining its endpoints.

**Circumcircle of a triangle.** The circle circumscribed about the triangle; its center is called the circumcenter of the triangle.

**Complete graph.** A collection of  $n$  points, called vertices, and the line segments joining them two by two, called edges, is the complete graph on  $n$  vertices.

**Complete set of residues.** The integers  $a_1, a_2, \dots, a_n$  form a complete set of residues modulo  $n$  if the difference of no two of them is divisible by  $n$ .

**Continued fraction.** If  $q_0, q_1, q_2, \dots, q_n$  are the repeated quotients in the Euclidean algorithm for the positive integers  $a$  and  $b$ , then

$$\frac{a}{b} = q_0 + \frac{1}{q_1 + \frac{1}{q_2 + \frac{1}{q_3 + \frac{1}{q_4 + \frac{1}{q_5 + \frac{1}{q_6 + \frac{1}{q_7 + \frac{1}{q_8 + \frac{1}{q_9 + \frac{1}{q_{10}}}}}}}}}}}} = [q_0; q_1, \dots, q_{n-1}, q_n].$$

This representation is called the expansion of  $a/b$  as a continued fraction.

**Convex function.** A real-valued function  $f$  defined on an interval  $I$  of real numbers is convex if, for any  $x, y$  in  $I$  and any nonnegative numbers  $\alpha, \beta$  with sum 1,

$$f(\alpha x + \beta y) \leq \alpha f(x) + \beta f(y).$$

**Convex set.** A set in the plane or in three-dimensional space is convex if it contains all points of the line segment connecting any two of its points.

**Cyclic permutation.** Any arrangement of the elements of a finite set about a circle. The number of the cyclic permutations of an  $n$ -element set is  $(n-1)!$ .

**Cyclic polygon.** A polygon that can be inscribed in a circle.

**Determinant.** The determinant of an  $n \times n$  matrix  $A = (a_{ij})$  is the algebraic sum

$$\sum (-1)^{[i_1 i_2 \dots i_n]} a_{1i_1} a_{2i_2} \dots a_{ni_n},$$

where the summation ranges over all permutations  $(i_1, i_2, \dots, i_n)$  of  $\{1, 2, \dots, n\}$ , and where  $[i_1 i_2 \dots i_n]$  is 0 or 1, according as the permutation  $(i_1, i_2, \dots, i_n)$  is even or odd.

**Dilatation.** Let  $O$  be a fixed point in the plane or in three-dimensional space, and let  $\lambda$  be a nonzero real number. The dilatation (also called homothety) with center  $O$  and coefficient (ratio)  $\lambda$  is the transformation that carries each point  $X$  to the point  $X'$  defined by

$$\overrightarrow{OX'} = \lambda \overrightarrow{OX}.$$

**Dot product.** If  $\mathbf{x}$  and  $\mathbf{y}$  are vectors in the plane or in three-dimensional space, then their dot product  $\mathbf{x} \cdot \mathbf{y}$  is defined geometrically as the product of

their lengths multiplied by the cosine of the angle between them. Assuming  $\mathbf{x}$  and  $\mathbf{y}$  lying in the plane and having coordinates  $(x_1, x_2)$  and  $(y_1, y_2)$  with respect to some Cartesian coordinate system, we have

$$\mathbf{x} \cdot \mathbf{y} = x_1 y_1 + x_2 y_2.$$

This formula has a natural analog in three-dimensional space.

**Euclidean algorithm.** A process of repeated divisions yielding the greatest common divisor of two positive integers  $a$  and  $b$ :

$$\begin{aligned} a &= q_0 b + r_1, & 0 < r_1 < b, \\ b &= q_1 r_1 + r_2, & 0 < r_2 < r_1, \\ &\vdots \\ r_{n-2} &= q_{n-1} r_{n-1} + r_n, & 0 < r_n < r_{n-1}, \\ r_{n-1} &= q_n r_n. \end{aligned}$$

The last nonzero remainder  $r_n$  is the greatest common divisor of  $a$  and  $b$ .

**Euler's formula.** If  $V$ ,  $E$ ,  $F$  denote, respectively, the number of vertices, edges and faces of a convex polyhedron, then

$$V - E + F = 2.$$

**Fermat's little theorem.** If  $p$  is a prime and  $a$  is not divisible by  $p$ , then

$$a^{p-1} \equiv 1 \pmod{p}.$$

**Fibonacci sequence.** The sequence defined by  $F_1 = F_2 = 1$  and

$$F_{n+1} = F_n + F_{n-1} \quad \text{for } n \geq 2.$$

**Generating function.** In this book the generating function  $f(x)$  of a set  $A$  of nonnegative integers is defined by

$$f(x) = \sum_{a \in A} x^a.$$

This is a polynomial if  $A$  is finite, and a (formal) power series if  $A$  is infinite.

**Hölder's inequality.** If  $a_{ij} \geq 0$ ,  $i = 1, 2, \dots, m$ ,  $j = 1, 2, \dots, n$  and  $\alpha_1, \alpha_2, \dots, \alpha_n$  are positive numbers such that  $\sum_{j=1}^n 1/\alpha_j = 1$ , then

$$\sum_{i=1}^m \prod_{j=1}^n a_{ij} \leq \prod_{j=1}^n \left( \sum_{i=1}^m a_{ij}^{\alpha_j} \right)^{1/\alpha_j}.$$

Most applications appeal to the special case  $\alpha_1 = \alpha_2 = \dots = \alpha_n = n$ , which yields

$$\left( \sum_{i=1}^m \prod_{j=1}^n a_{ij} \right)^n \leq \prod_{j=1}^n \sum_{i=1}^m a_{ij}^n.$$

**Identity theorem.** If a polynomial with real or complex coefficients of degree not exceeding  $n$  takes the value 0 at more than  $n$  distinct values of the indeterminate, then this polynomial is identically zero, that is, all its coefficients are zero.

**Incircle of a triangle.** The circle inscribed in the triangle; its center is called the incenter of the triangle.

**Injective function.** A function  $f: A \rightarrow B$  is injective (one-to-one) if  $x, y \in A$  and  $x \neq y$  imply  $f(x) \neq f(y)$ .

**Lagrange's interpolation formula.** Let  $x_0, x_1, \dots, x_n$  be distinct real numbers and  $y_0, y_1, \dots, y_n$  arbitrary real numbers. Then there exists a unique polynomial  $P(x)$  of degree at most  $n$  such that  $P(x_i) = y_i$  for each  $i = 0, 1, \dots, n$ . This is the polynomial given by

$$P(x) = \sum_{i=0}^n y_i \frac{(x - x_0) \cdots (x - x_{i-1})(x - x_{i+1}) \cdots (x - x_n)}{(x_i - x_0) \cdots (x_i - x_{i-1})(x_i - x_{i+1}) \cdots (x_i - x_n)}.$$

**Matrix.** A rectangular array of numbers  $(a_{ij})$ , where  $a_{ij}$  is the entry in the  $i$ th row and the  $j$ th column.

**Nonsingular matrix.** A matrix whose determinant is different from 0.

**Orthocenter of a triangle.** The common point of the altitudes or their extensions.

**Orthogonal projection.** The orthogonal projection of a point onto a line in the plane is the foot of the perpendicular from the point to the line. Orthogonal projections are defined in a similar manner in three-dimensional space.

**Perfect power.** A number of the form  $m^n$ , where  $m$  and  $n$  are integers greater than 1.

**Periodic function.** A function  $f: \mathbf{R} \rightarrow \mathbf{R}$  for which there exists a nonzero number  $T$  such that

$$f(x + T) = f(x) \quad \text{for all real } x.$$

**Permutation.** Any arrangement of the elements of a finite set. The number of the permutations of an  $n$ -element set is  $n!$ .

If  $(i_1, i_2, \dots, i_n)$  is a permutation of the set  $\{1, 2, \dots, n\}$ , then  $i_k$  and  $i_l$  are said to form an inversion if  $k < l$  and  $i_k > i_l$ . A permutation is called even (odd) if it has an even (odd) number of inversions.

**Pigeonhole principle (Dirichlet's principle).** If  $n$  objects are distributed among  $k$  boxes and  $n > k$ , then there is a box that contains at least two objects.

**Power of a point.** For a circle  $c$  with radius  $R$  and a point  $P$  whose distance to the center of the circle is  $d$ , the power of  $P$  with respect to  $c$  is the number  $d^2 - R^2$ .

For any two points  $A$  and  $B$  that are on the circle and collinear with  $P$ , one has

$$\overline{PA} \cdot \overline{PB} = d^2 - R^2.$$

Here  $\overline{XY}$  denotes the signed length of the line segment  $XY$ . This statement is referred to as the power-of-a-point theorem.

**Radical axis of two nonconcentric circles.** The locus of points of equal powers with respect to the two circles.

If the circles intersect, then their radical axis is the line containing their common chord. If they are tangent, then the radical axis coincides with their common tangent. In all cases, the radical axis of the circles is a line perpendicular to the line that connects their centers.

**Radical center of three circles with noncollinear centers.** The common point of the three radical axes of the three pairs of circles.

**Rational-root theorem.** If a rational number  $m/n$ , where  $m$  and  $n$  are relatively prime, is a root of a polynomial with integer coefficients, then  $m$  divides the constant term of this polynomial and  $n$  divides its leading coefficient.

**Root mean square inequality.** If  $a_1, a_2, \dots, a_n$  are nonnegative numbers, then

$$\sqrt{\frac{a_1^2 + a_2^2 + \cdots + a_n^2}{n}} \geq \frac{a_1 + a_2 + \cdots + a_n}{n},$$

with equality if and only if  $a_1 = a_2 = \cdots = a_n$ .

**Root of unity.** A solution of the equation  $x^n - 1 = 0$ .

**Surjective function.** A function  $f: A \rightarrow B$  is surjective (onto) if for any  $y \in B$  there exists an  $x \in A$  such that  $y = f(x)$ .

**Weighted arithmetic mean–geometric mean inequality.** For any nonnegative numbers  $a_1, a_2, \dots, a_n$ , if  $w_1, w_2, \dots, w_n$  are nonnegative numbers (weights) with sum 1, then

$$w_1 a_1 + w_2 a_2 + \cdots + w_n a_n \geq a_1^{w_1} a_2^{w_2} \cdots a_n^{w_n},$$

with equality if and only if  $a_1 = a_2 = \cdots = a_n$ .



## About the Authors

**Svetoslav Savchev** was born in Gabrovo, Bulgaria, and graduated from Sofia University in 1980. He has been working for the mathematical journal *Matematika* since 1981, and is currently its vice editor-in-chief. A number of books and articles were authored or co-authored by him, including a nontraditional geometry textbook in Bulgarian and a book of a similar flavor in Spanish. Engaged in diverse activities for gifted students, he was a jury member at national and regional mathematics contests in Bulgaria and Latin America for several years, and is an active problem proposer. He also served on the Problem Selection Committees at the International Mathematical Olympiads in Argentina (1997), Taiwan (1998) and Republic of Korea (2000).

Svetoslav is known for his contacts with mathematically inclined students in his country and abroad. Many young minds were influenced by the spirit of his seminars.

**Titu Andreescu** is the Director of the American Mathematics Competitions, a program of the Mathematical Association of America (MAA). He also serves as Chair of the USA Mathematical Olympiad Committee, Head Coach of the USA International Mathematical Olympiad Team, and Director of the Mathematical Olympiad Summer Program.

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